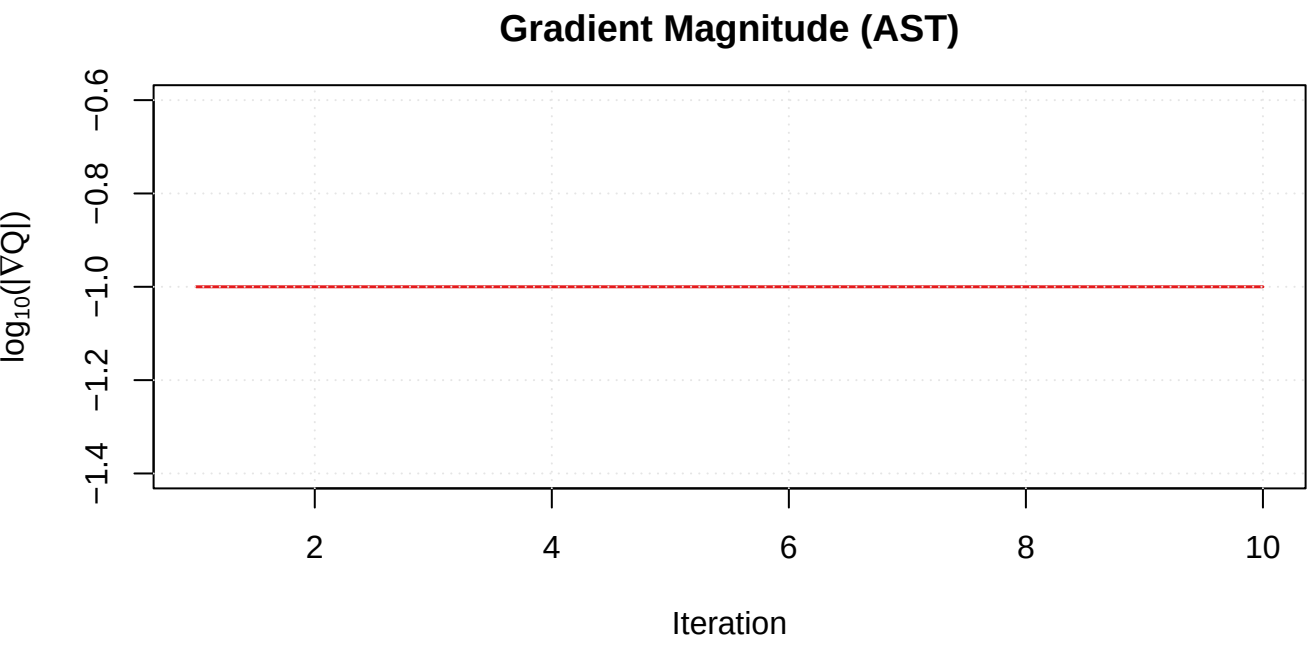
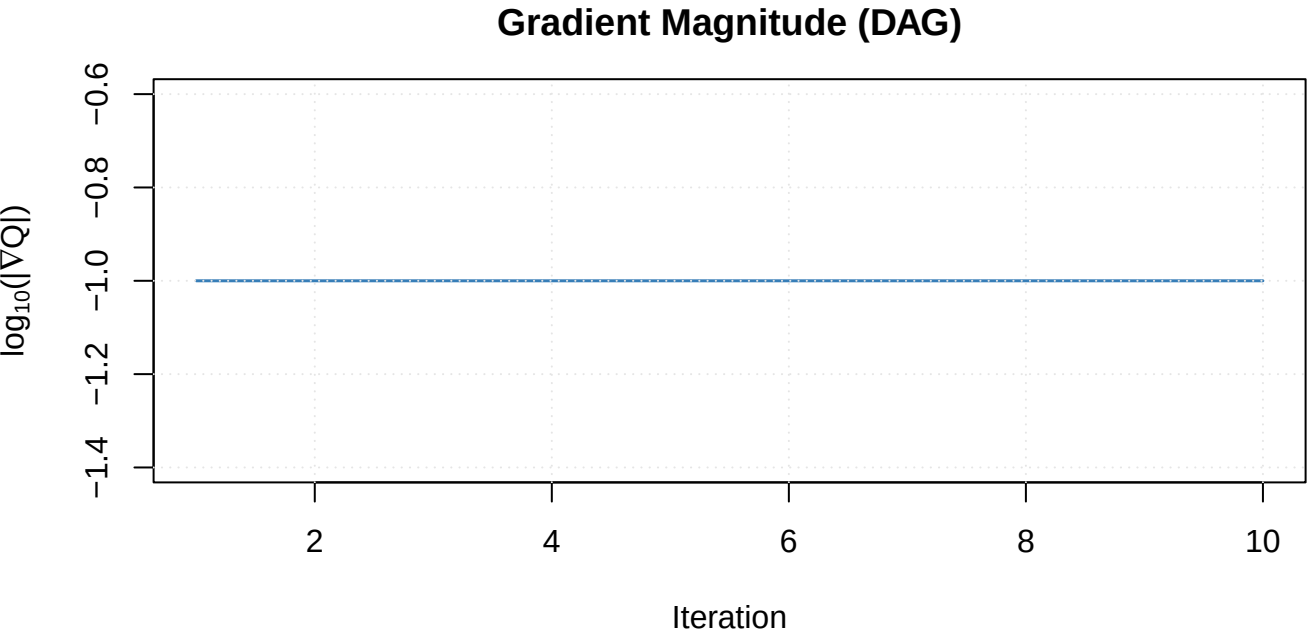
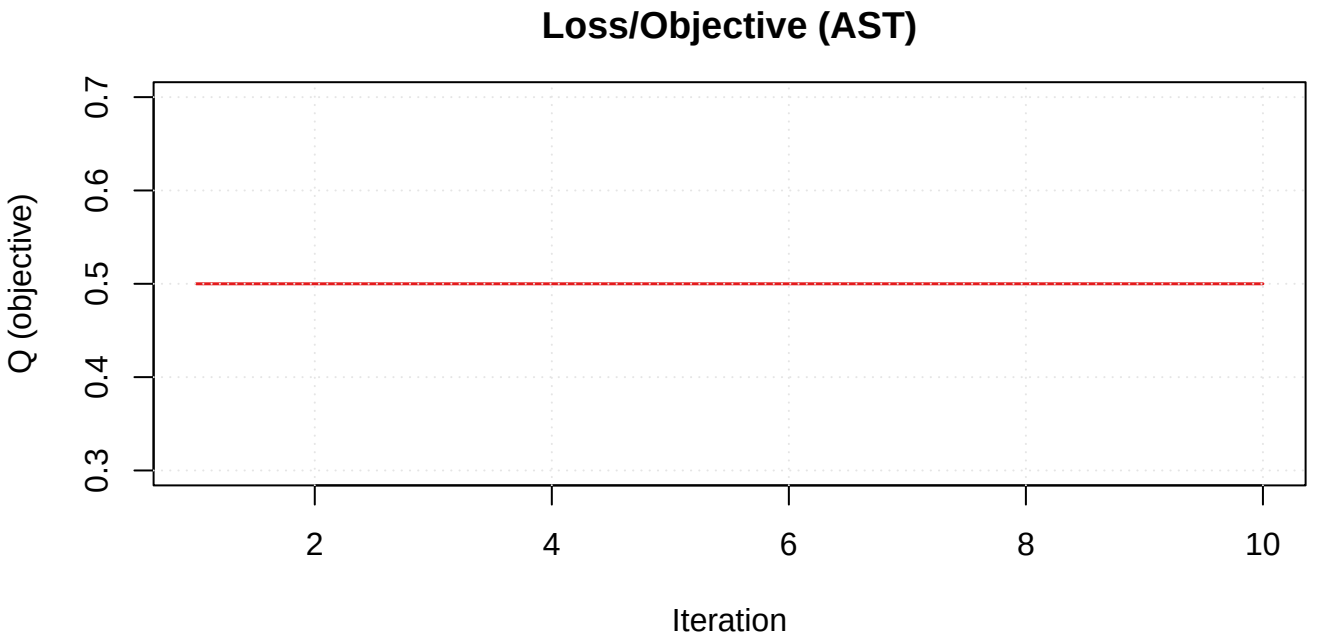
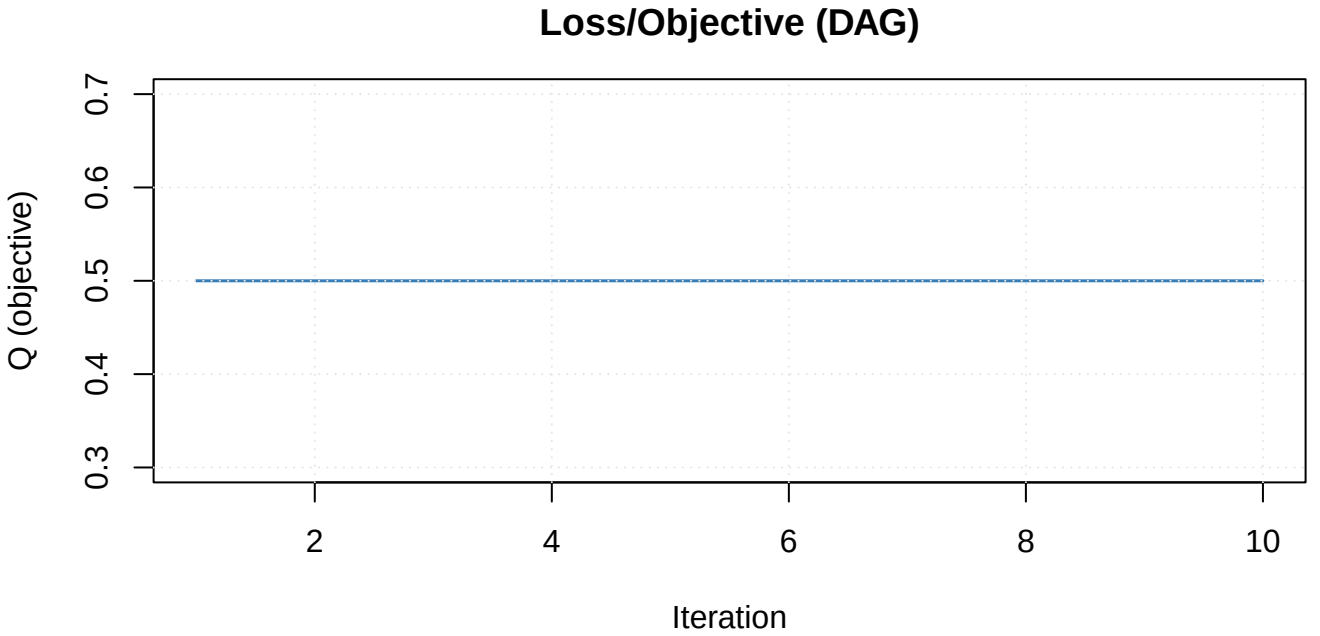
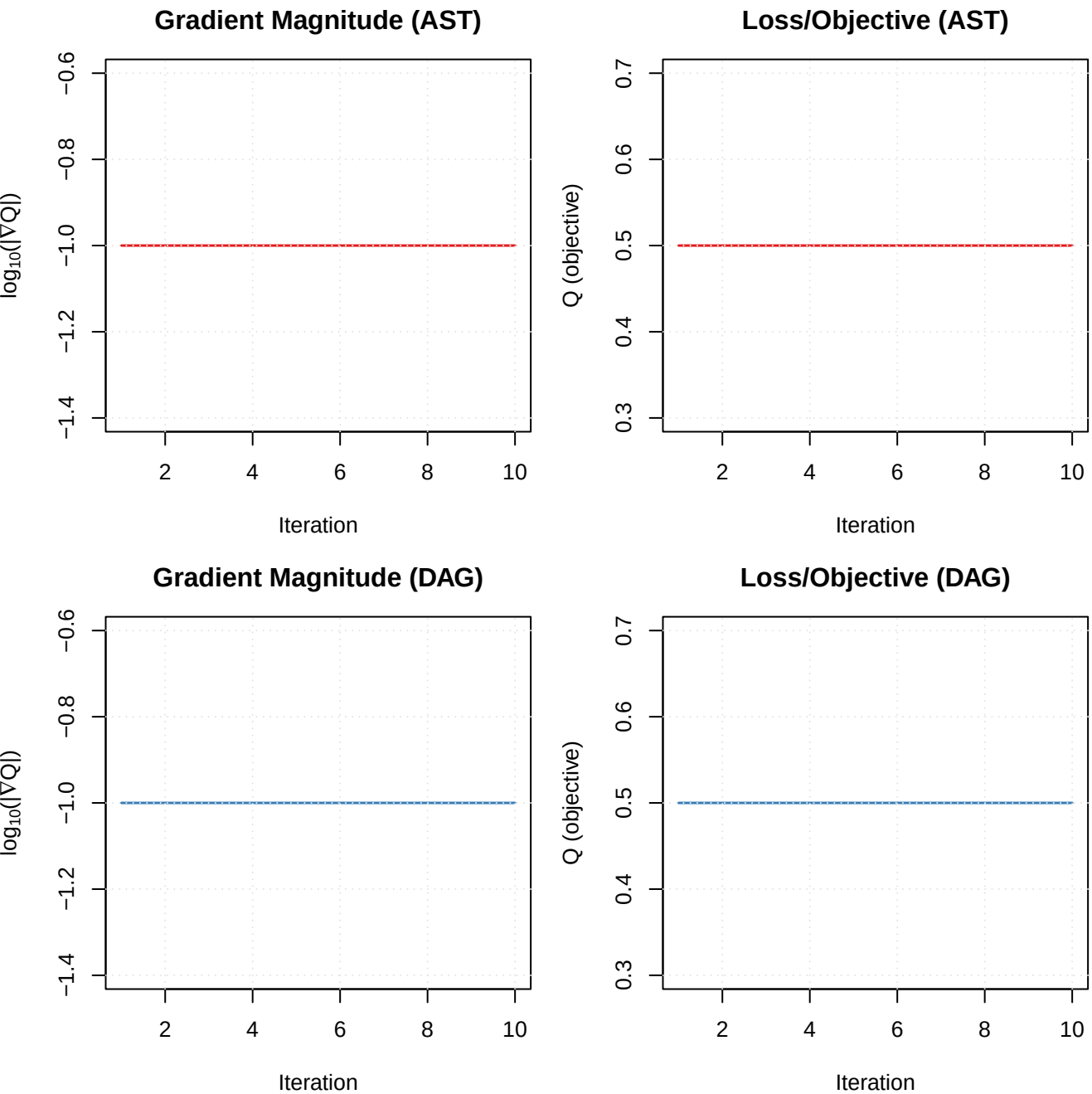
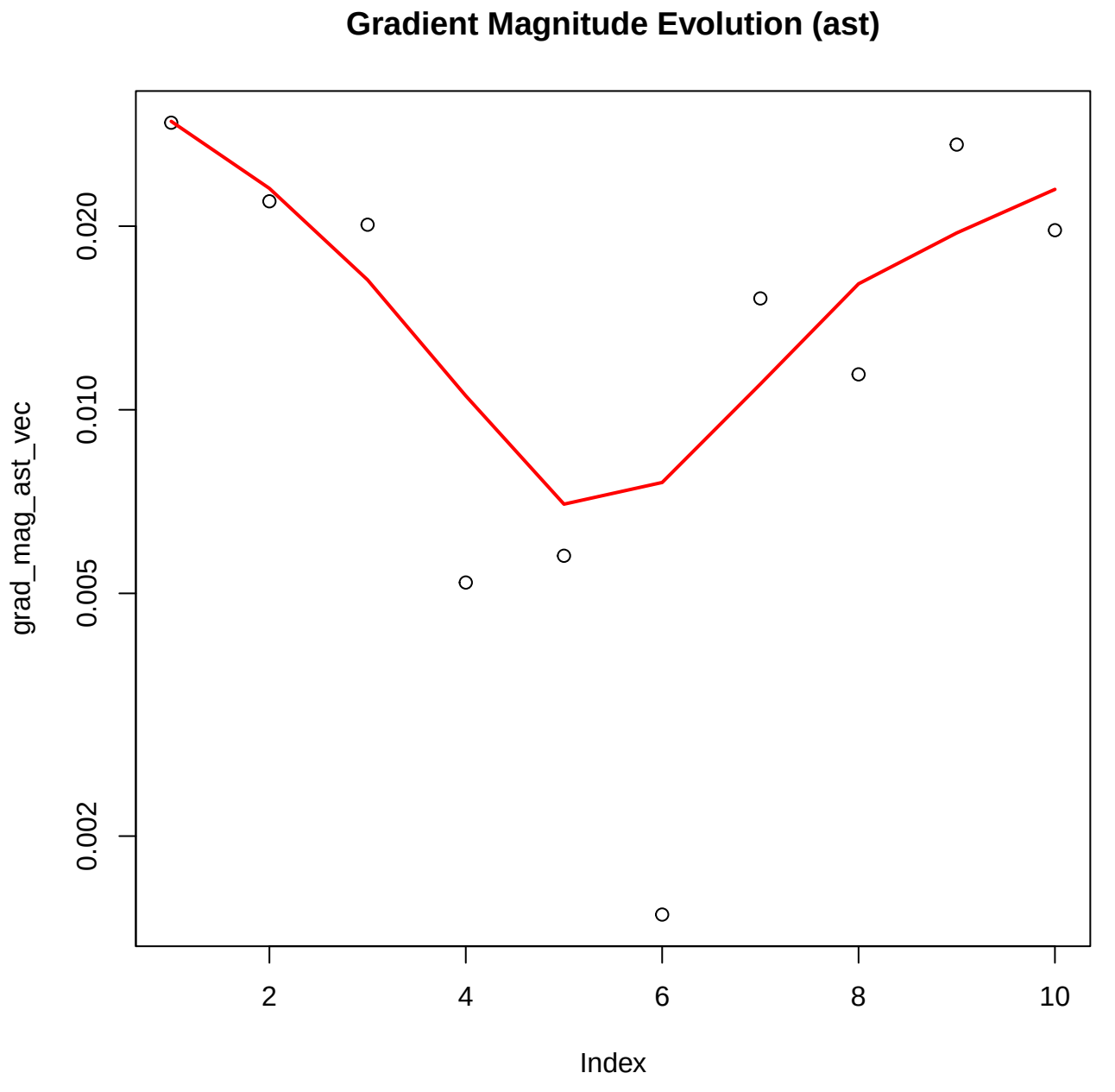

Gradient Magnitude (AST)
log10(|∇Q|)
−0.6
−0.8
−1.0
−1.2
−1.4
2
4
6
8
10
Iteration

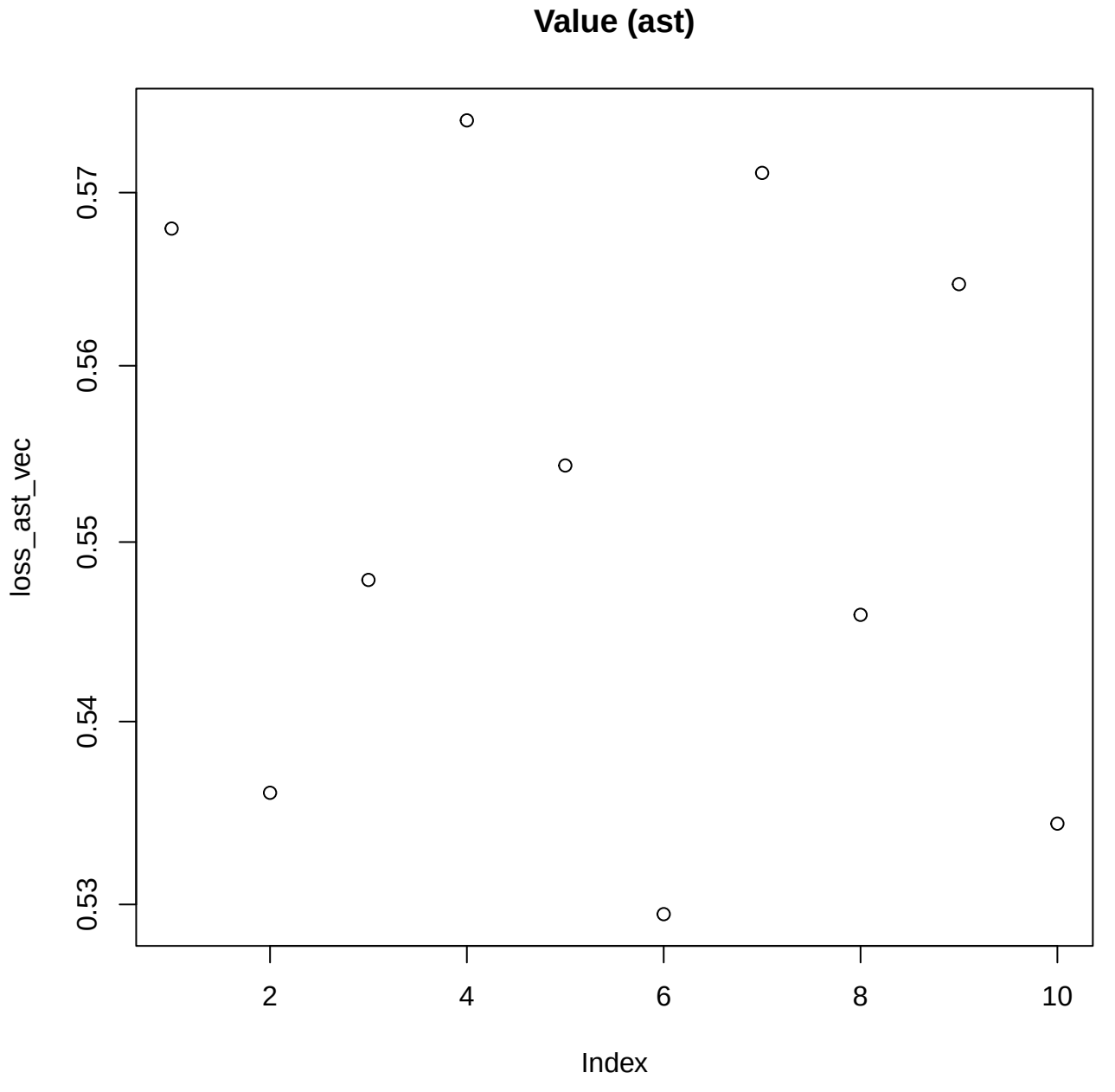

Gradient Magnitude (DAG)
log10(|∇Q|)
−0.6
−0.8
−1.0
−1.2
−1.4
2
4
6
8
10
Iteration

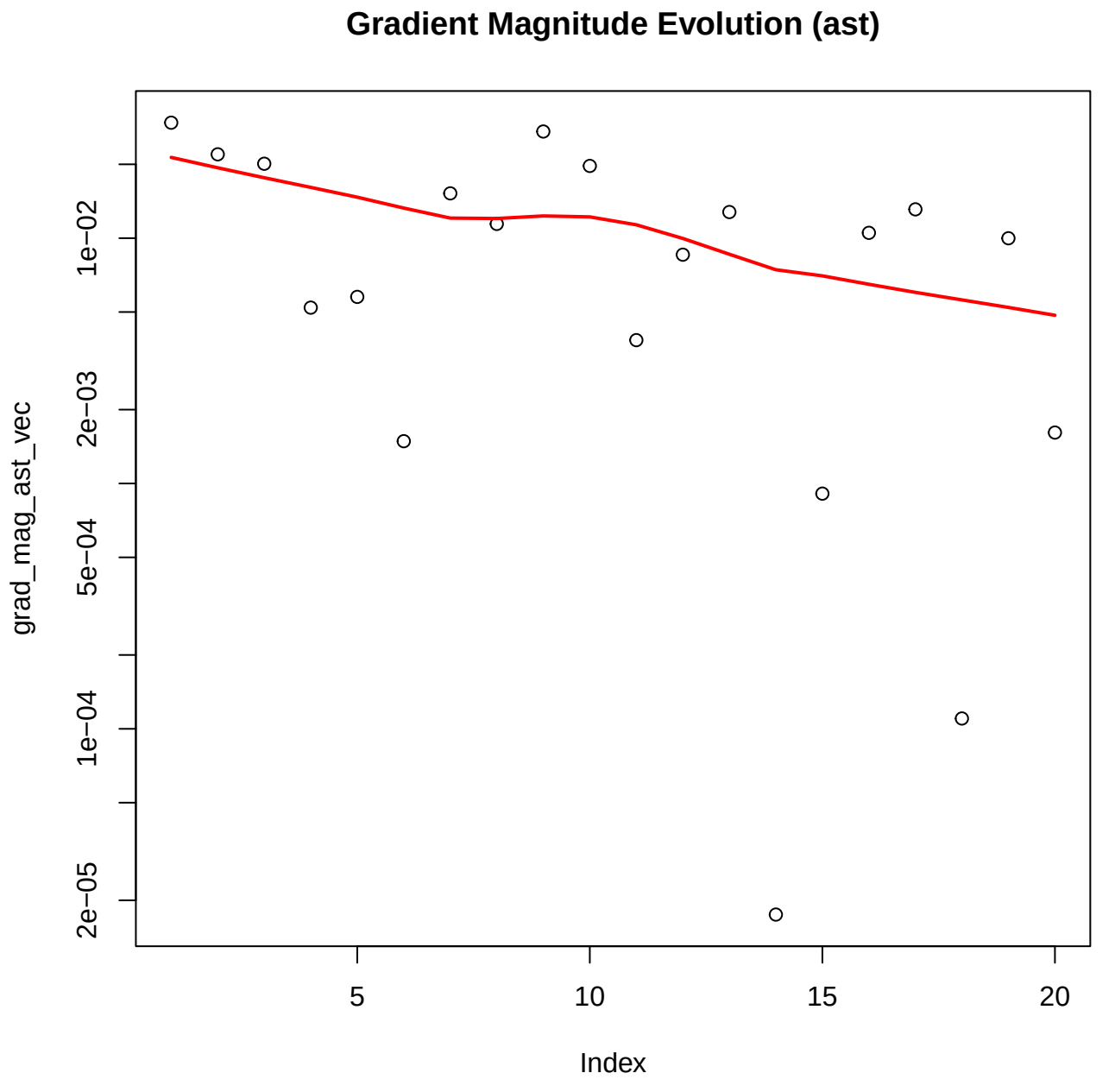

Loss/Objective (AST)
Q (objective)
0.3
0.4
0.5
0.6
0.7
2
4
6
8
10
Iteration

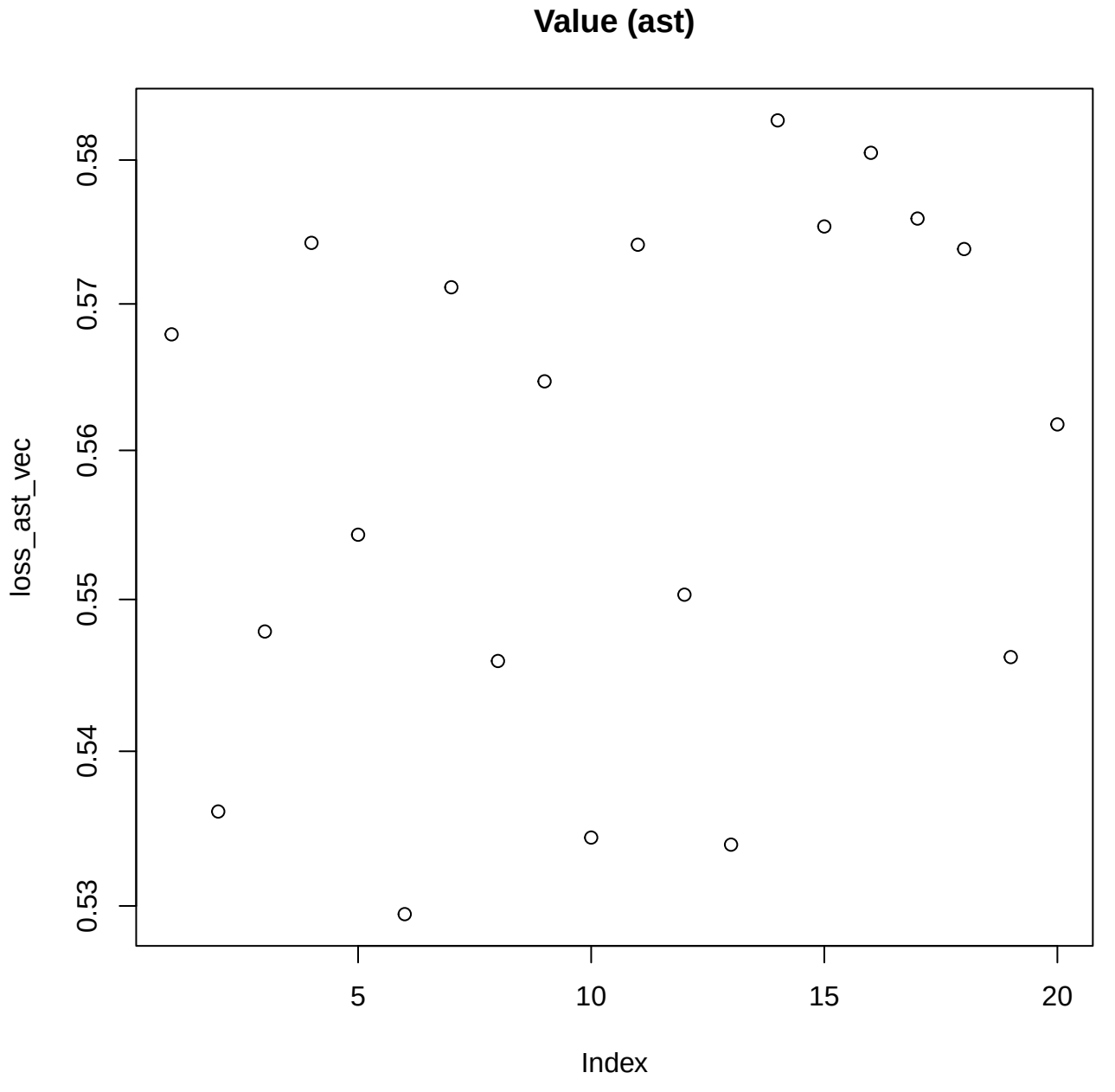

Loss/Objective (DAG)
Q (objective)
0.3
0.4
0.5
0.6
0.7
2
4
6
8
10
Iteration

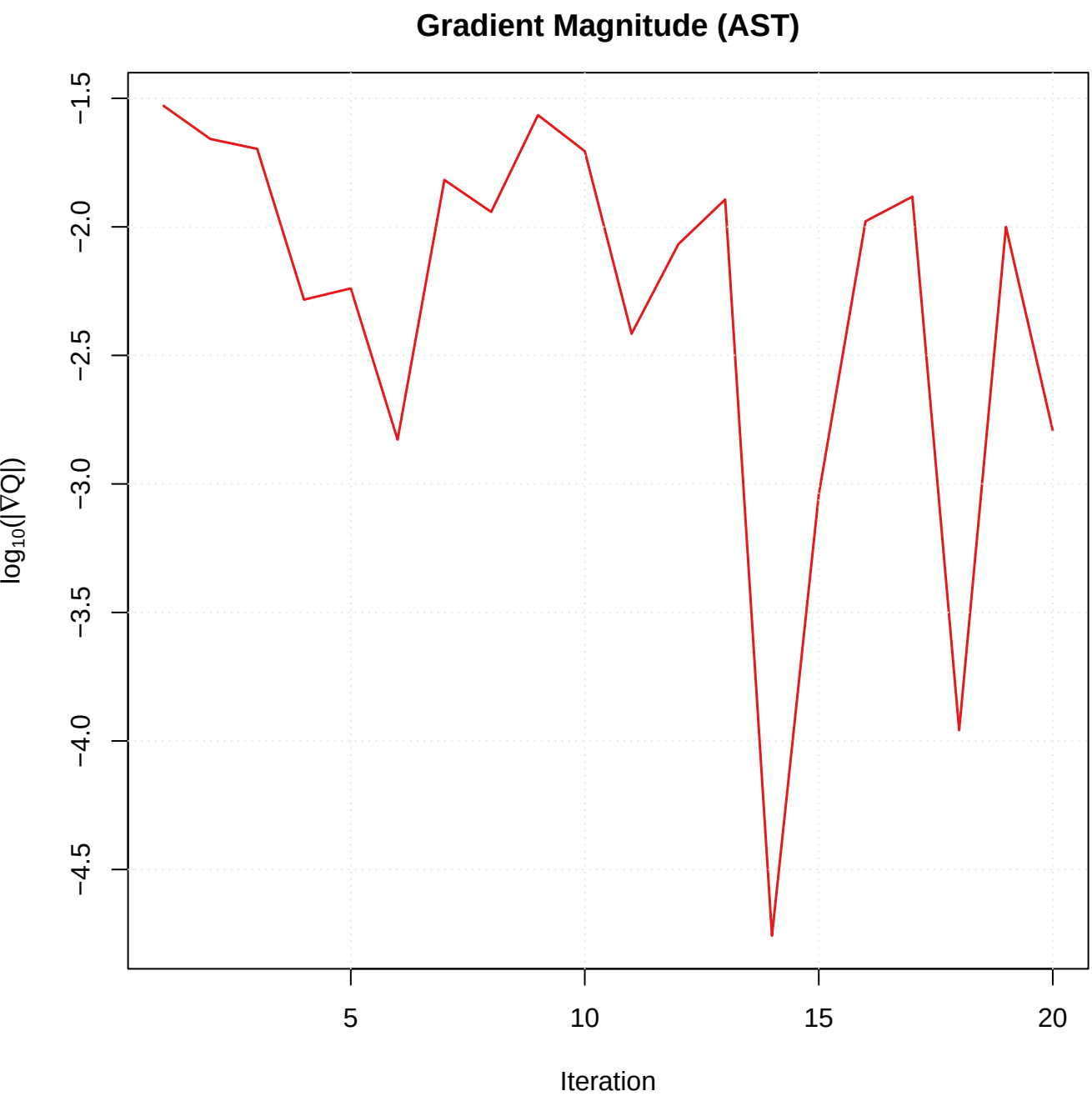

Gradient Magnitude (AST)
log10(|∇Q|)
−0.6
−0.8
−1.0
−1.2
−1.4
2
4
6
8
10
Iteration
Loss/Objective (AST)
Q (objective)
0.7
0.6
0.5
0.4
0.3
2
4
6
8
10
Iteration
Gradient Magnitude (DAG)
log10(|∇Q|)
−0.6
−0.8
−1.0
−1.2
−1.4
2
4
6
8
10
Iteration
Loss/Objective (DAG)
Q (objective)
0.7
0.6
0.5
0.4
0.3
2
4
6
8
10
Iteration

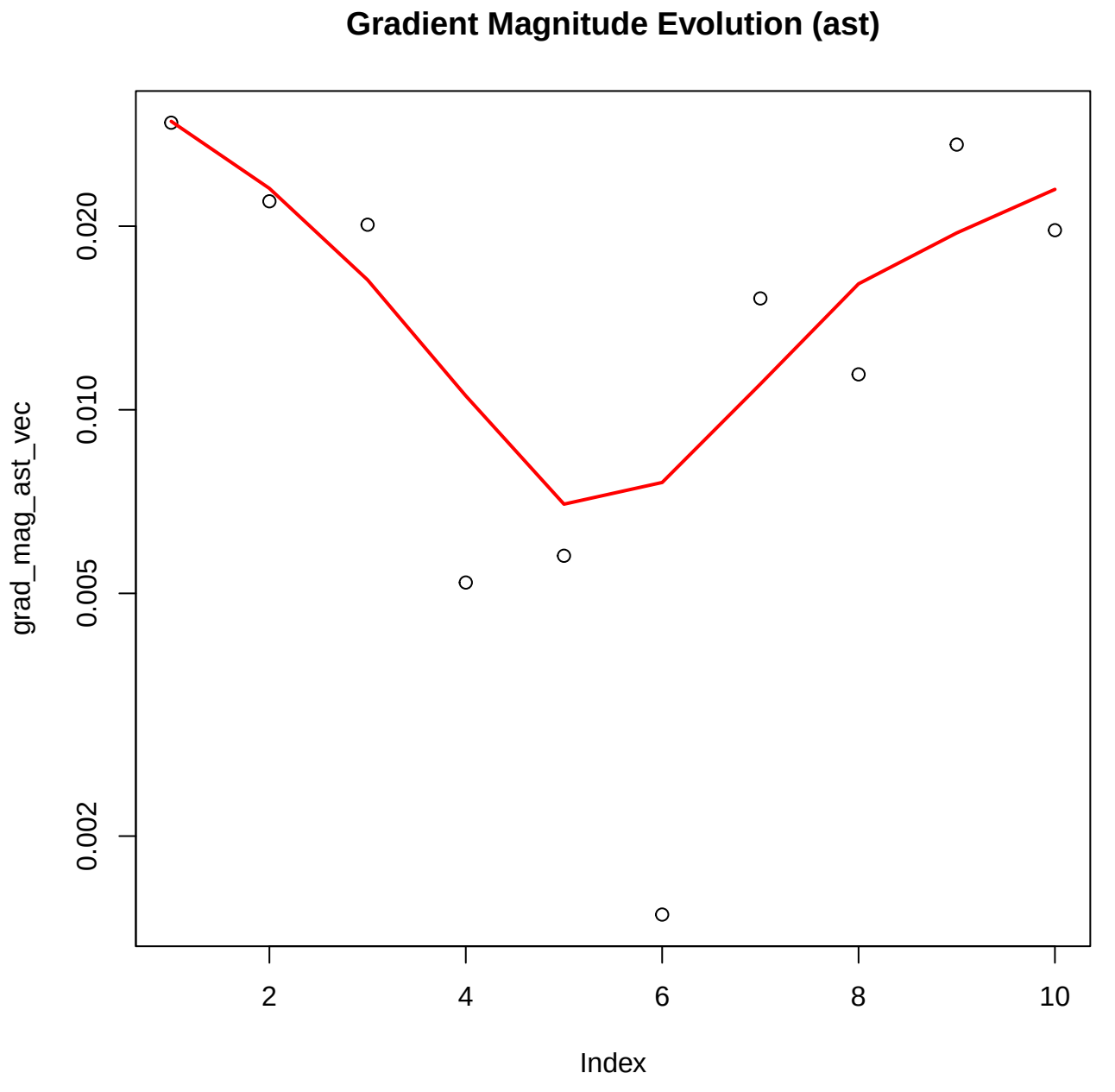

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.002
0.005
0.010
0.020
2
4
6
8
10
Index

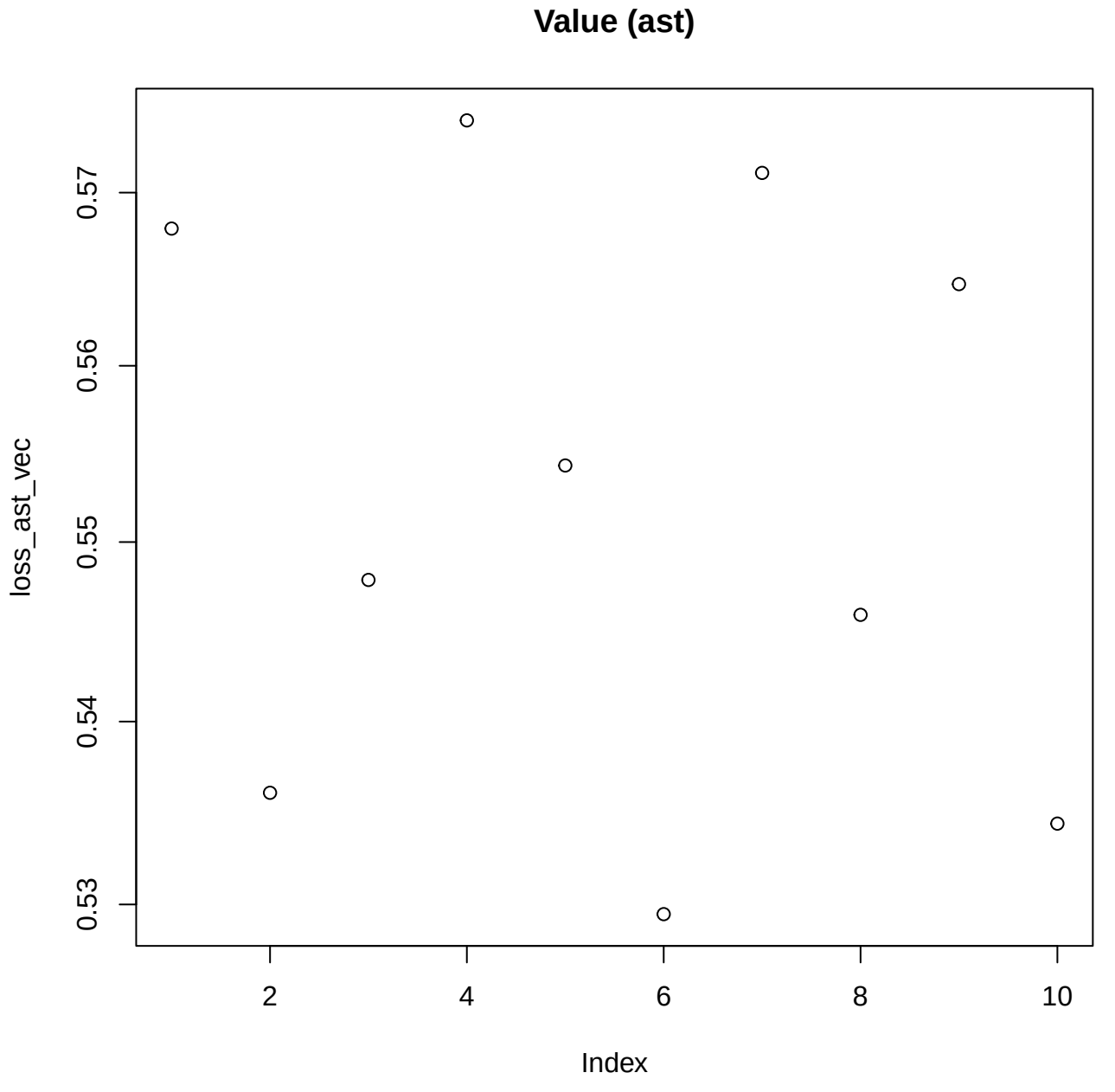

Value (ast)
loss_ast_vec
0.53
0.54
0.55
0.56
0.57
2
4
6
8
10
Index

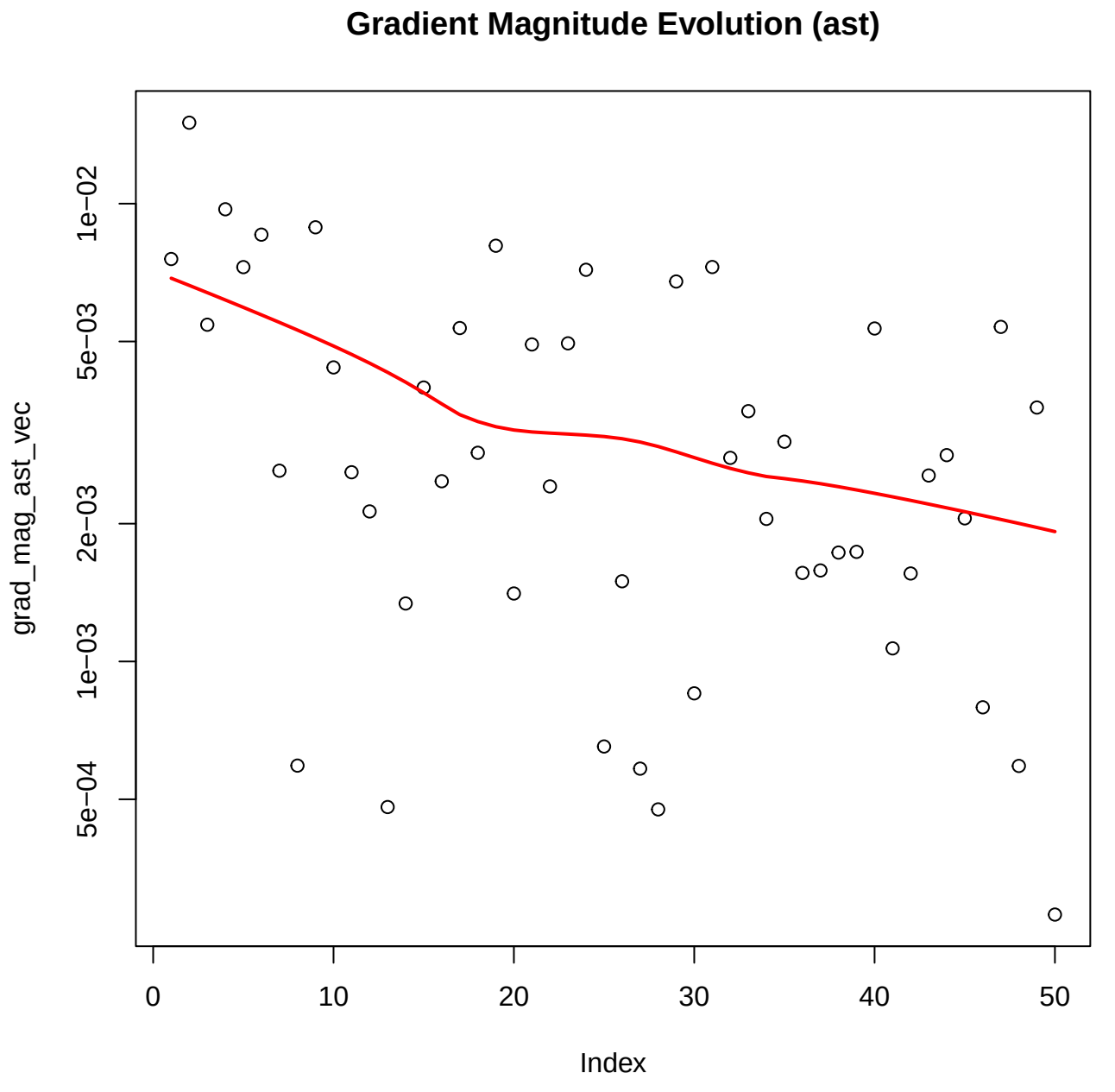

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e−02
2e−03
5e−04
1e−04
2e−05
5
10
15
20
Index

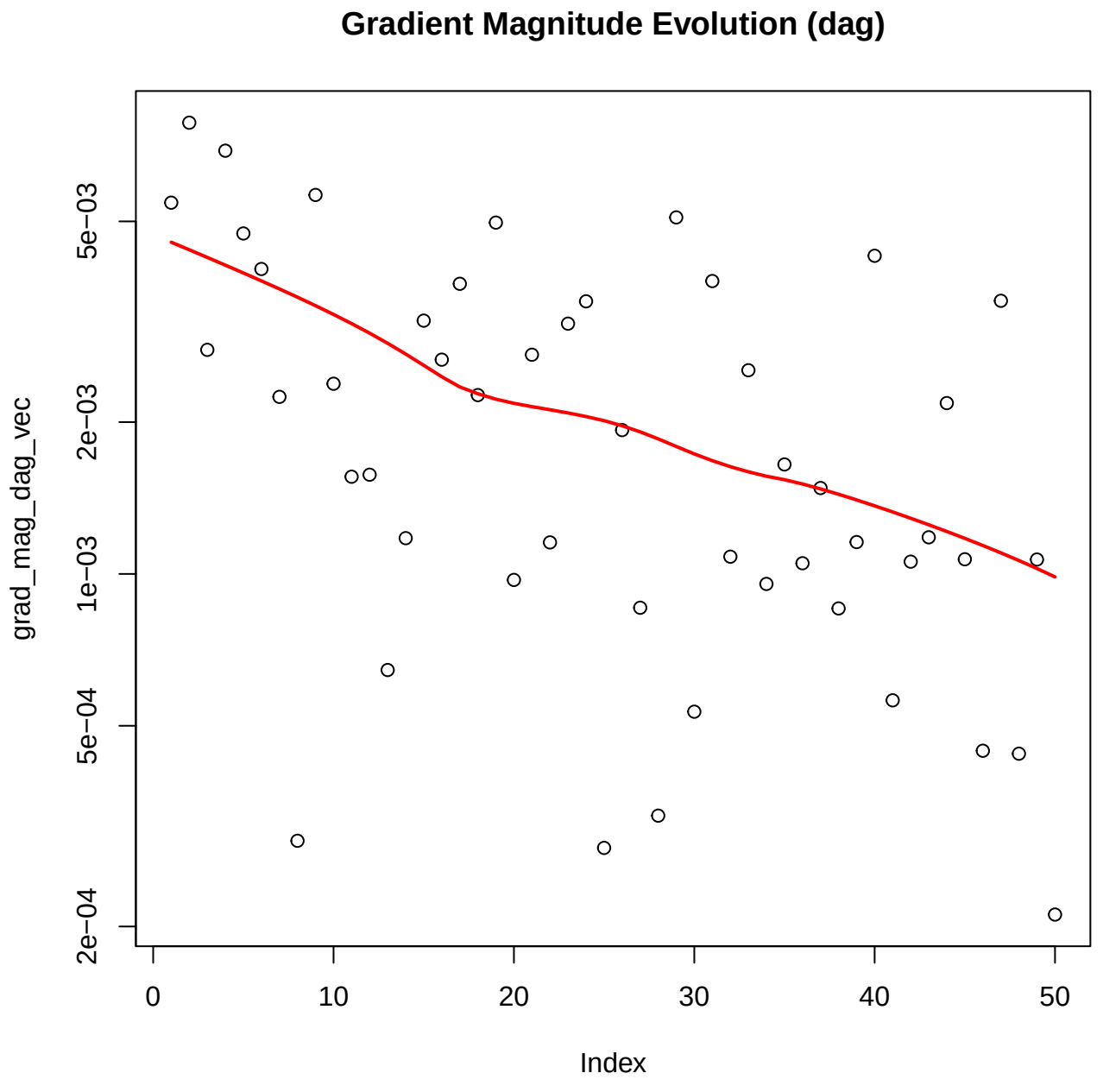

Value (ast)
loss_ast_vec
0.53
0.54
0.55
0.56
0.57
0.58
5
10
15
20
Index

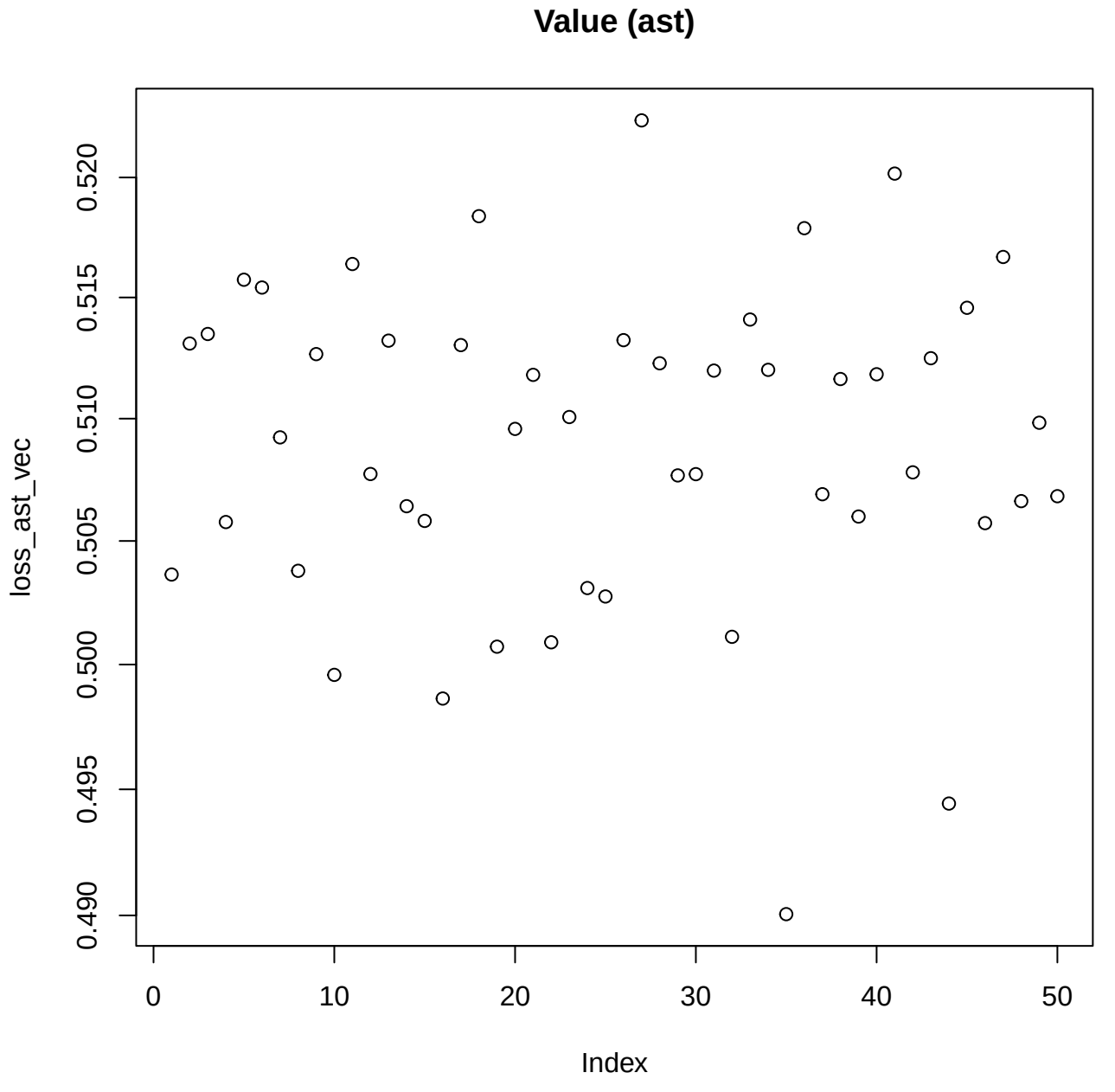

Gradient Magnitude (AST)
−1.5
−2.0
−2.5
−3.0
−3.5
−4.0
−4.5
log10(|∇Q|)
5
10
15
20
Iteration

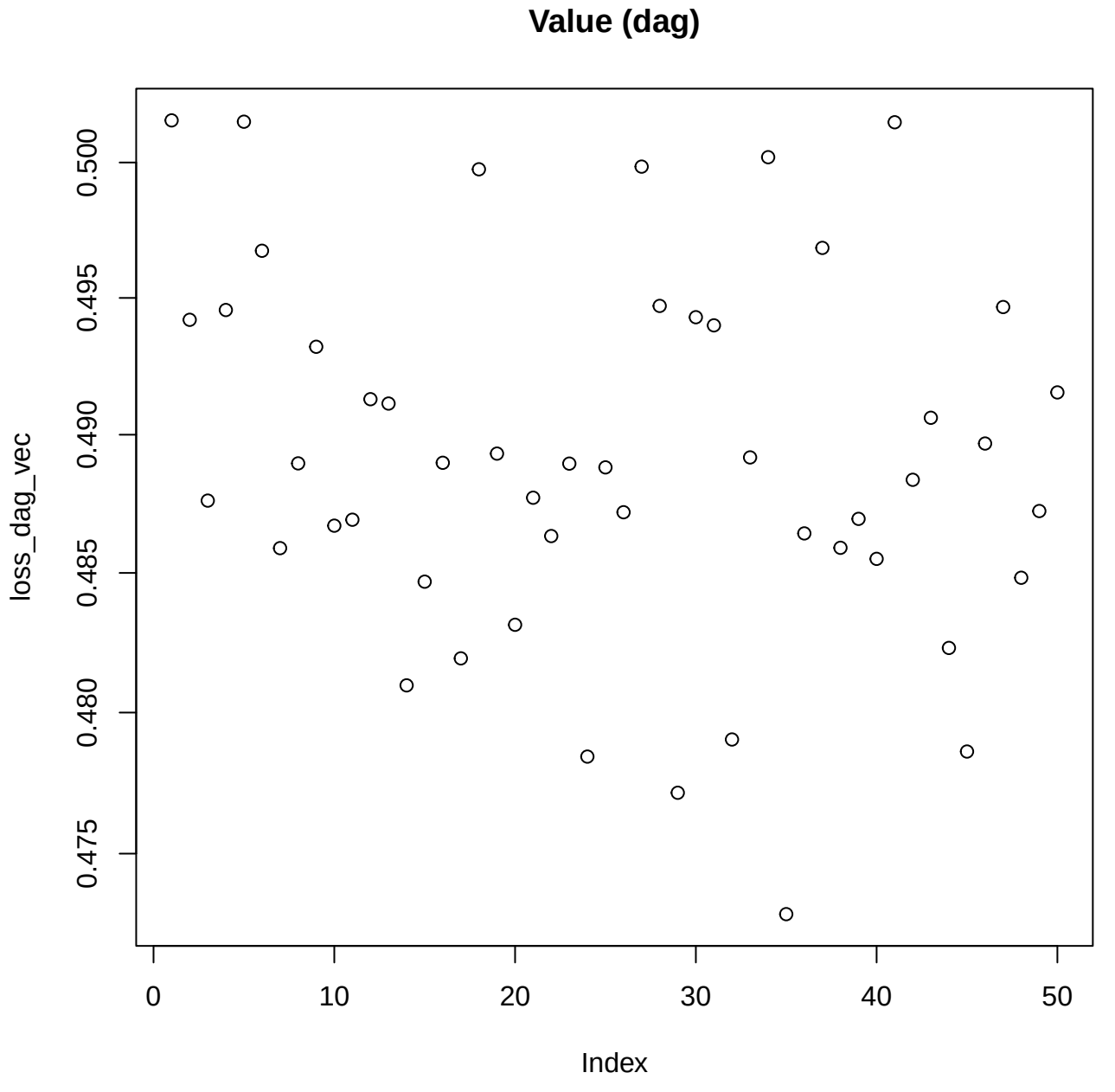

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.002
0.005
0.010
0.020
2
4
6
8
10
Index

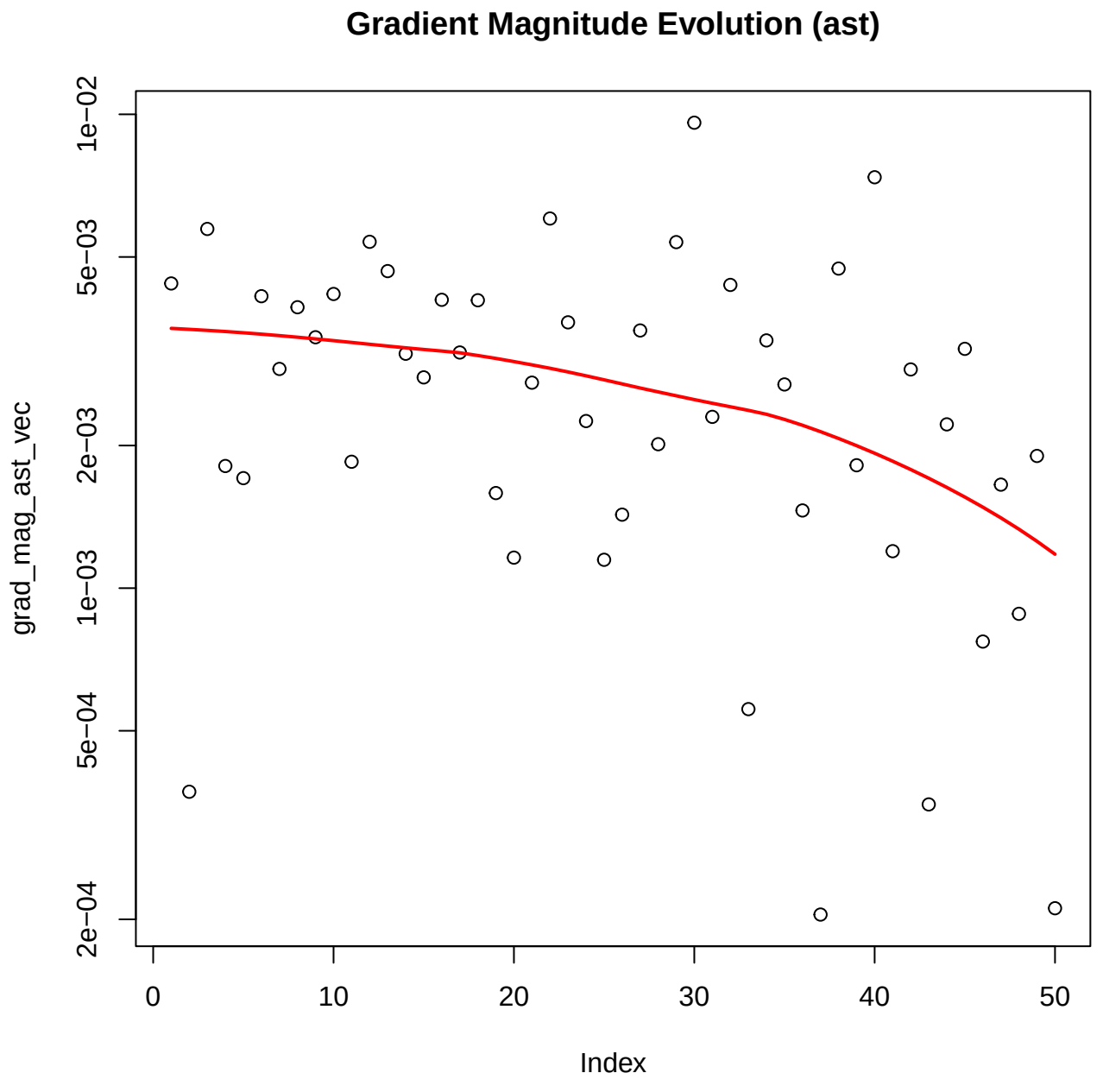

Value (ast)
loss_ast_vec
0.53
0.54
0.55
0.56
0.57
2
4
6
8
10
Index

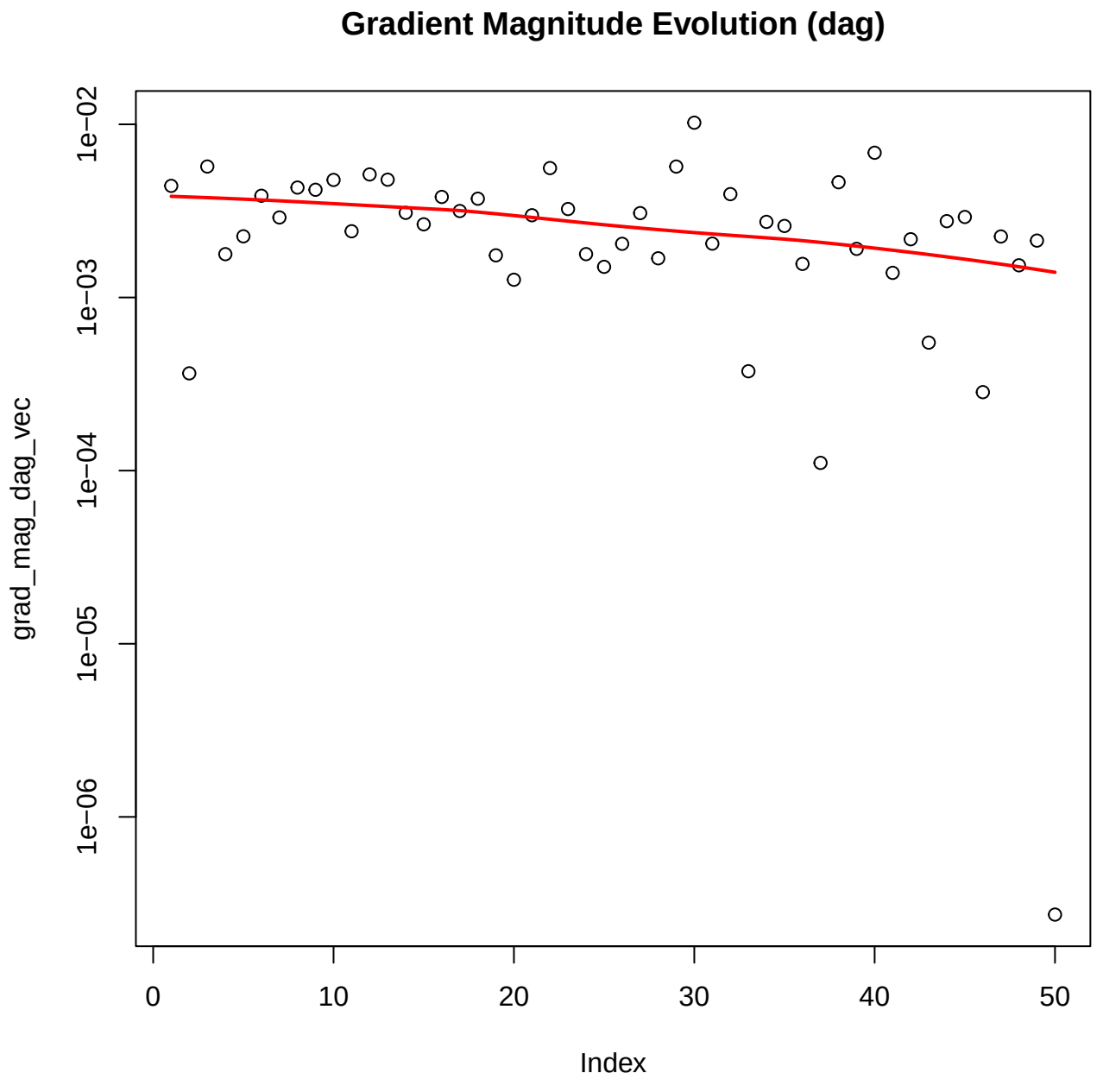

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e−02
5e−03
2e−03
1e−03
5e−04
0
10
20
30
40
50
Index

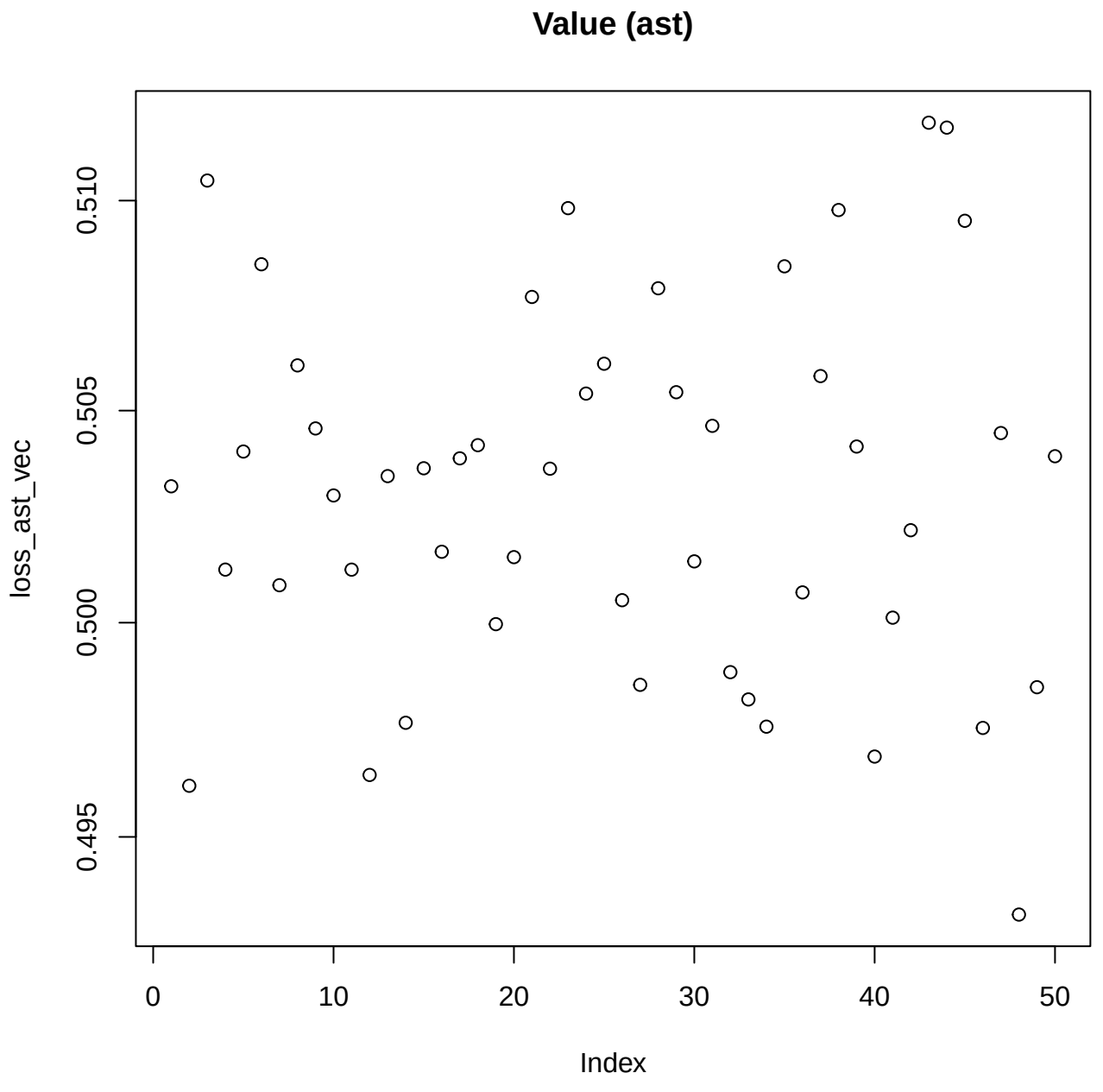

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
5e−03
2e−03
1e−03
5e−04
2e−04
0
10
20
30
40
50
Index

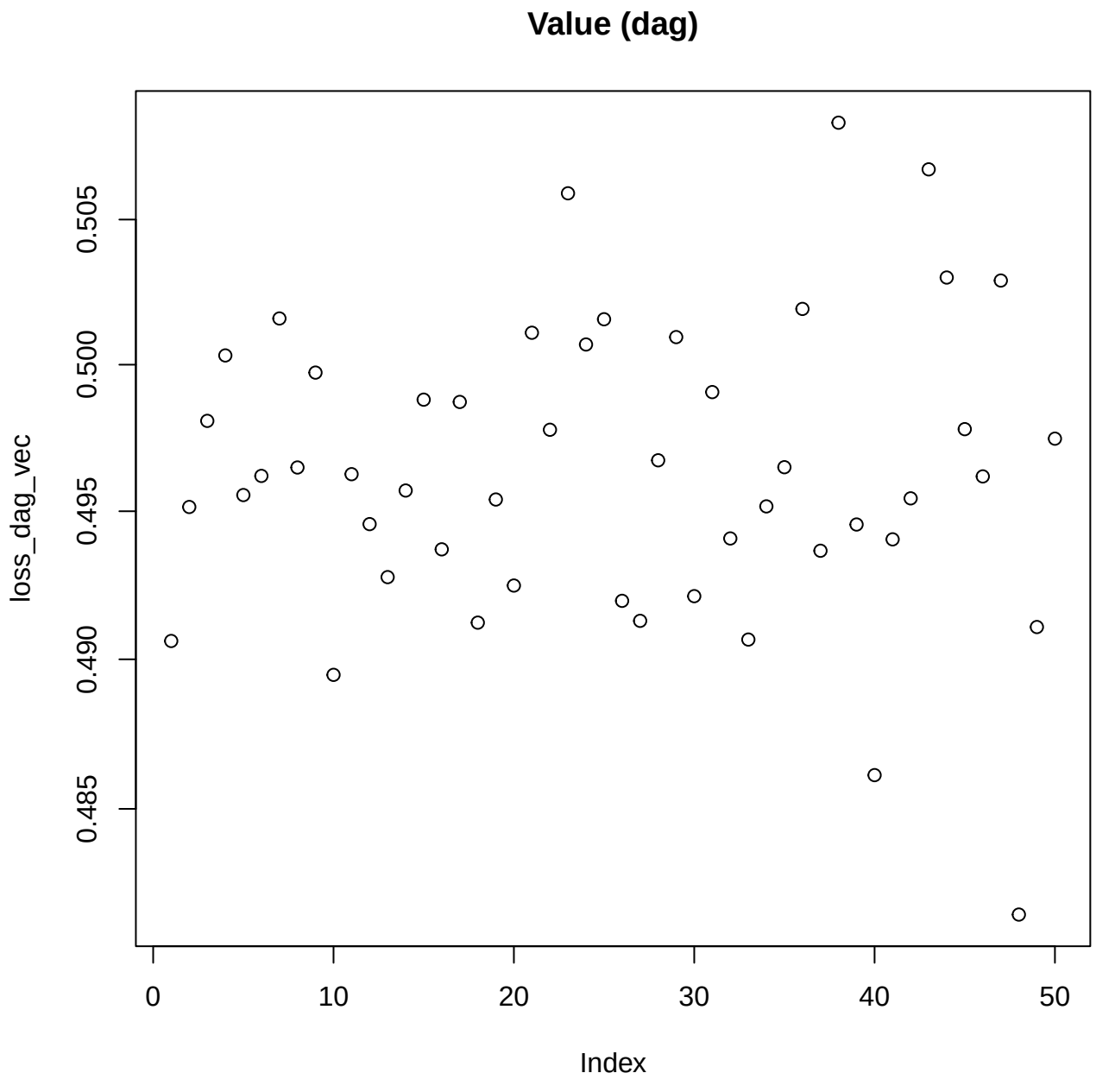

Value (ast)
loss_ast_vec
0.490
0.495
0.500
0.505
0.510
0.515
0.520
0
10
20
30
40
50
Index

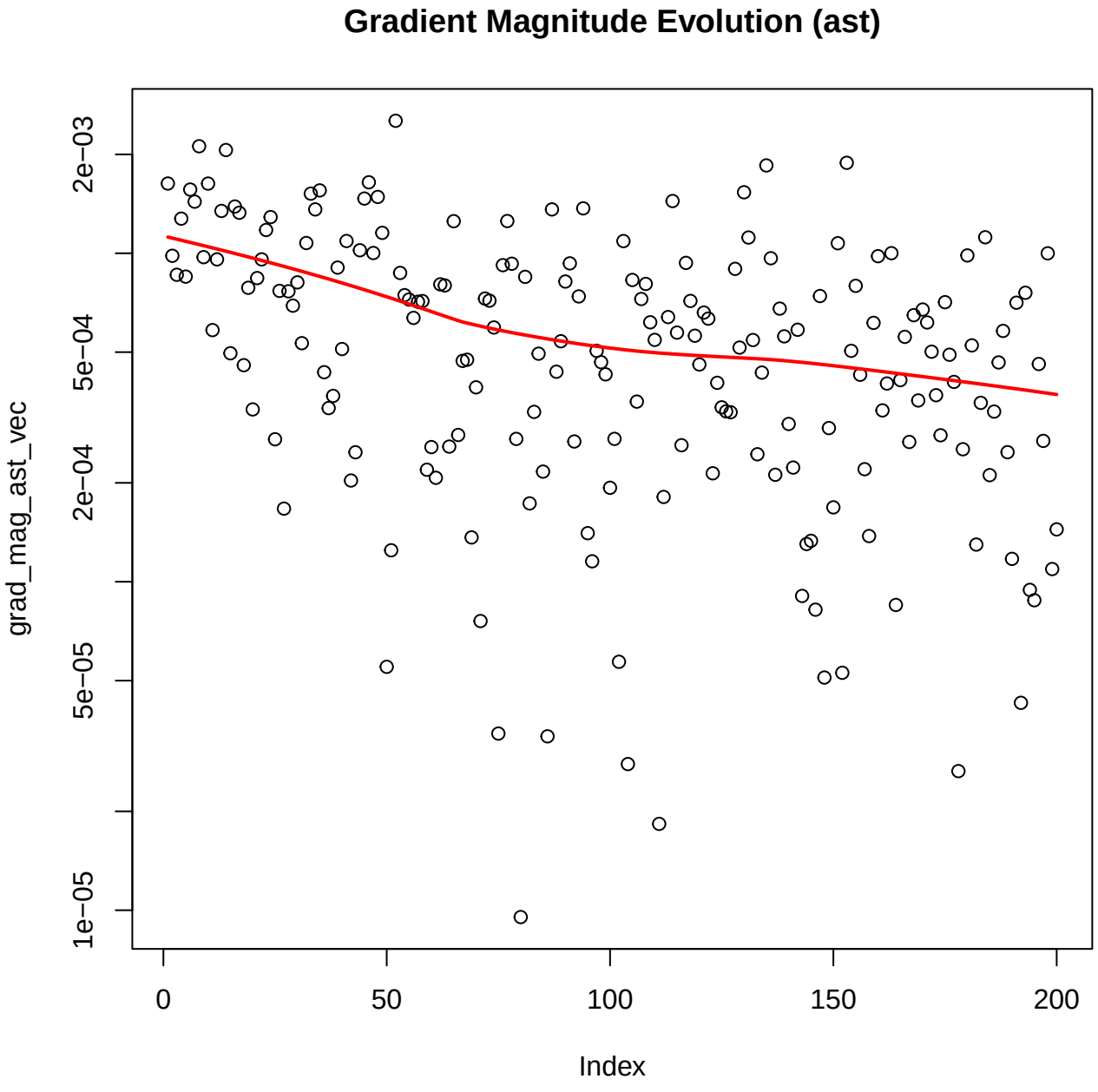

Value (dag)
loss_dag_vec
0.475
0.480
0.485
0.490
0.495
0.500
0
10
20
30
40
50
Index

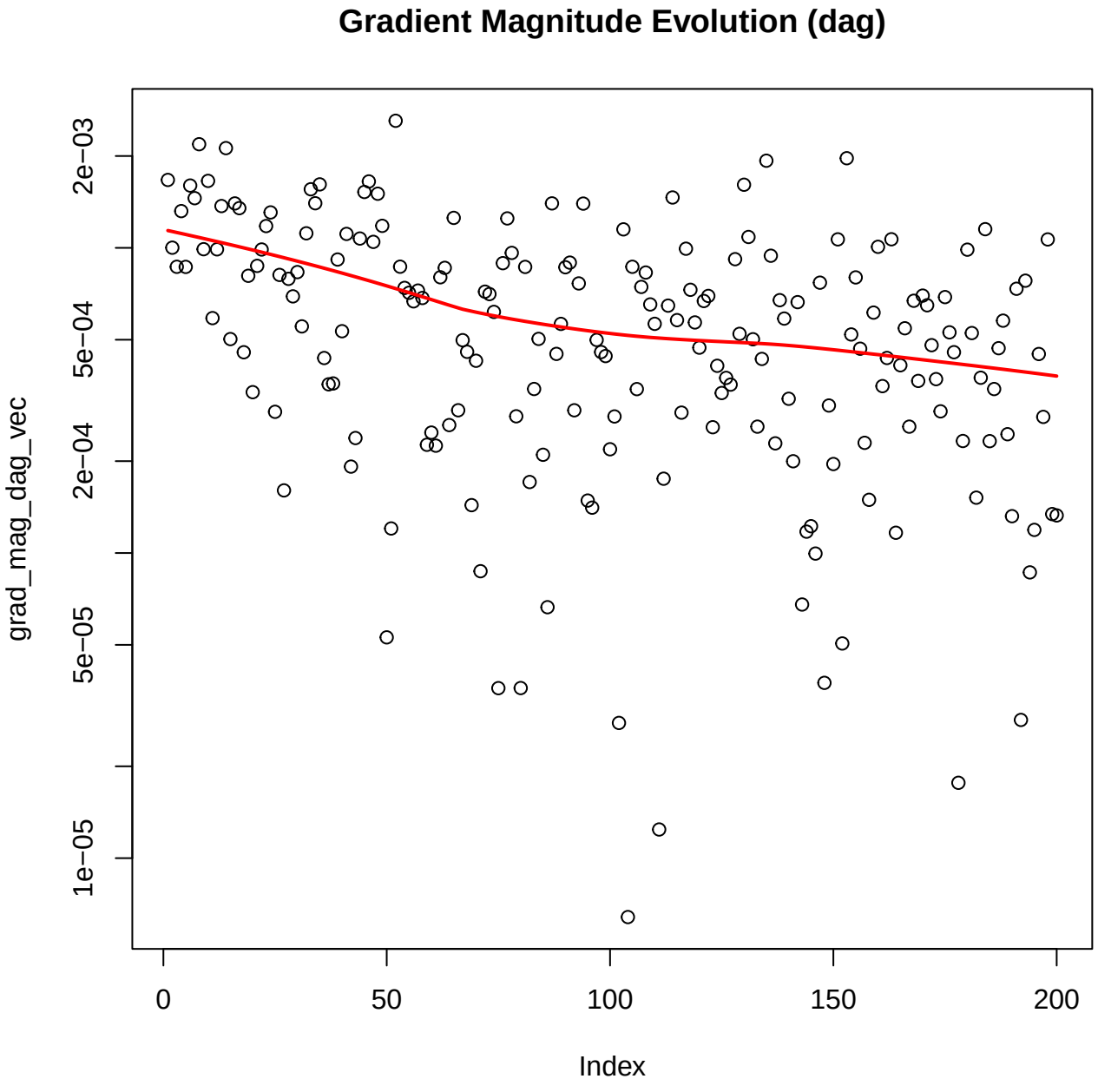

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e−02
5e−03
2e−03
1e−03
5e−04
2e−04
0
10
20
30
40
50
Index

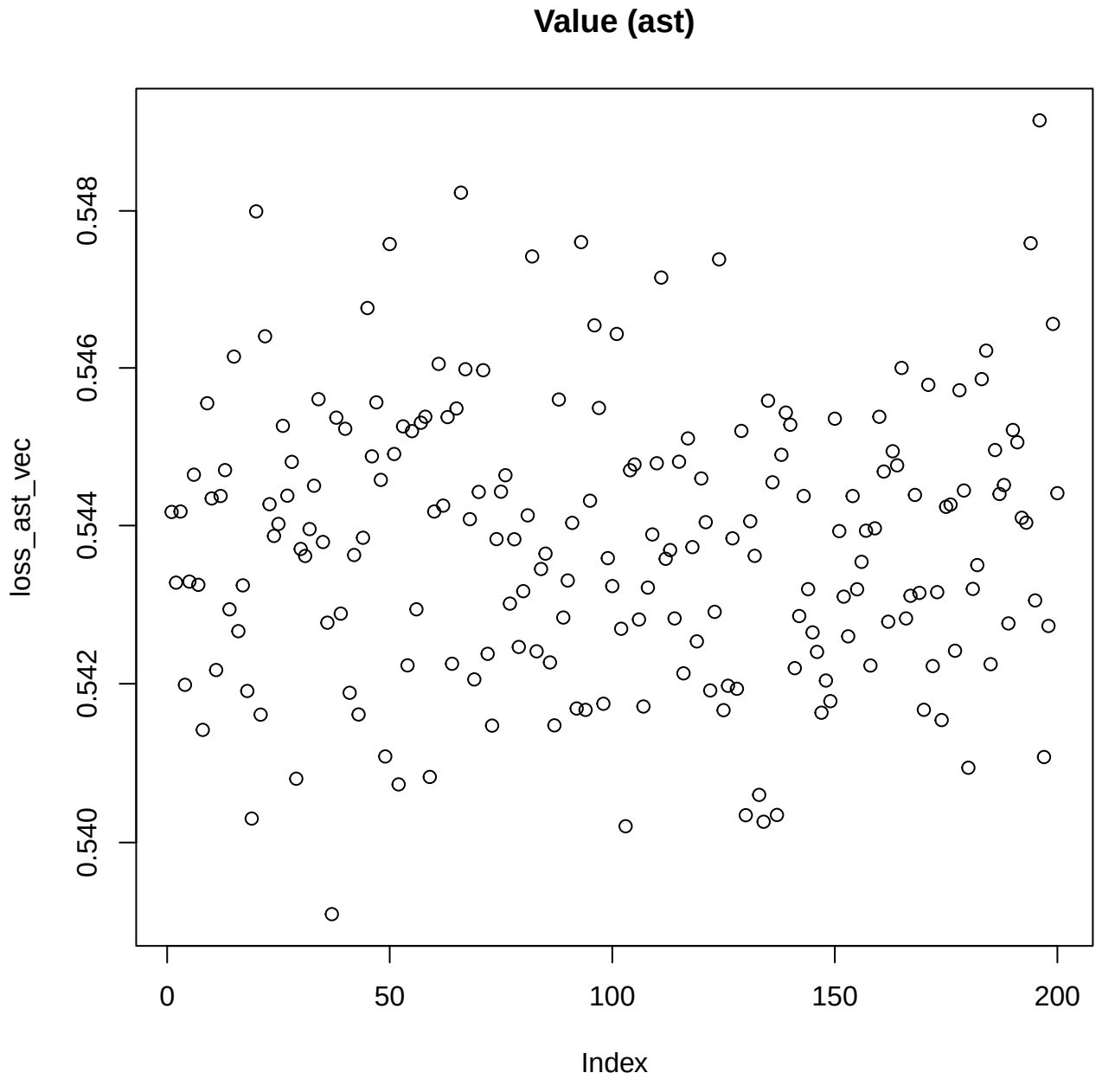

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
1e−02
1e−03
1e−04
1e−05
1e−06
0
10
20
30
40
50
Index

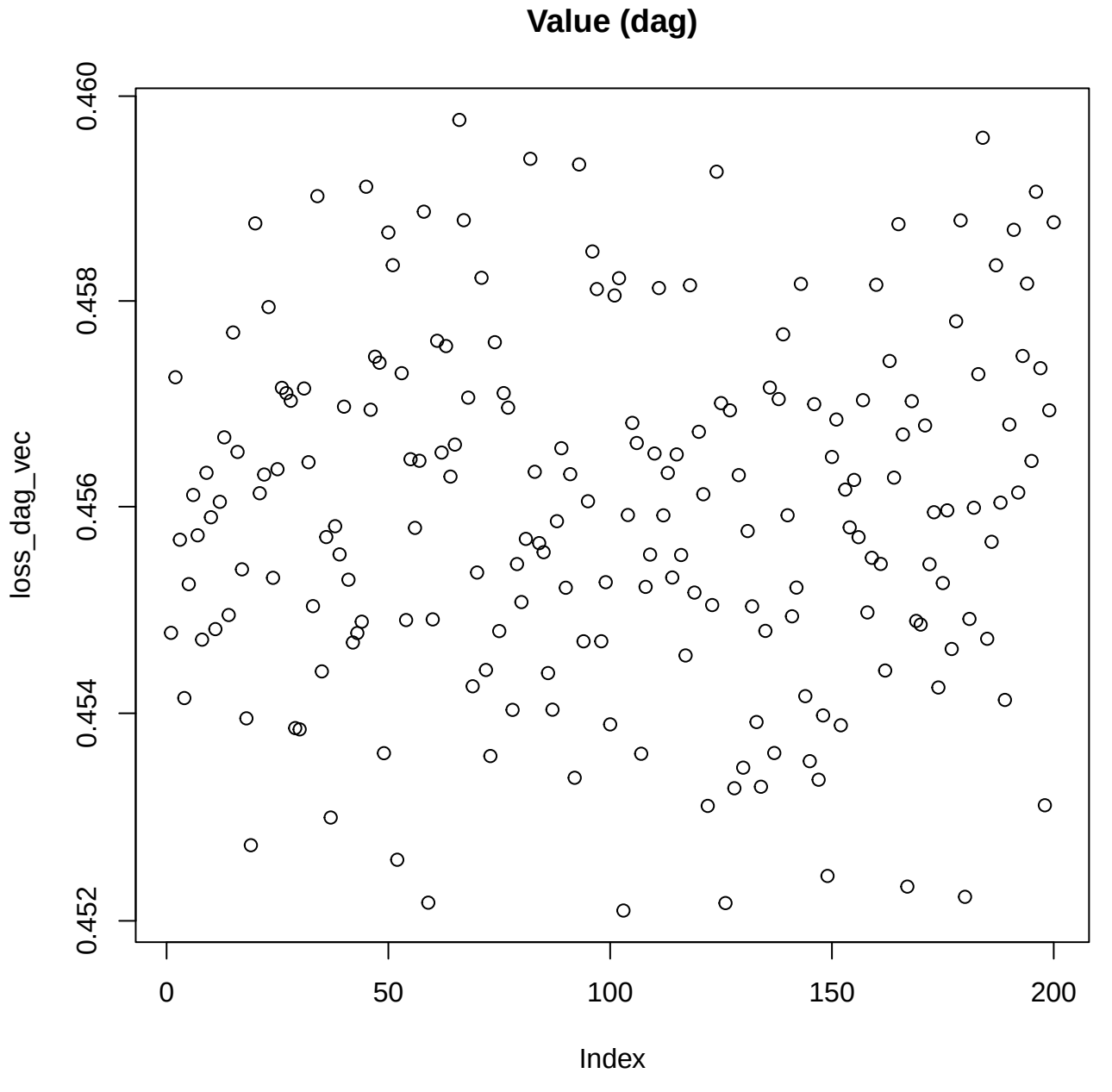

Value (ast)
loss_ast_vec
0.495
0.500
0.505
0.510
0
10
20
30
40
50
Index

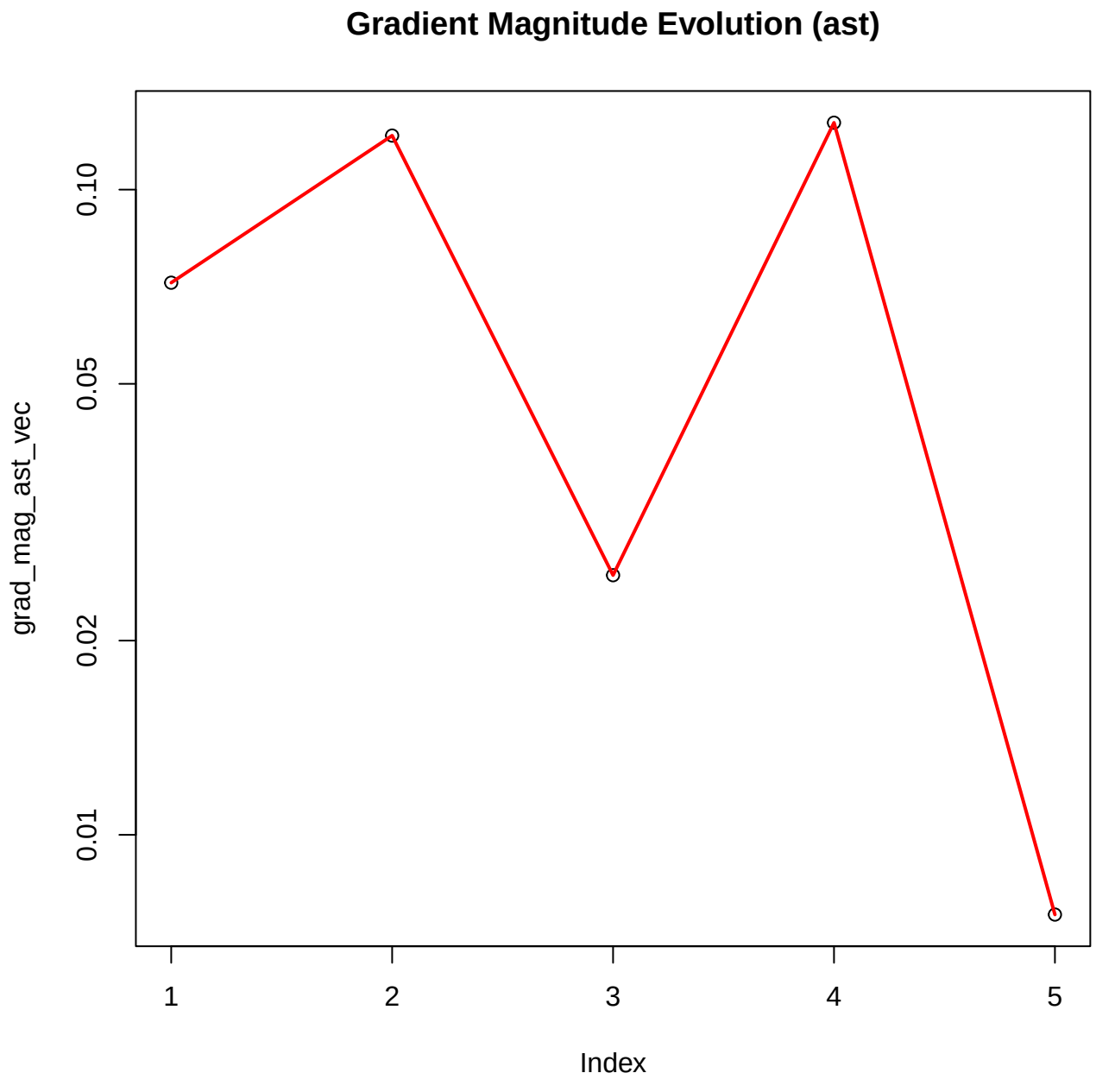

Value (dag)
loss_dag_vec
0.485
0.490
0.495
0.500
0.505
0
10
20
30
40
50
Index

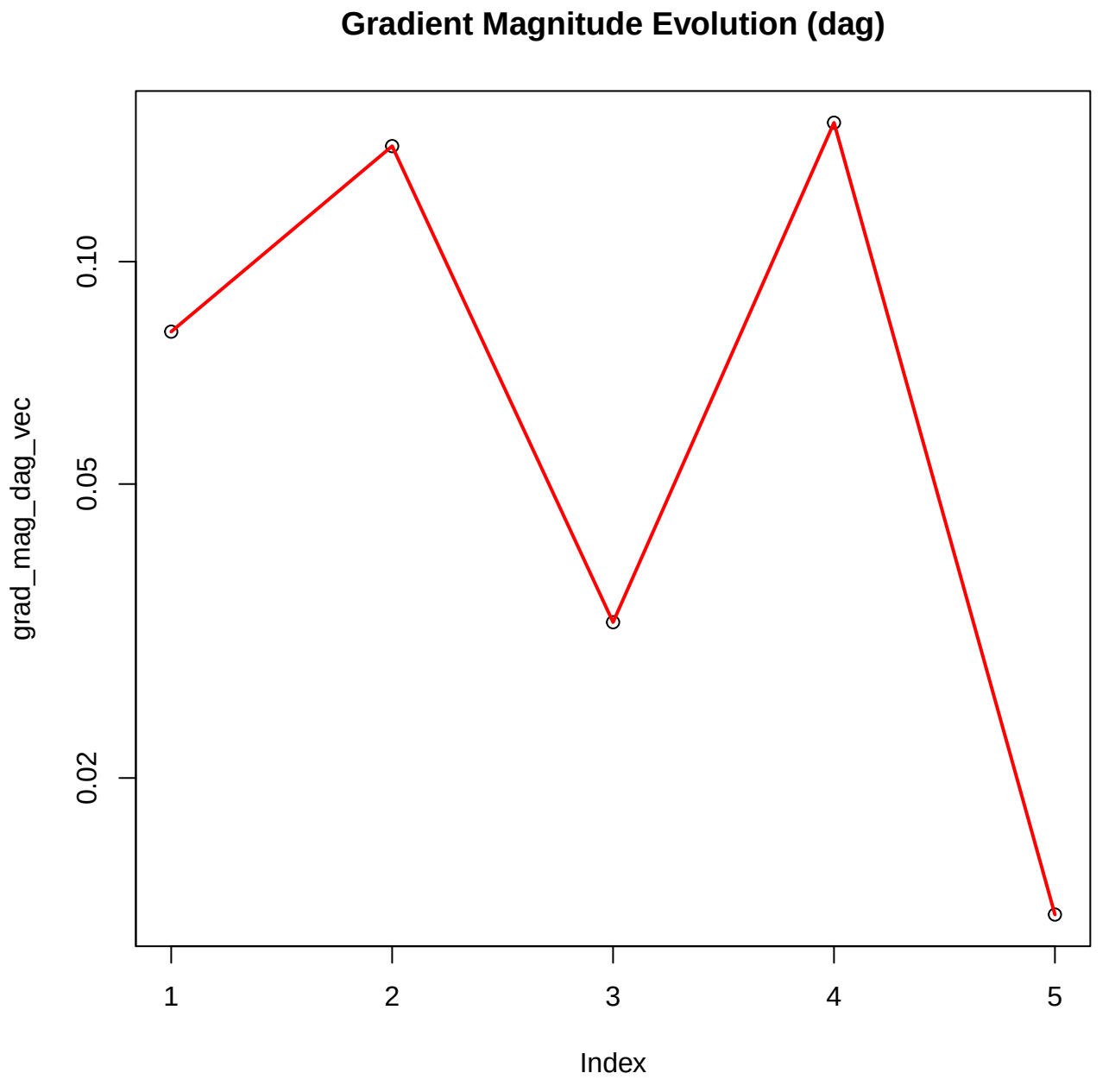

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
2e−03
5e−04
2e−04
5e−05
1e−05
0
50
100
150
200
Index

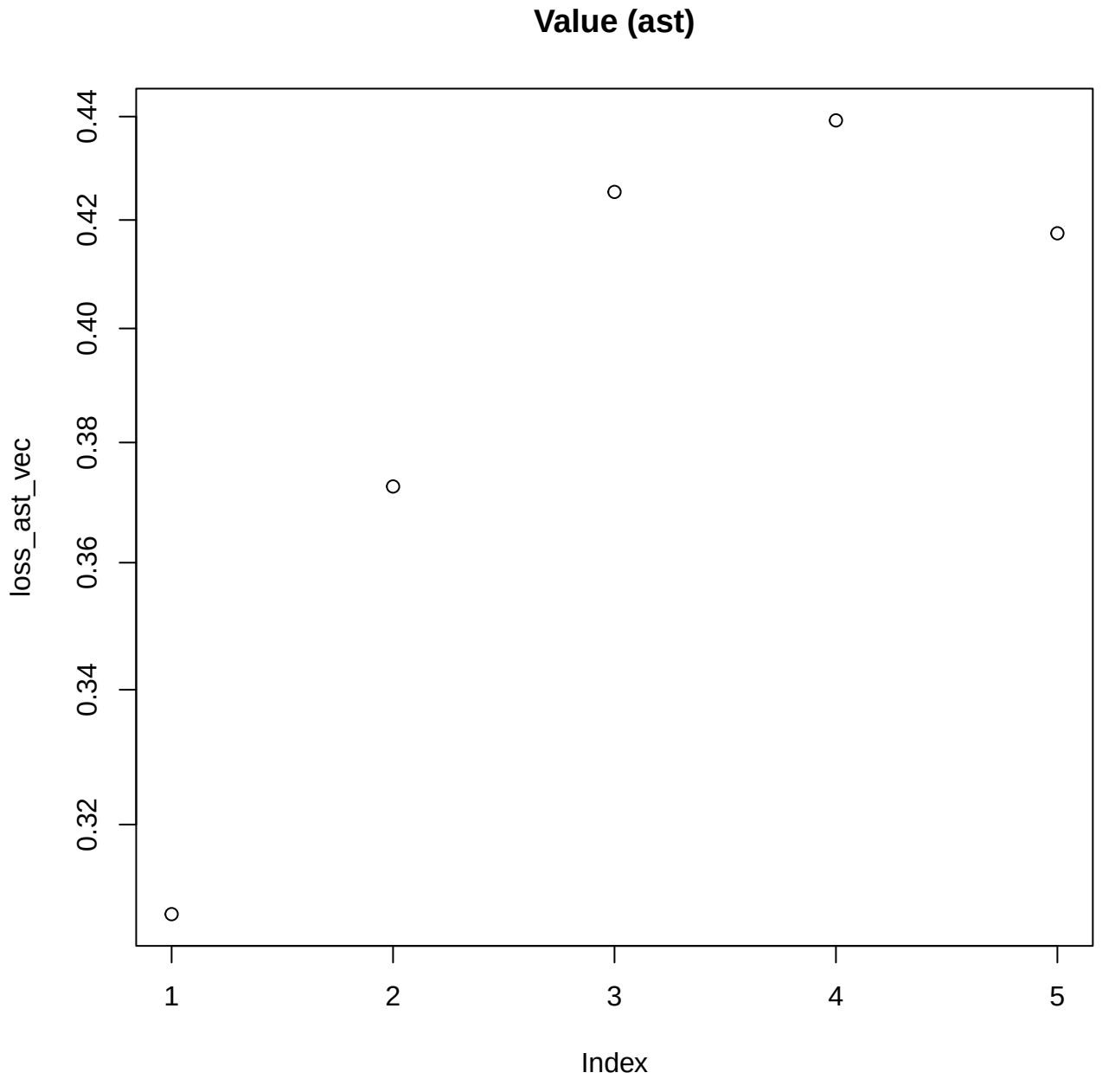

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
2e−03
5e−04
2e−04
5e−05
1e−05
0
50
100
150
200
Index


Value (ast)
loss_ast_vec
0.540
0.542
0.544
0.546
0.548
0
50
100
150
200
Index


Value (dag)
loss_dag_vec
0.452
0.454
0.456
0.458
0.460
0
50
100
150
200
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.01
0.02
0.05
0.10
1
2
3
4
5
Index


Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.02
0.05
0.10
1
2
3
4
5
Index


Value (ast)
loss_ast_vec
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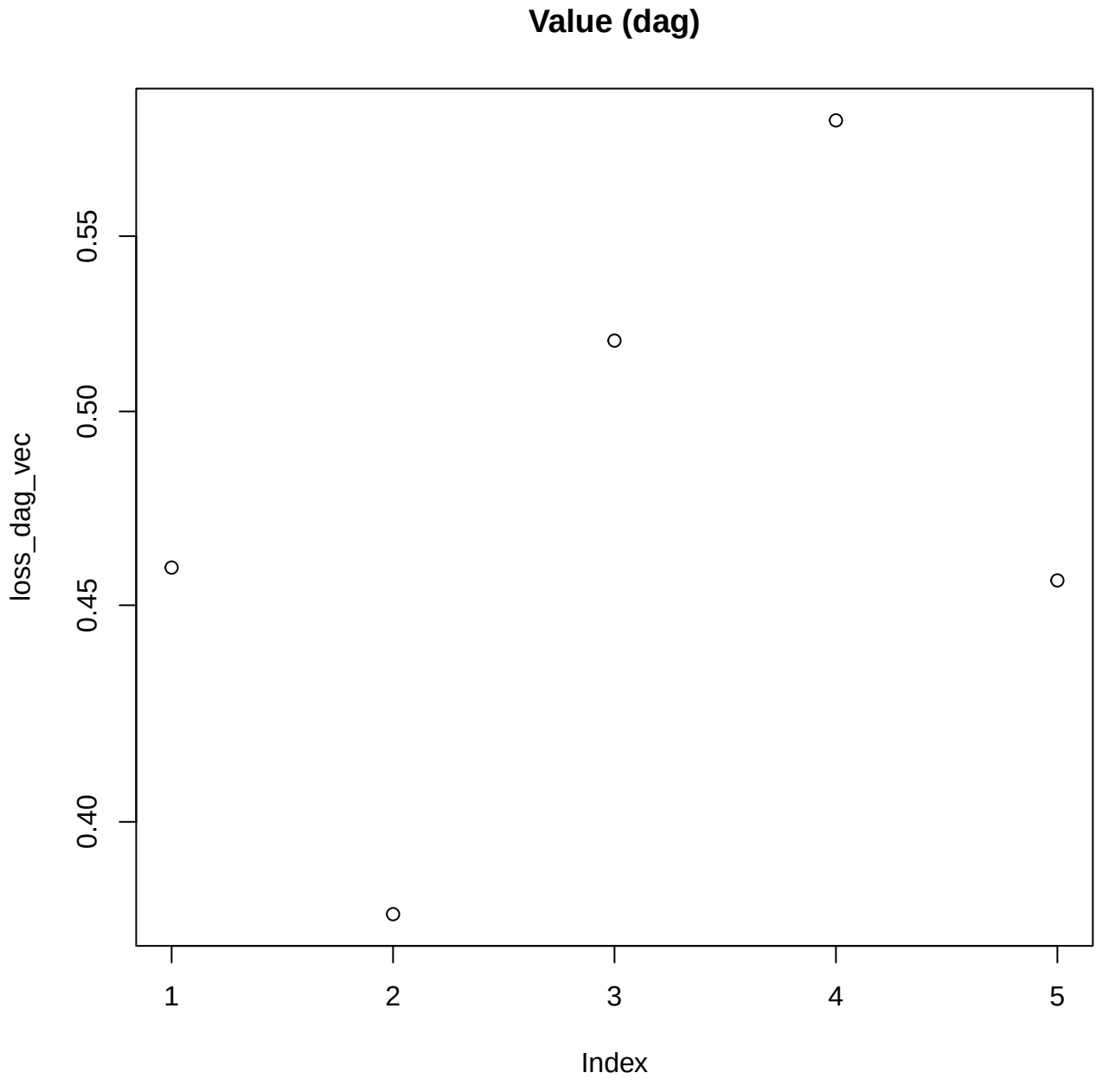
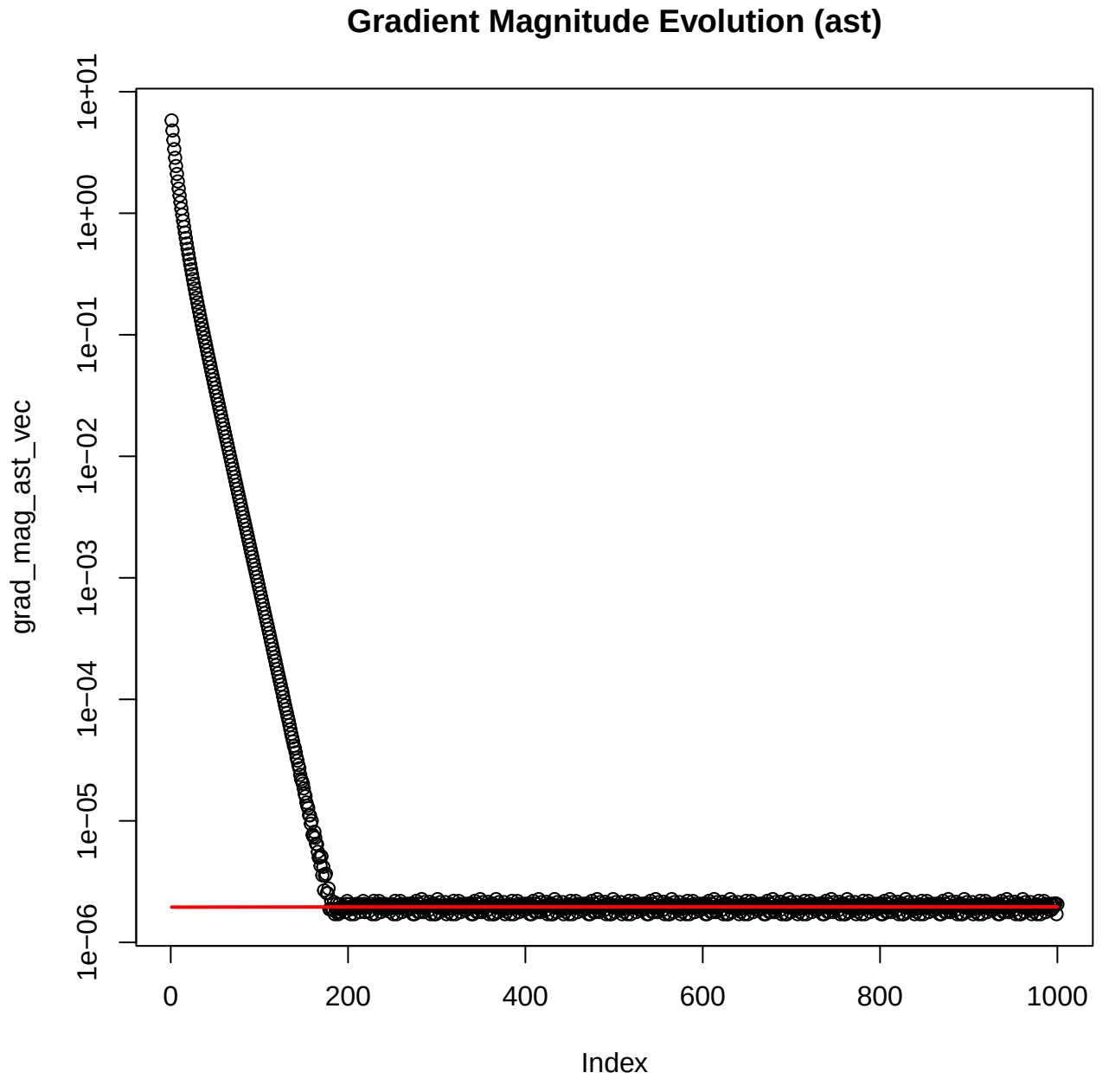
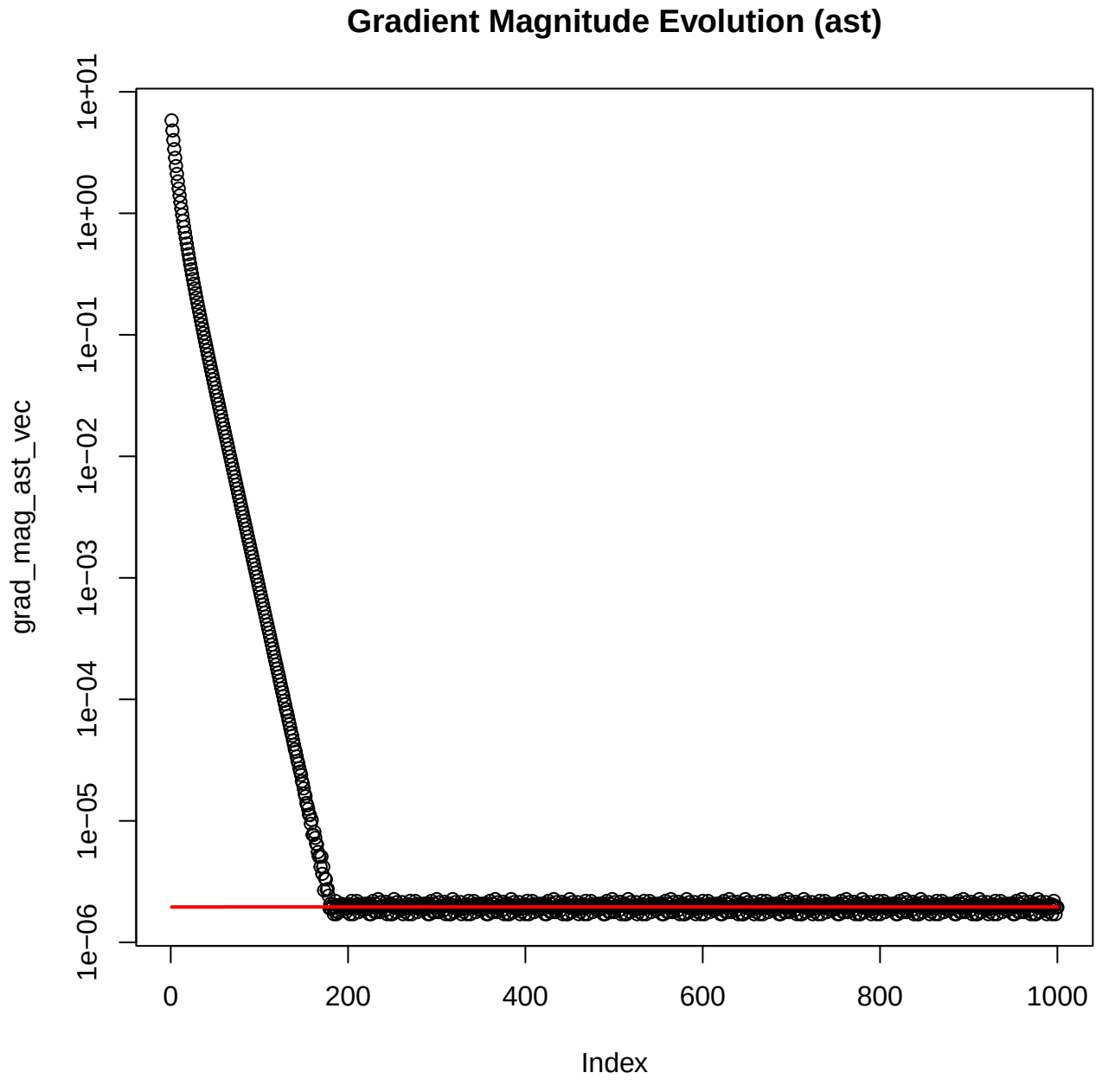
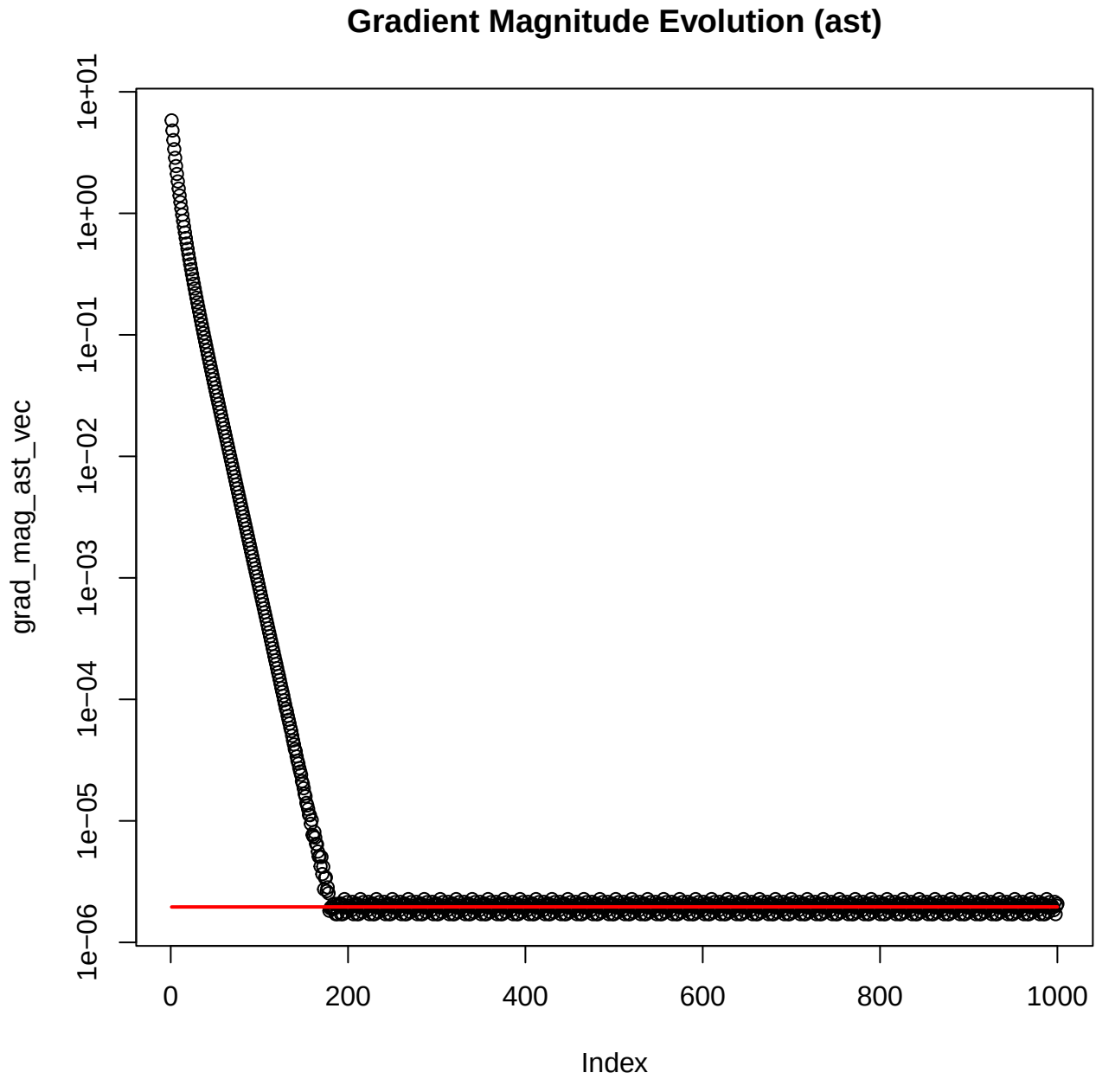
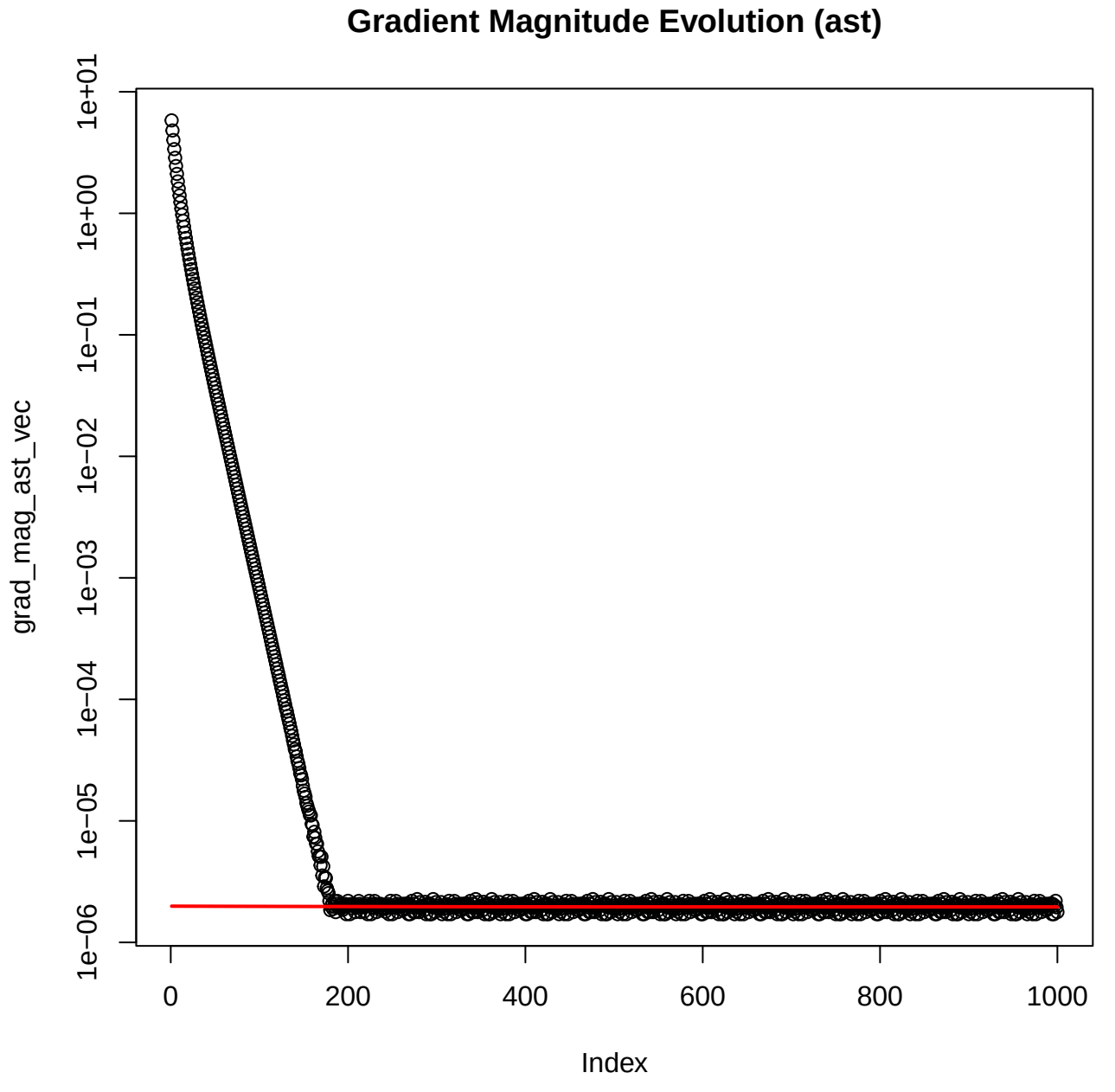
0.34
0.36
0.38
0.40
0.42
0.44
1
2
3
4
5
Index

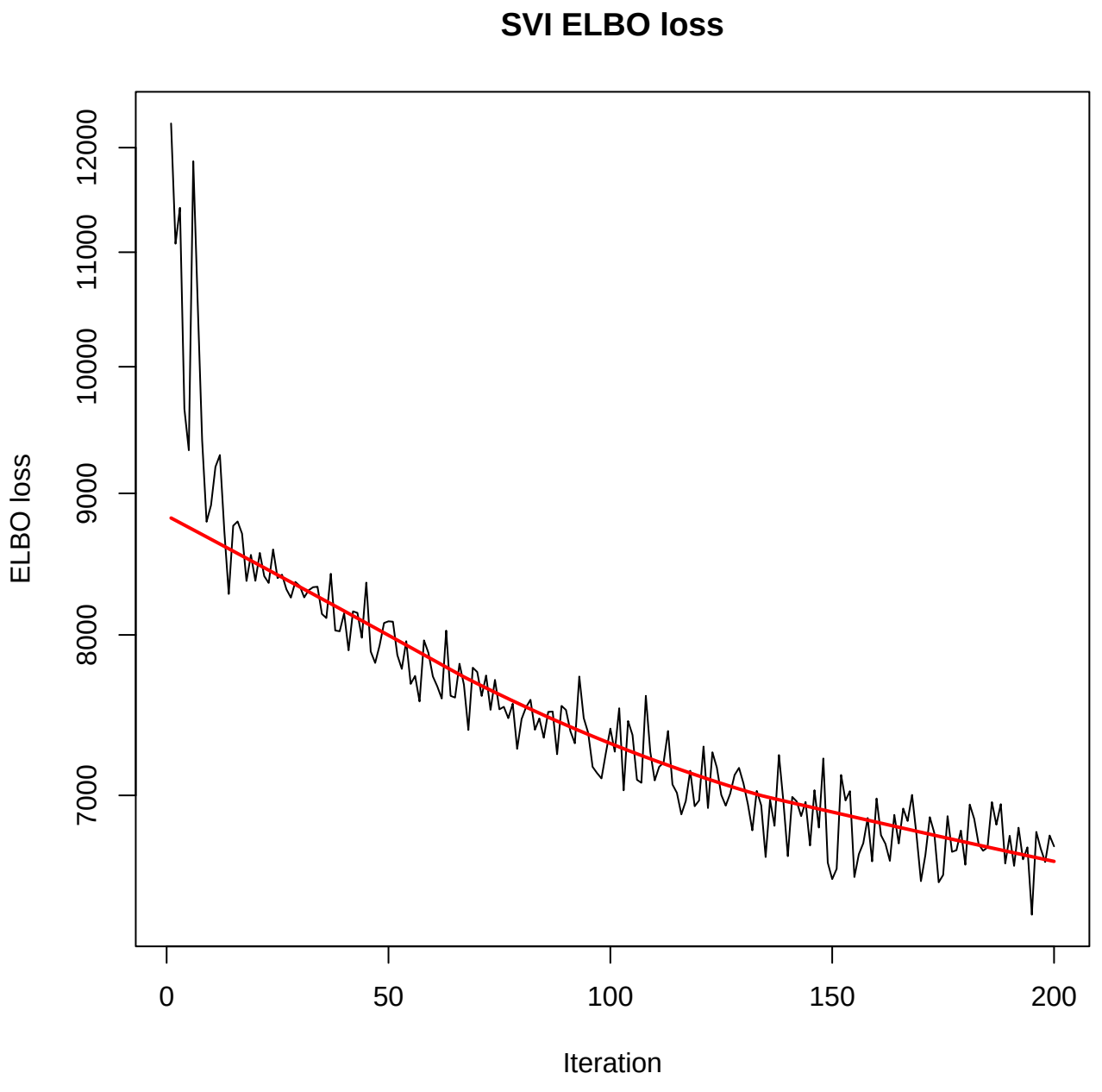

Value (dag)
loss_dag_vec
0.40
0.45
0.50
0.55
1
2
3
4
5
Index

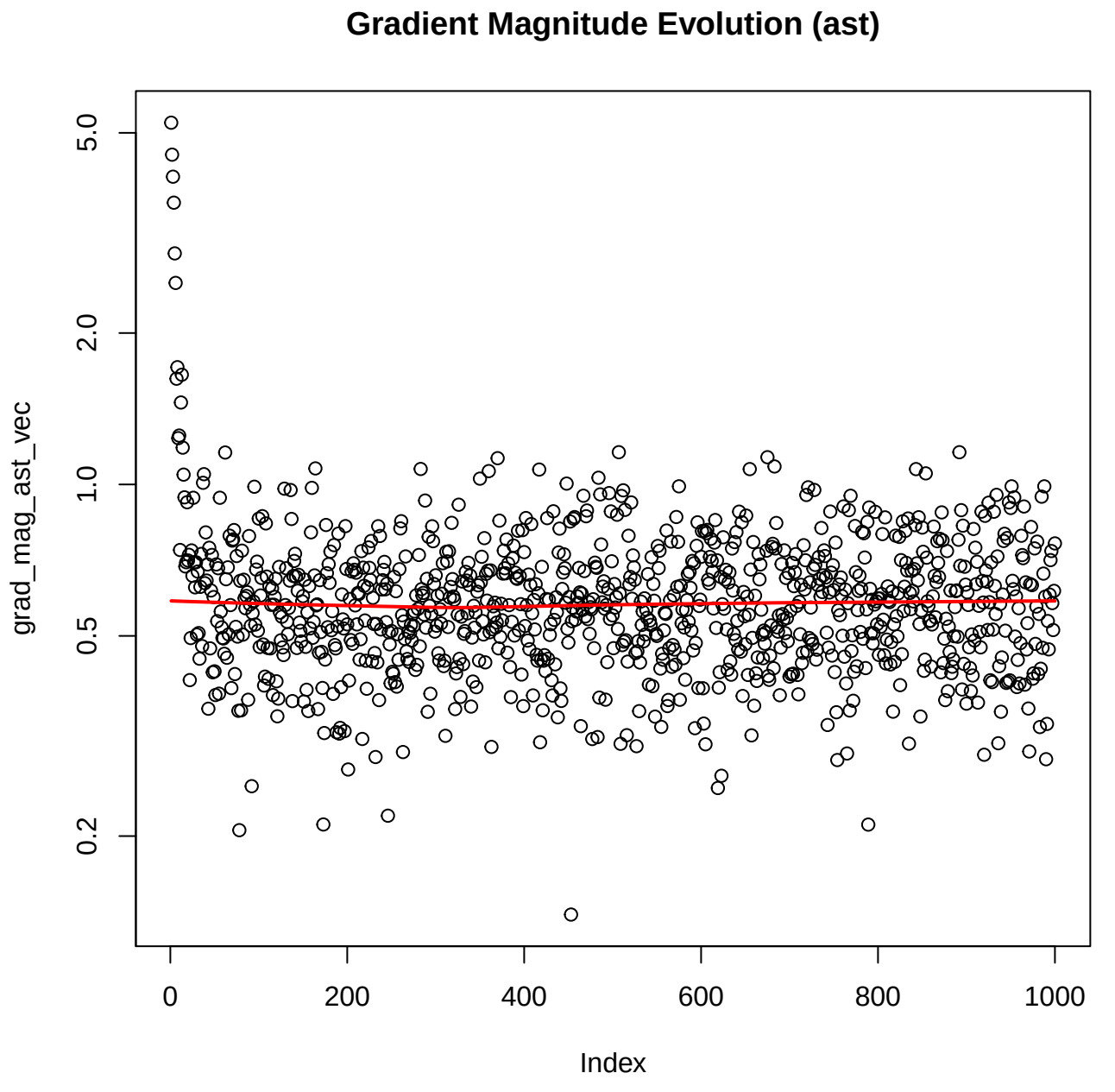

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

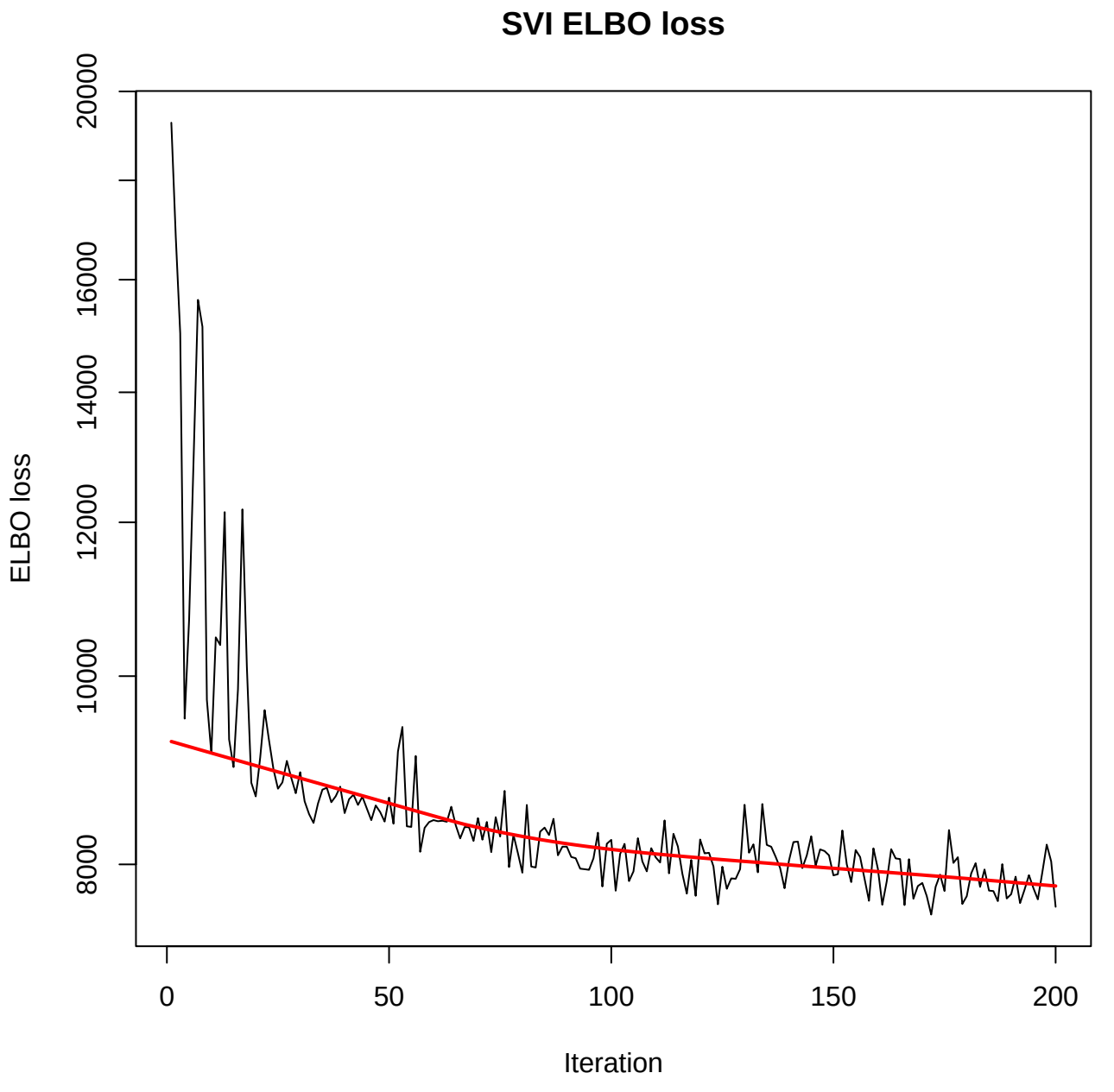

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

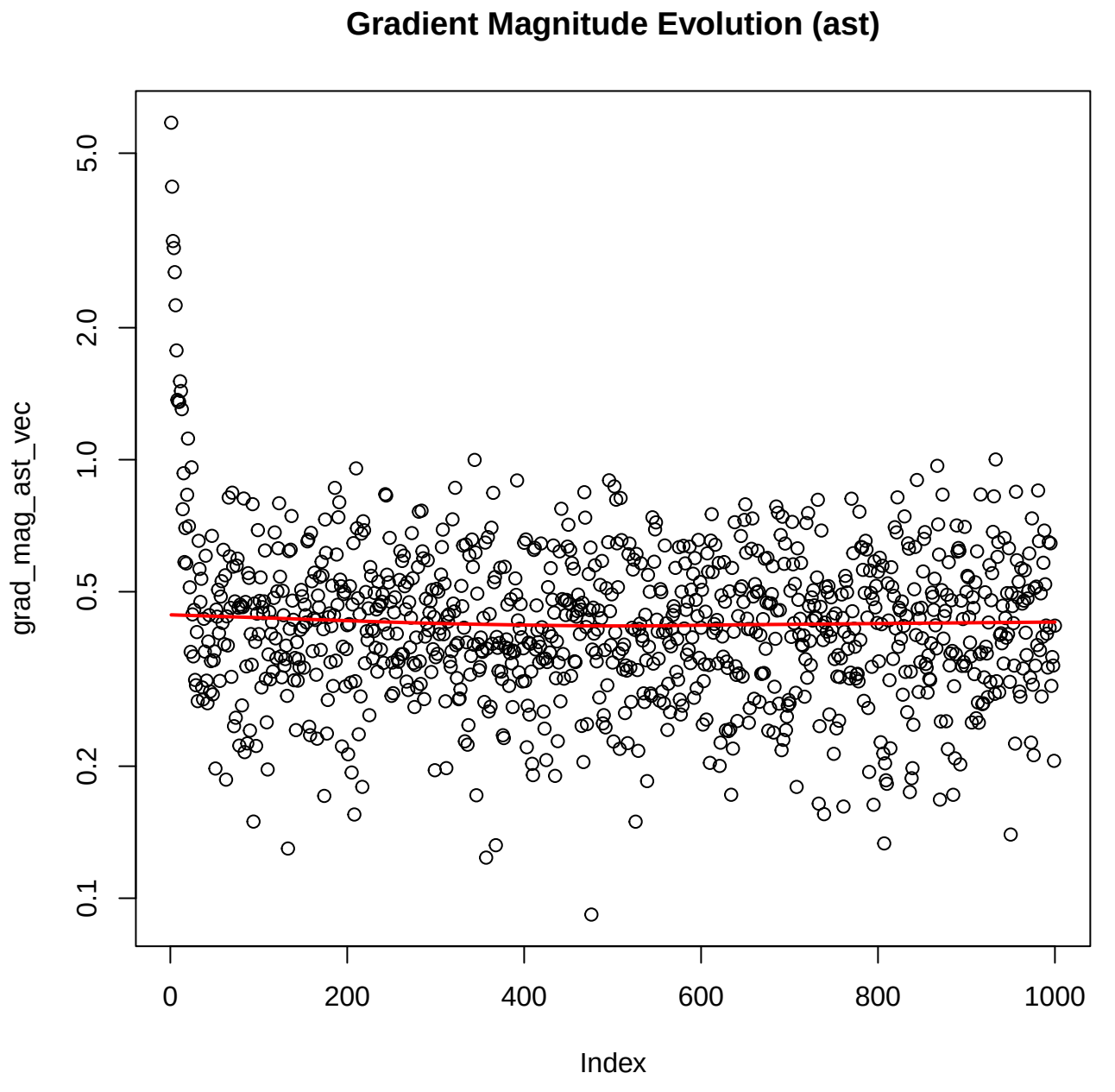

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

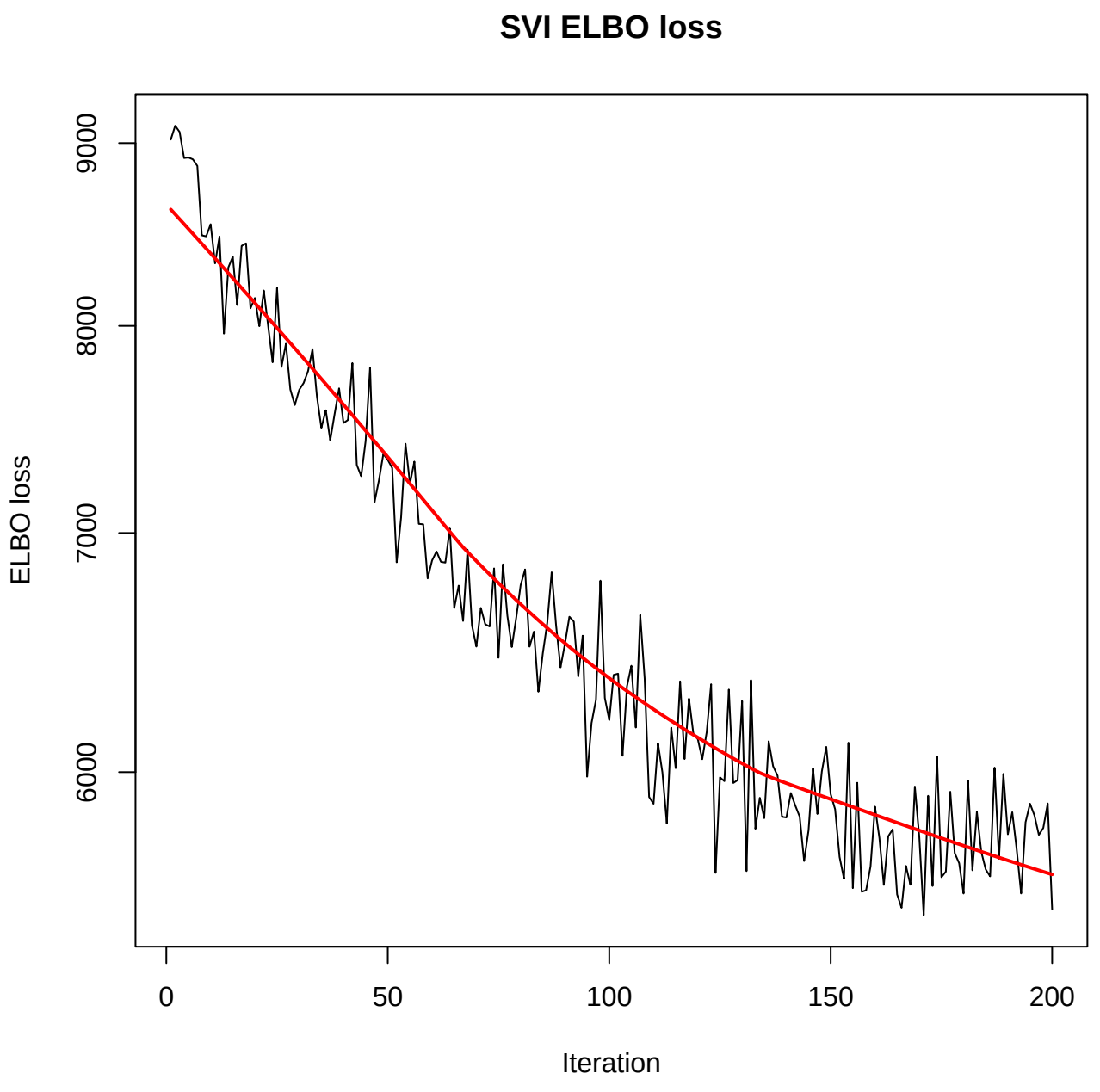

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

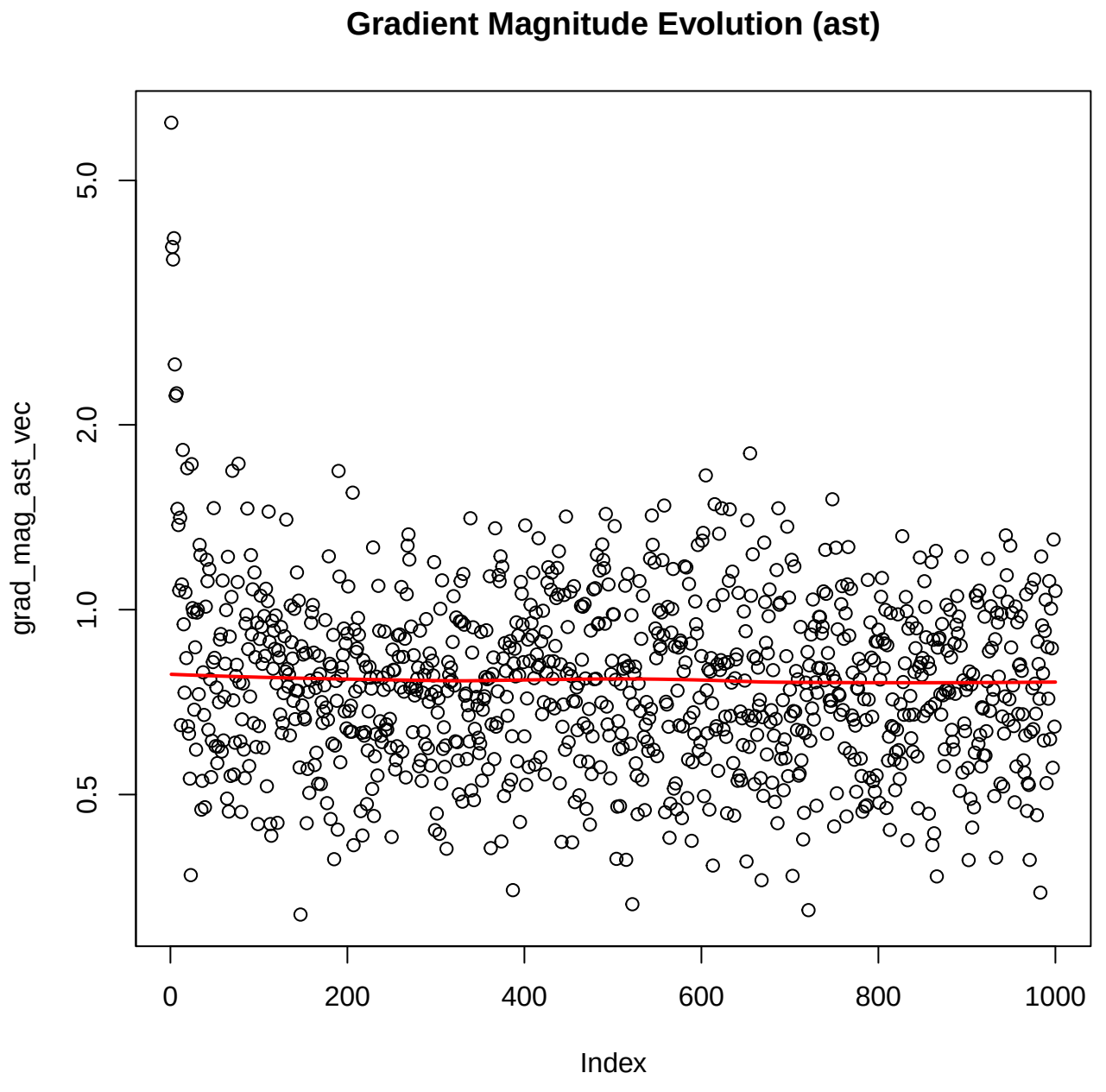

SVI ELBO loss
ELBO loss
12000
11000
10000
9000
8000
7000
0
50
100
150
200
Iteration

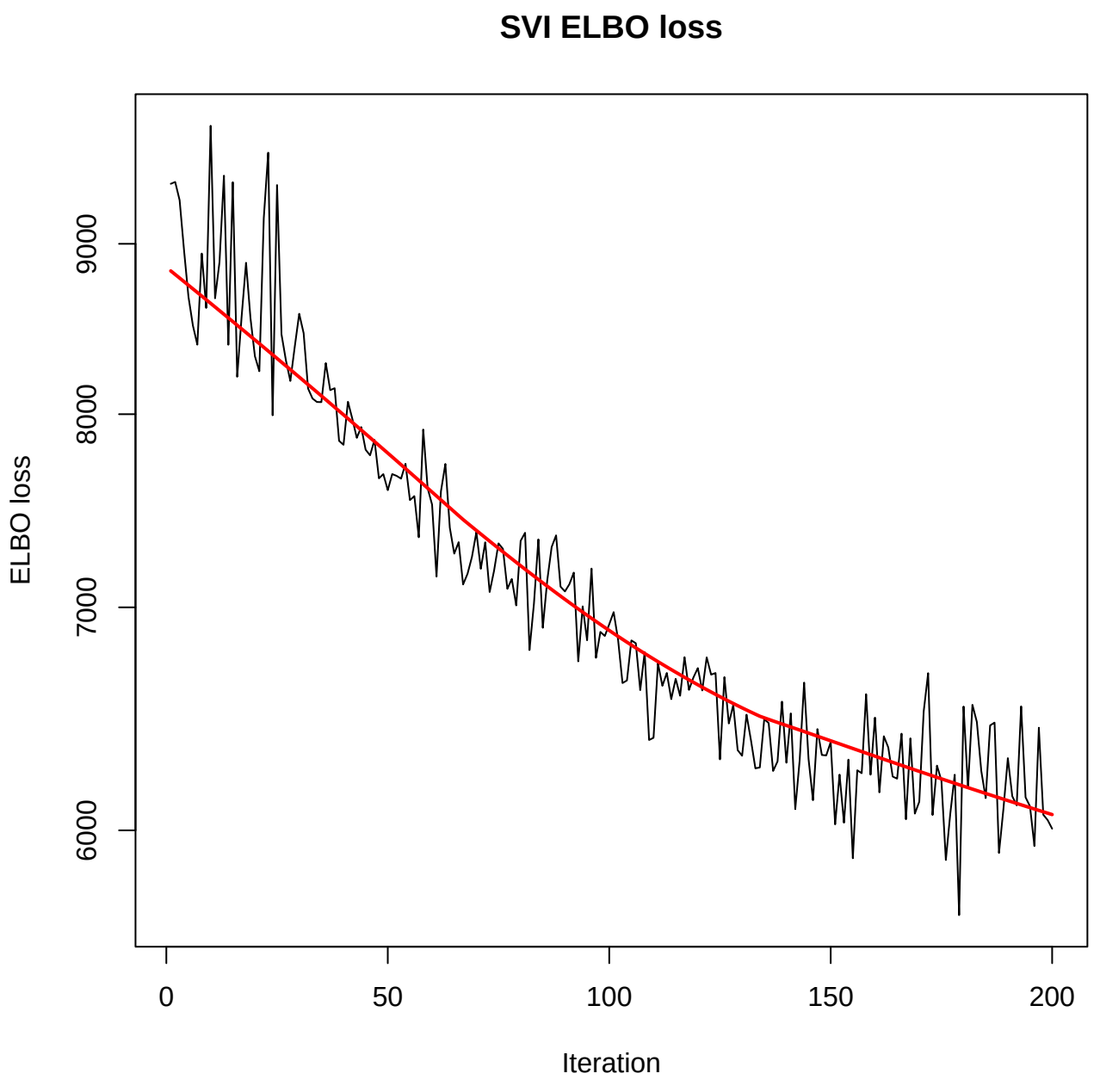

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
5.0
2.0
1.0
0.5
0.2
0
200
400
600
800
1000
Index


SVI ELBO loss
ELBO loss
20000
16000
14000
12000
10000
8000
0
50
100
150
200
Iteration


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
5.0
2.0
1.0
0.5
0.2
0.1
0
200
400
600
800
1000
Index


SVI ELBO loss
ELBO loss
9000
8000
7000
6000
0
50
100
150
200
Iteration


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.5
1.0
2.0
5.0
0
200
400
600
800
1000
Index


SVI ELBO loss
ELBO loss
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7000
6000
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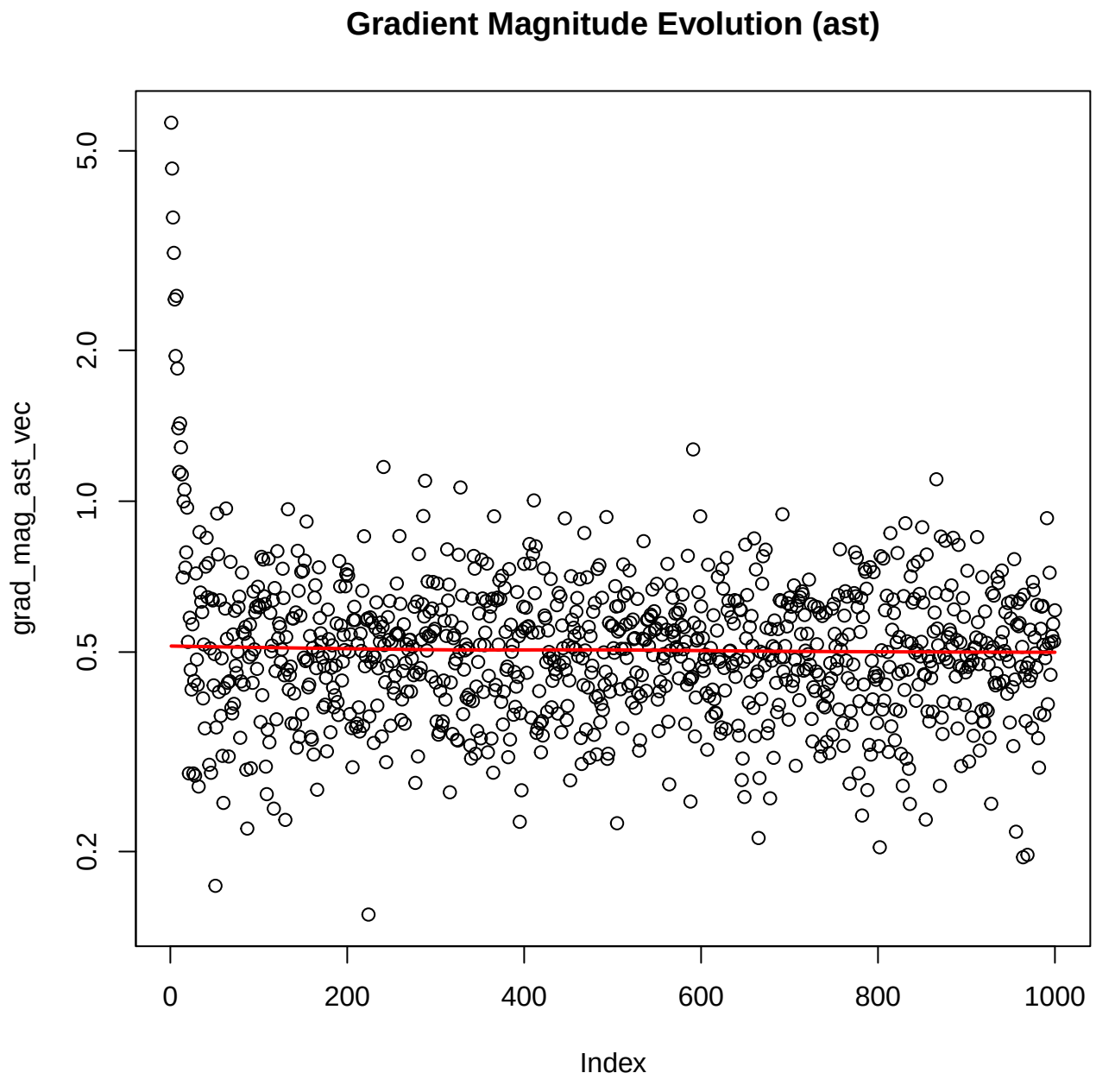
50
100
150
200
Iteration


Gradient Magnitude Evolution (ast)
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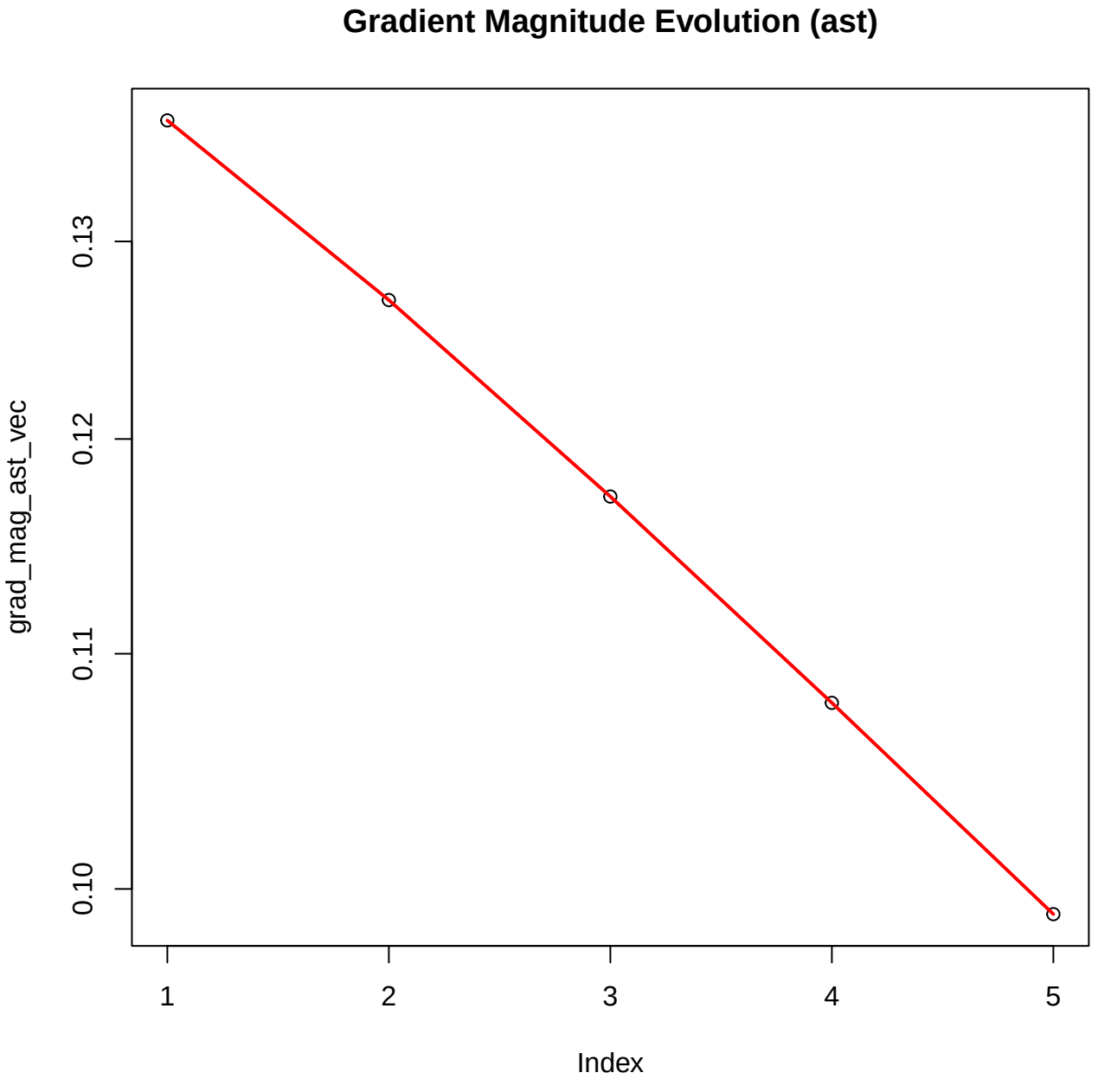
Gradient Magnitude Evolution (ast)
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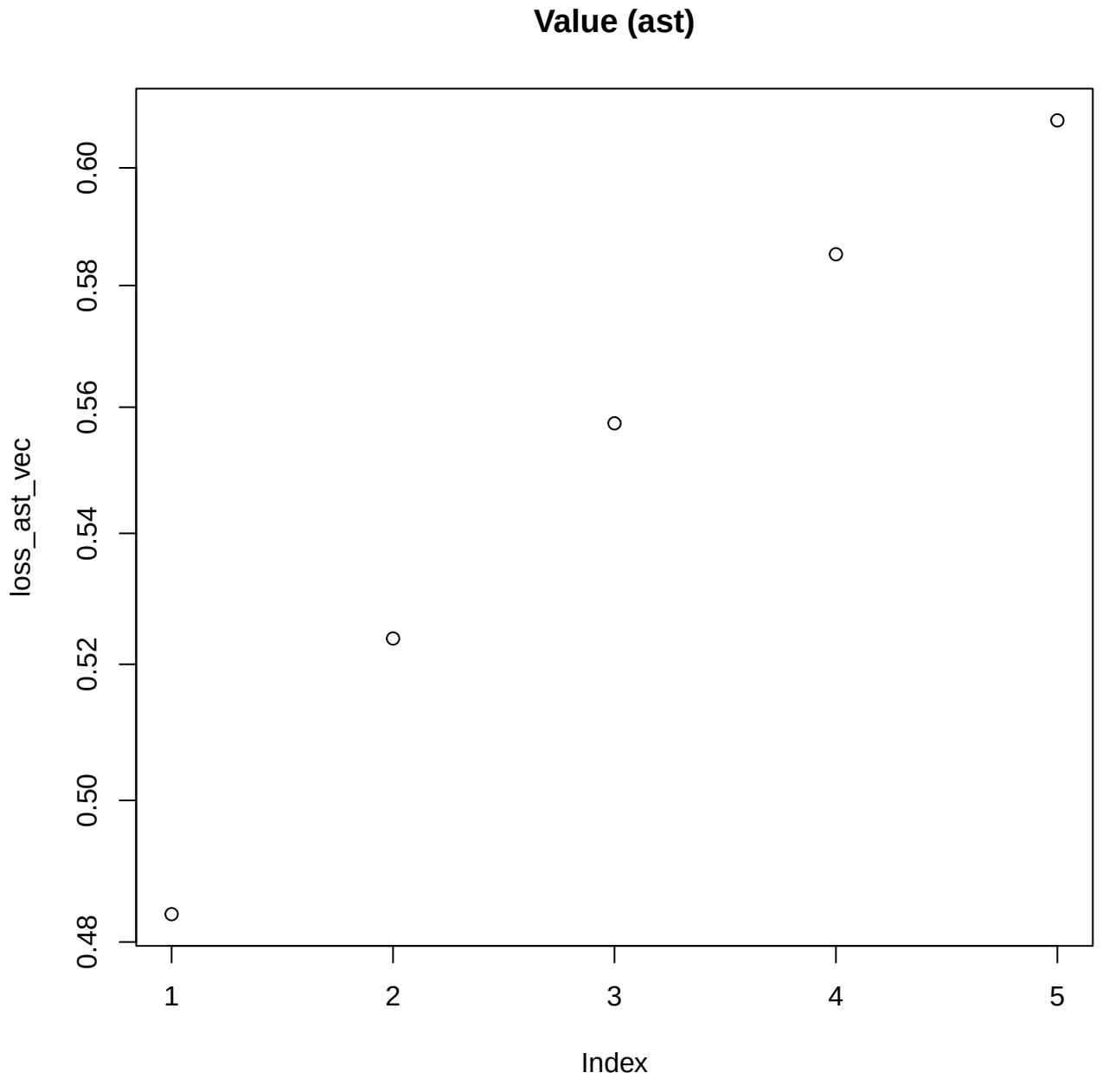
Value (ast)
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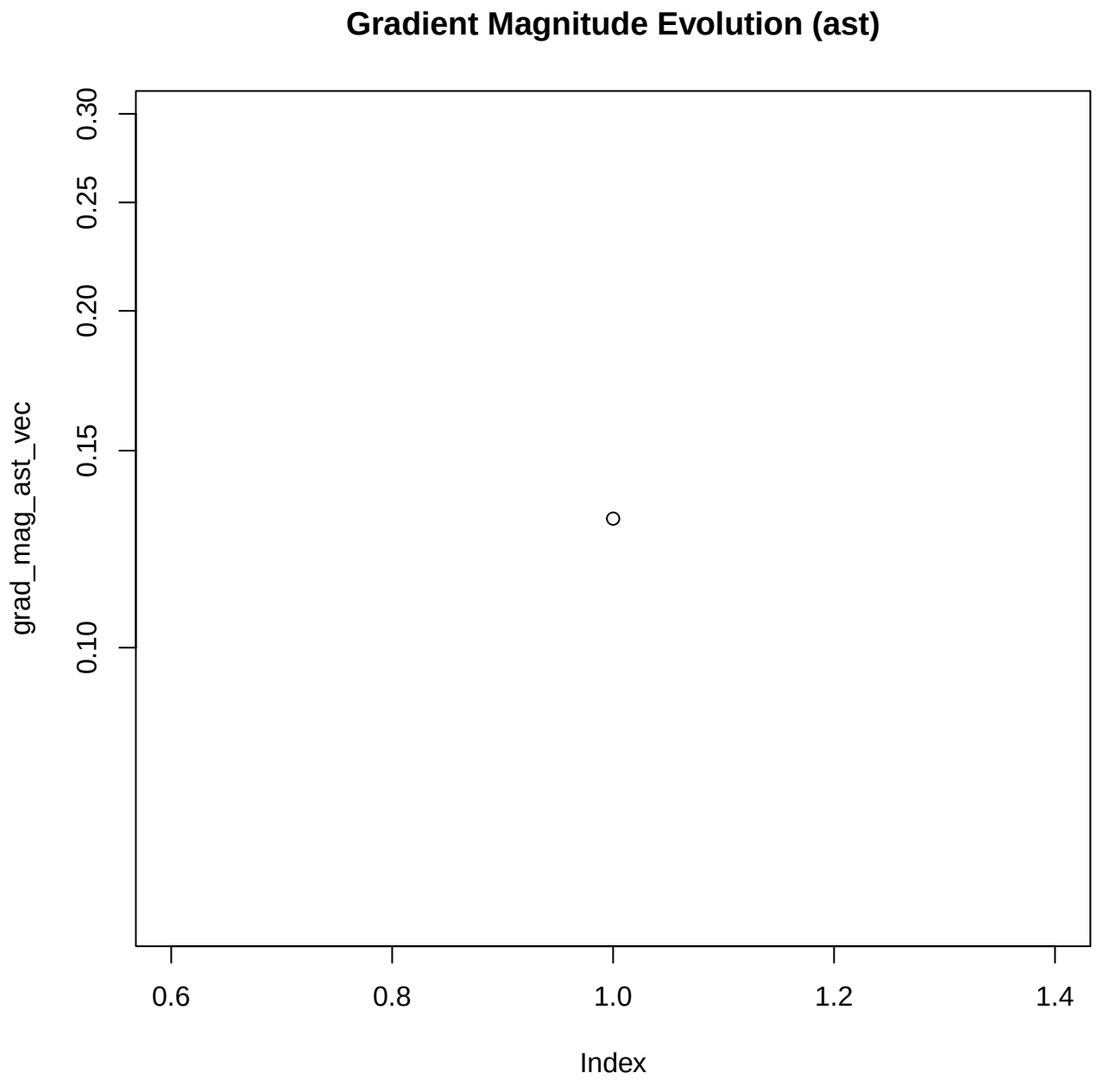
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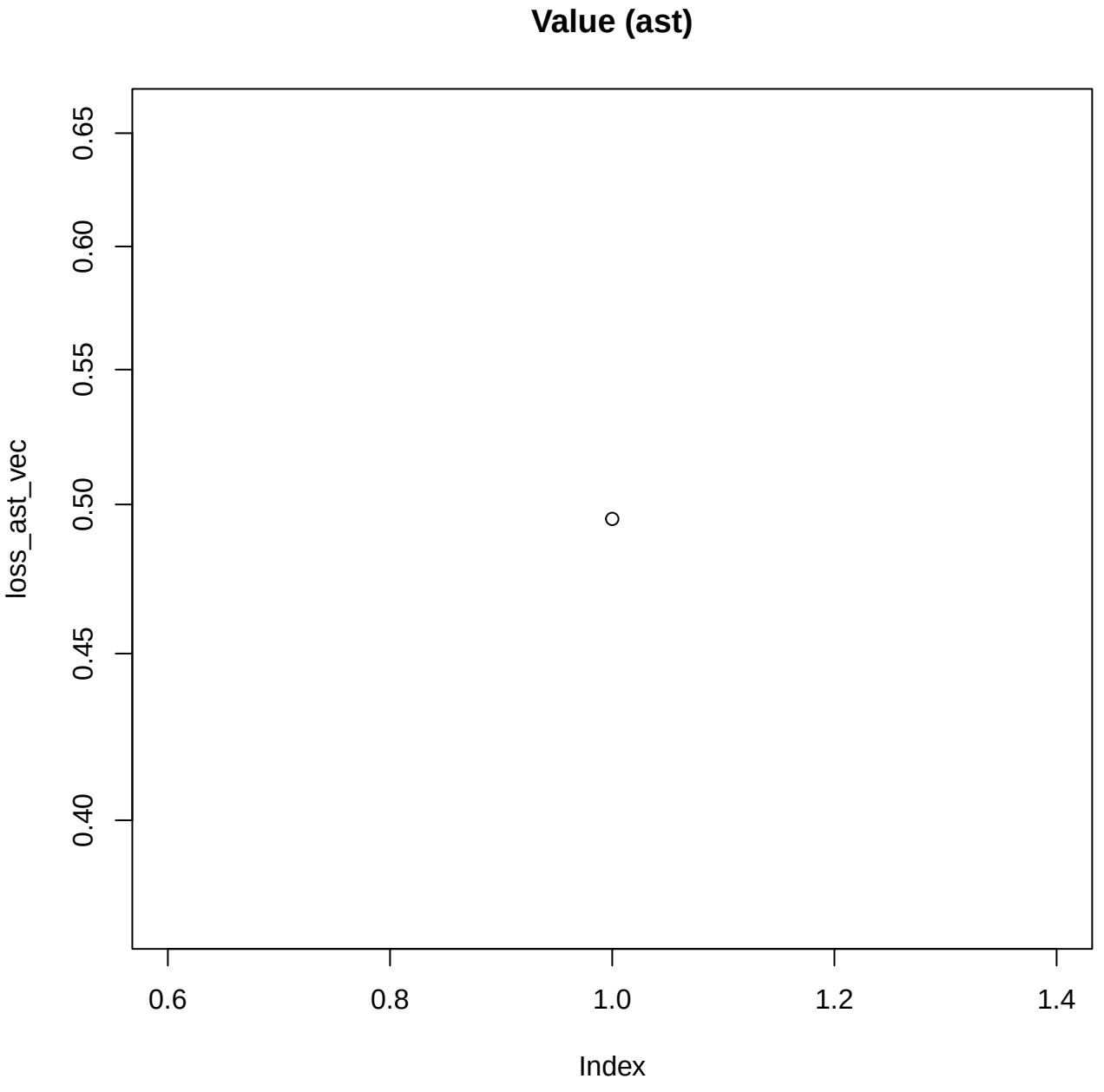
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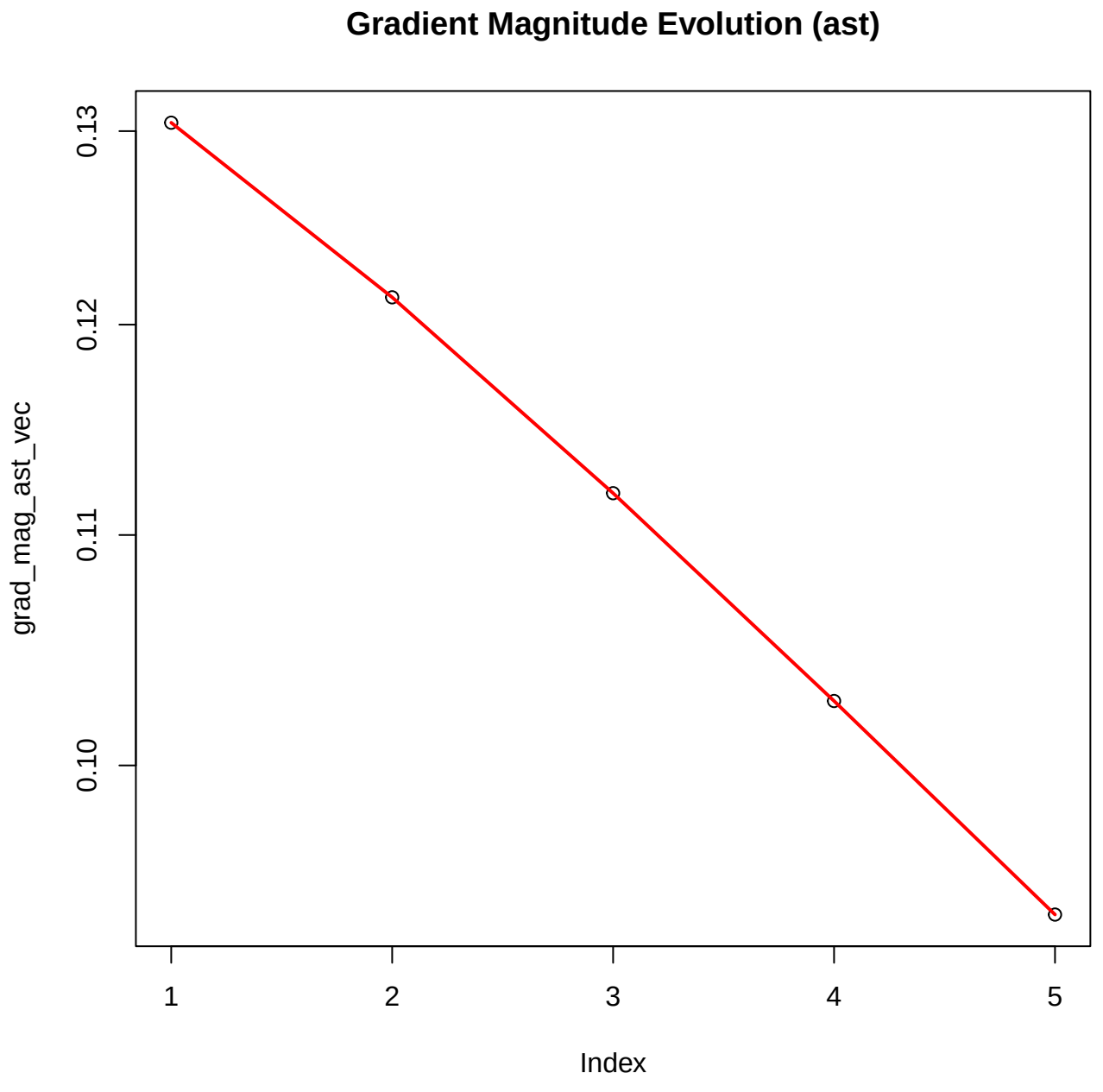
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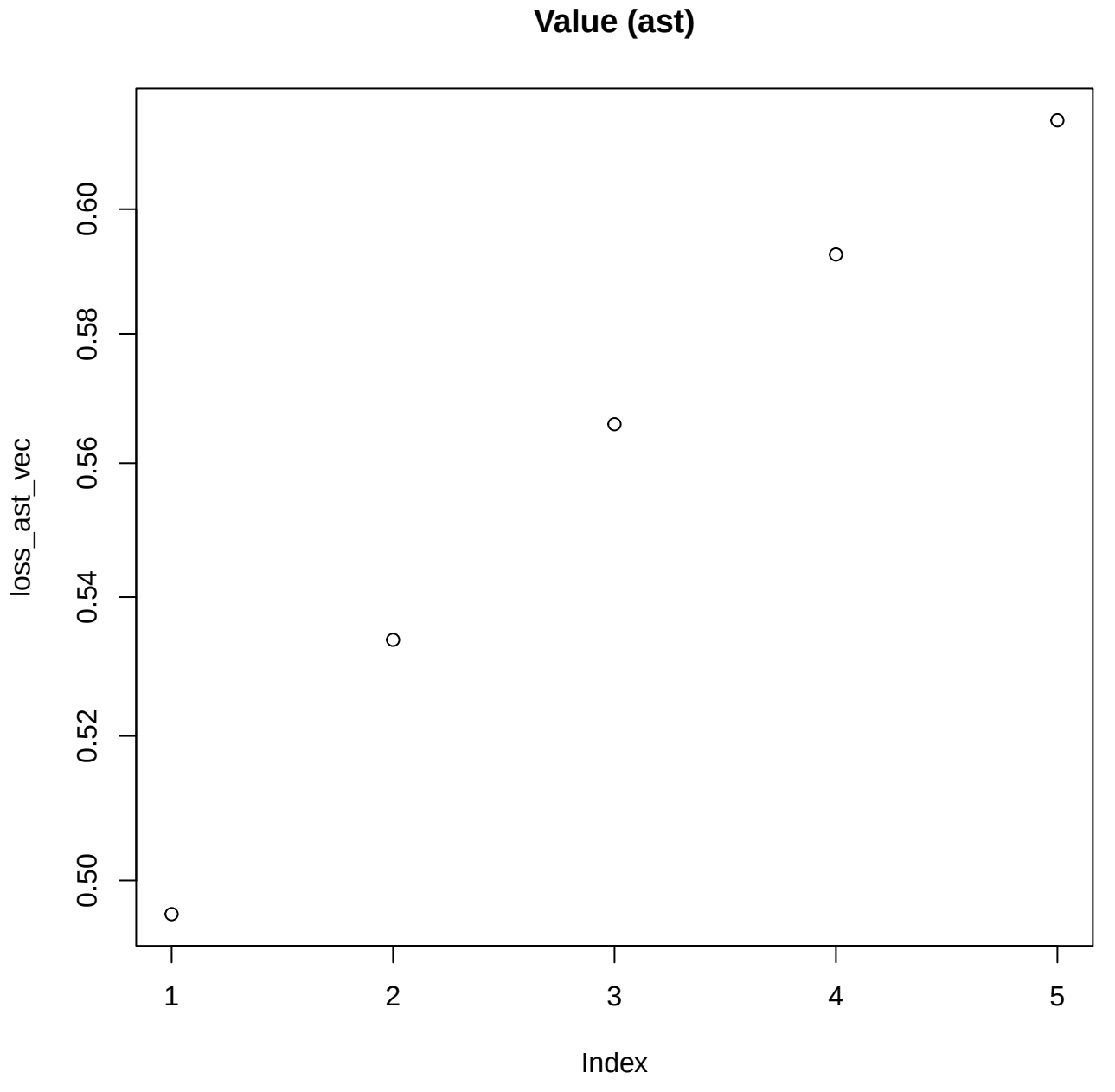
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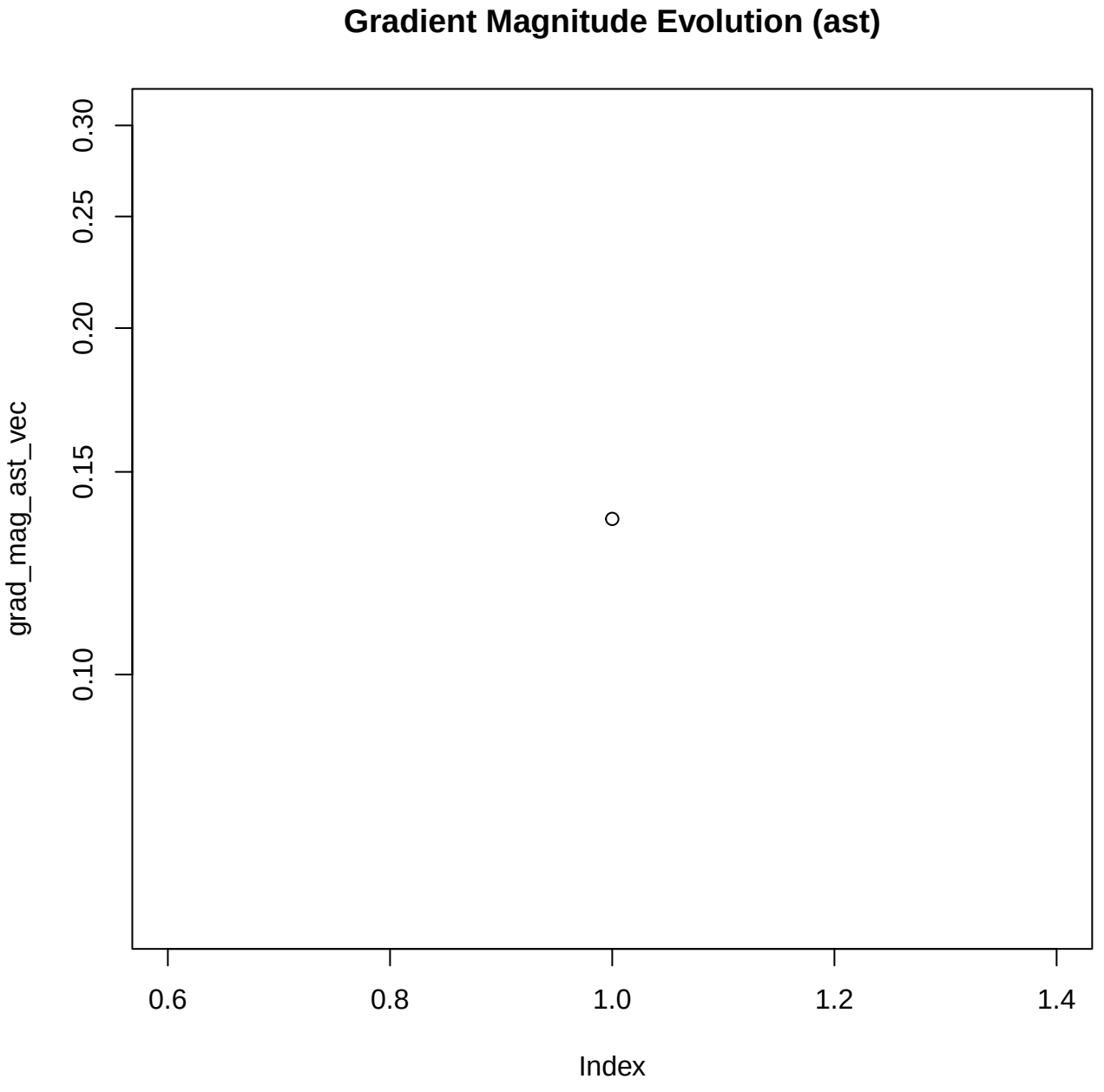
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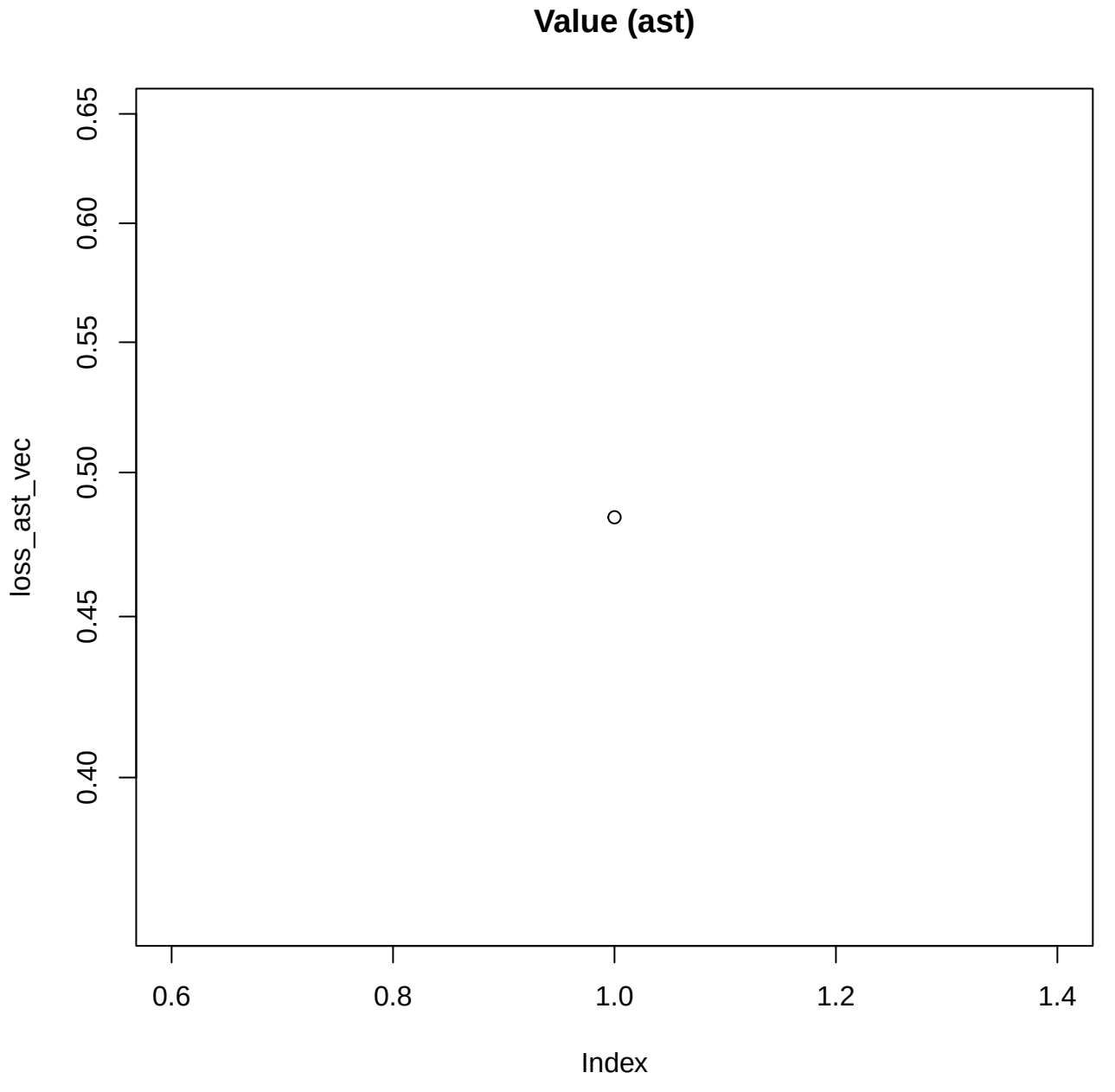
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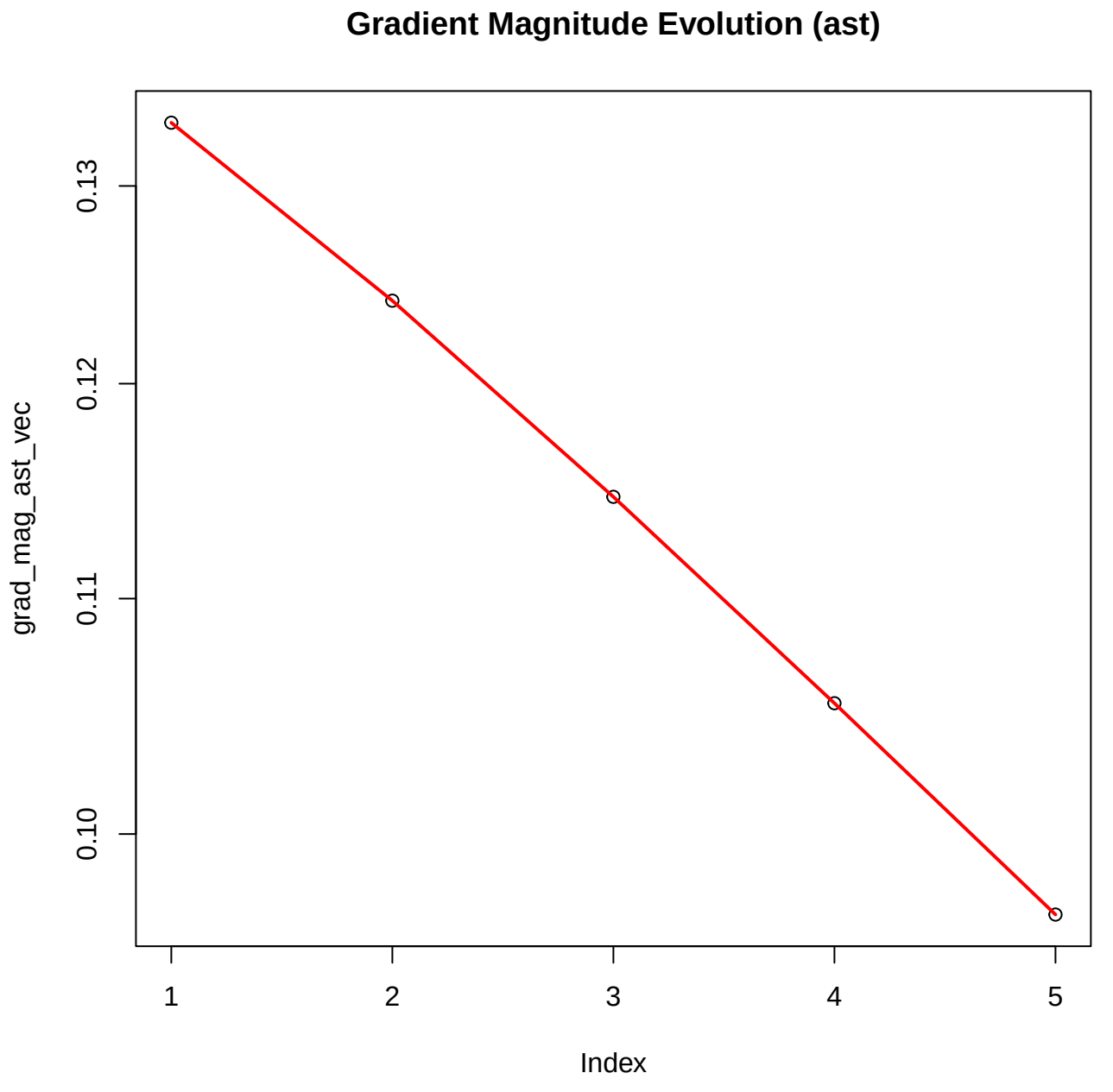
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grad_mag_ast_vec
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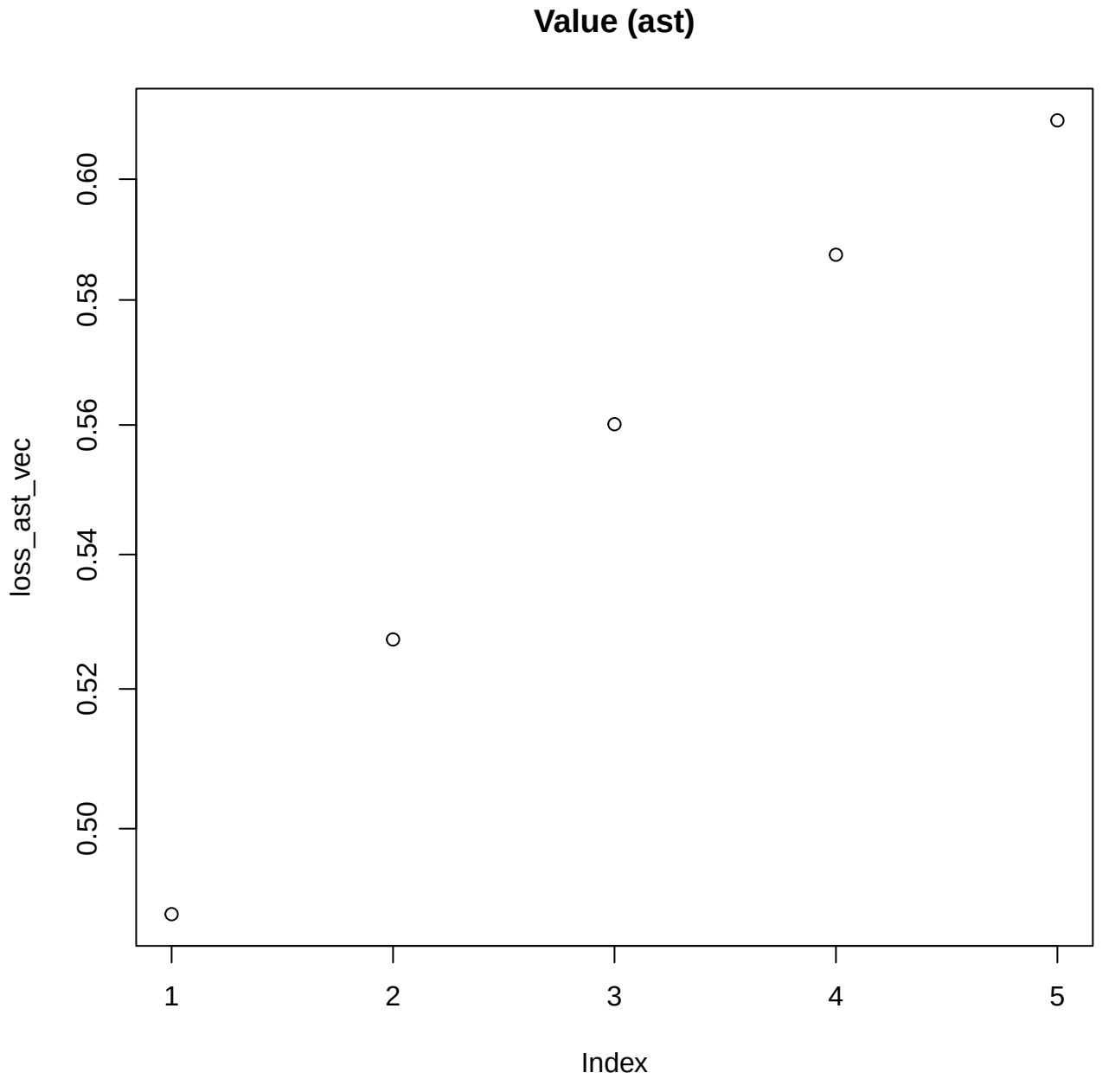
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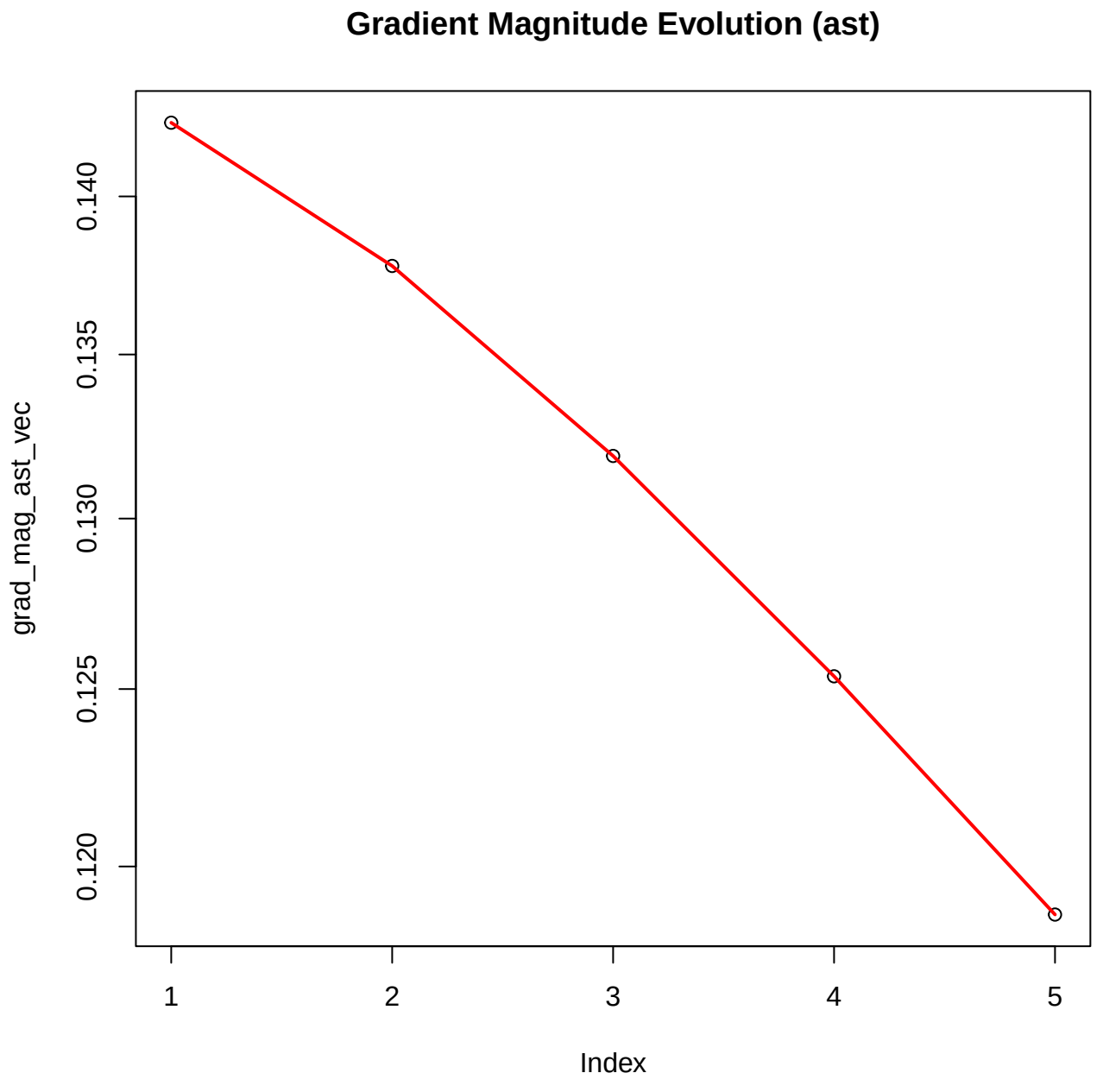
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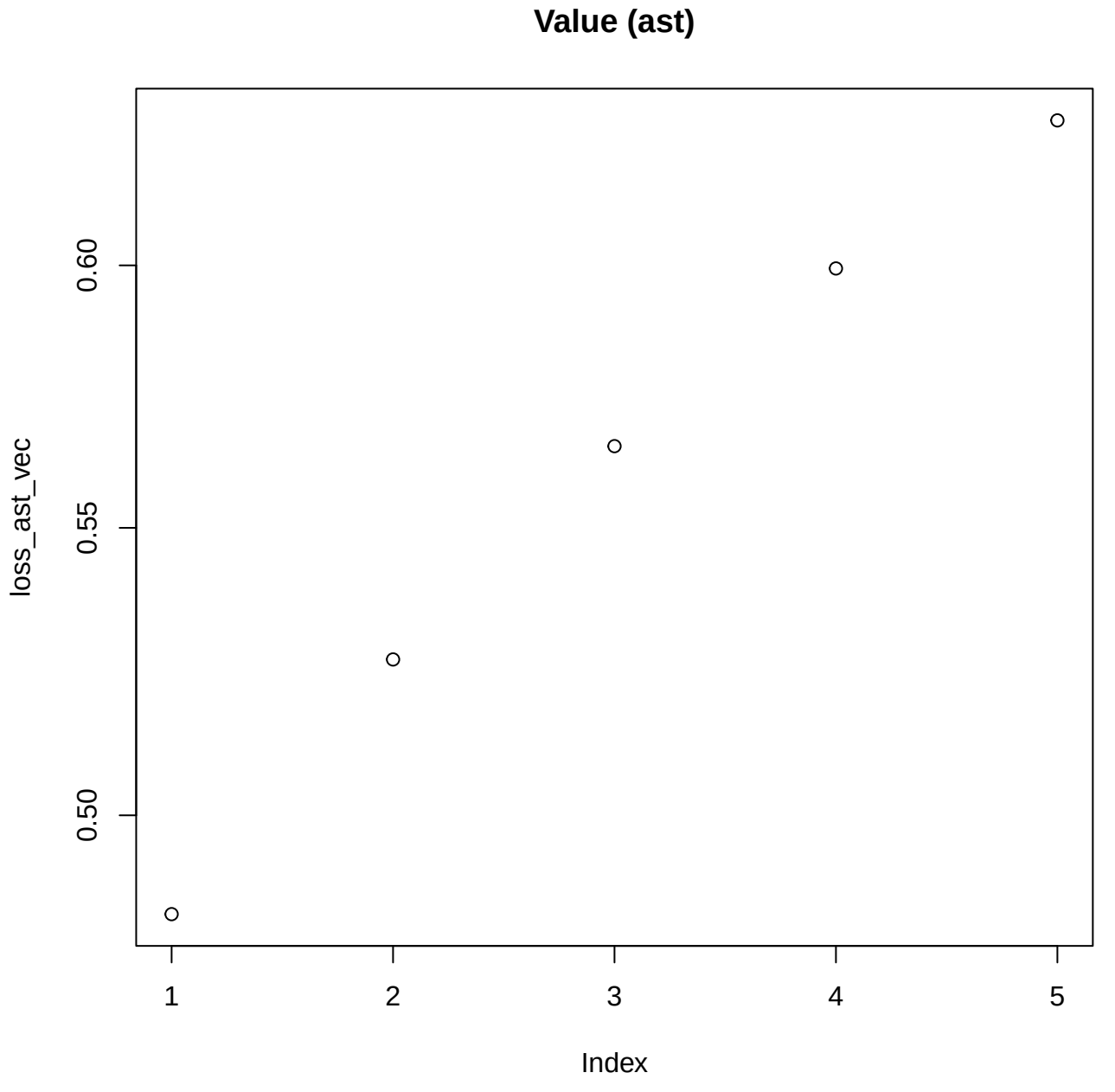
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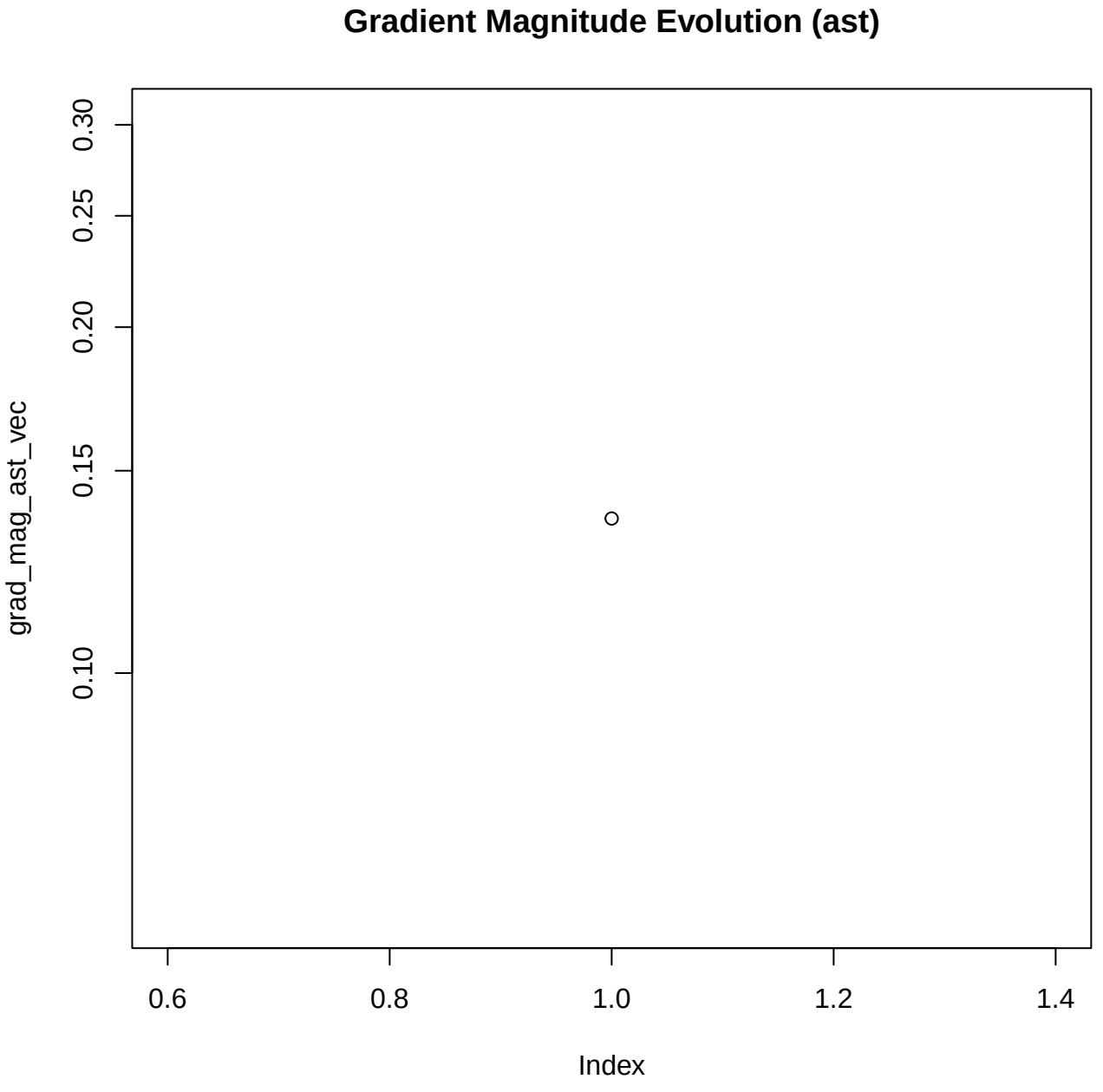
Gradient Magnitude Evolution (ast)
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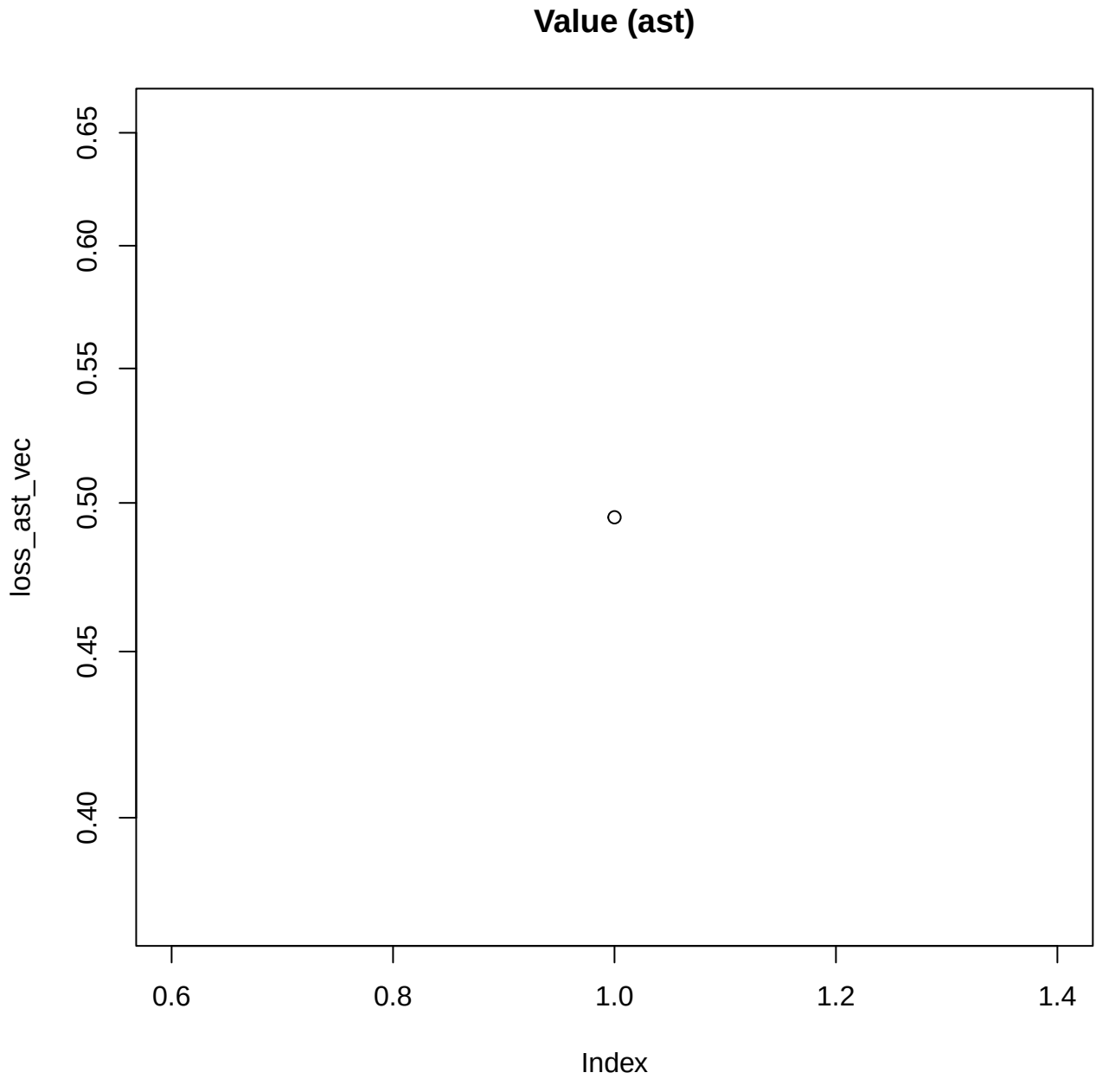
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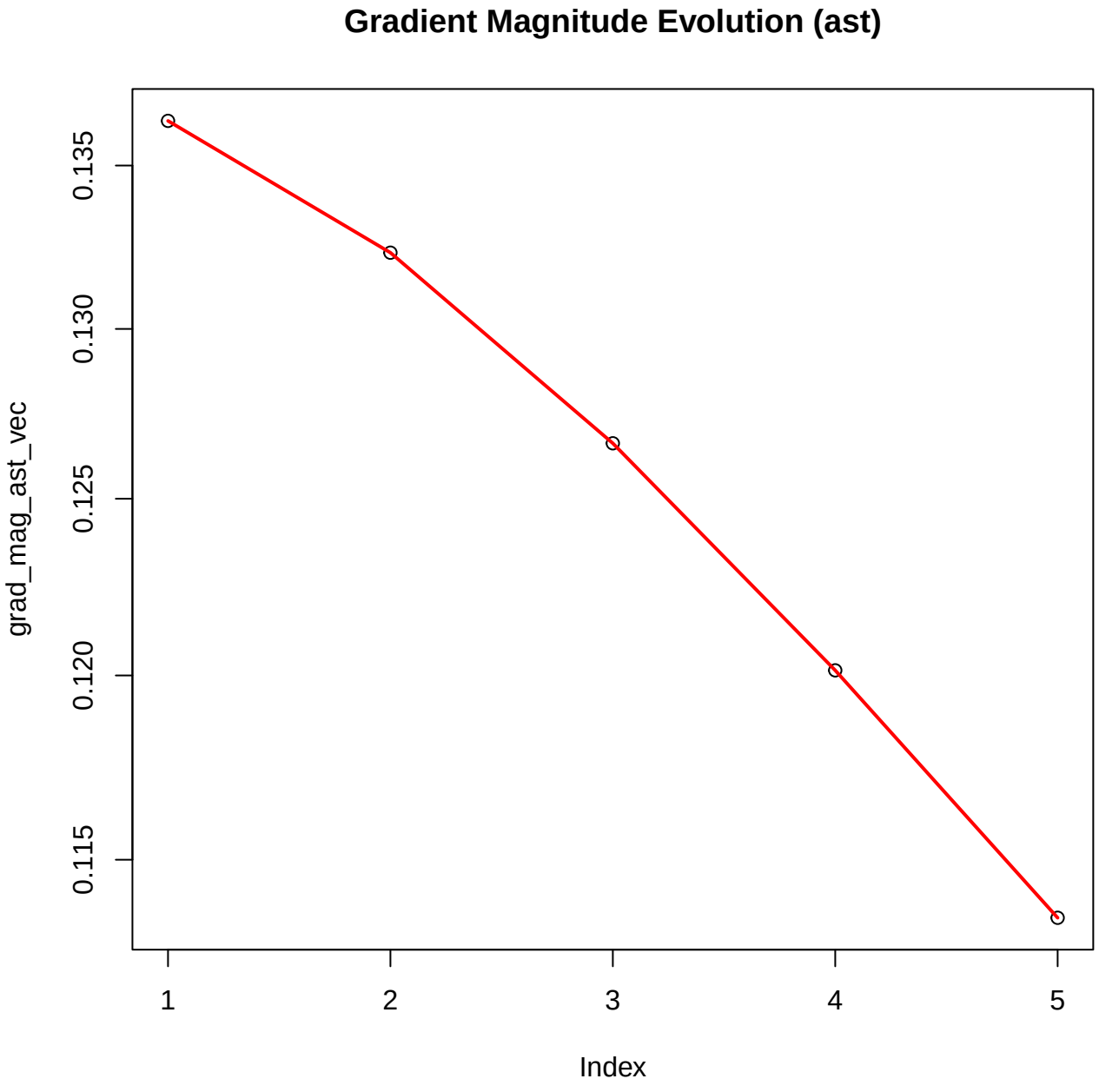
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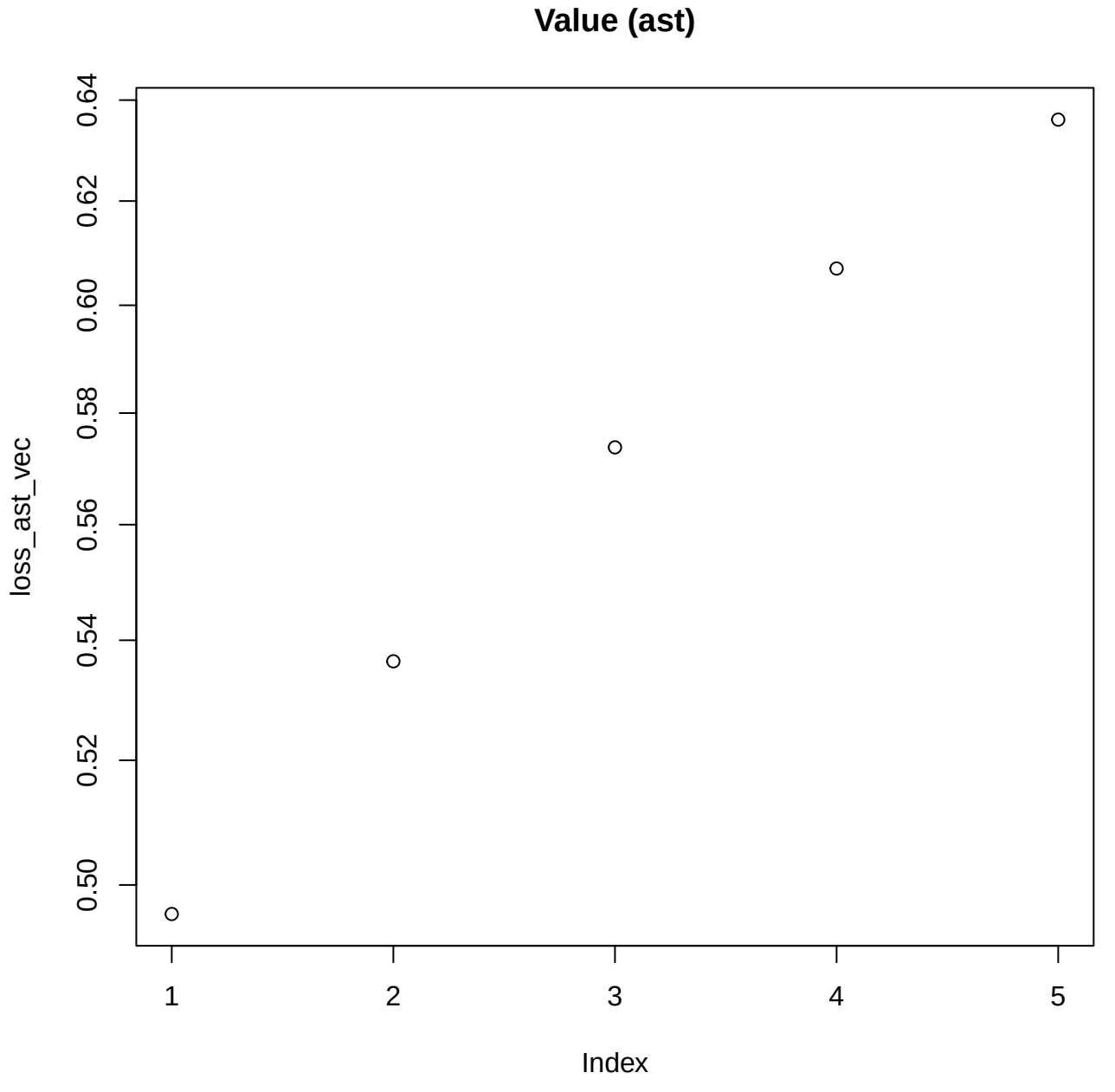
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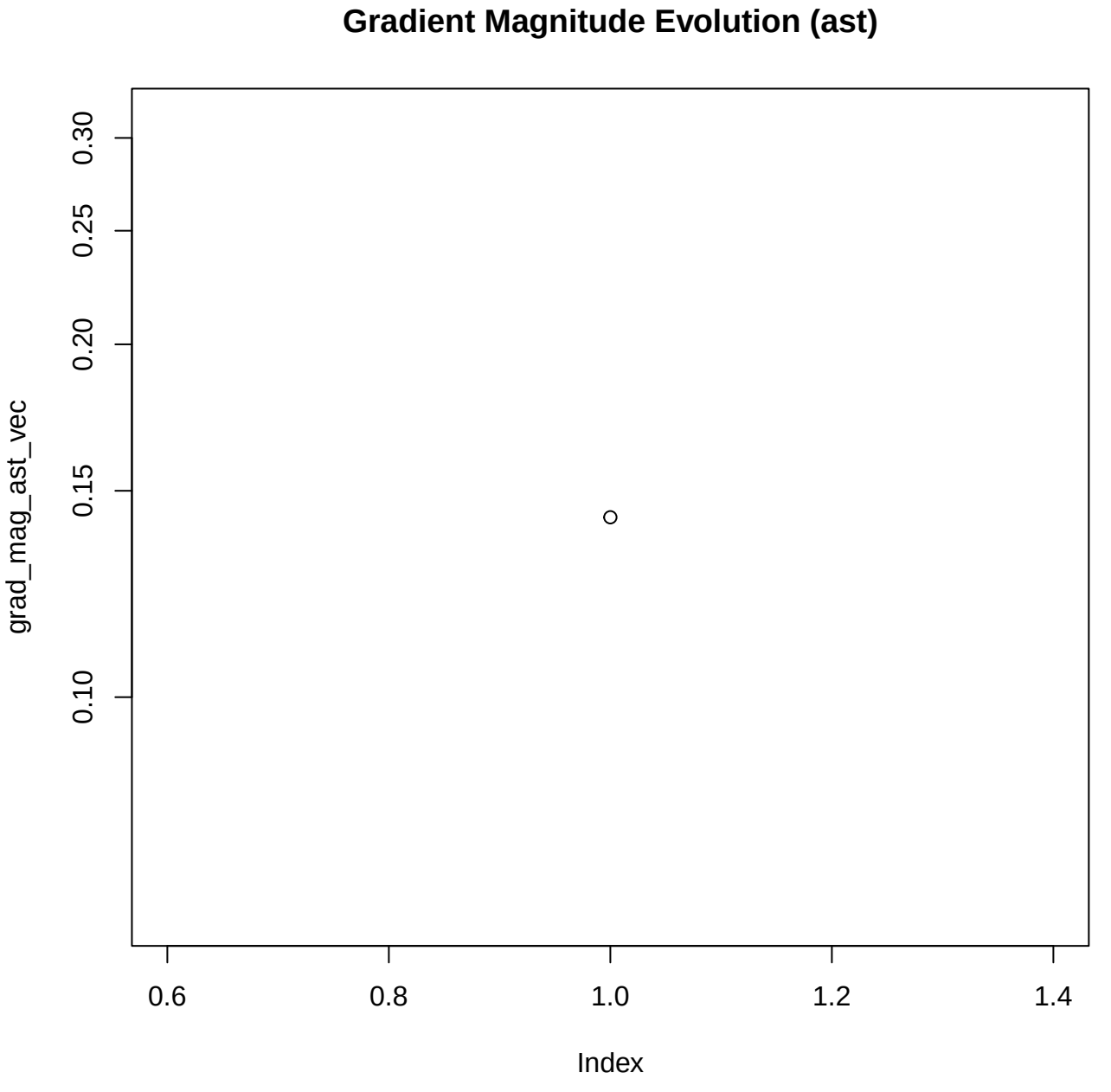
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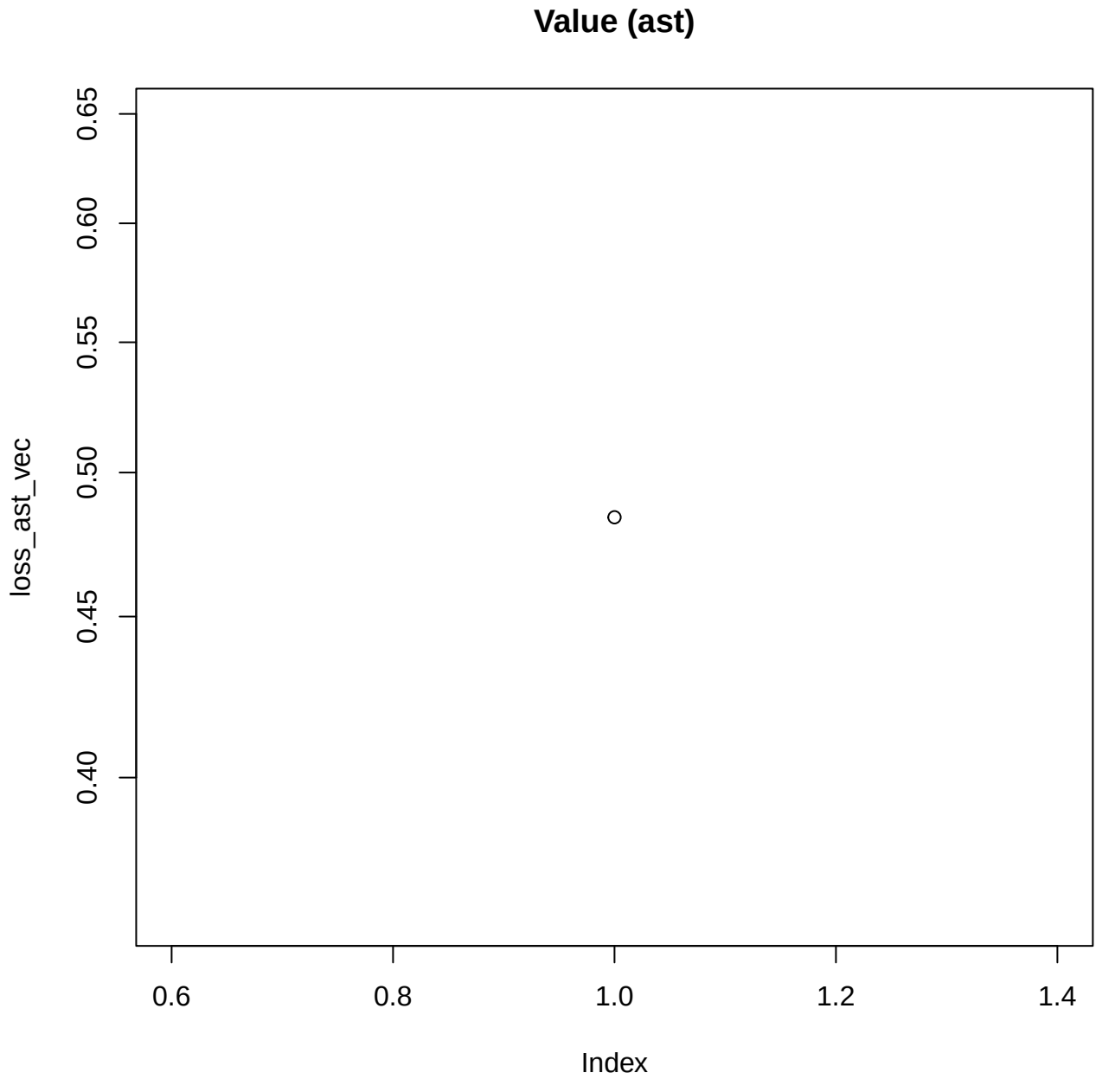
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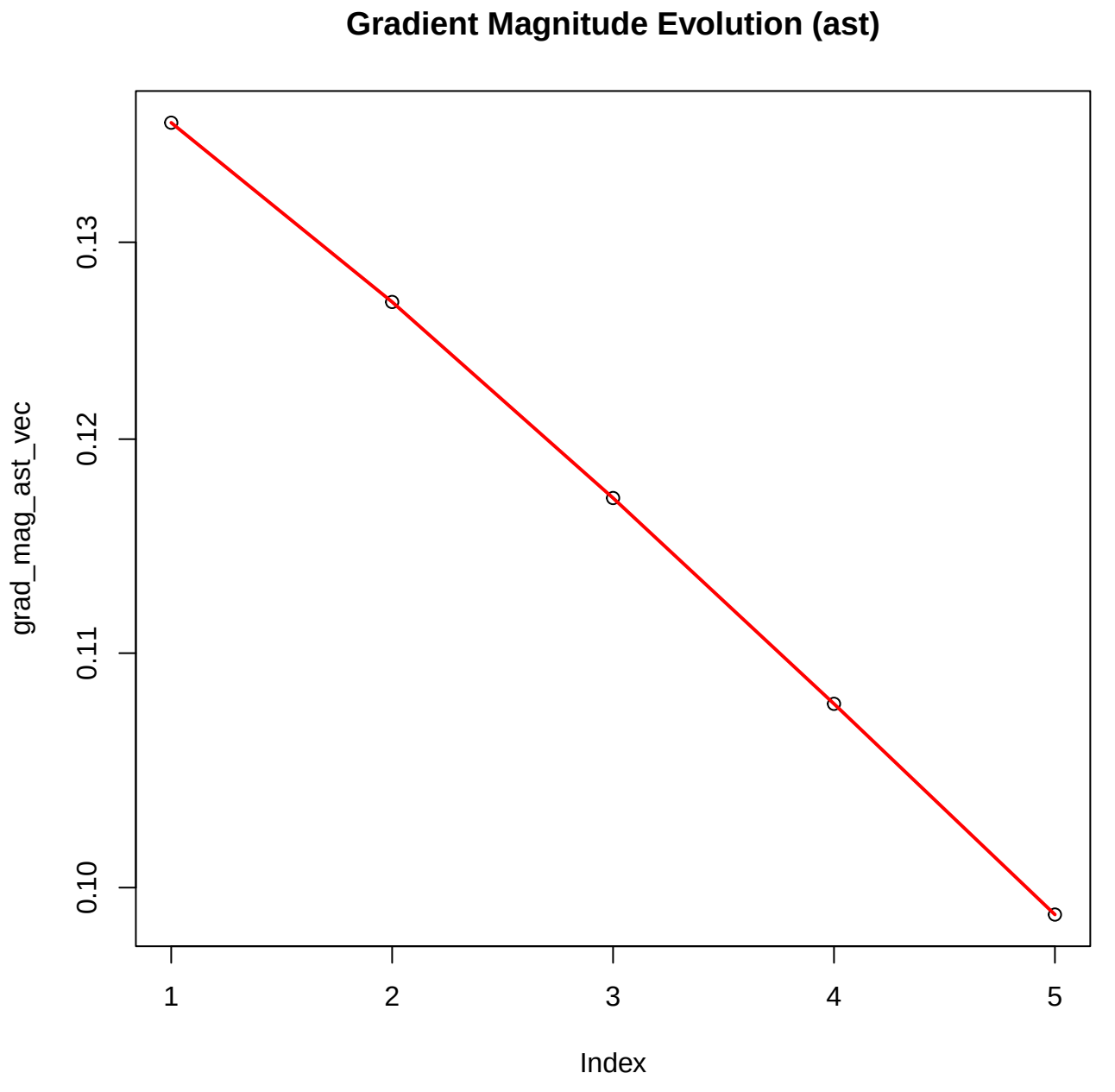
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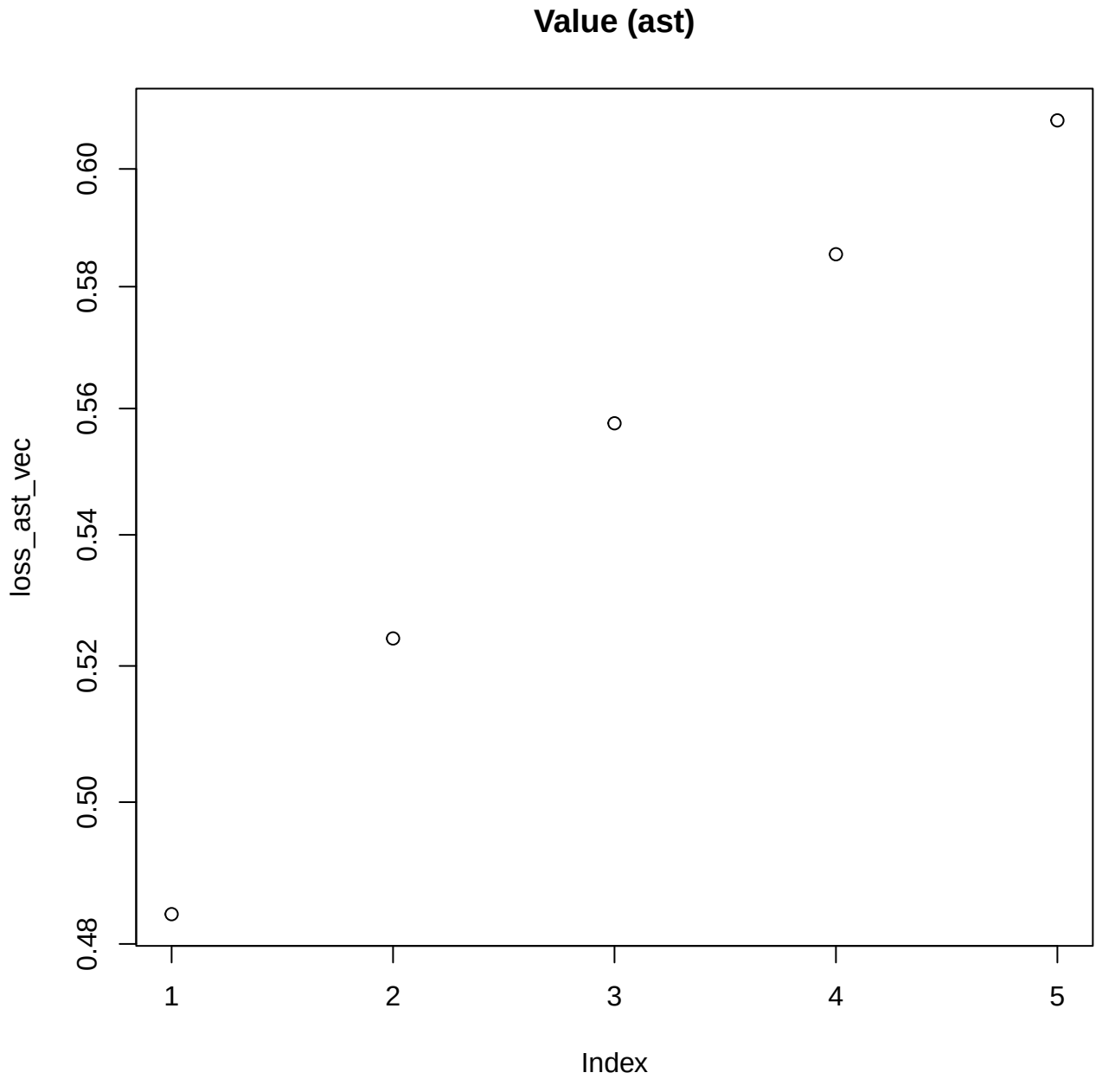
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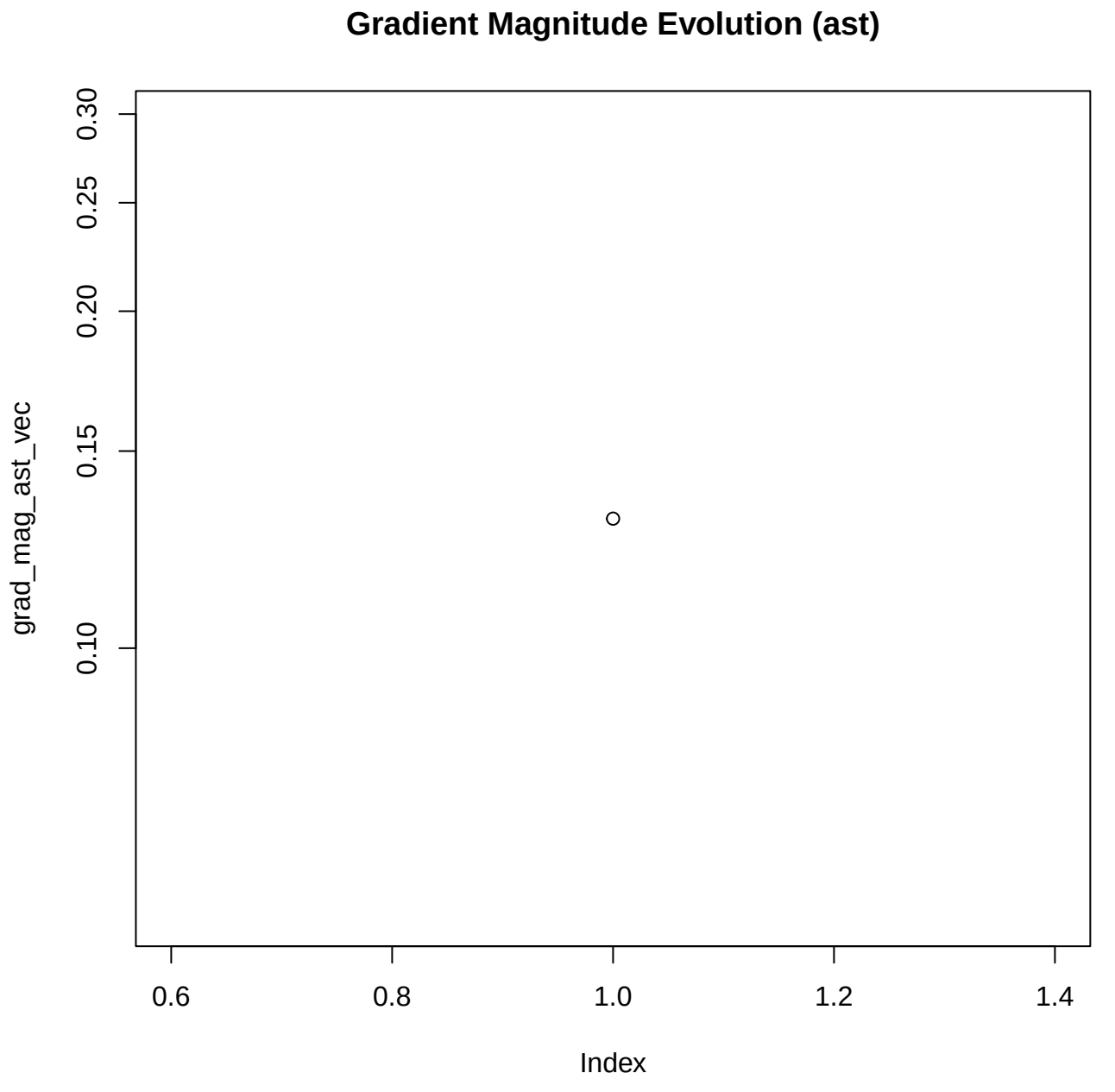
Gradient Magnitude Evolution (ast)
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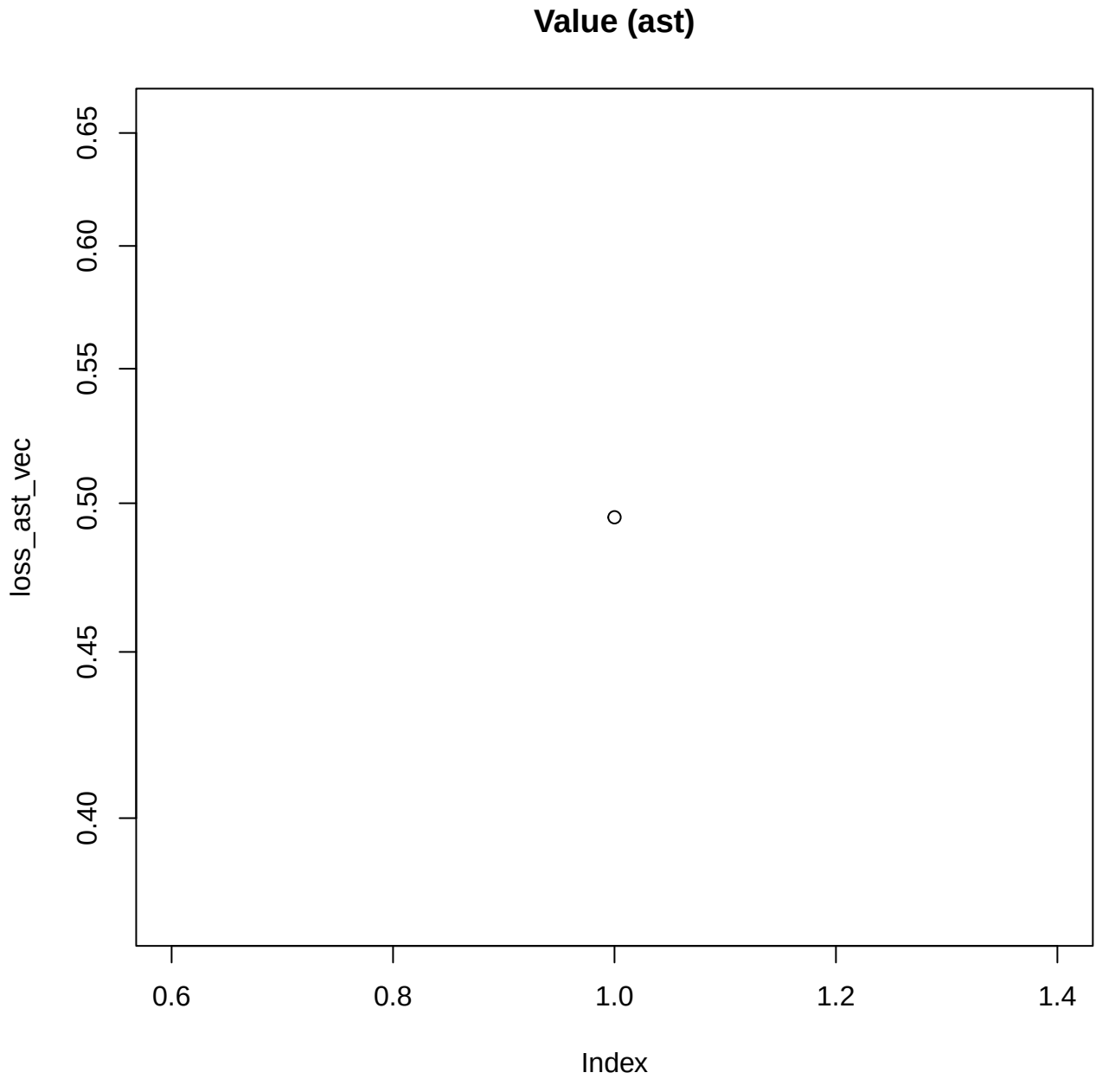
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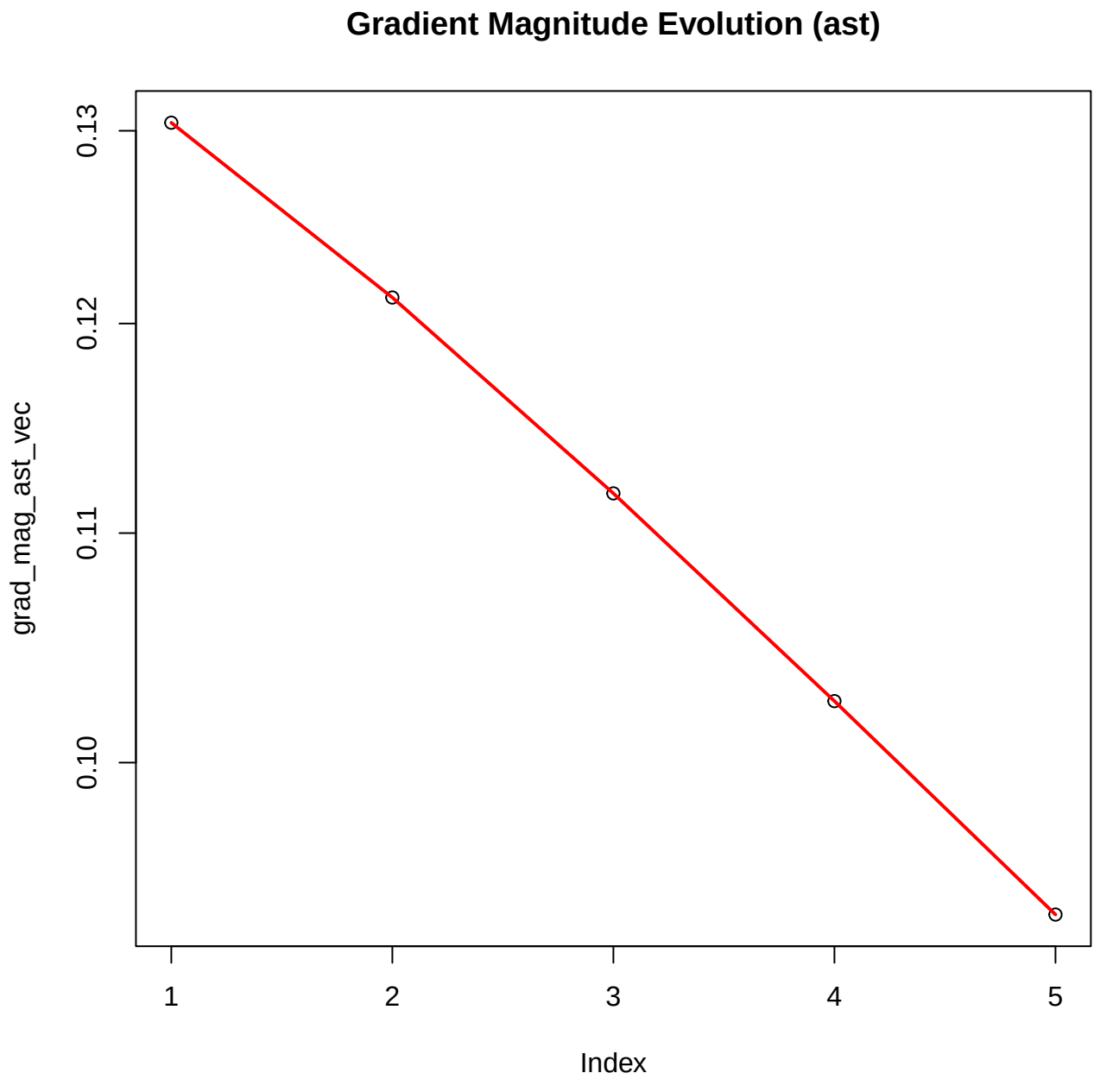
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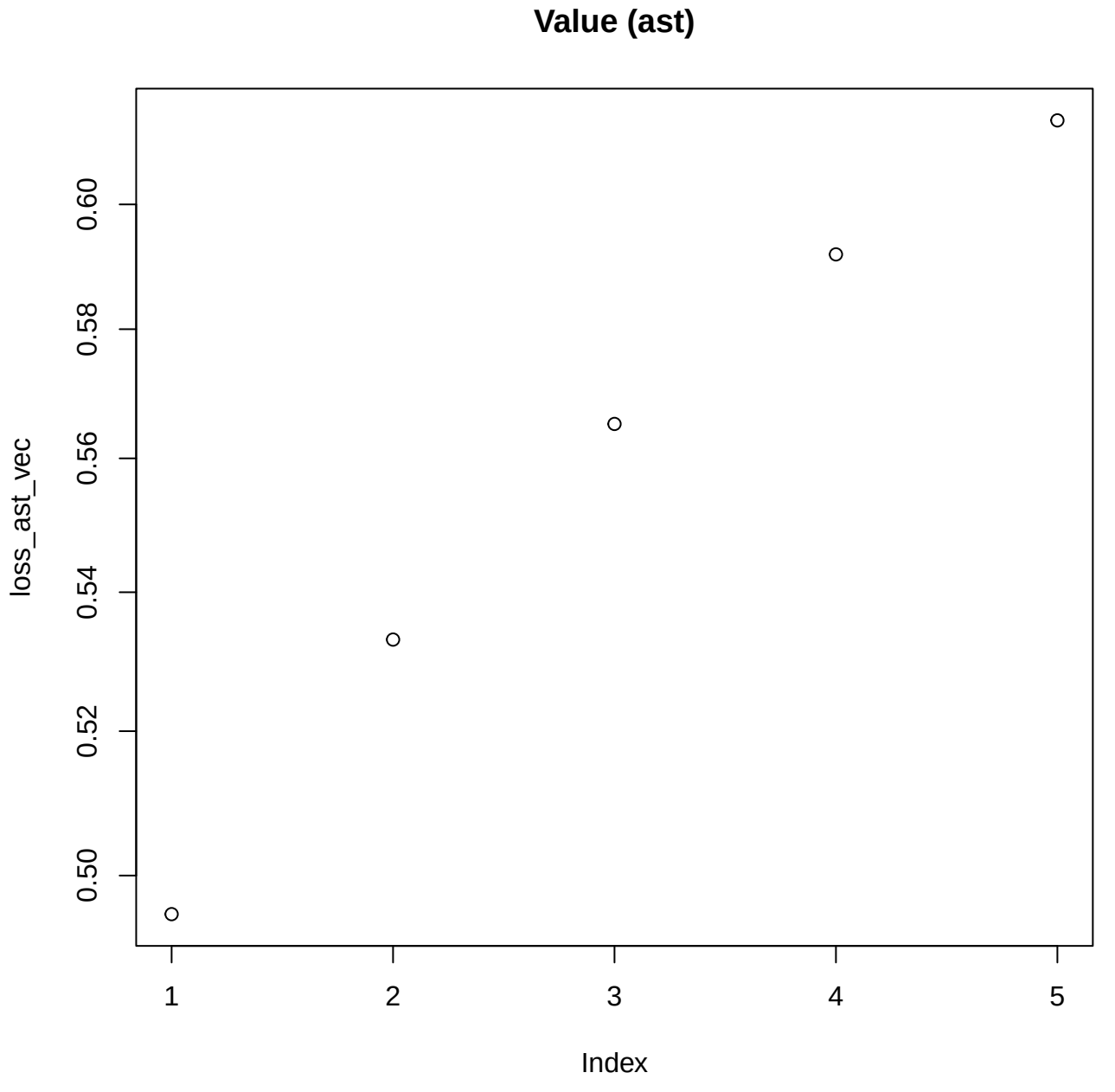
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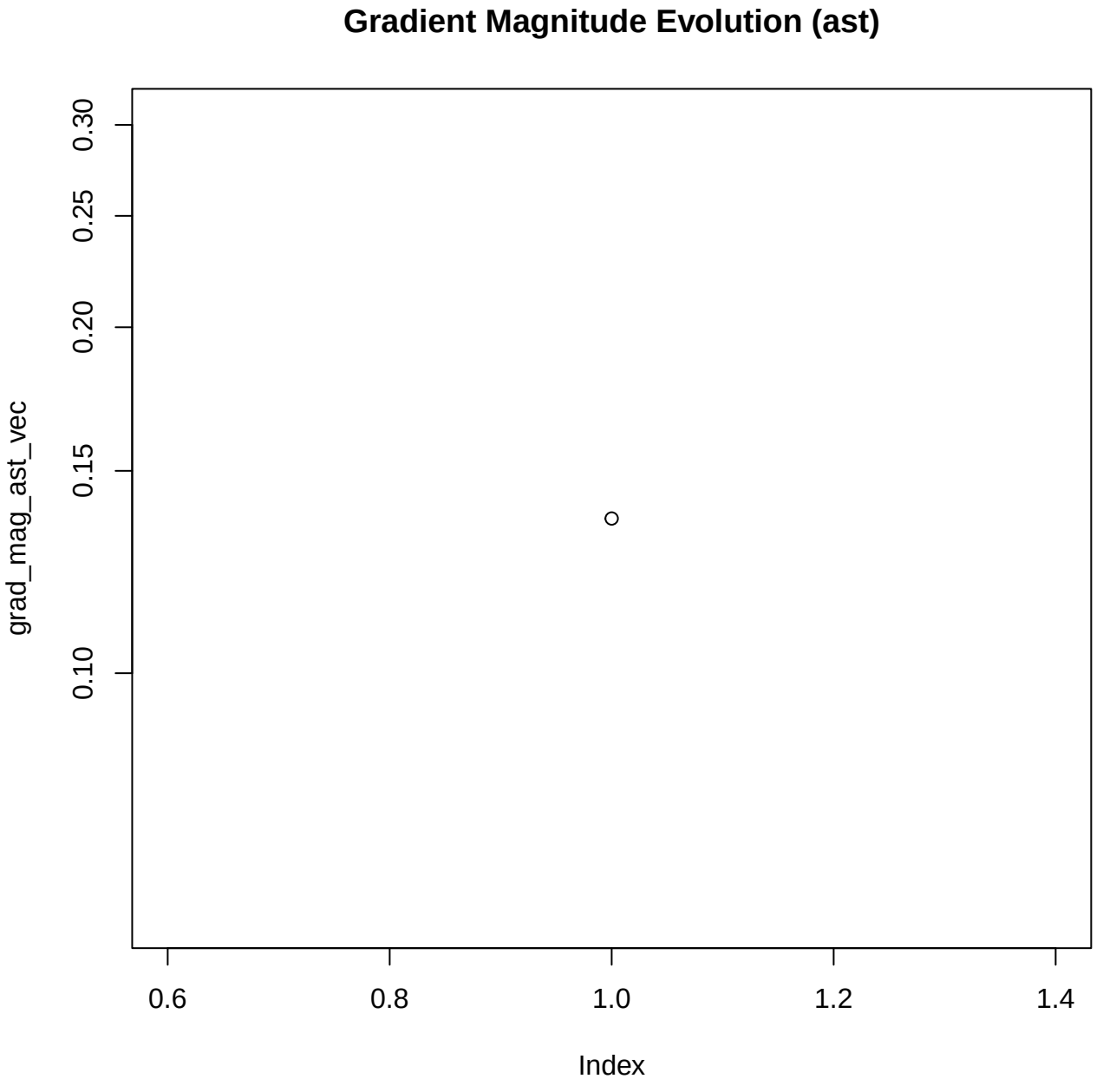
Gradient Magnitude Evolution (ast)
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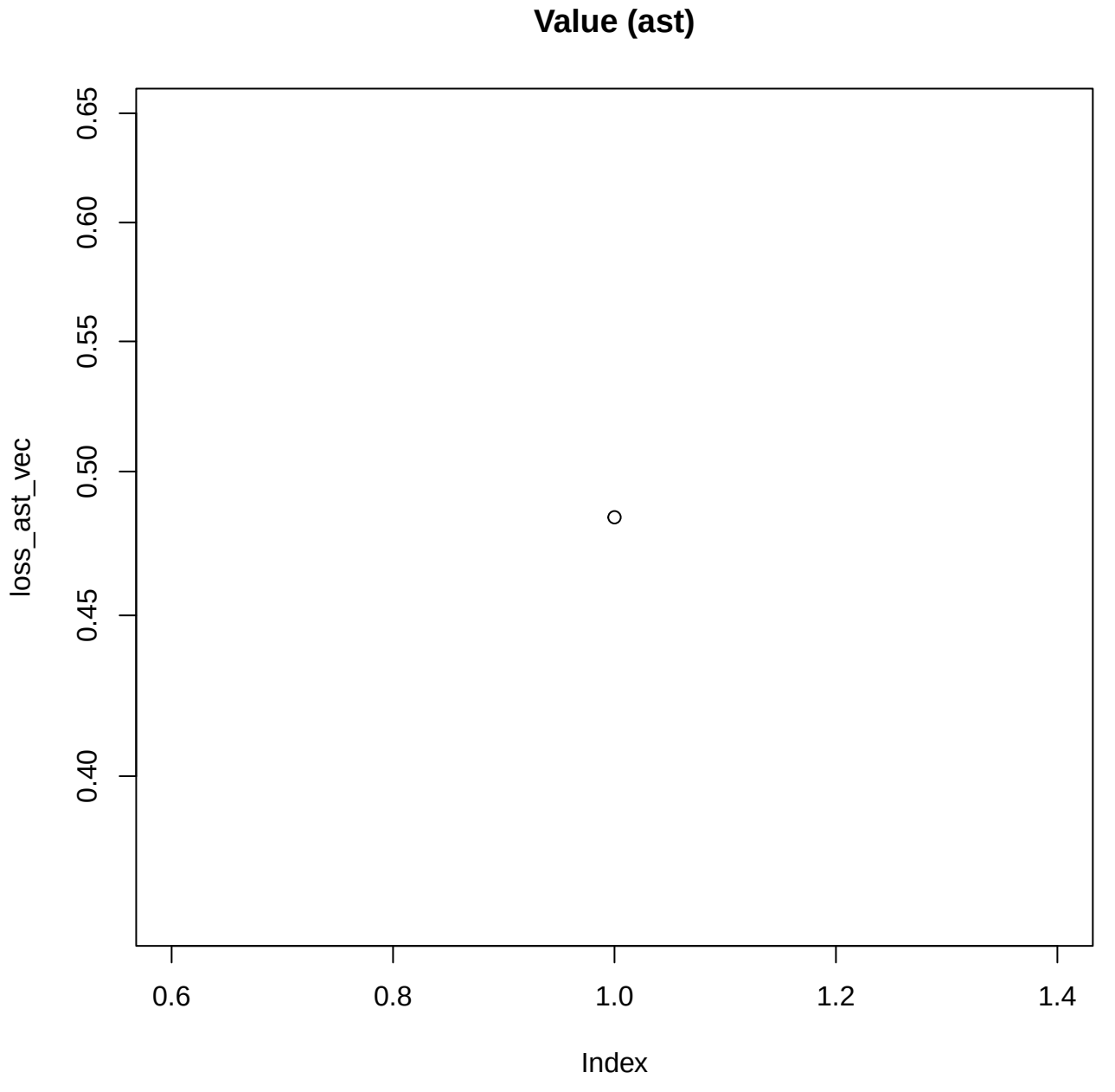
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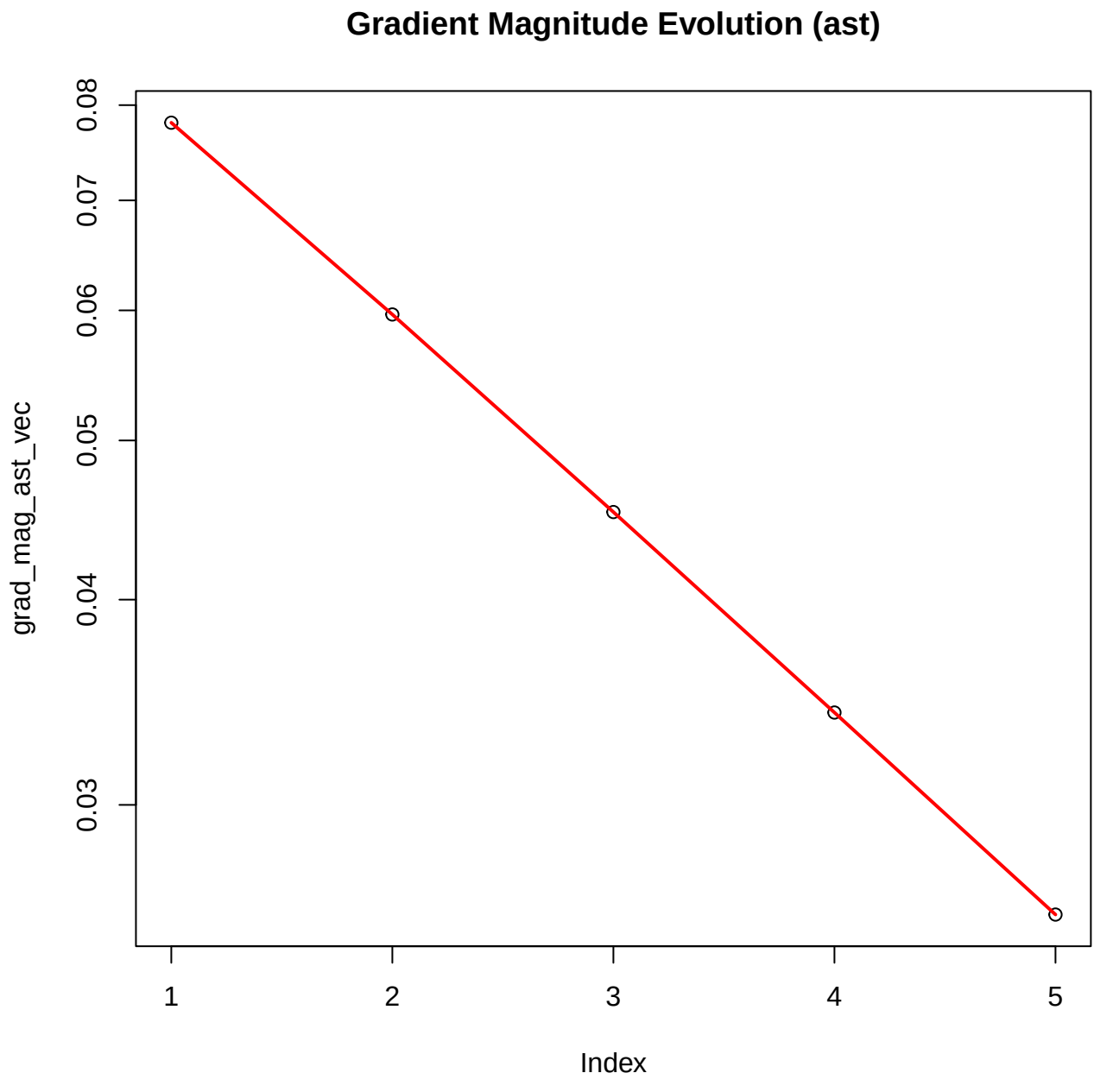
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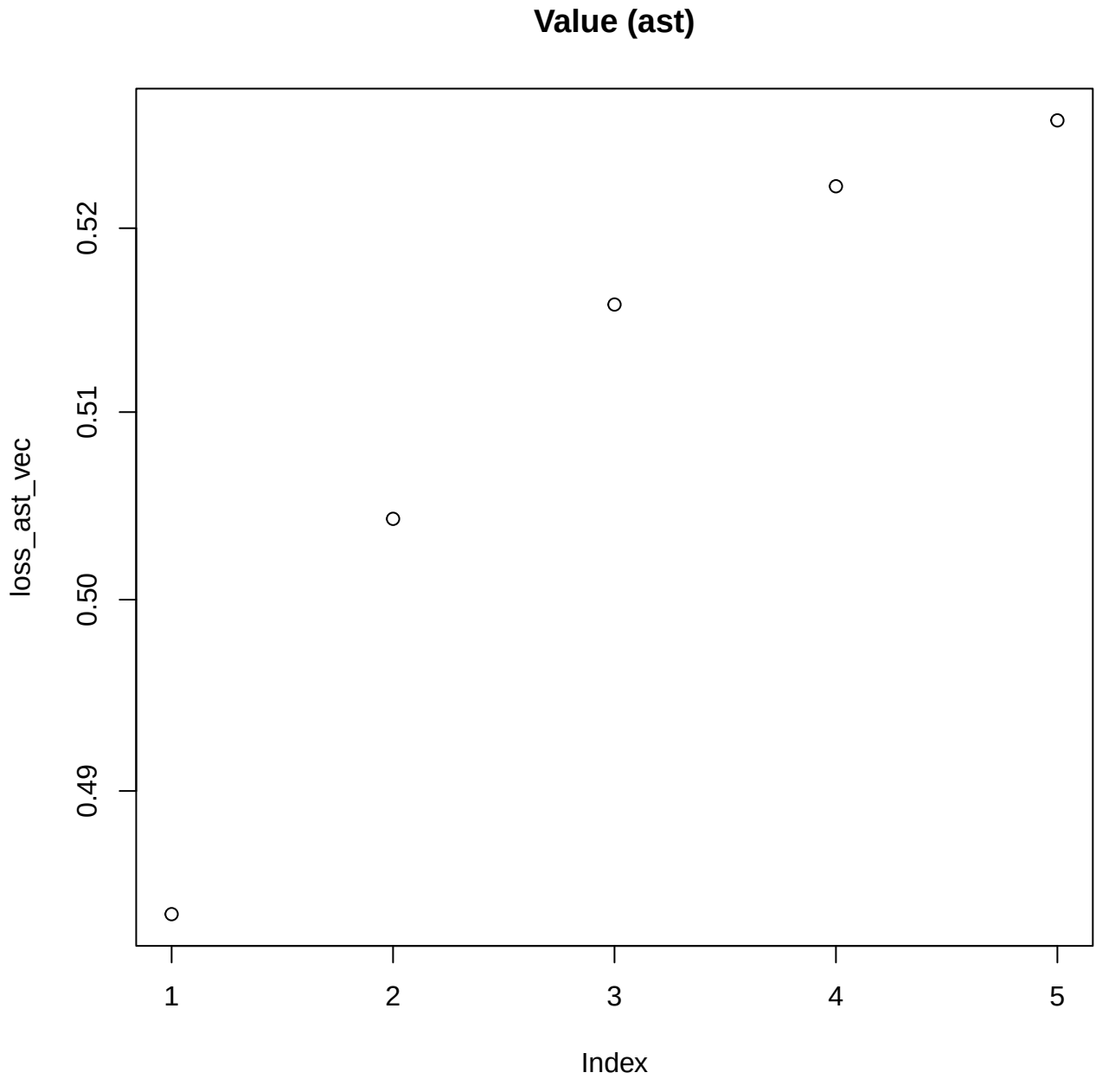
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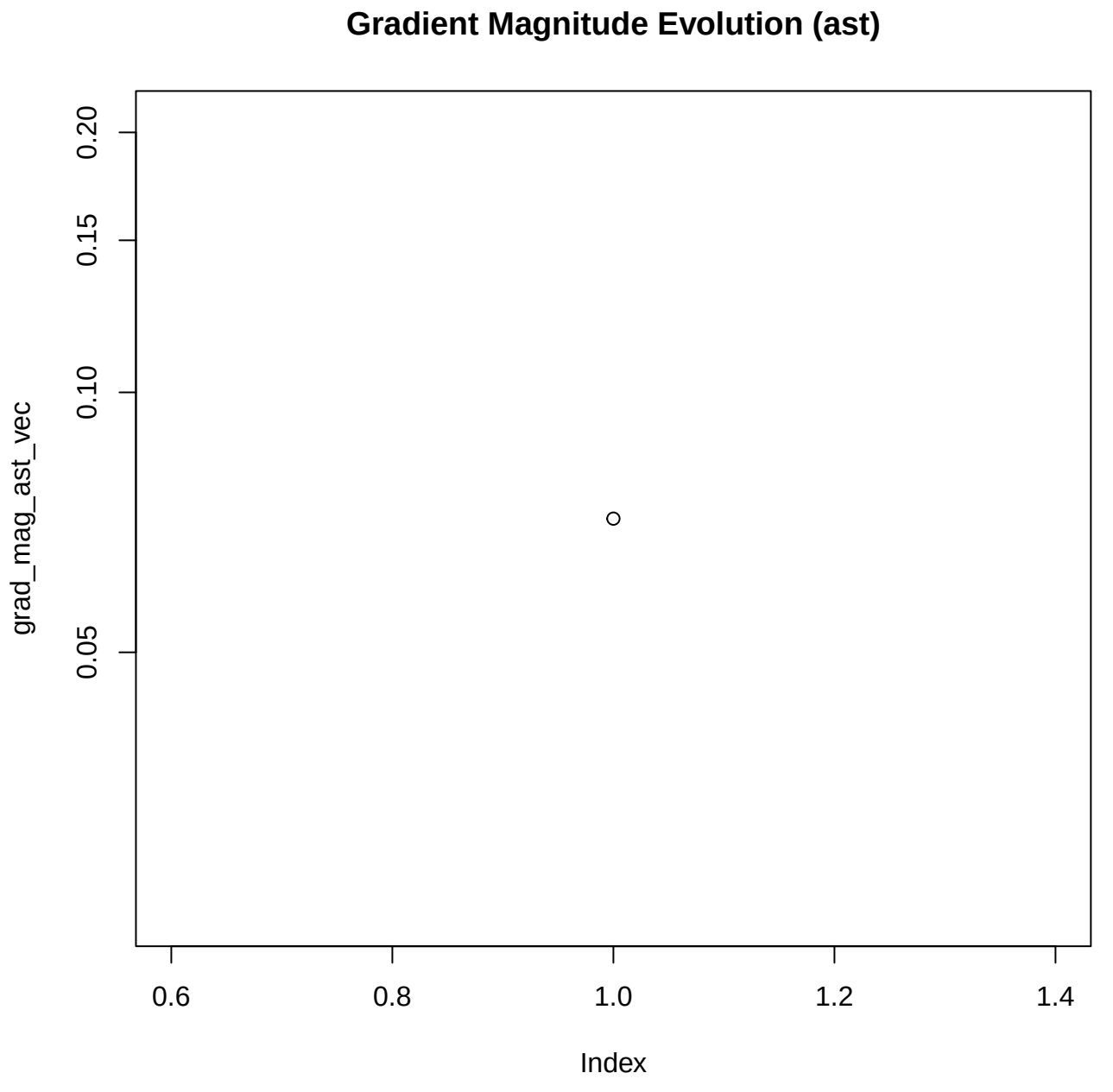
Gradient Magnitude Evolution (ast)
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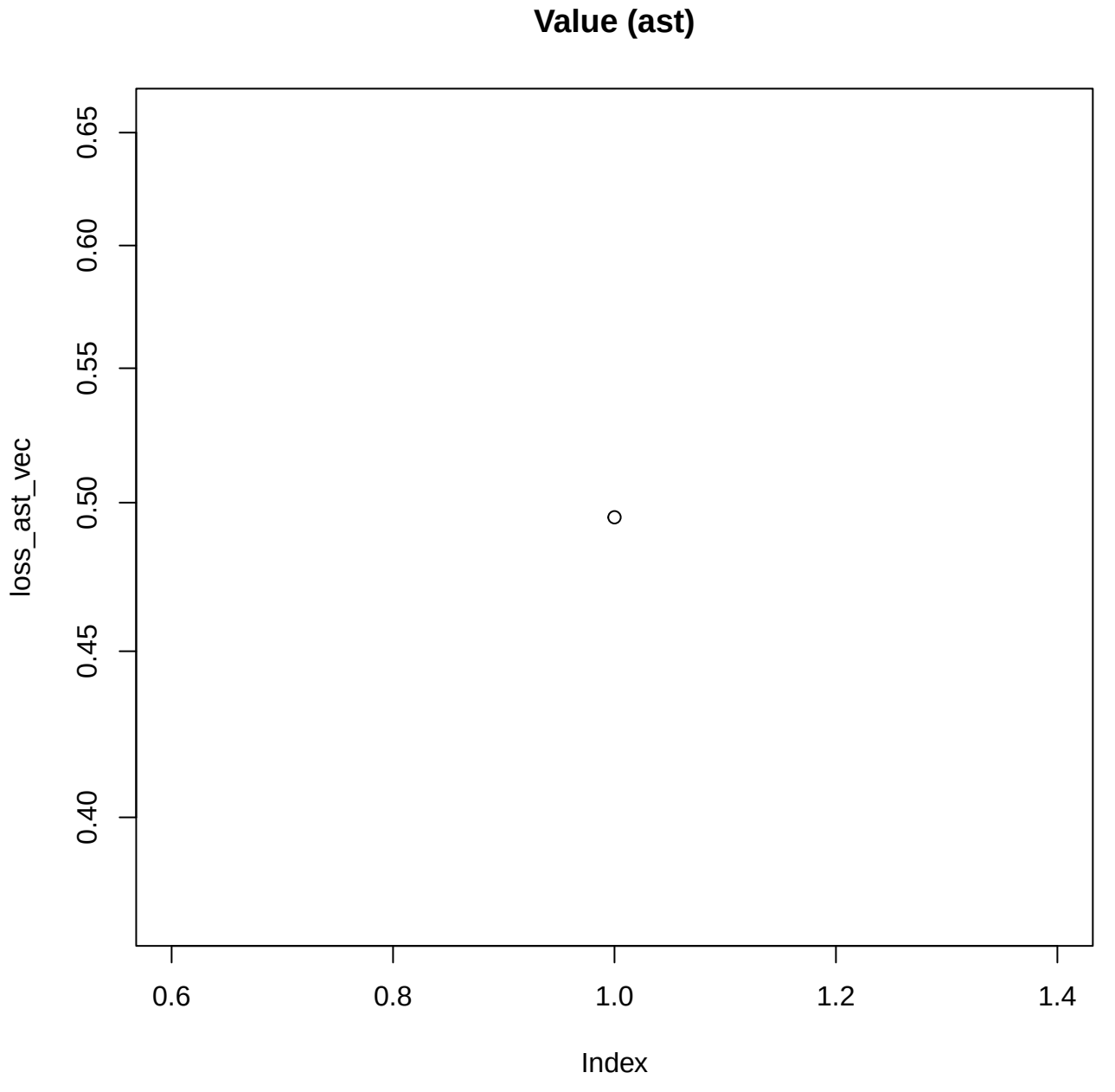
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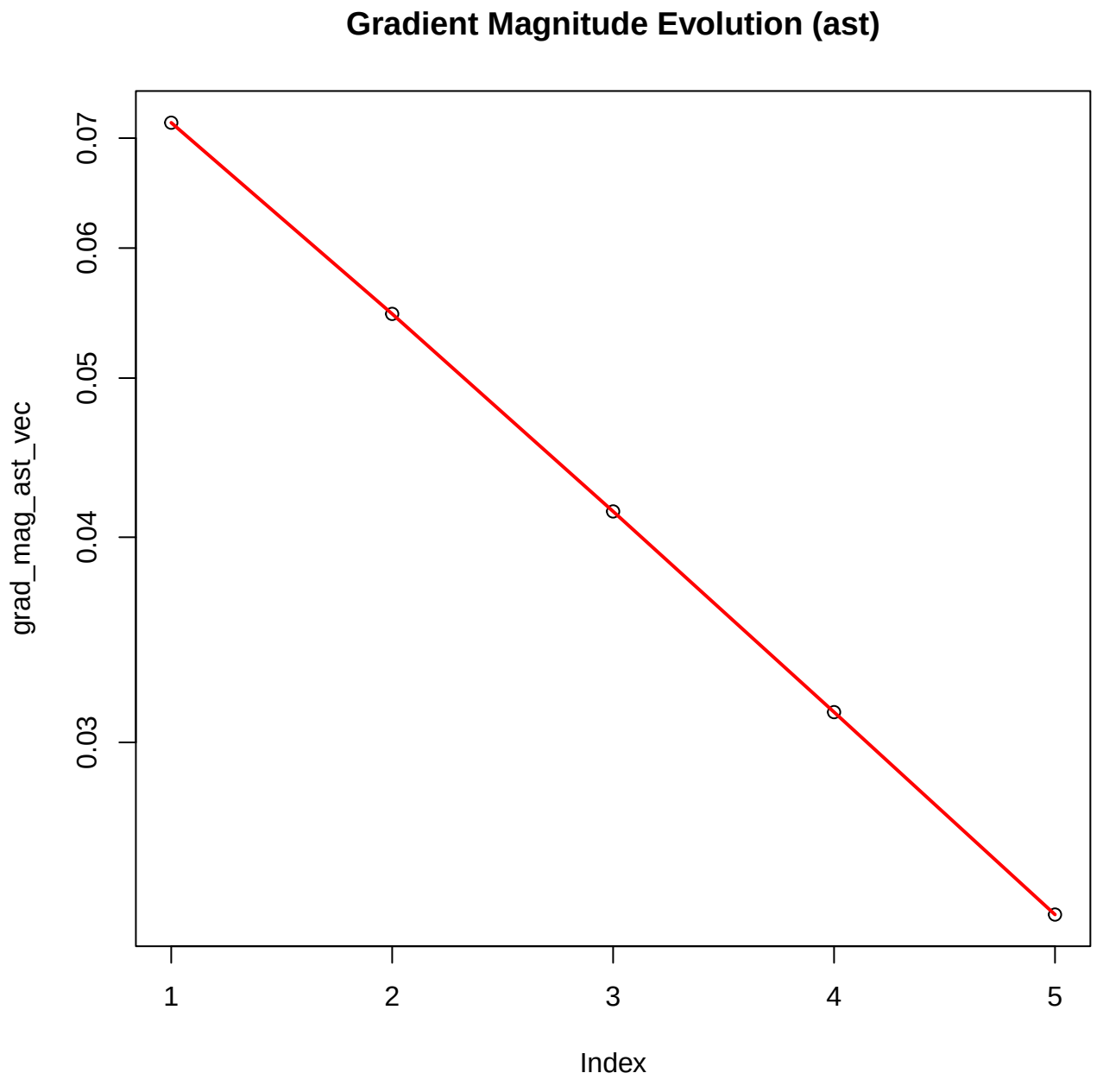
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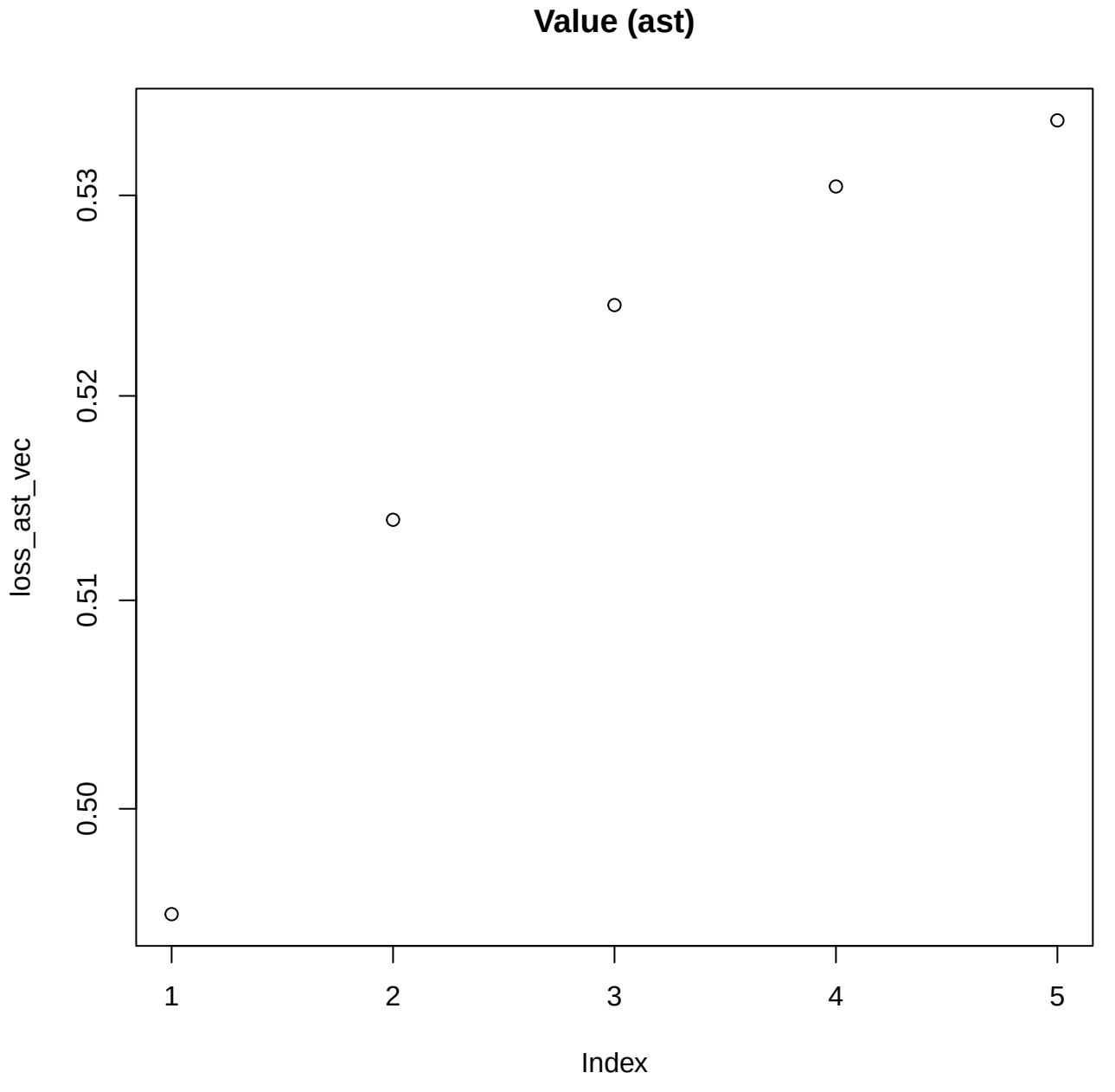
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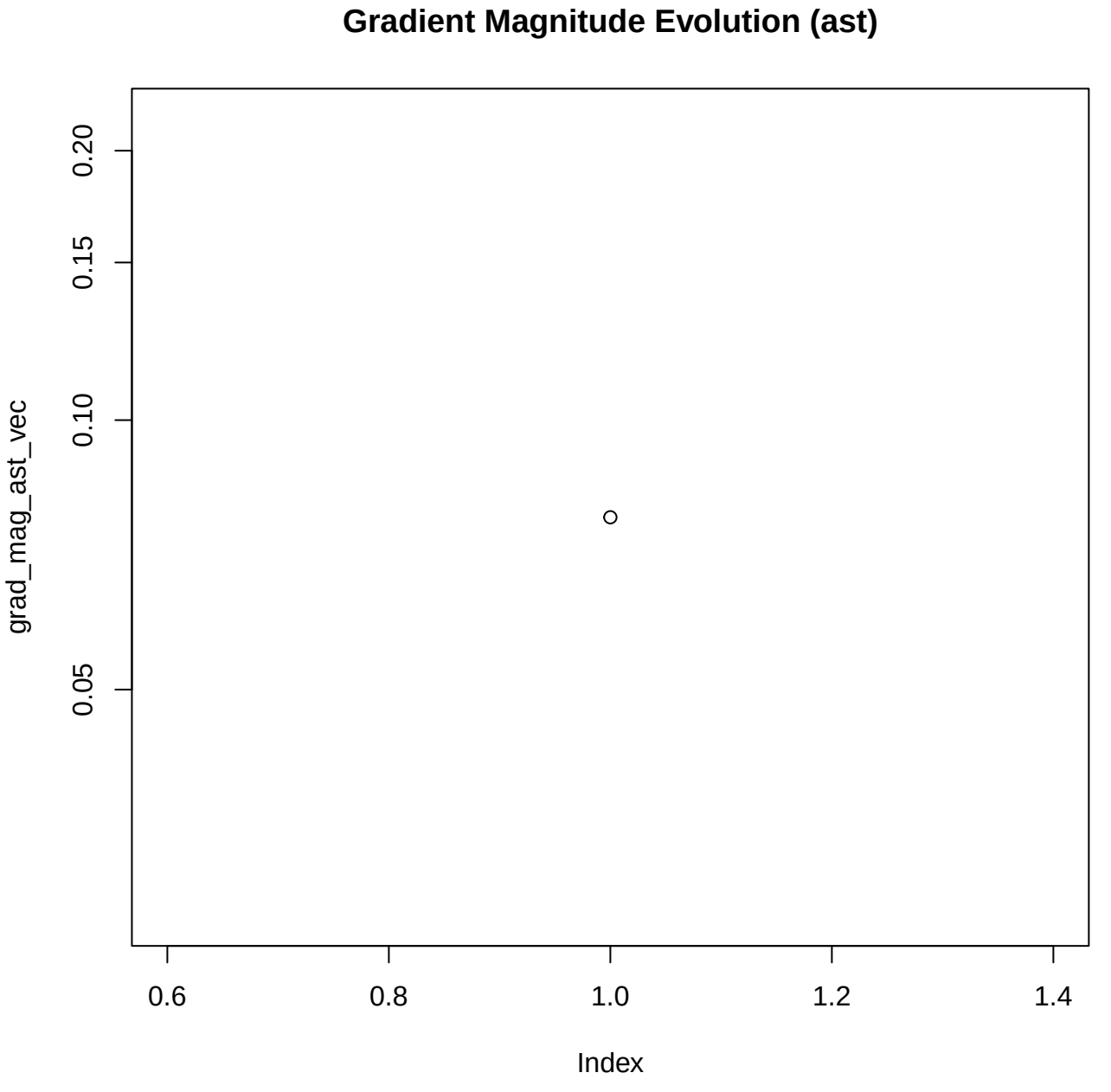
Gradient Magnitude Evolution (ast)
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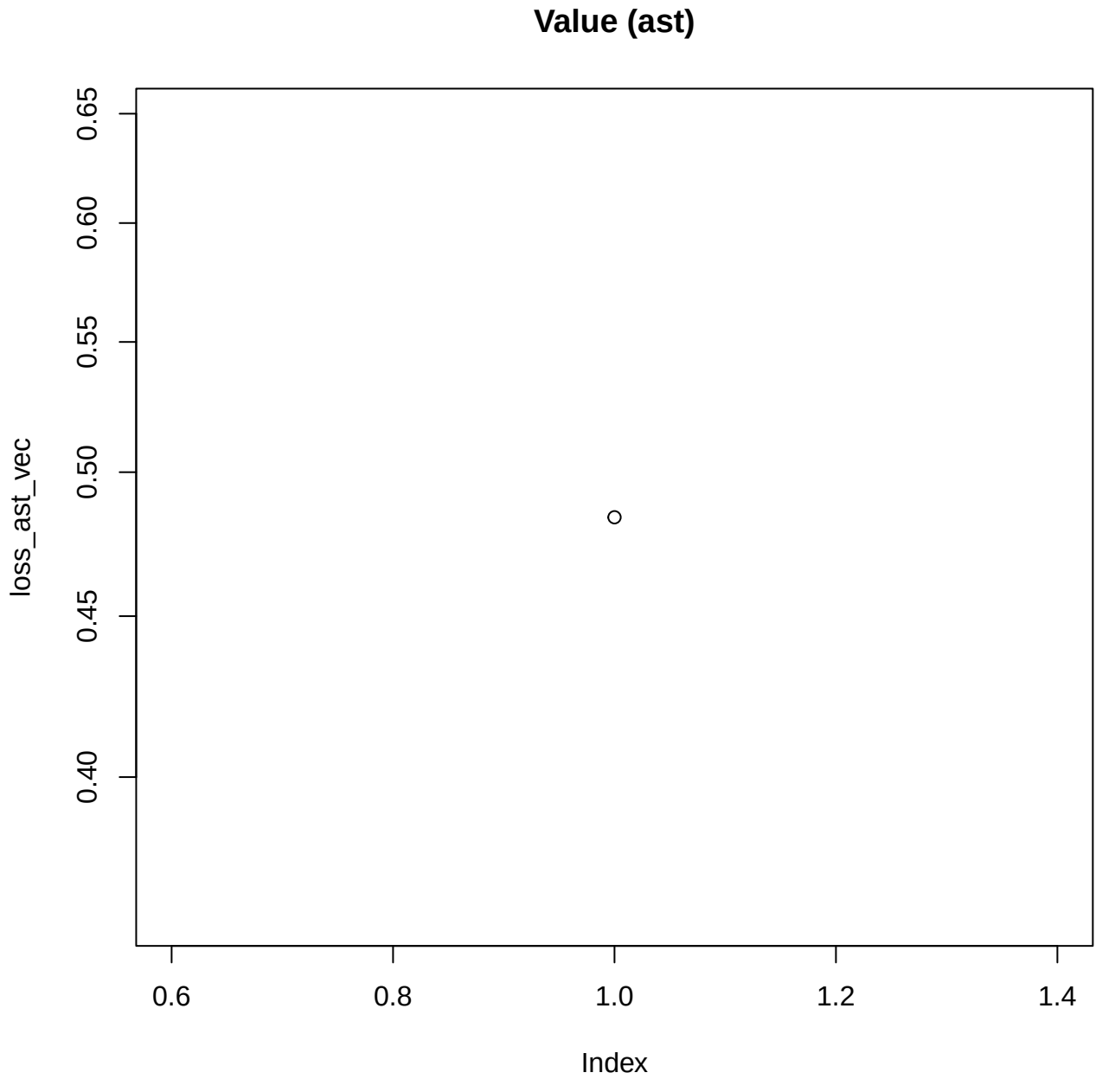
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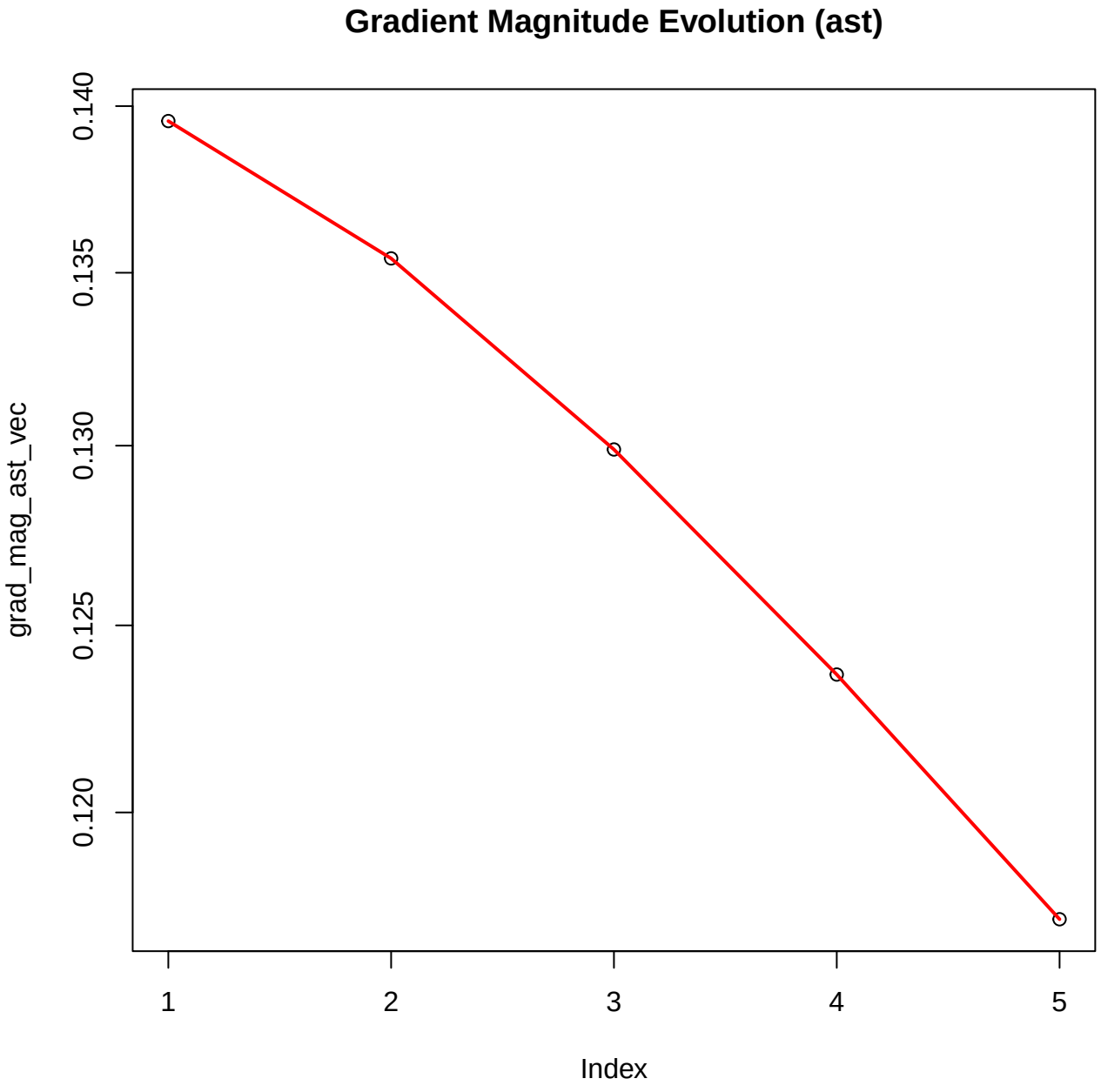
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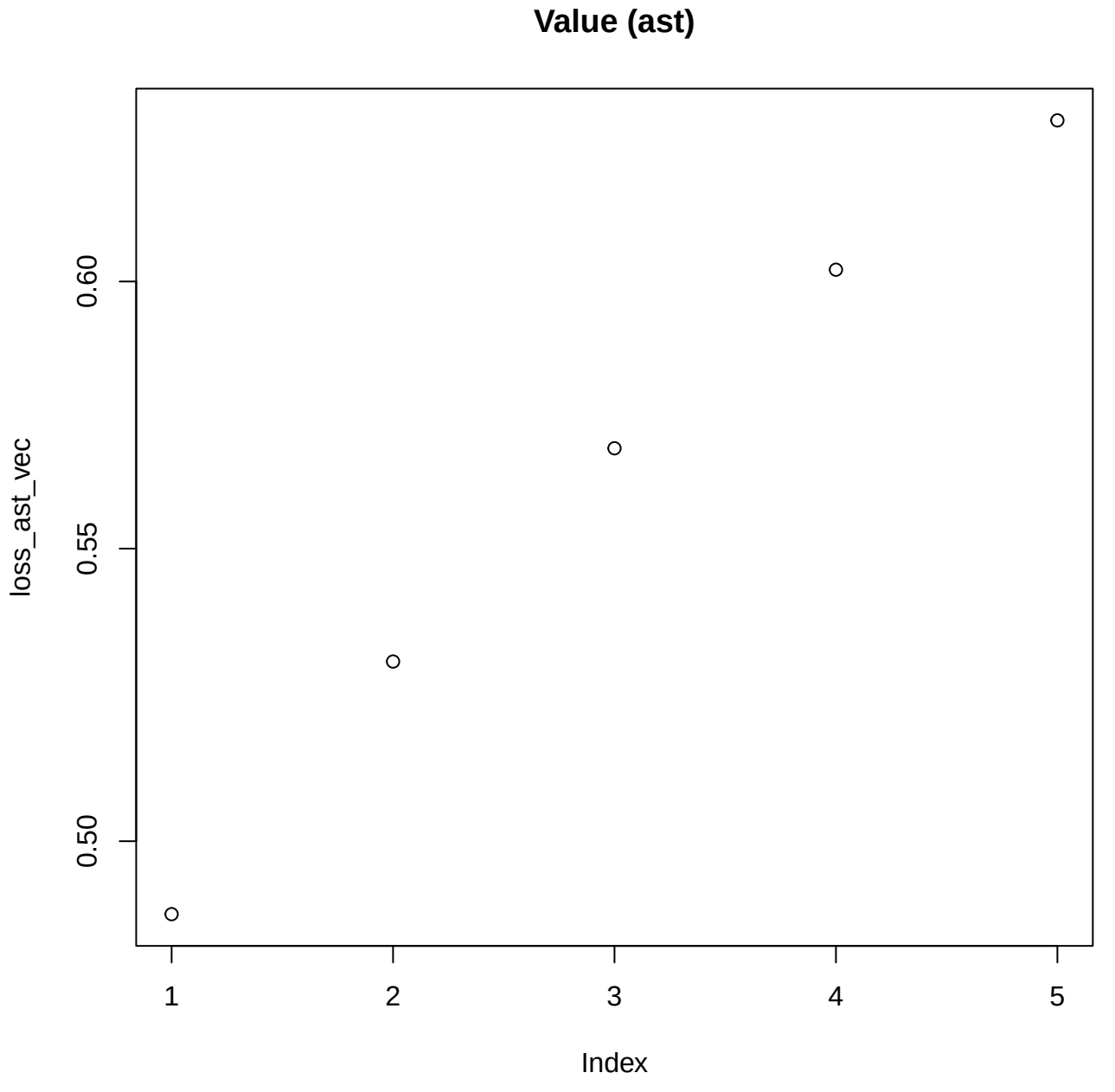
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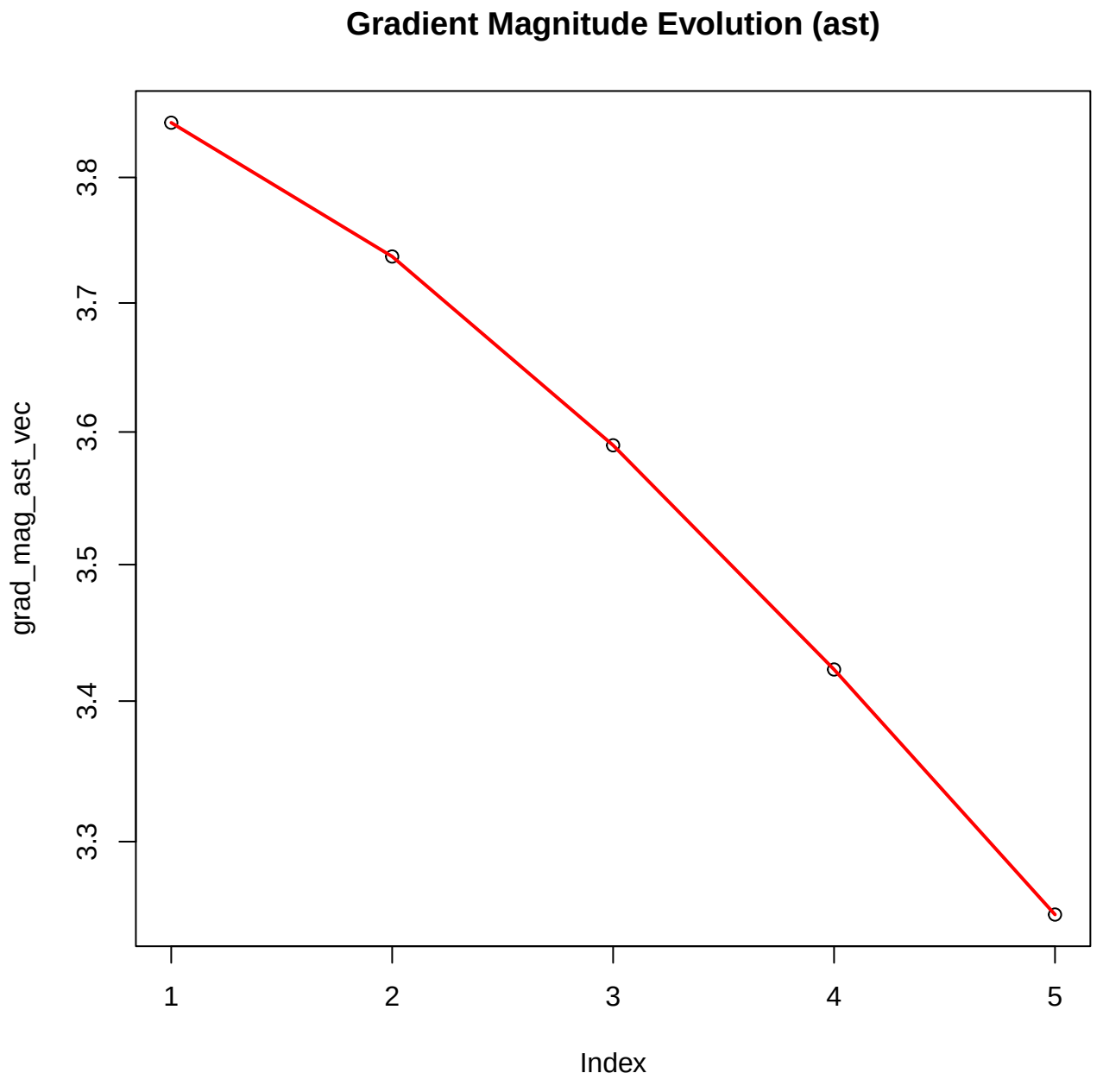
Gradient Magnitude Evolution (ast)
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3
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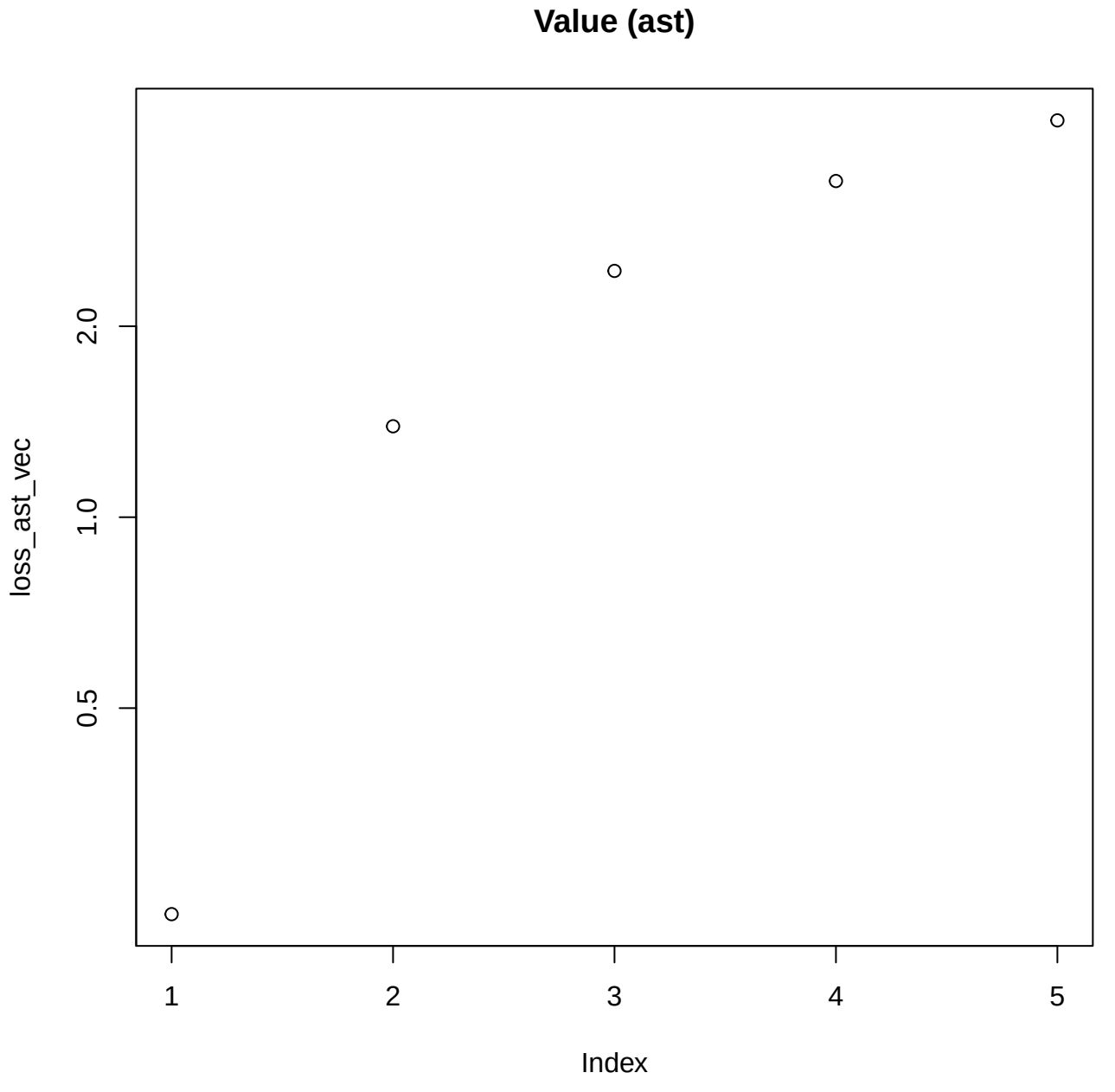
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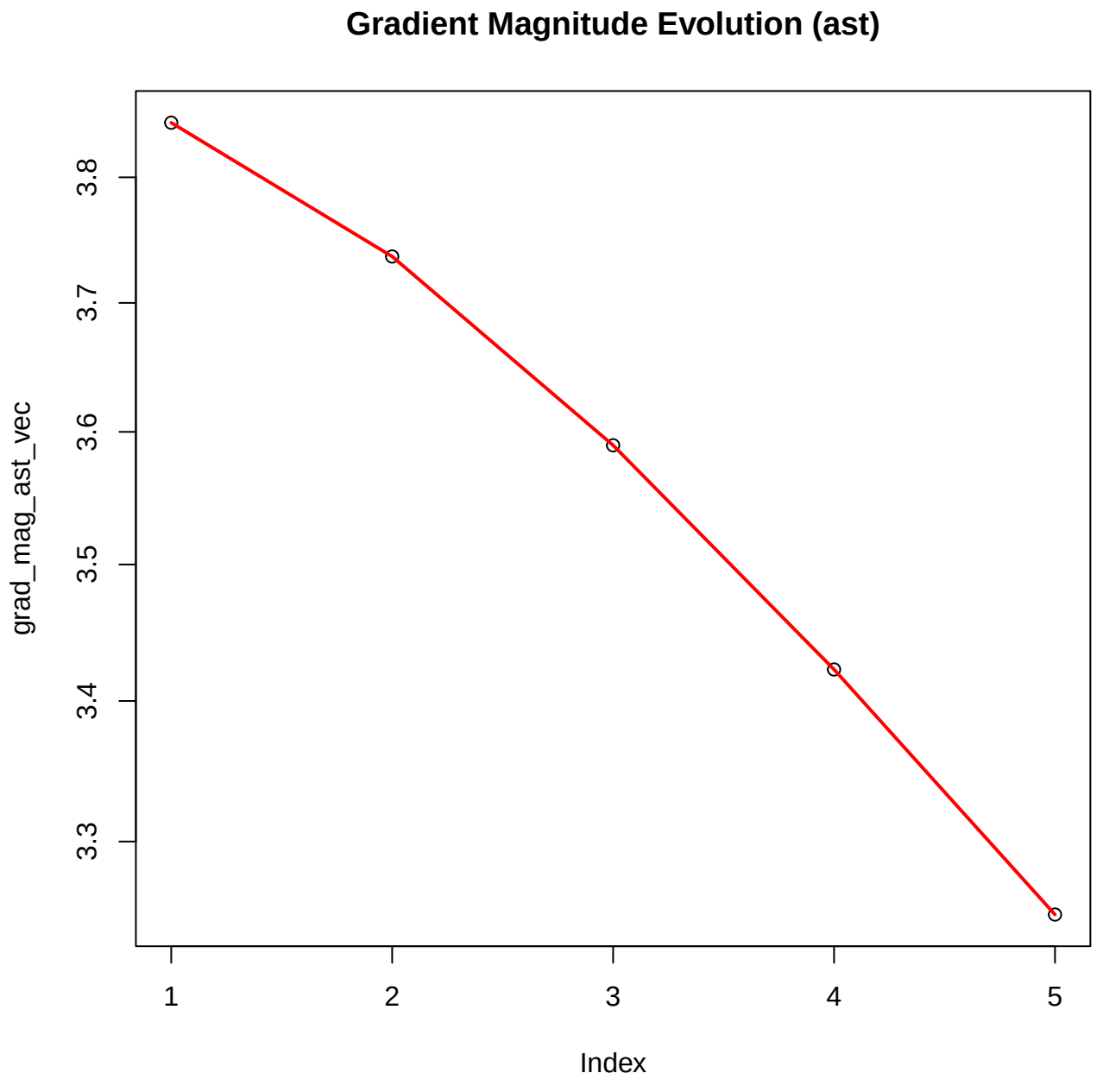
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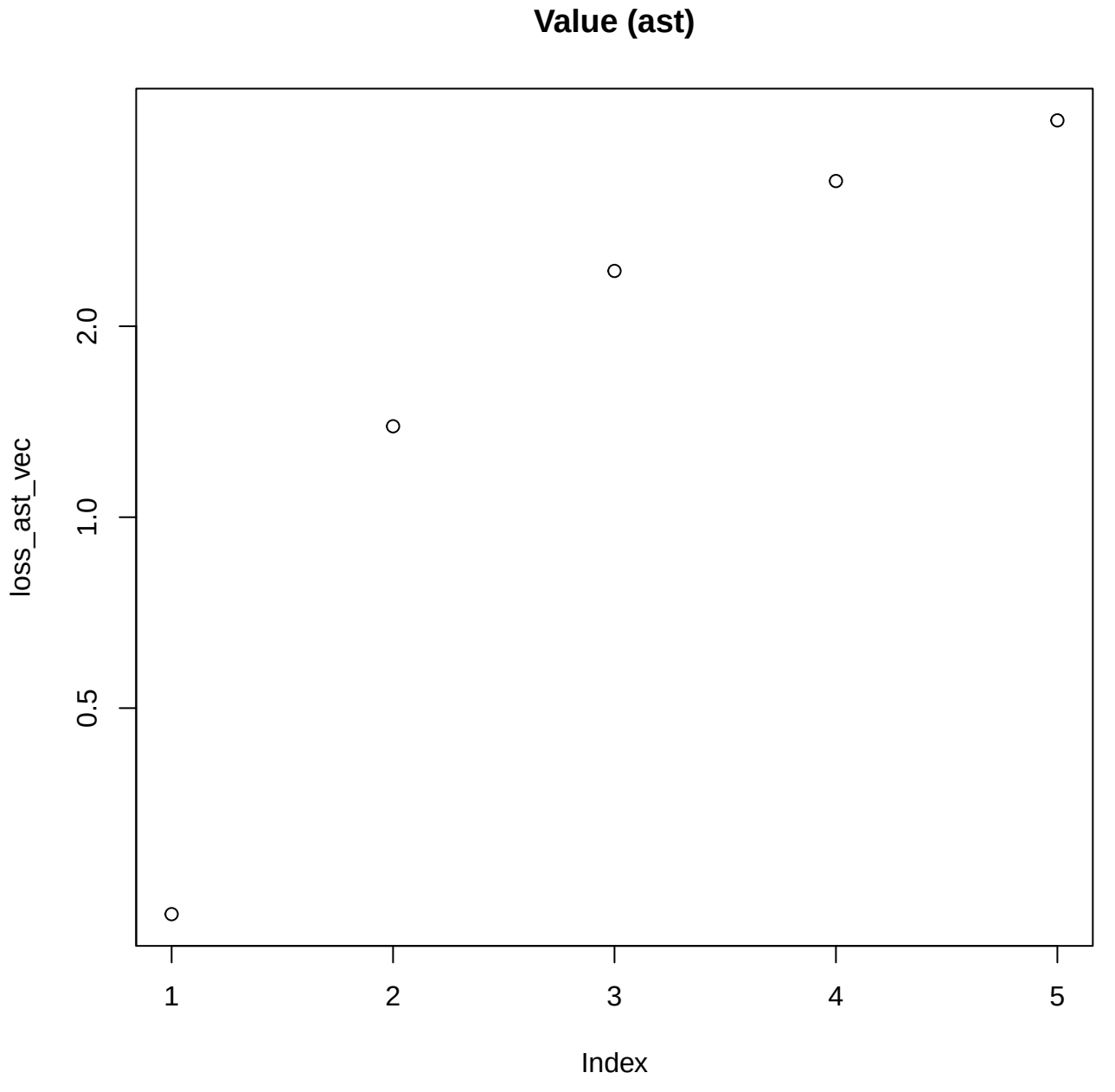
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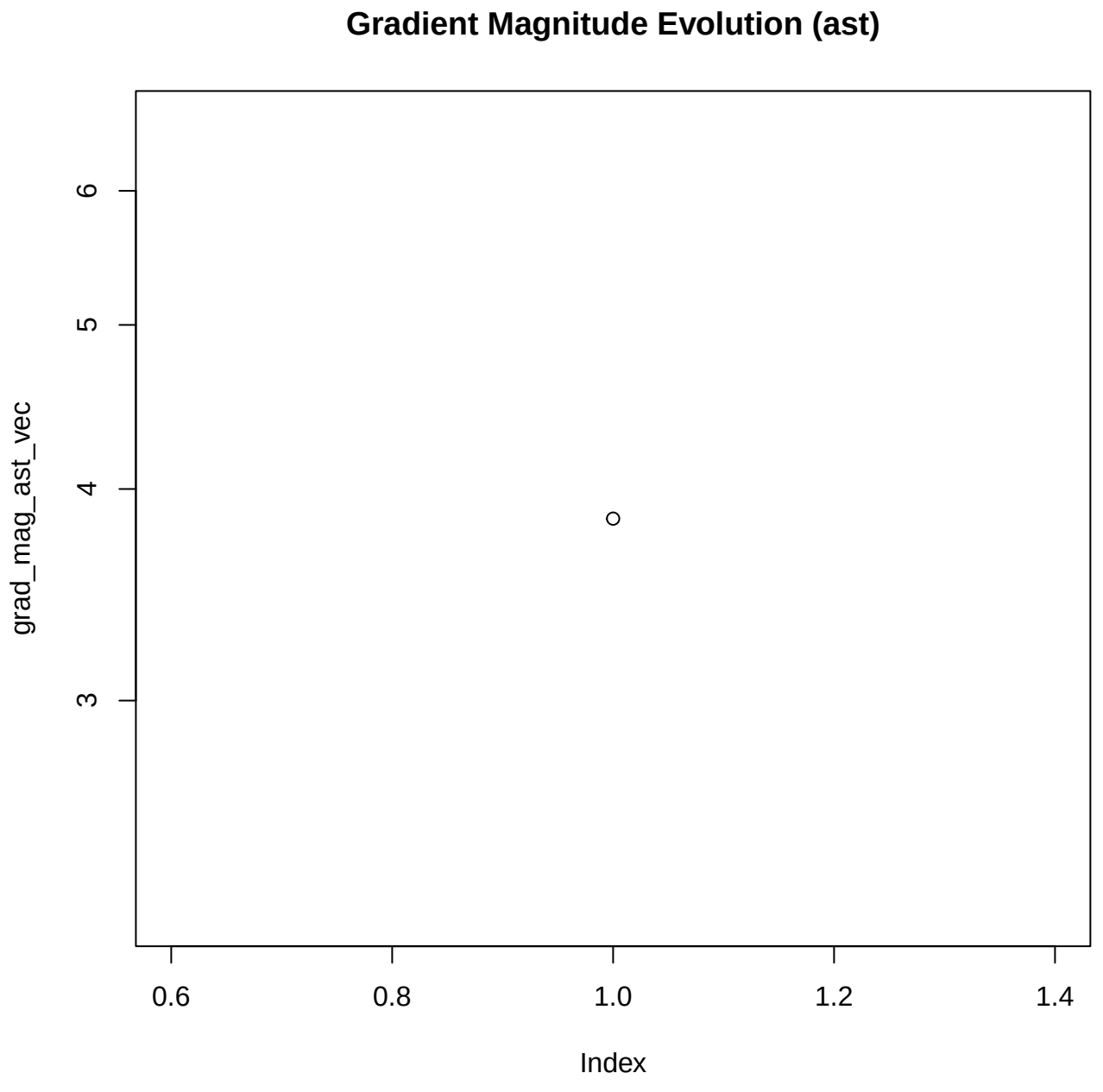
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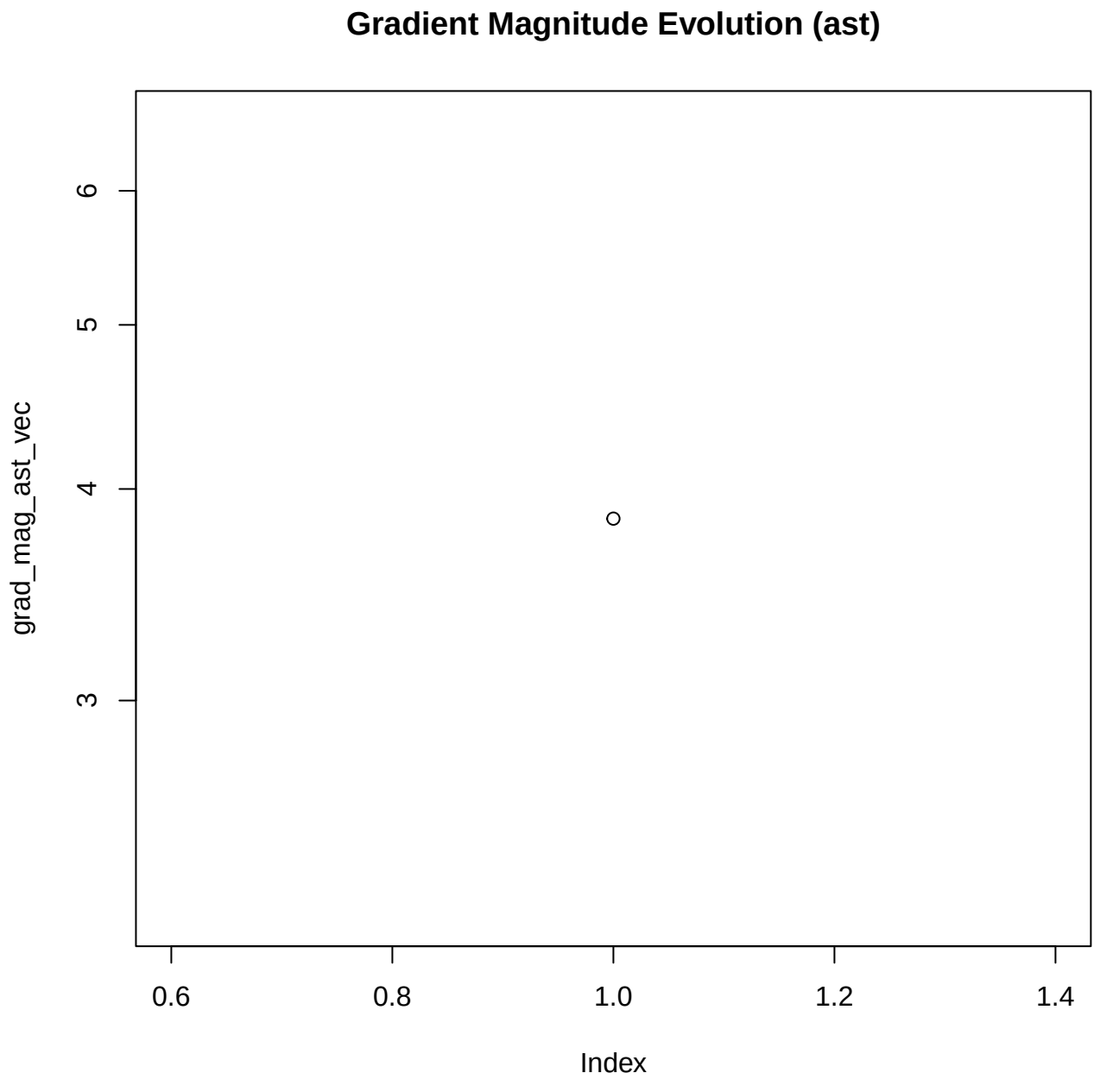
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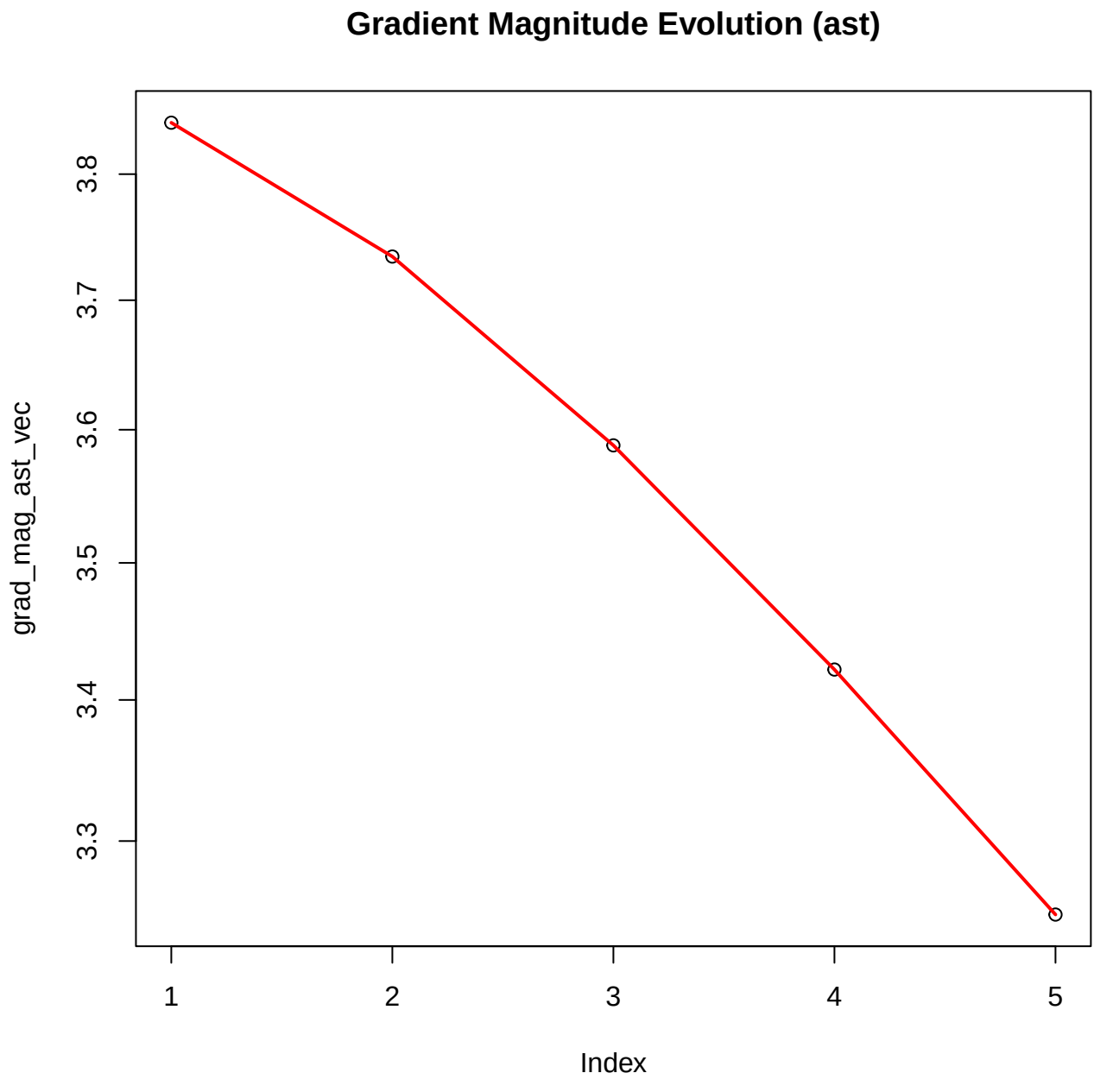
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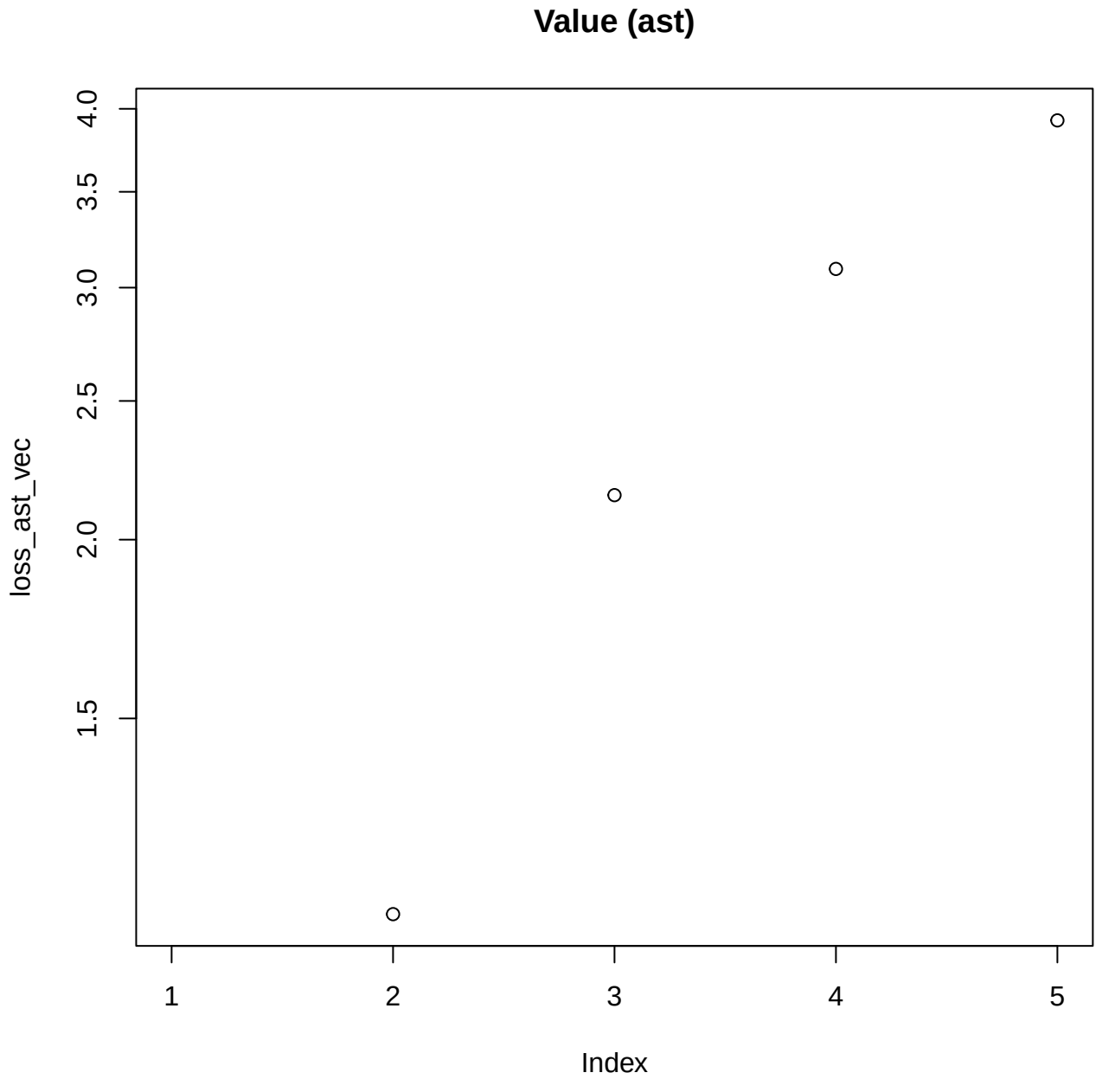
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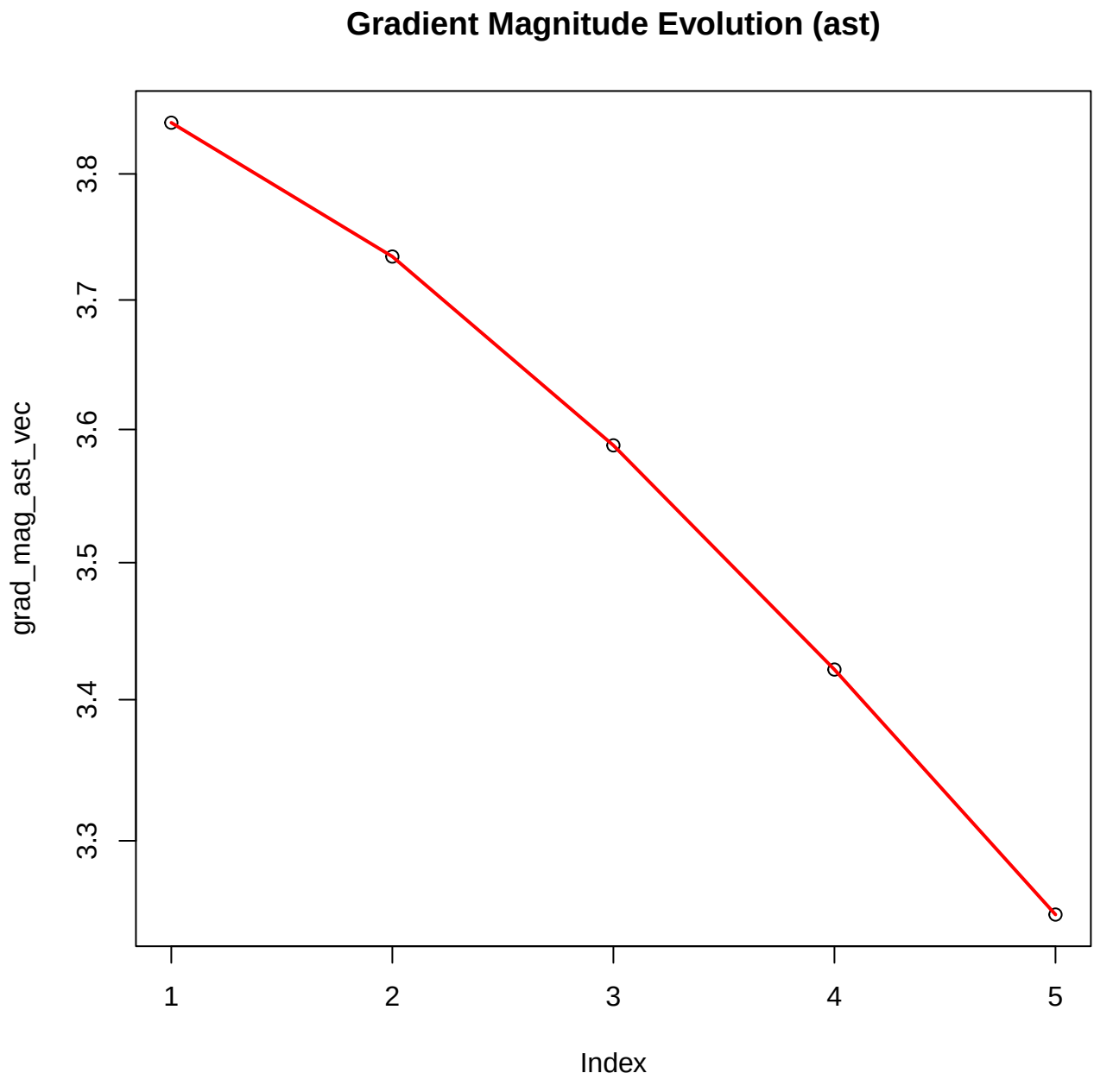
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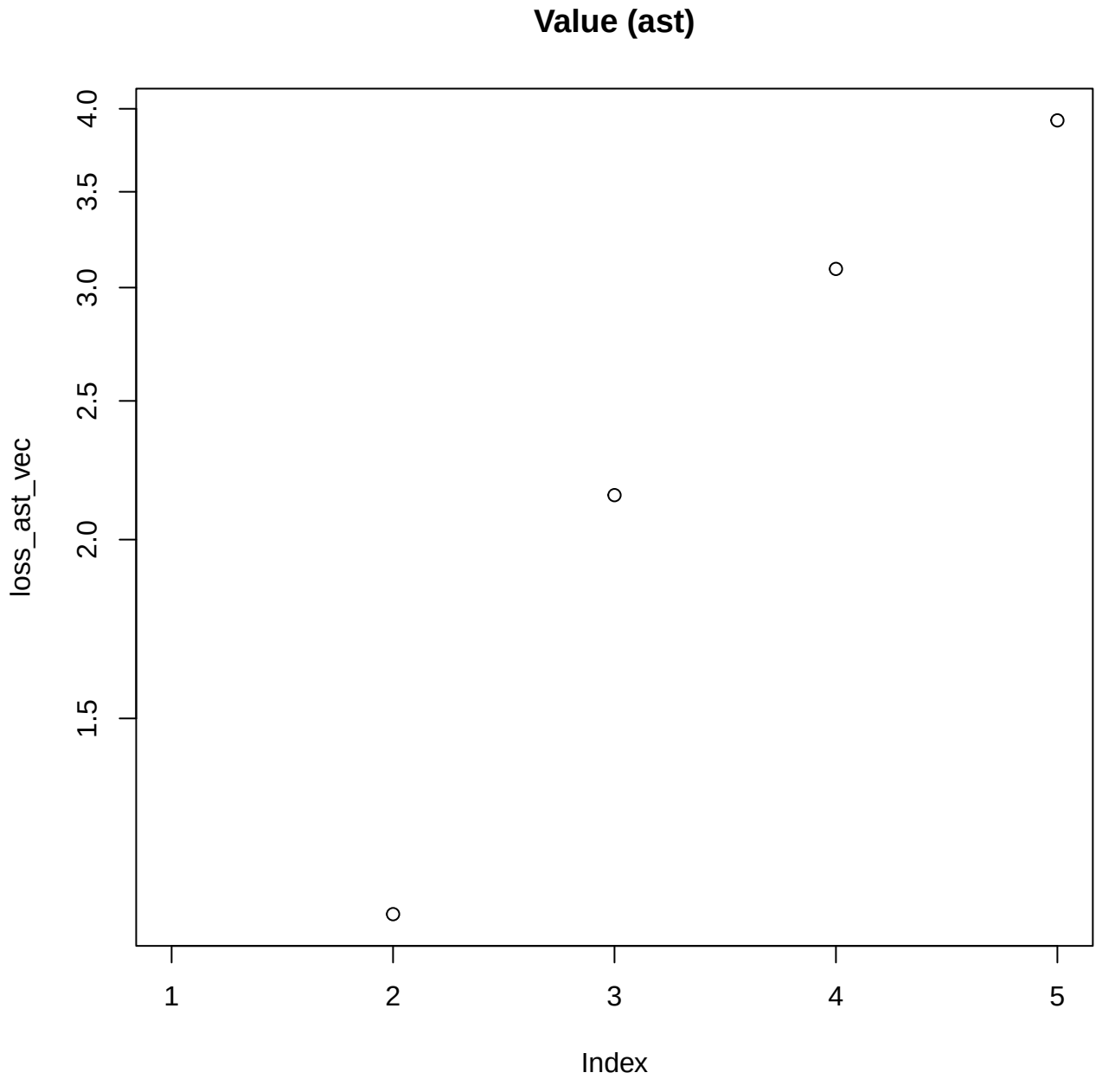
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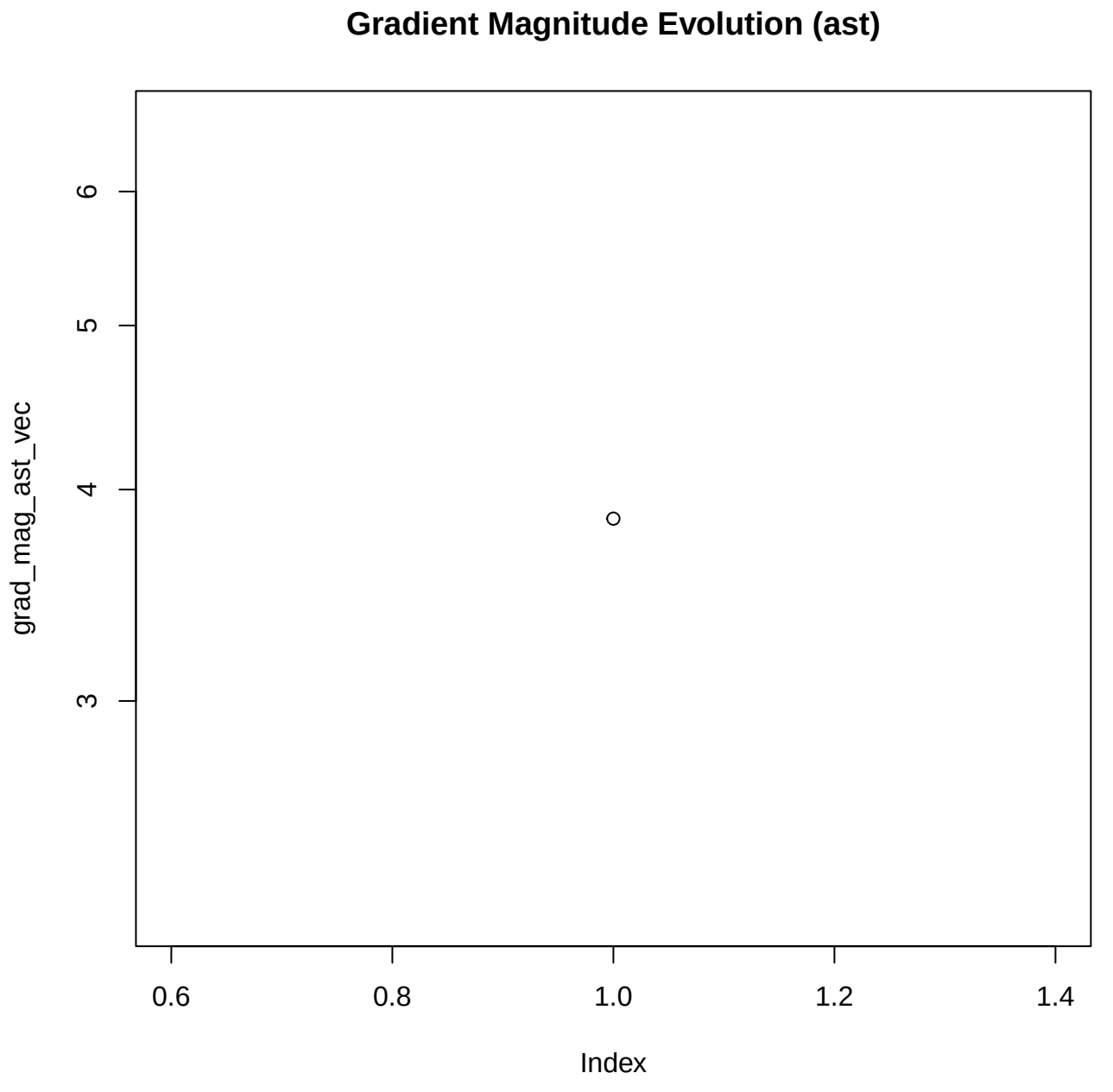
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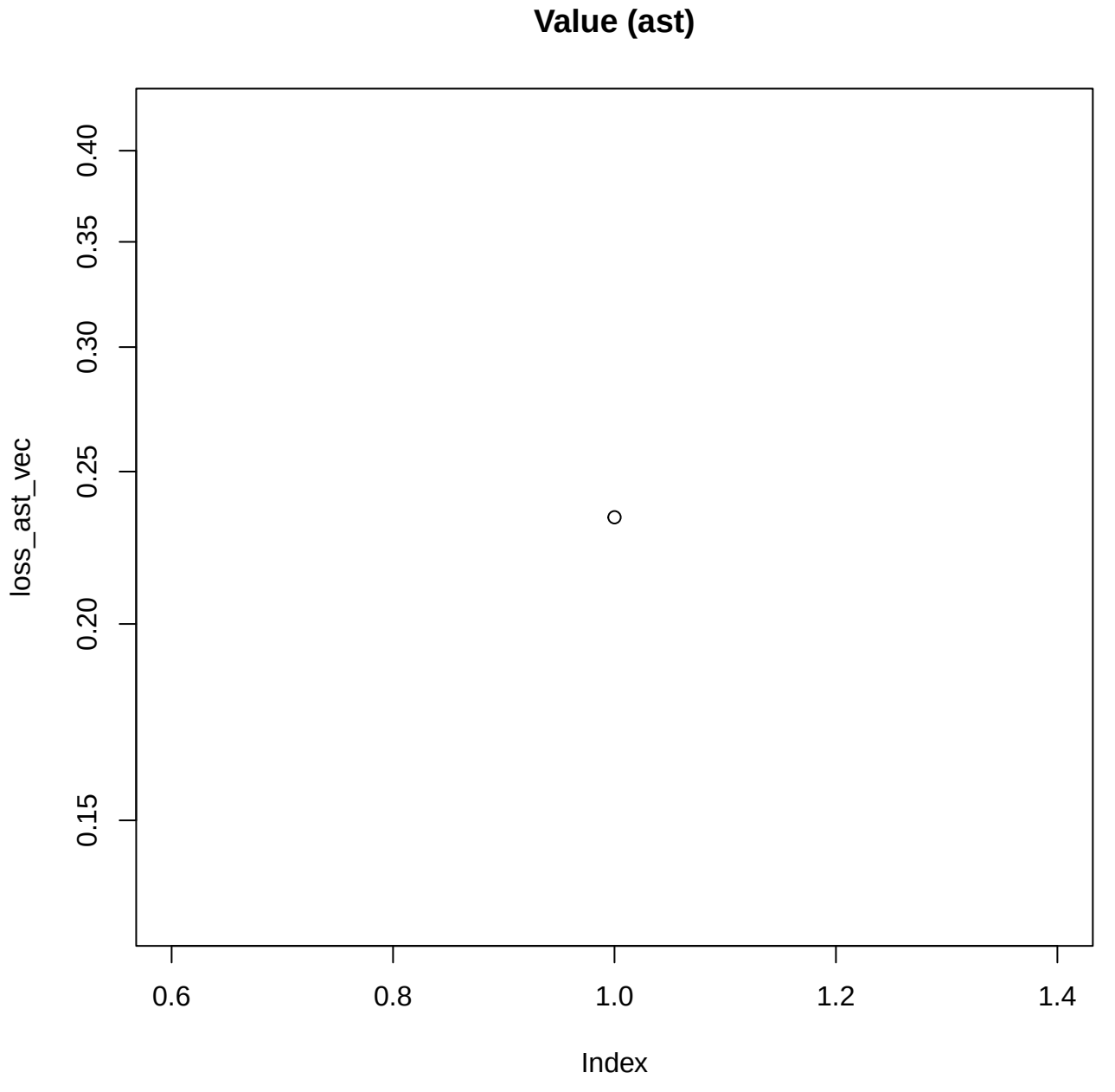
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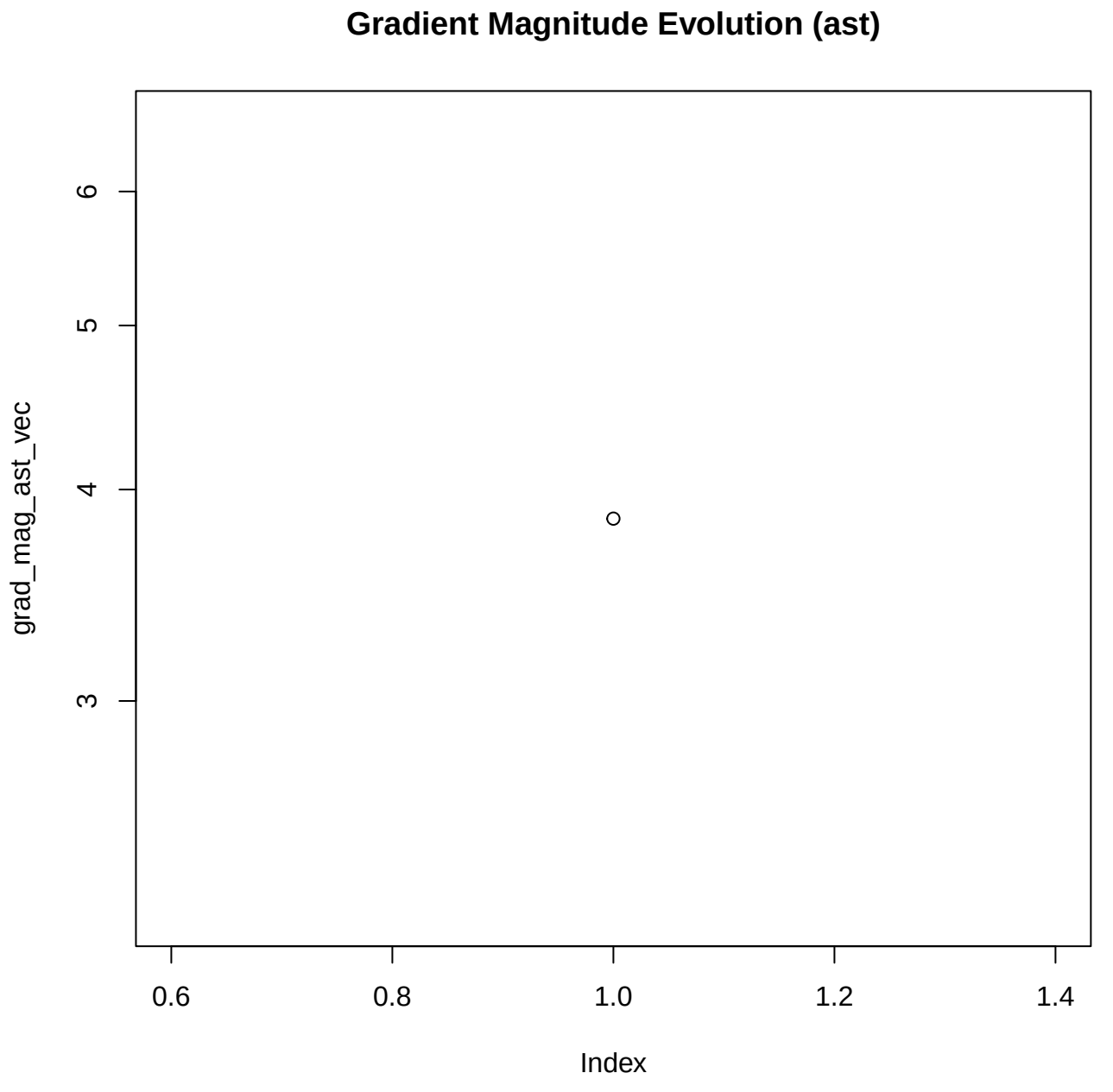
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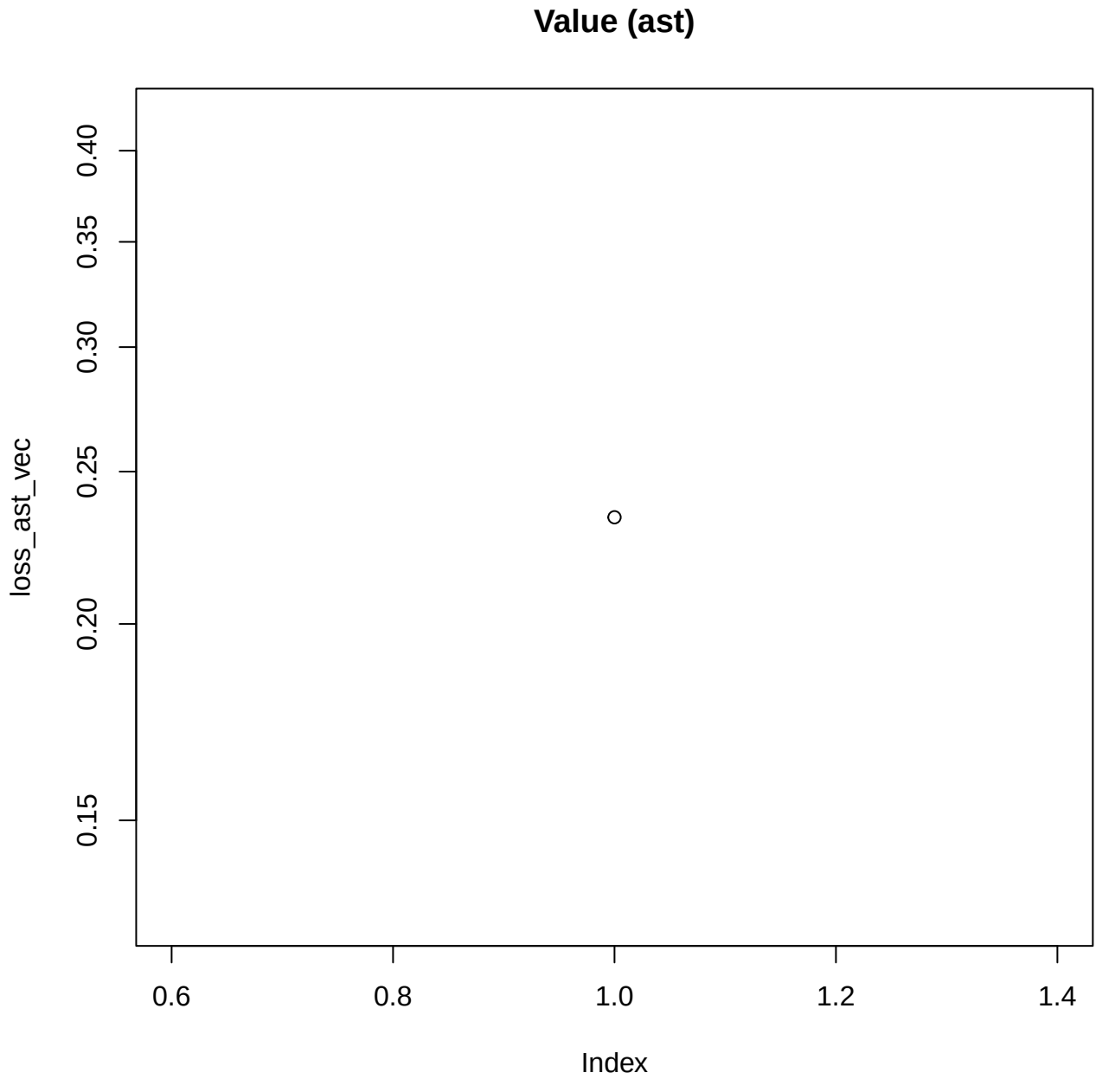
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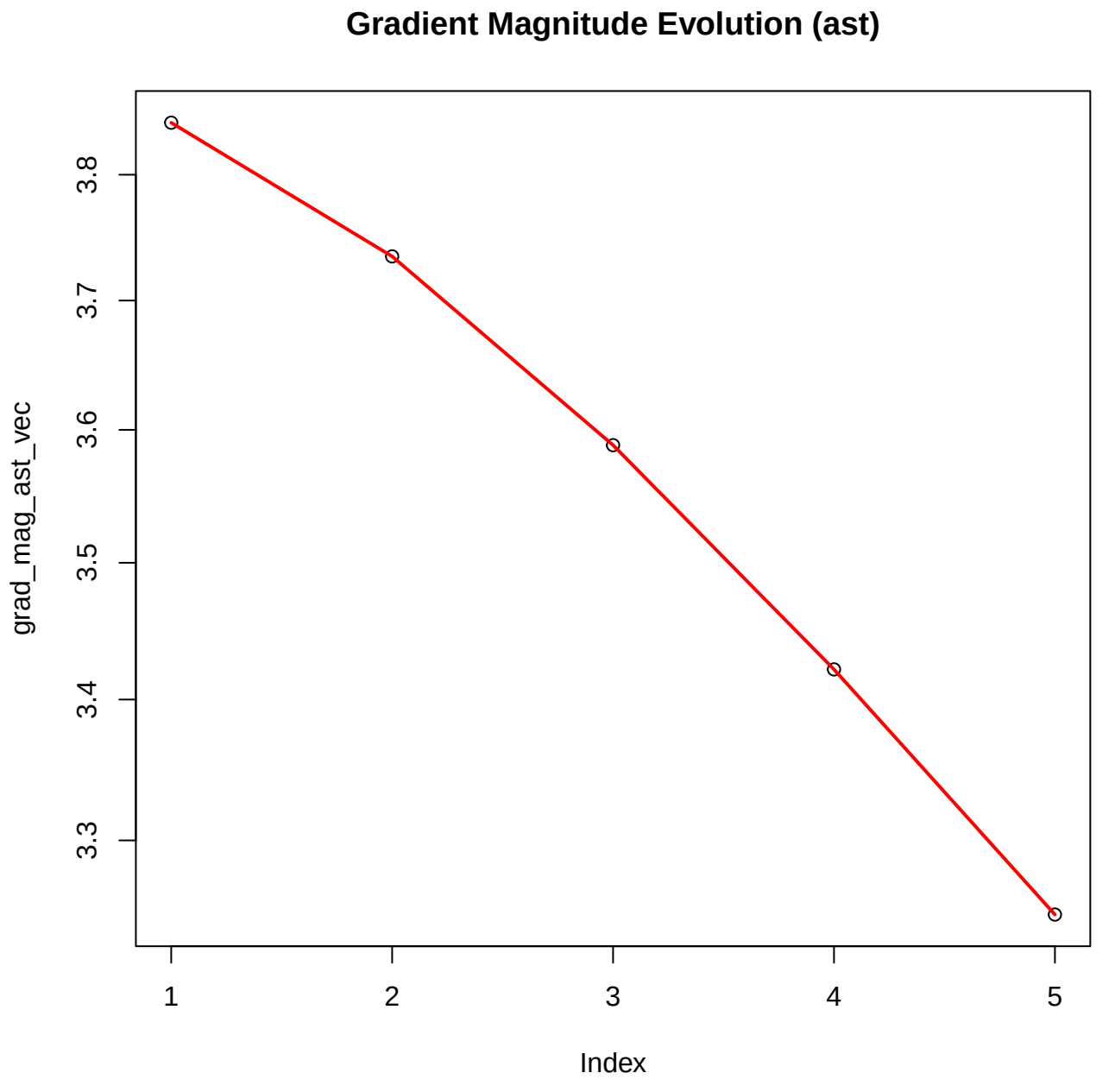
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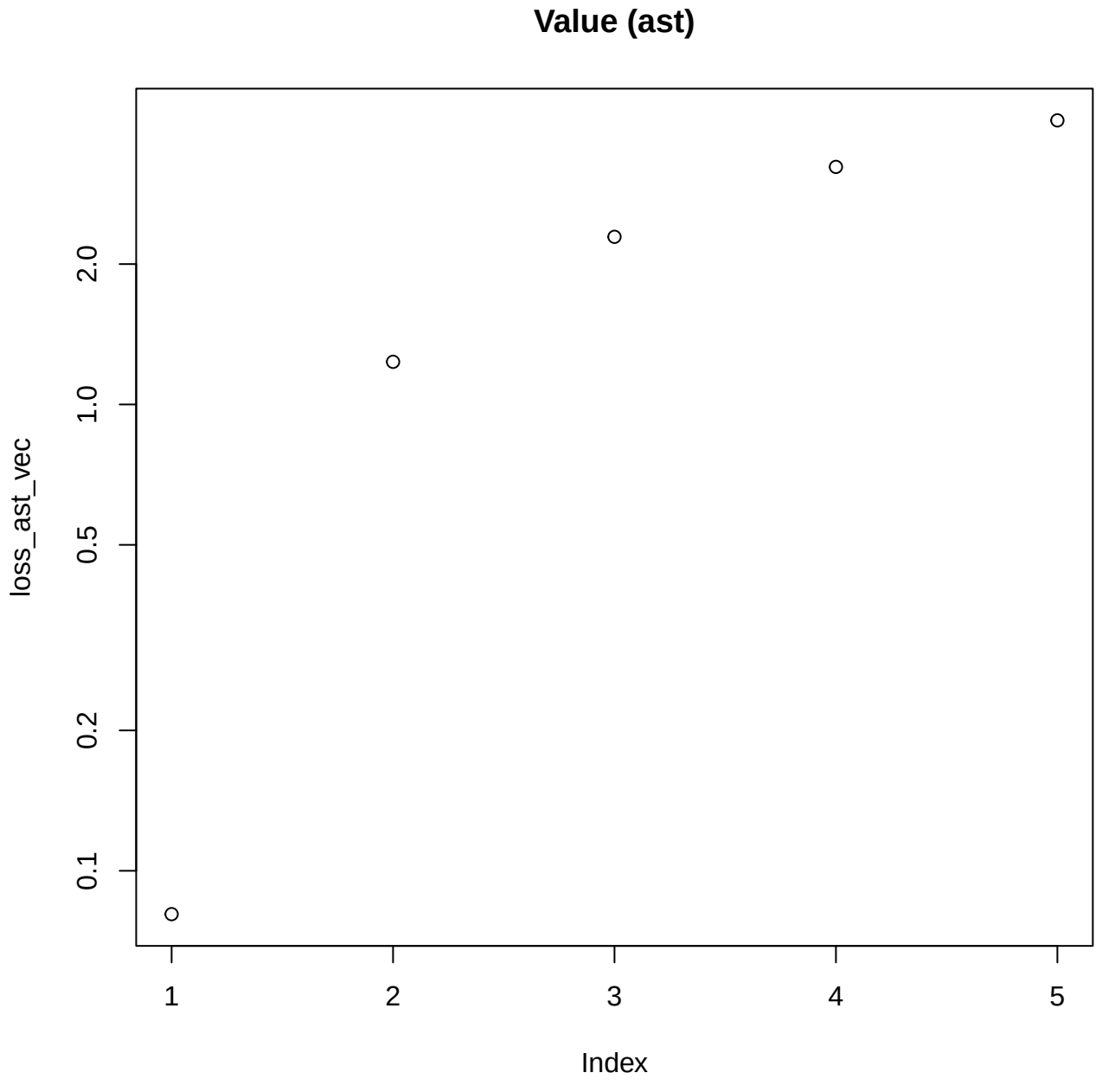
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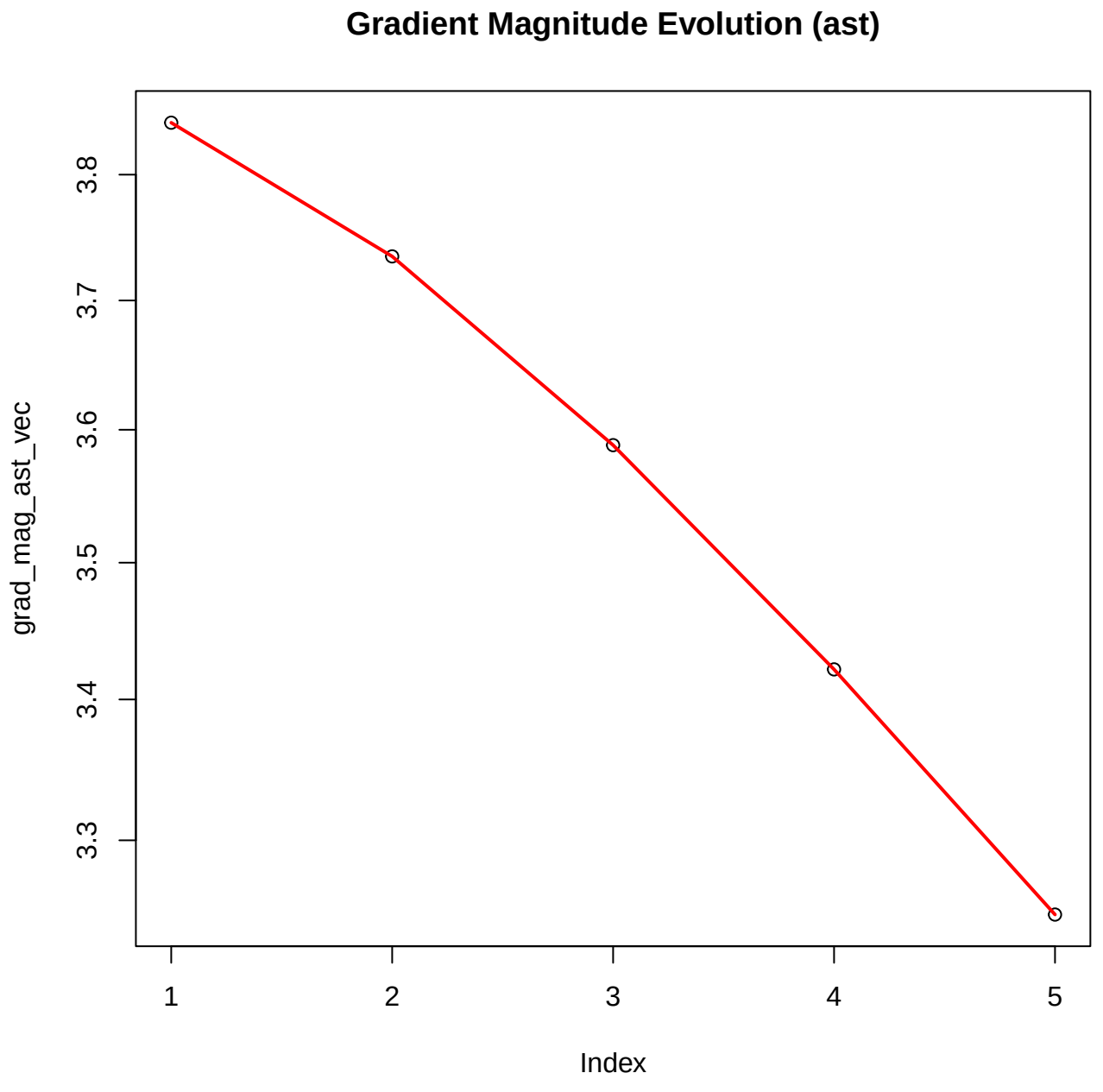
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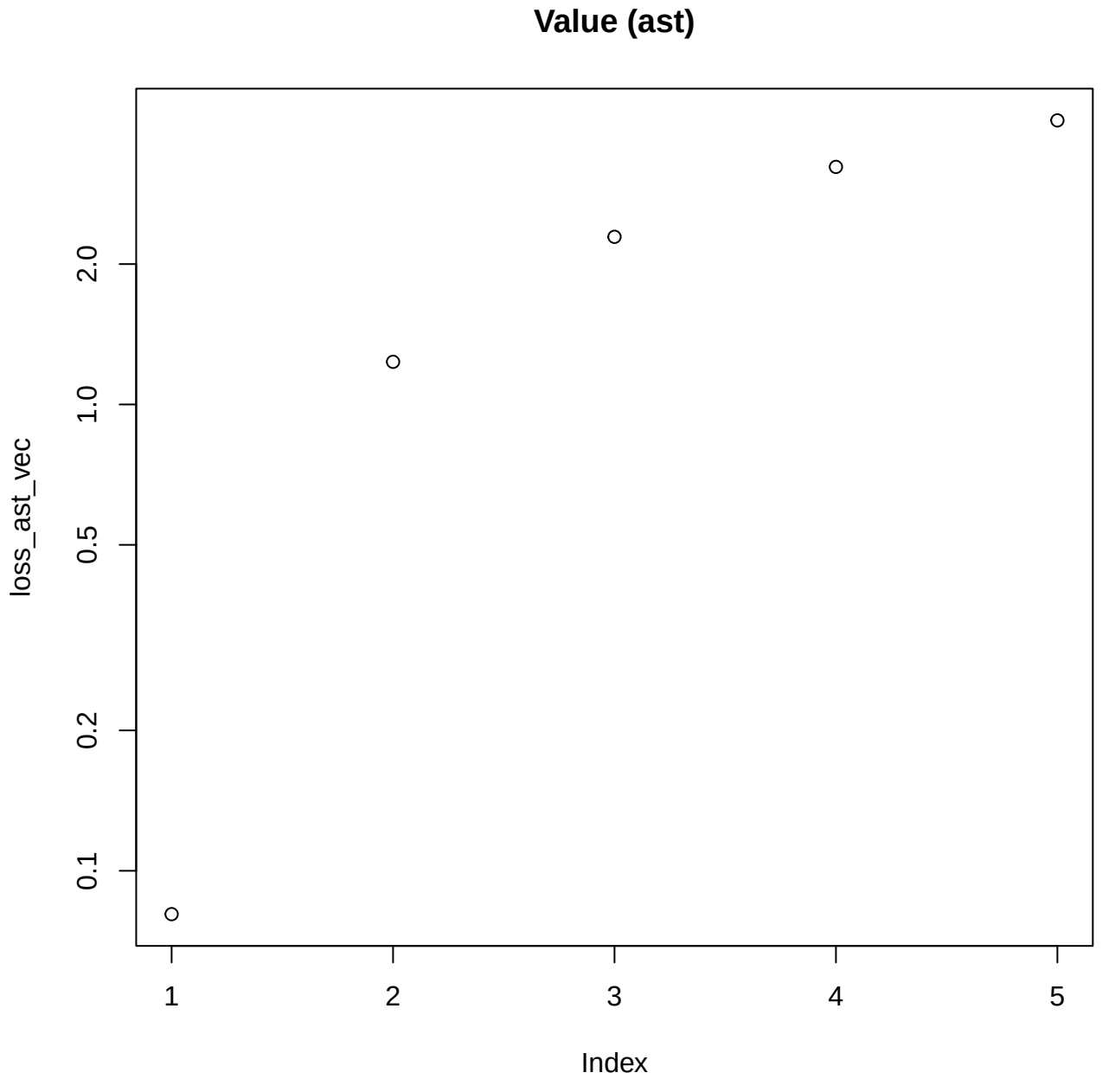
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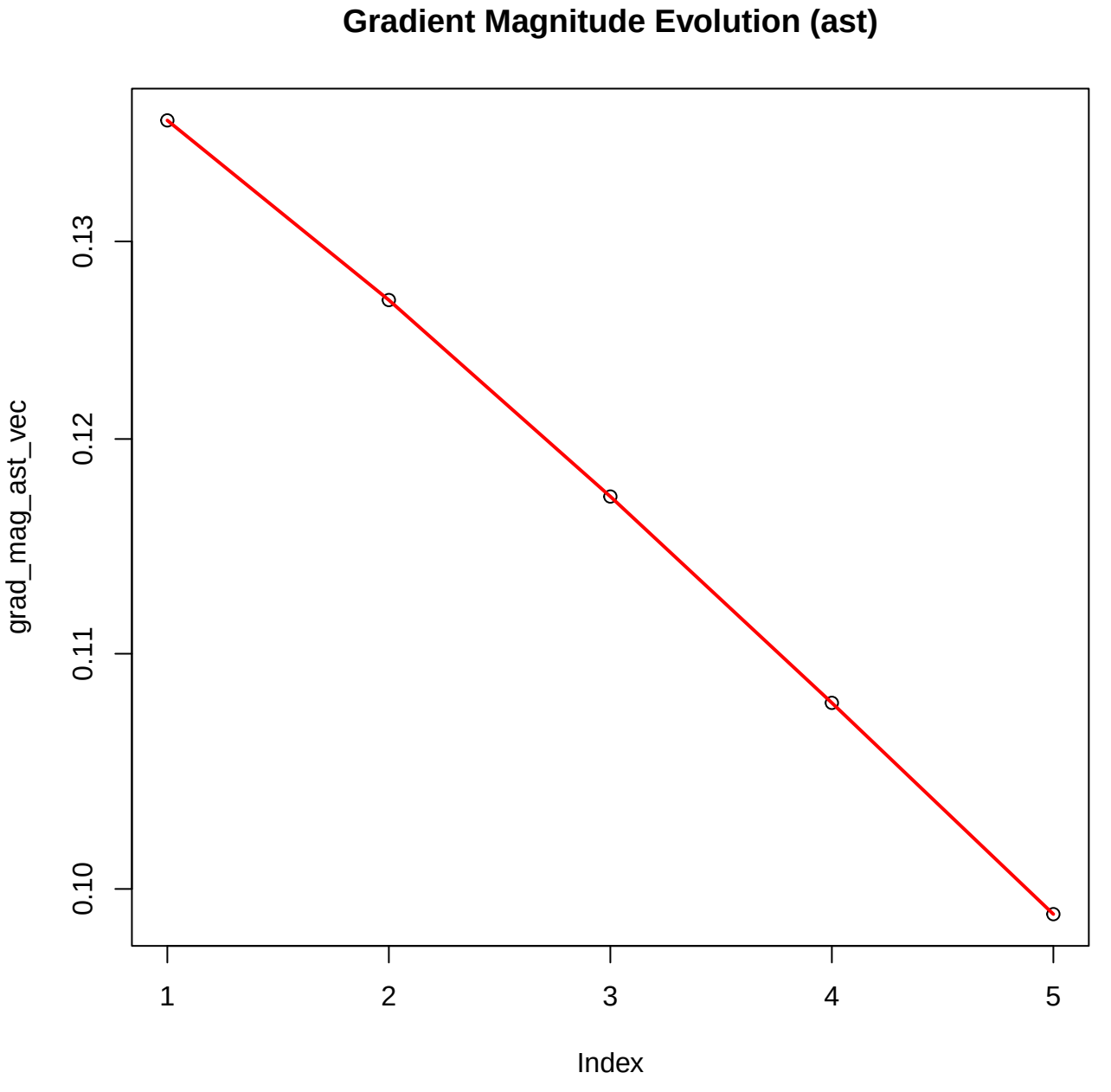
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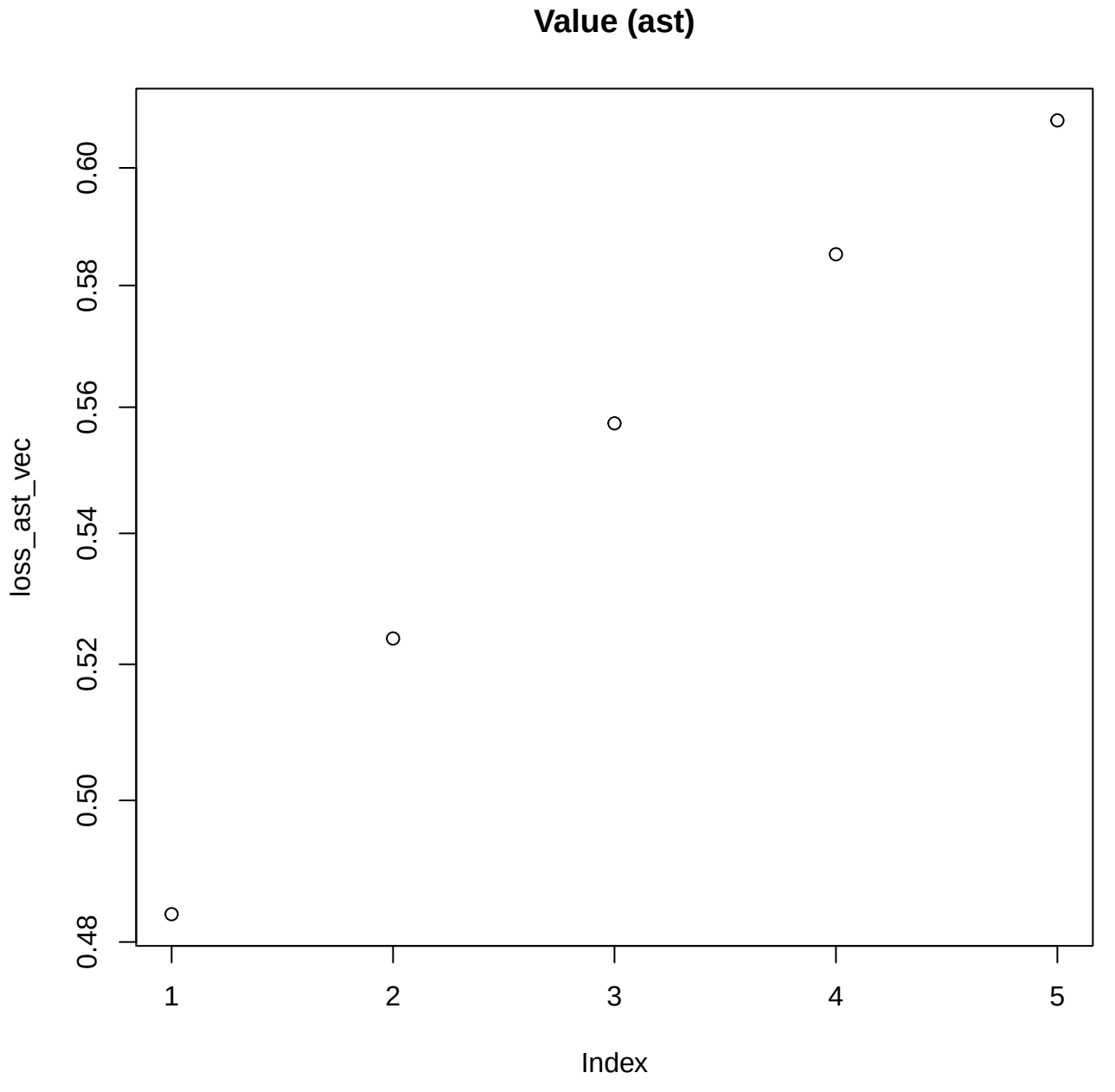
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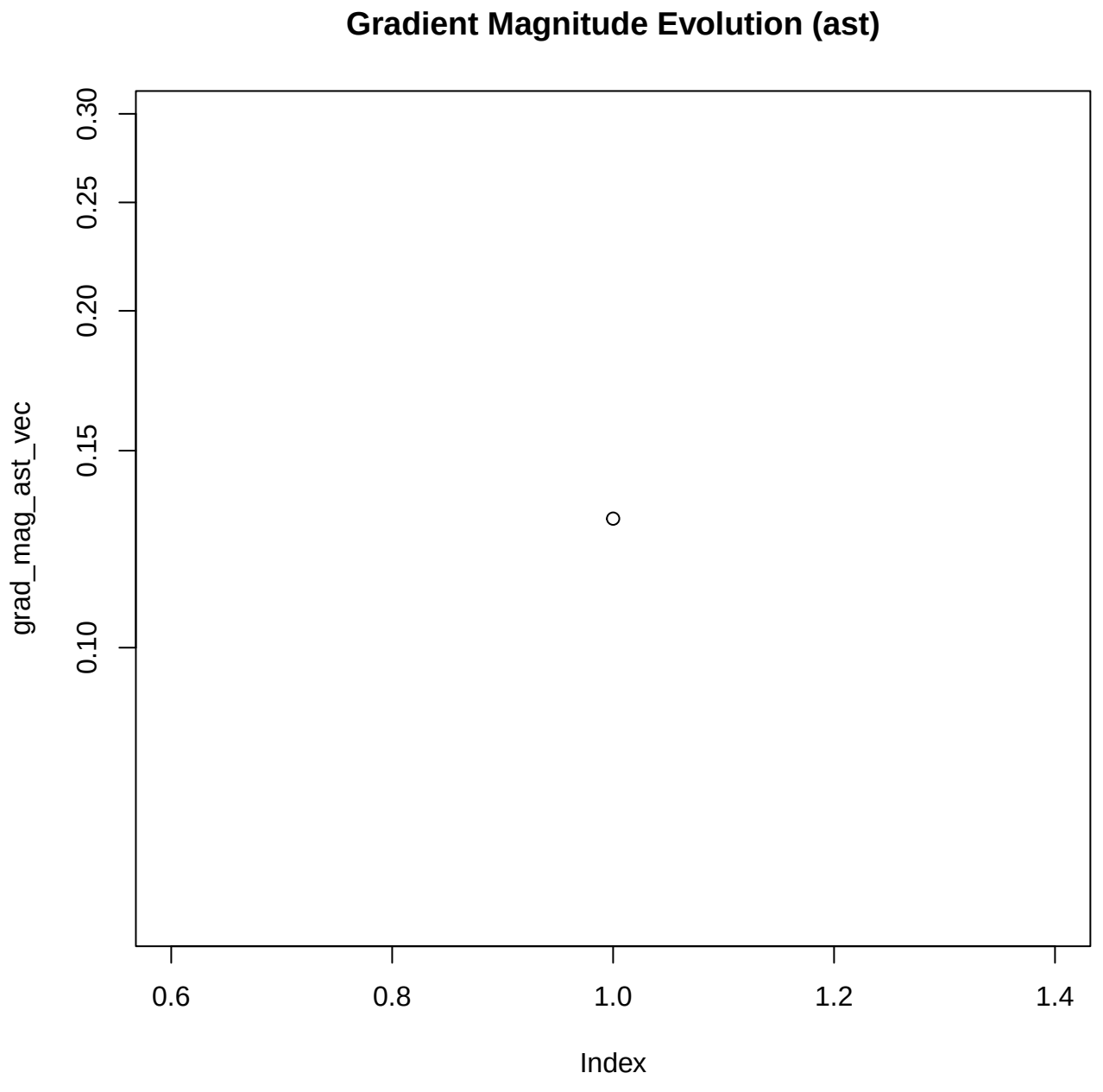
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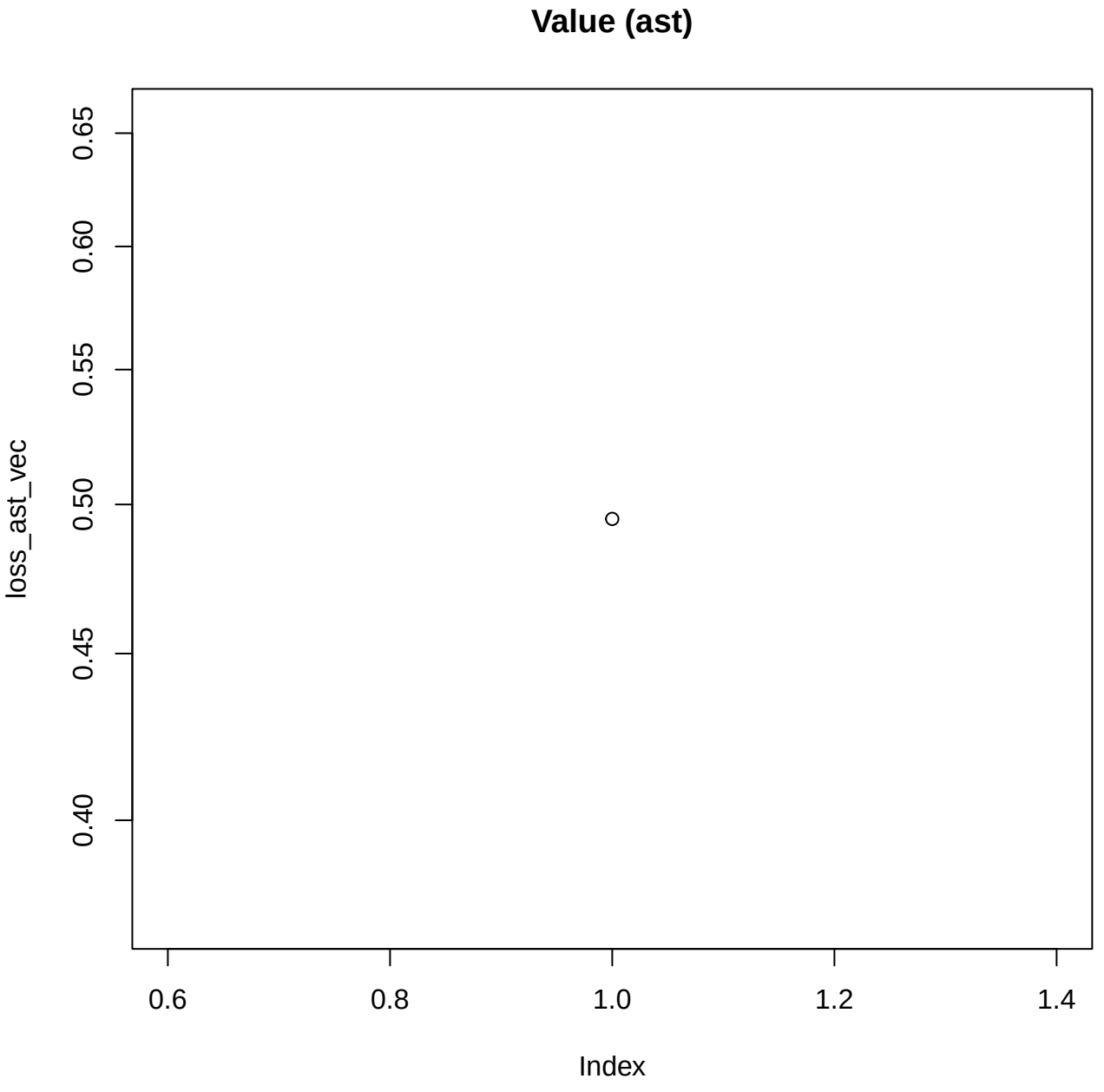
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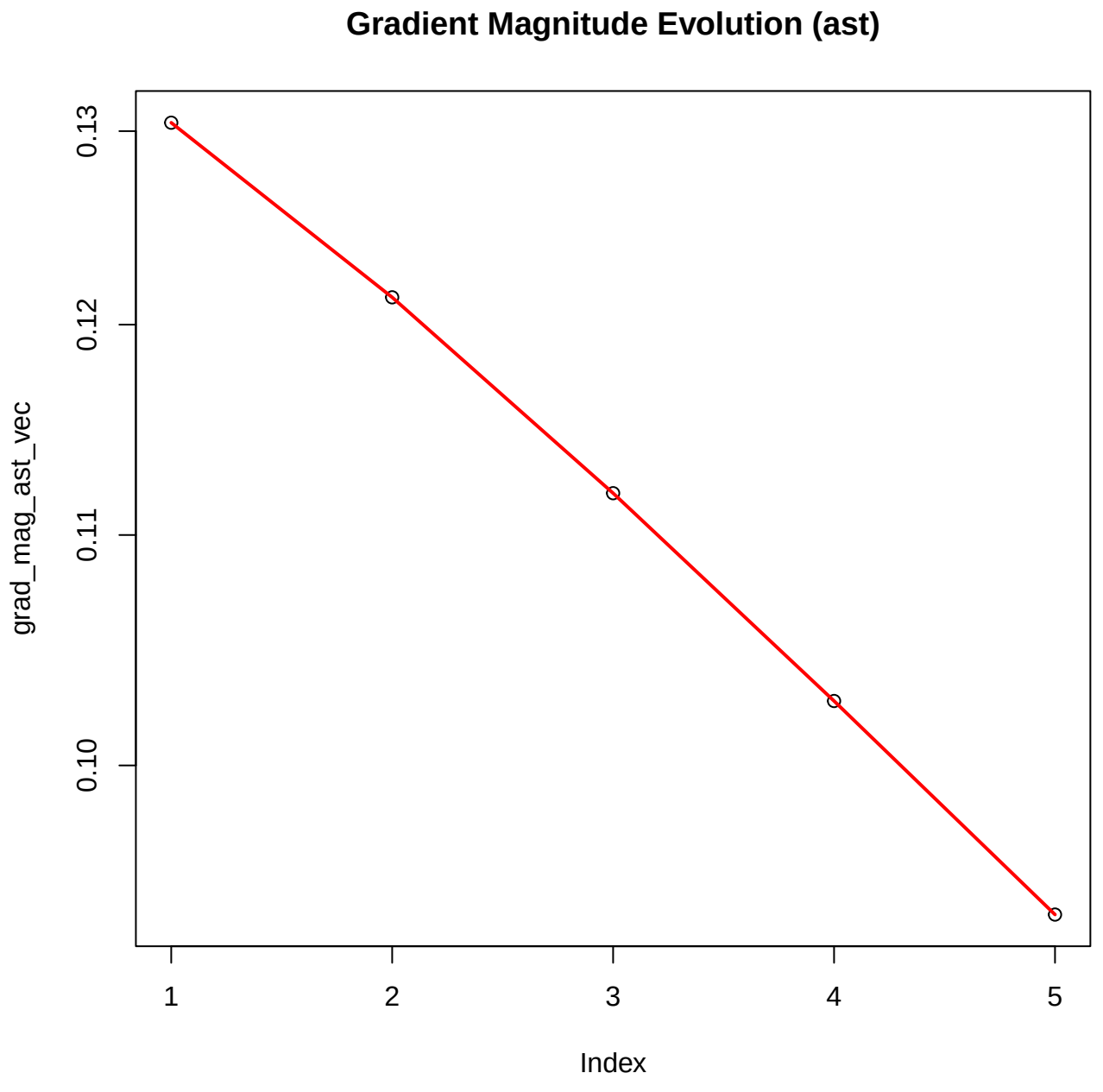
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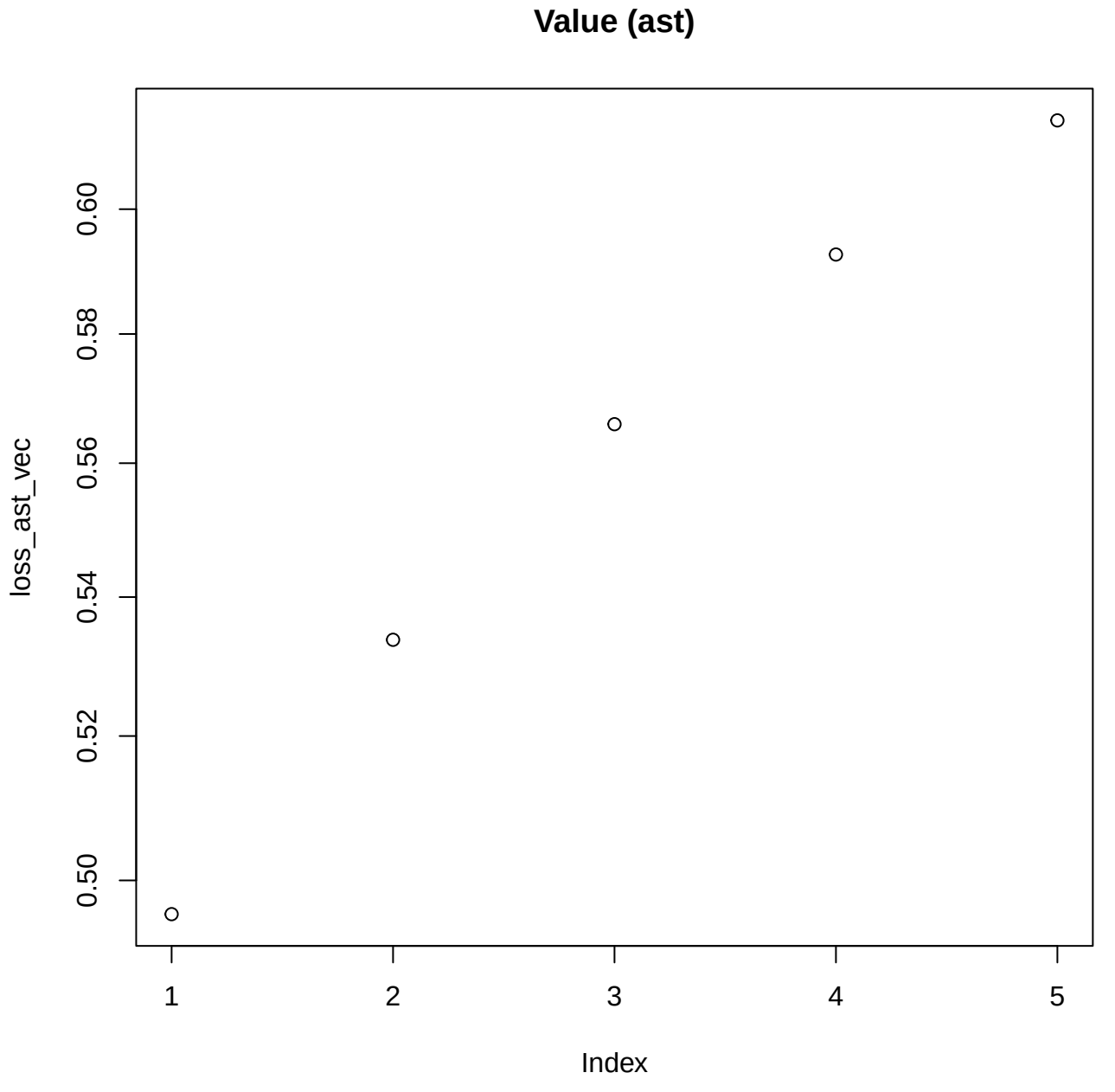
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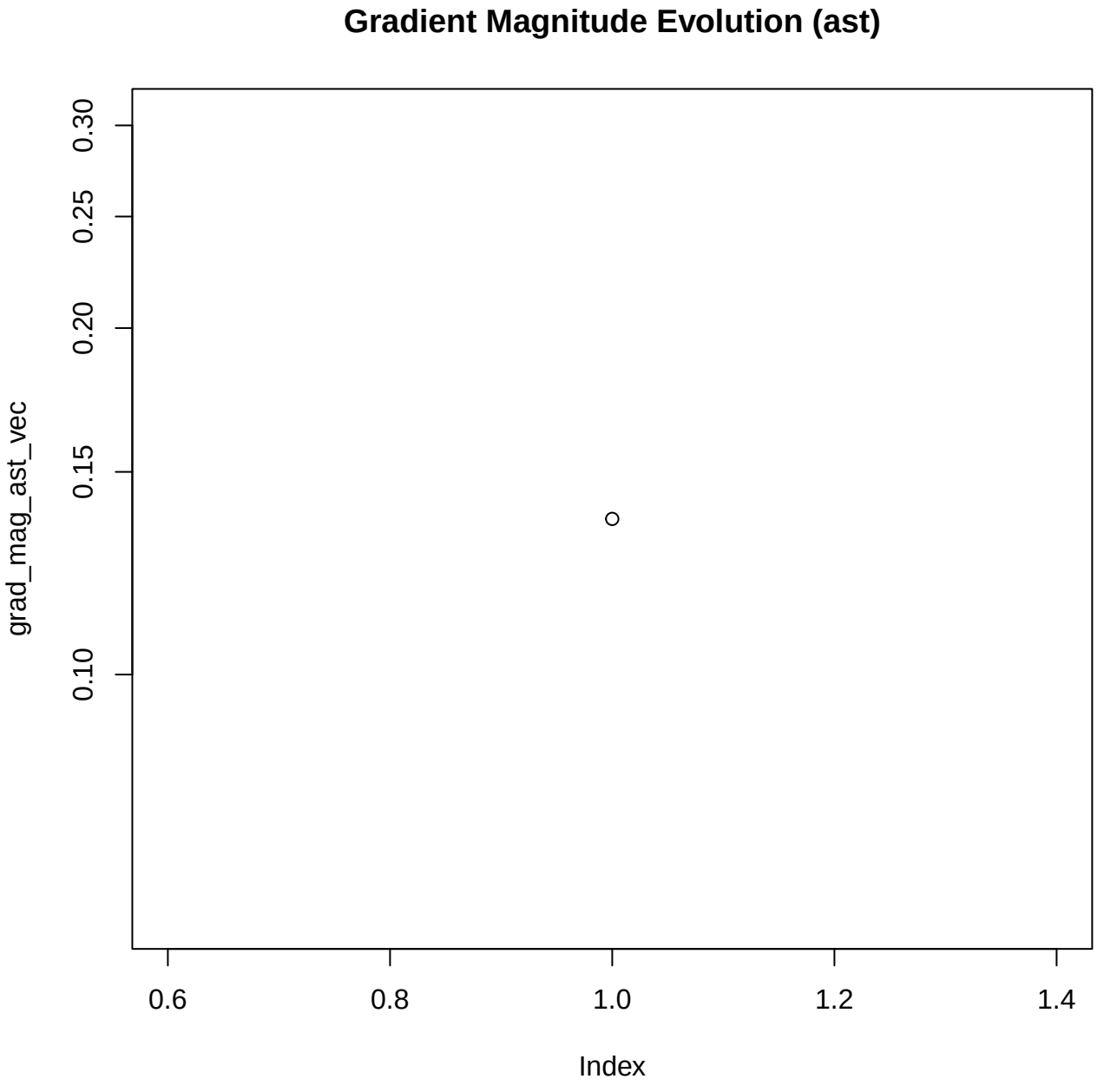
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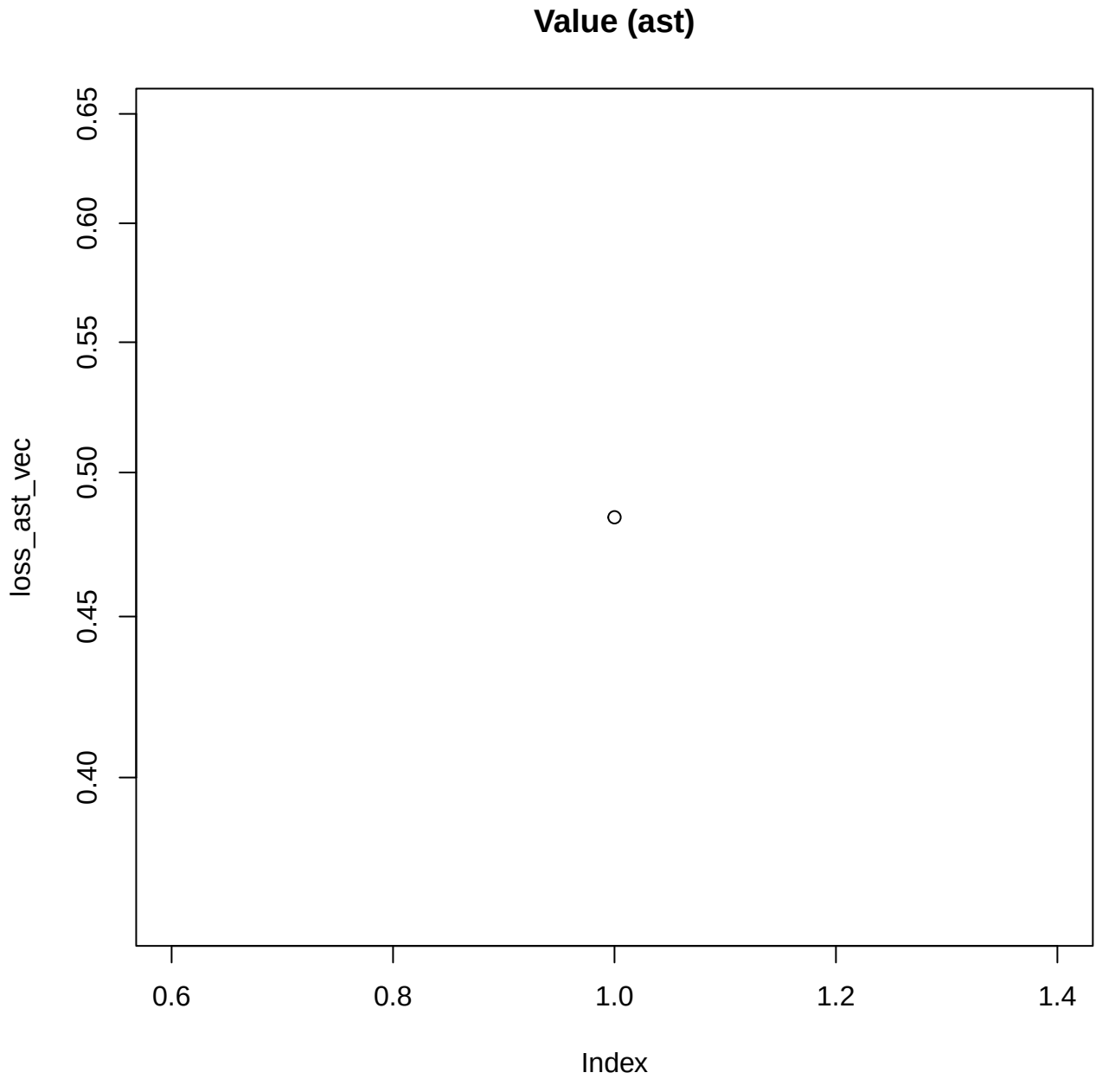
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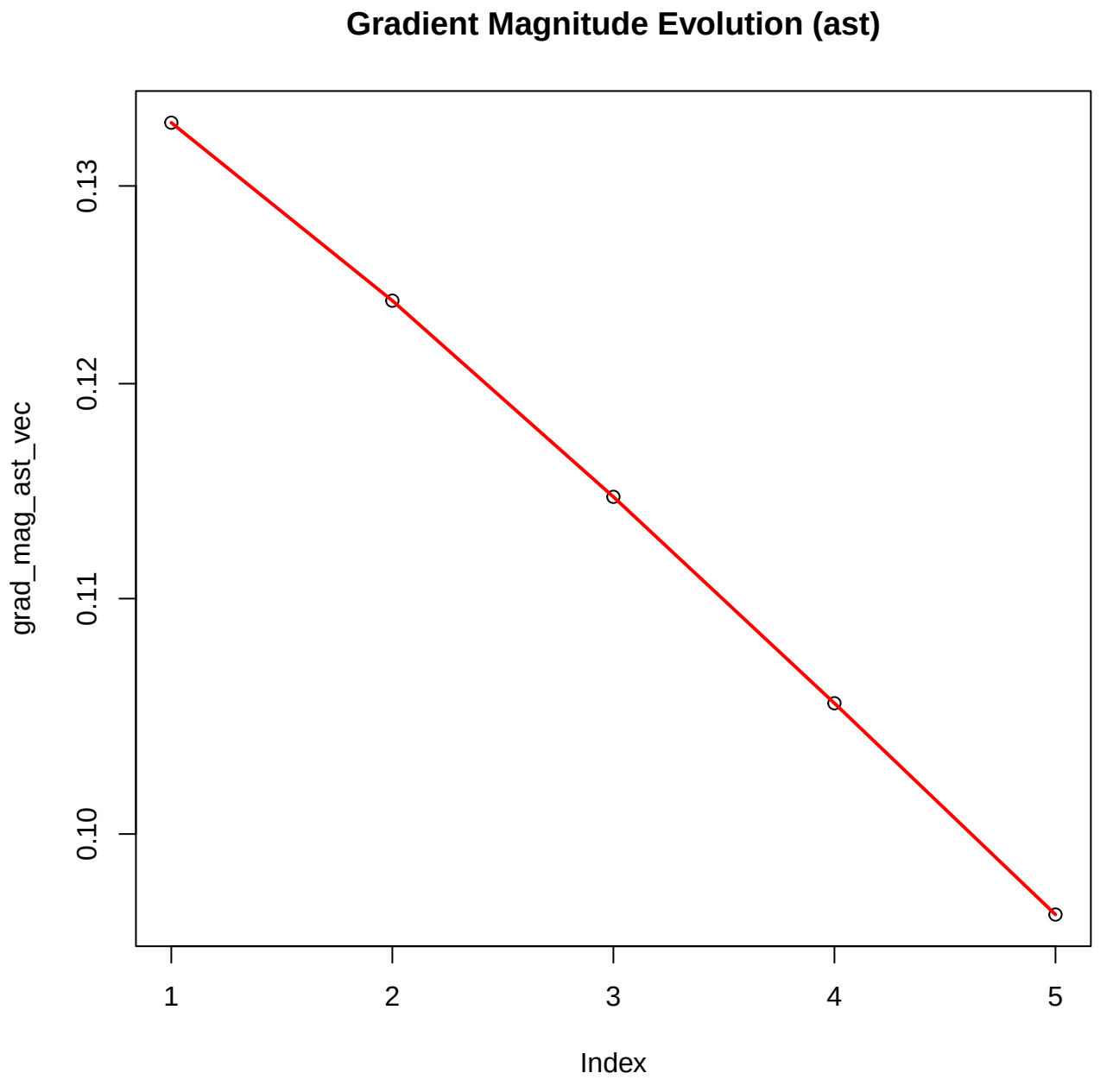
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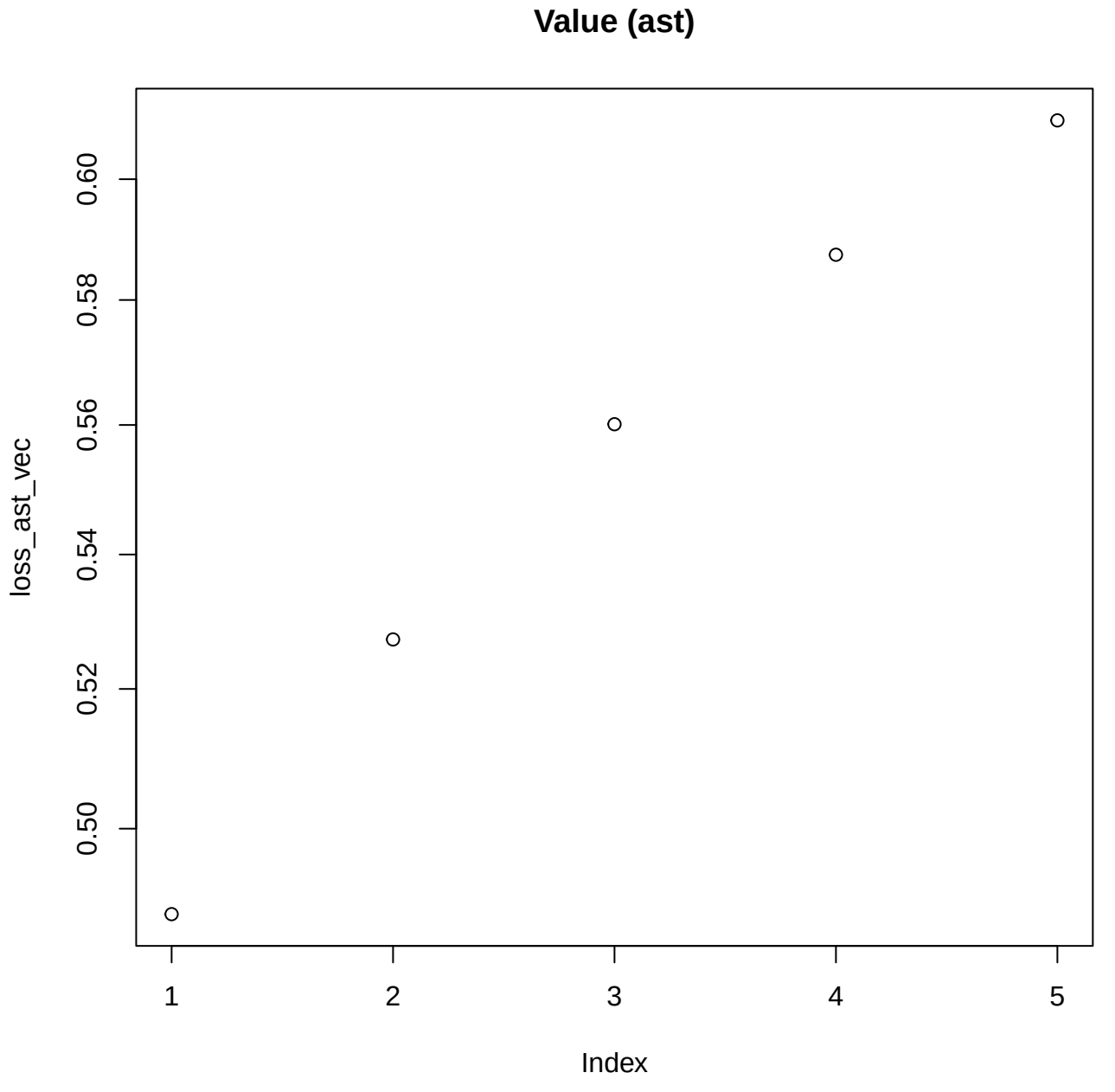
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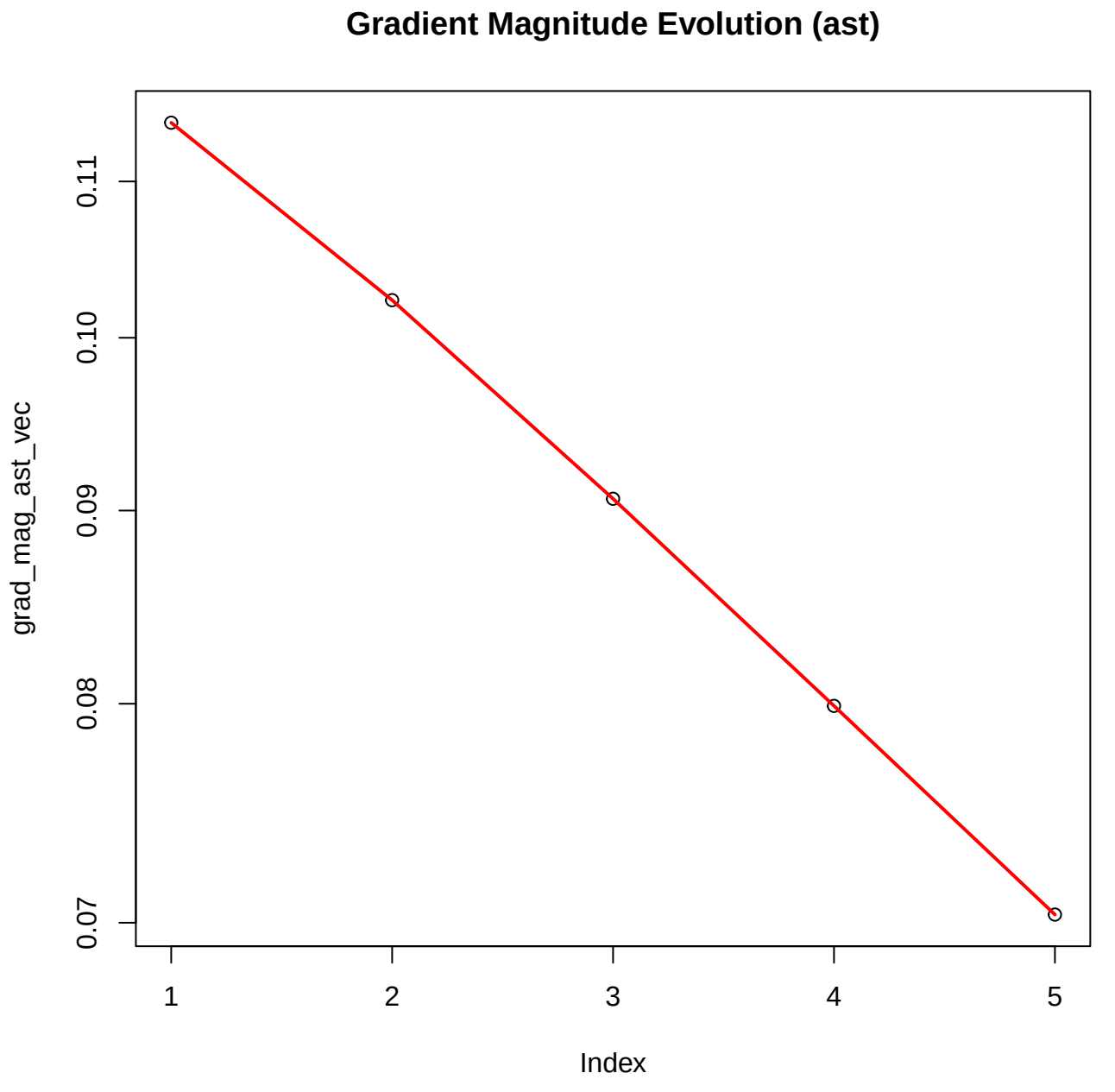
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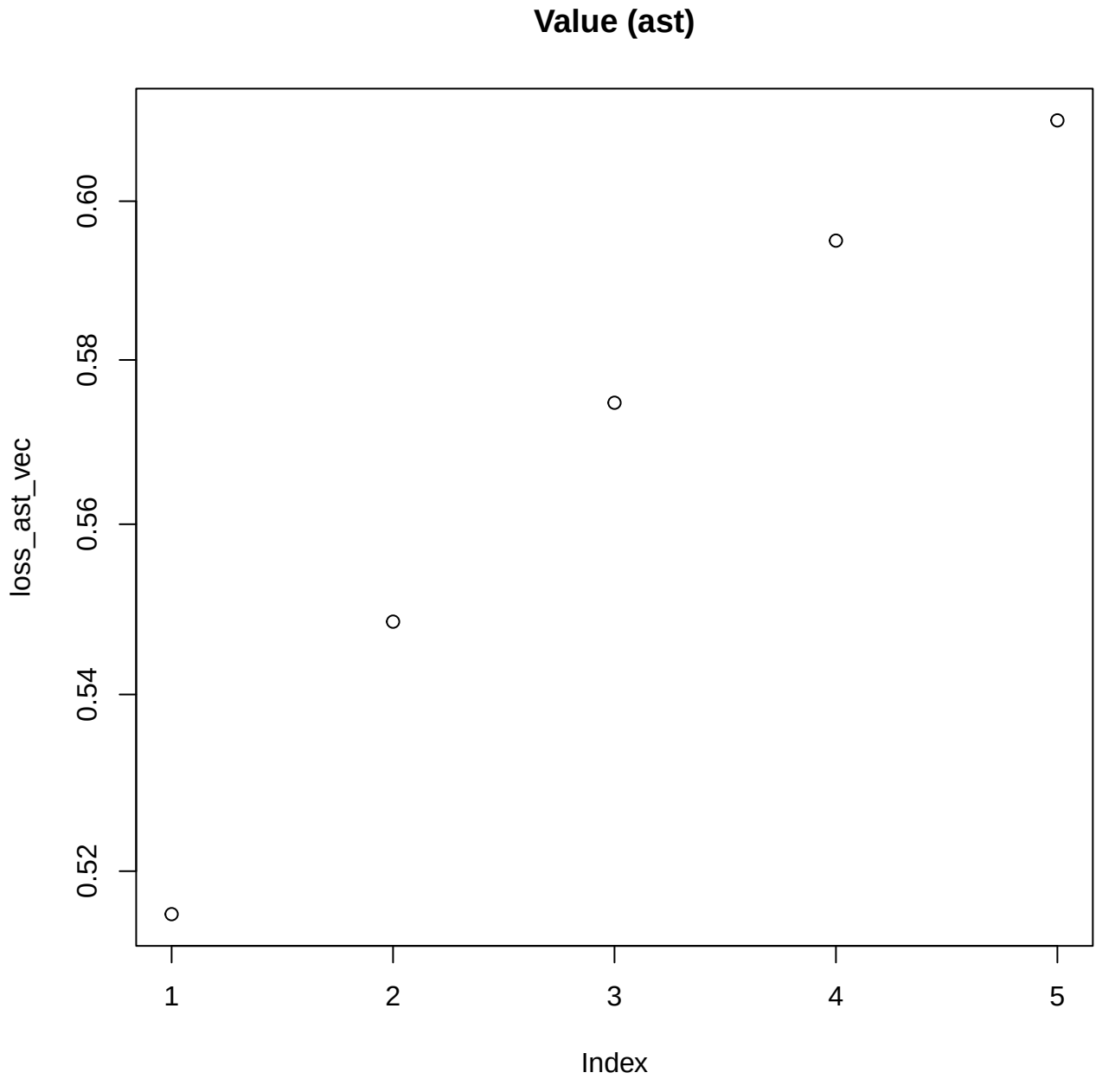
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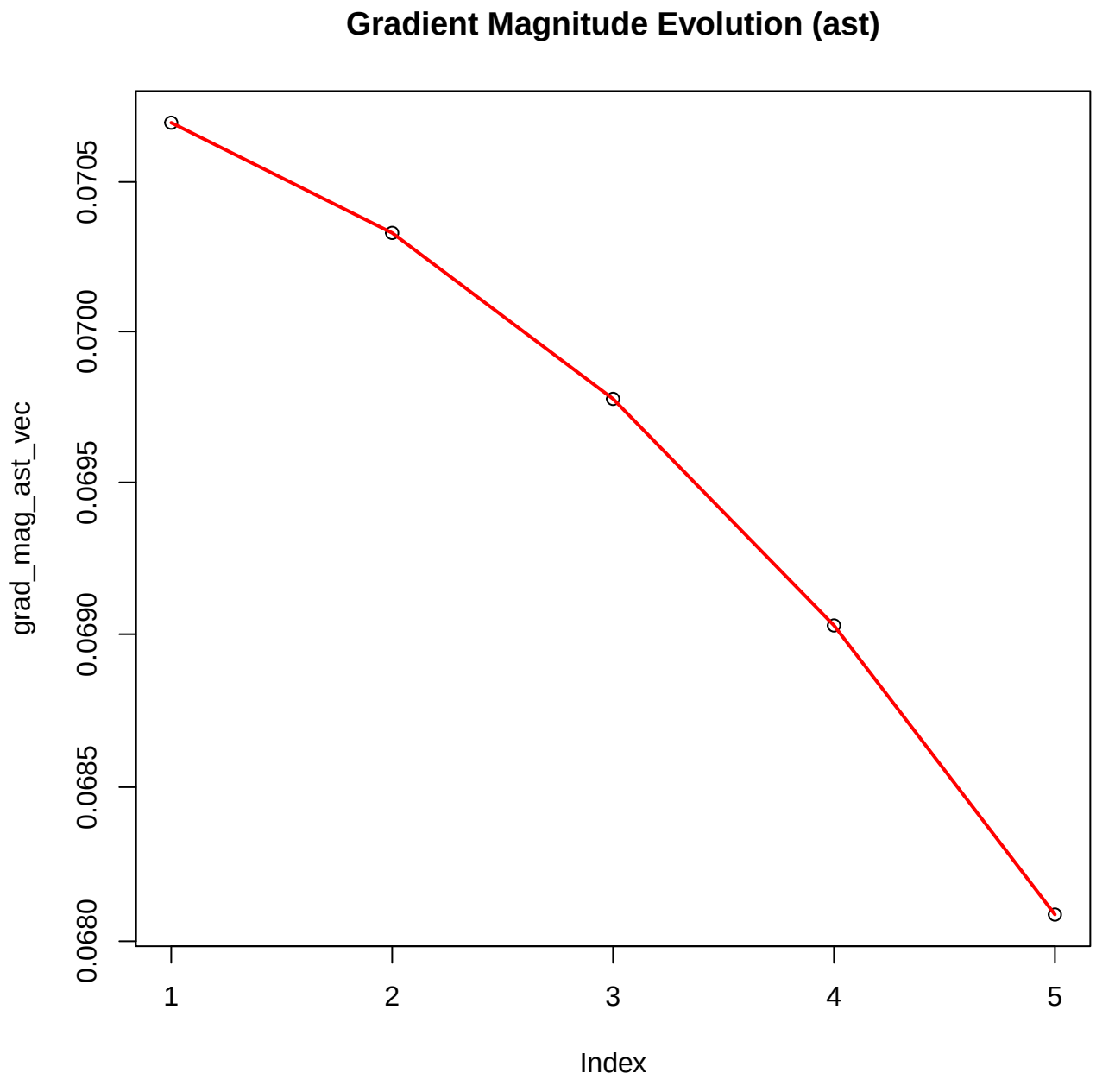
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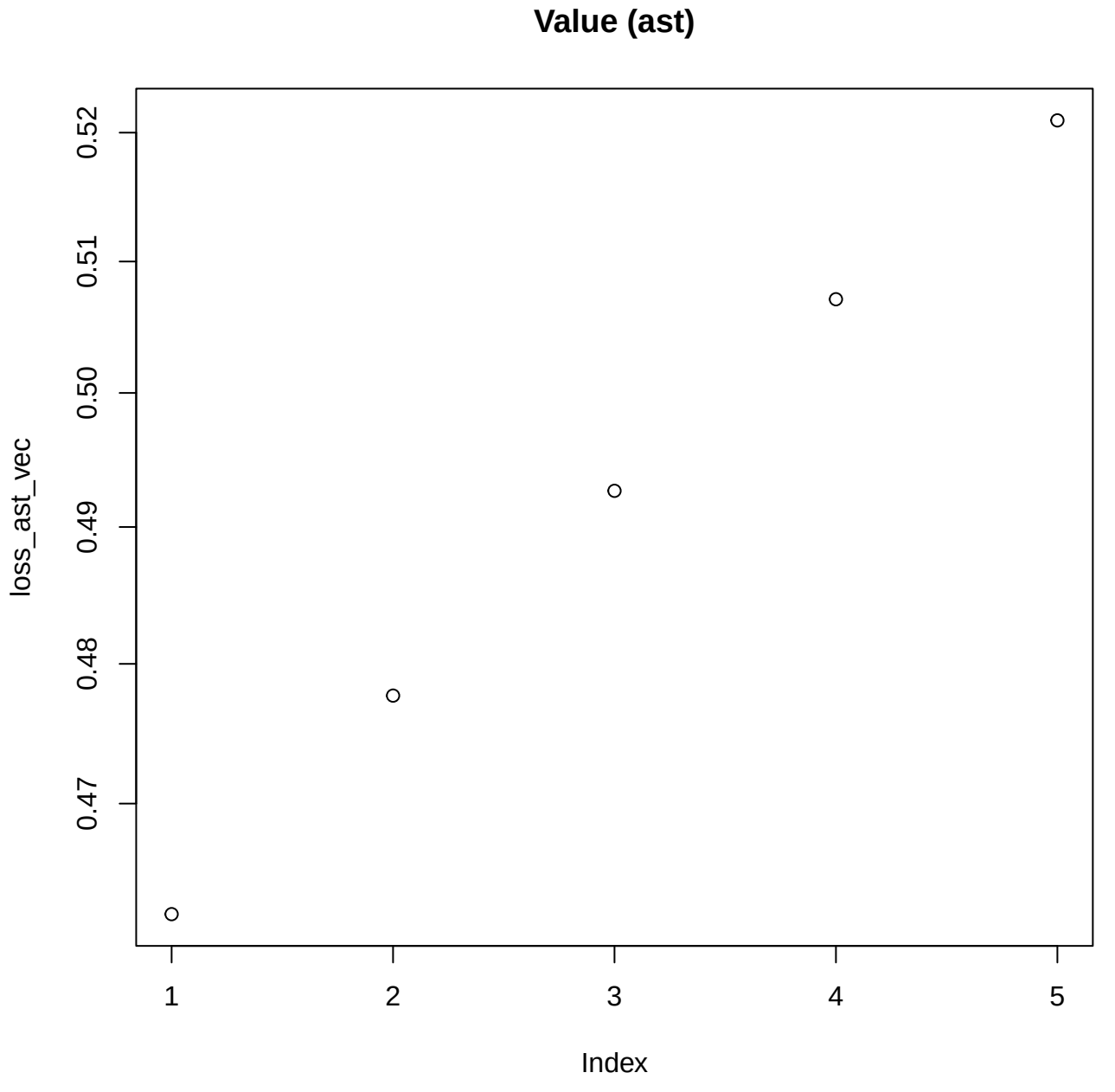
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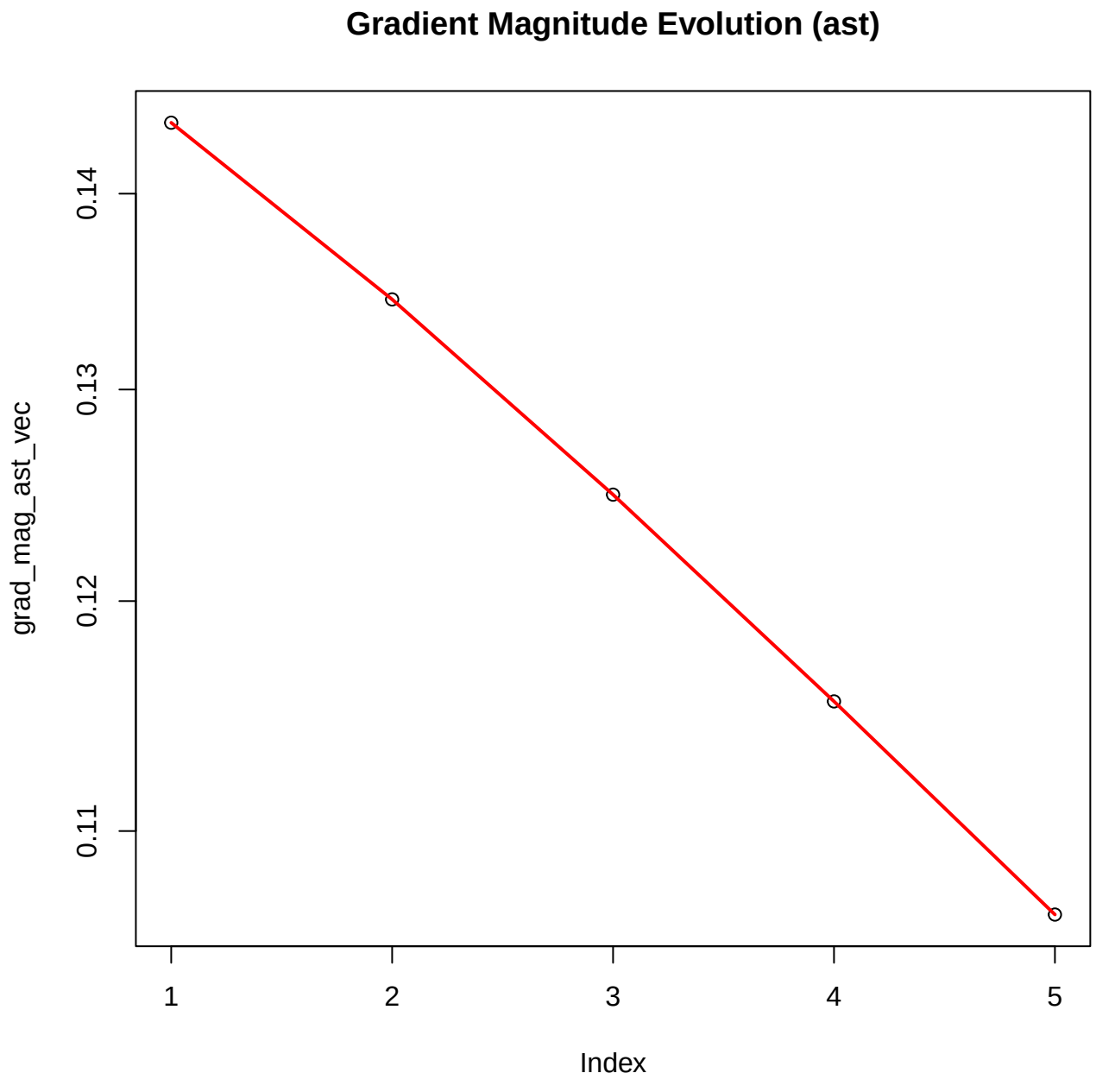
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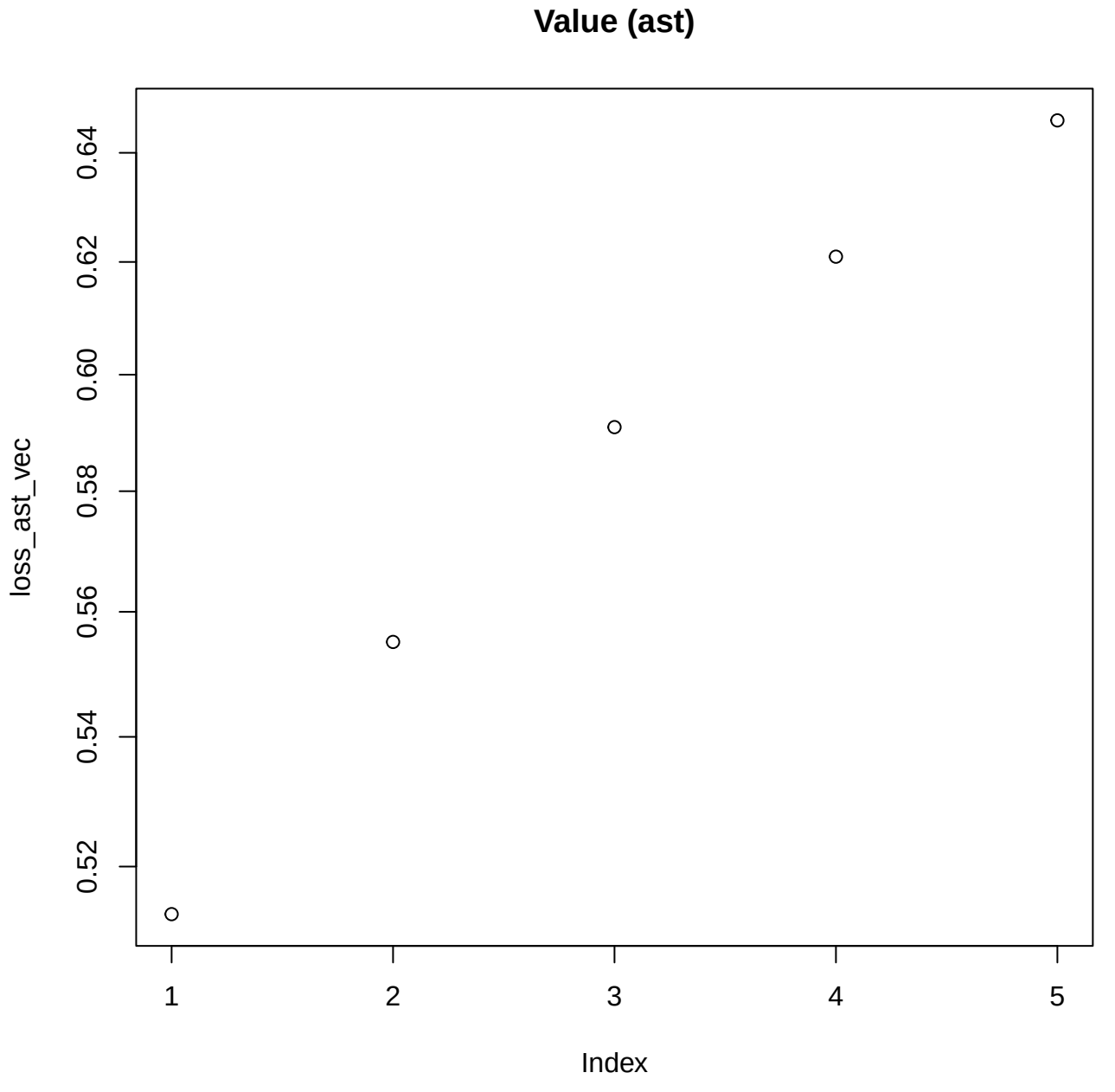
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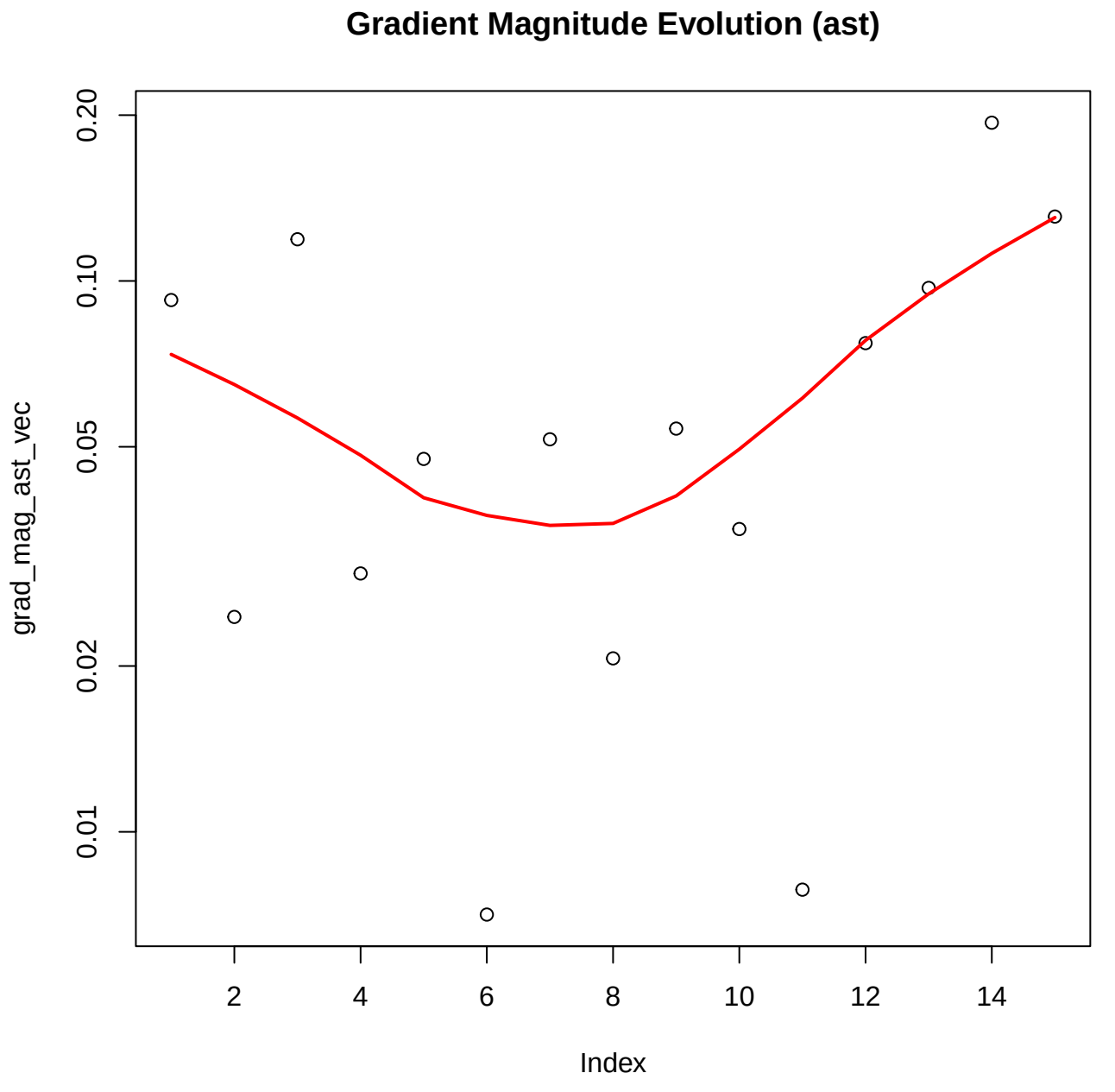
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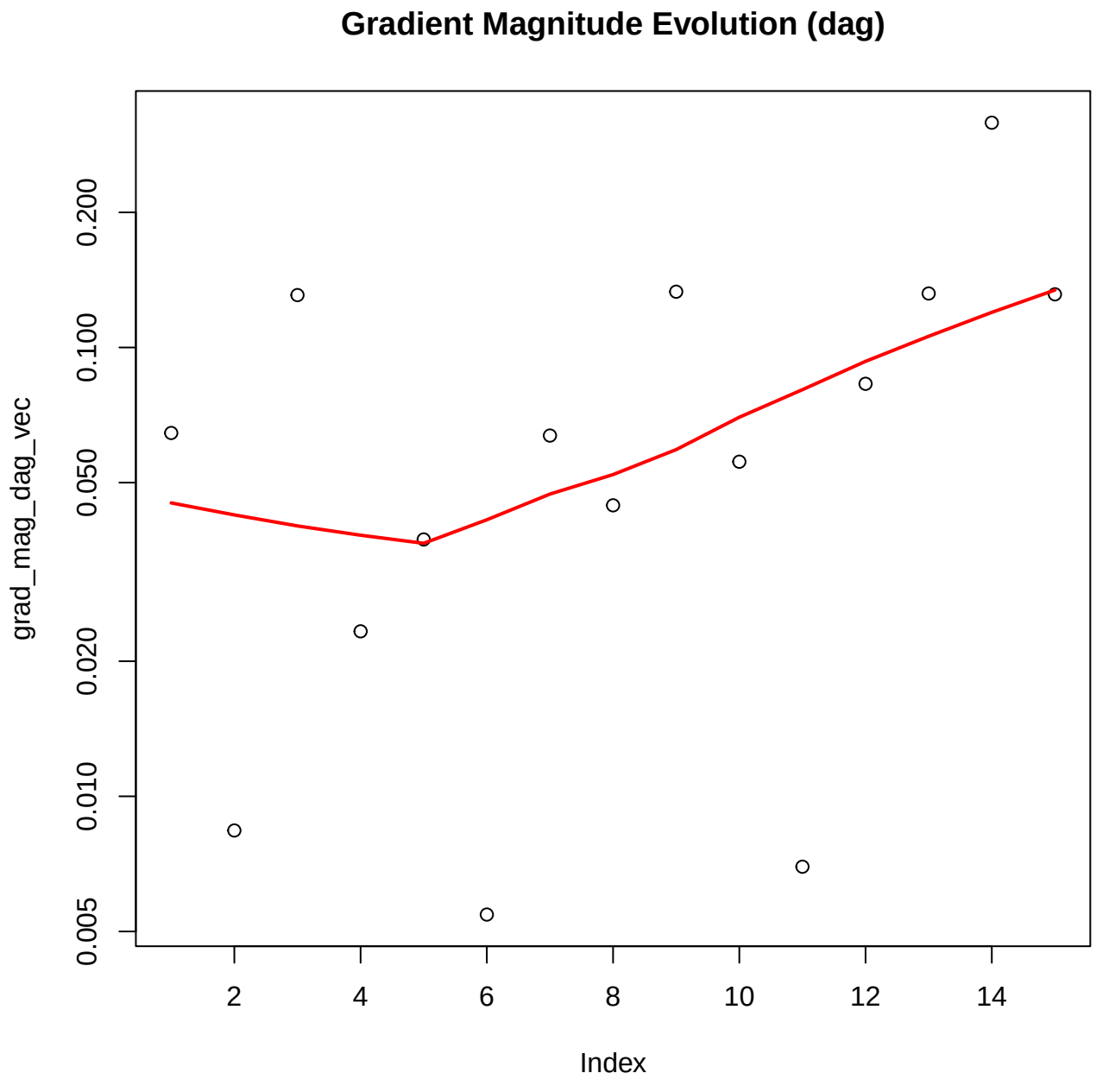
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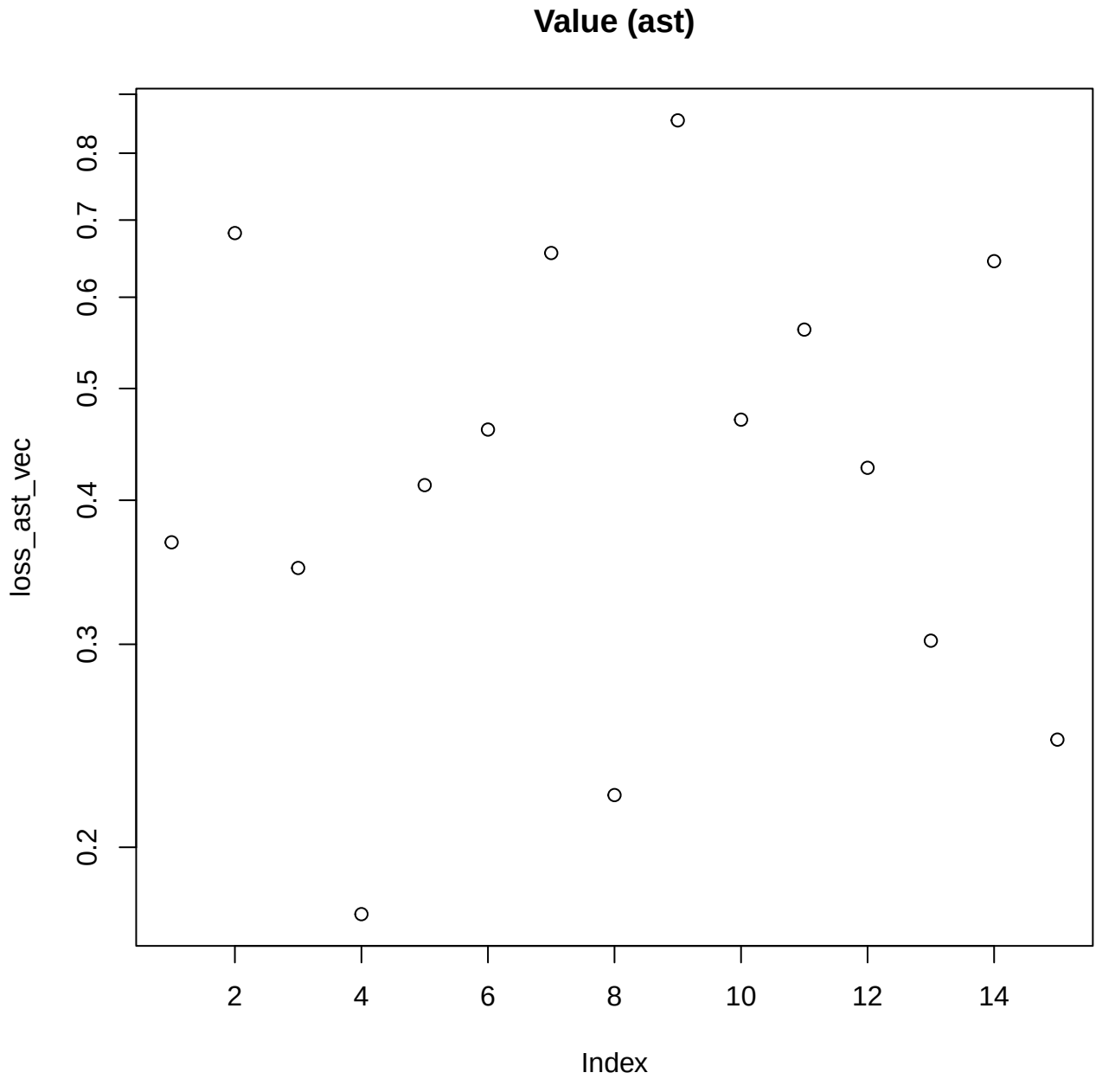
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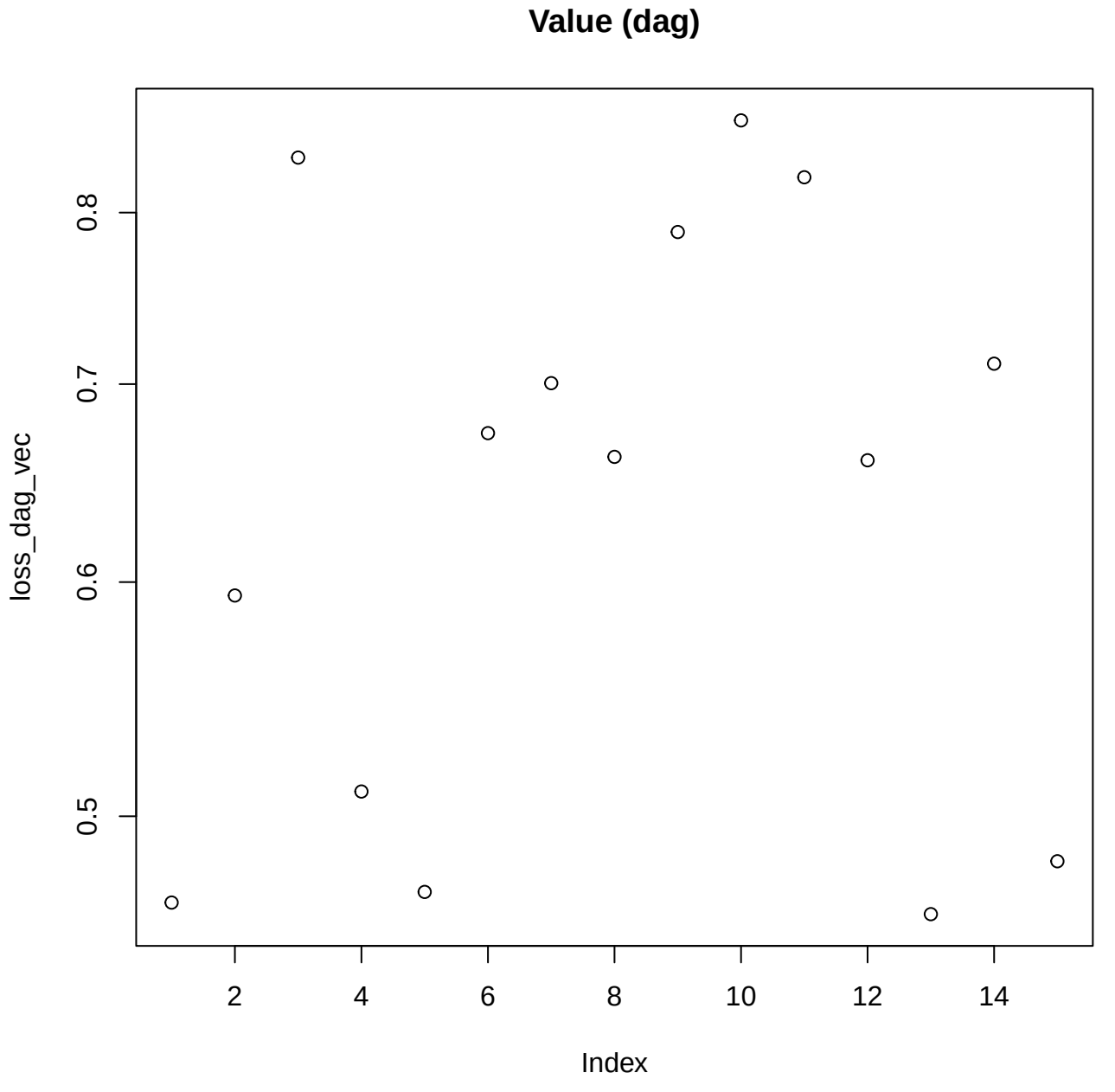
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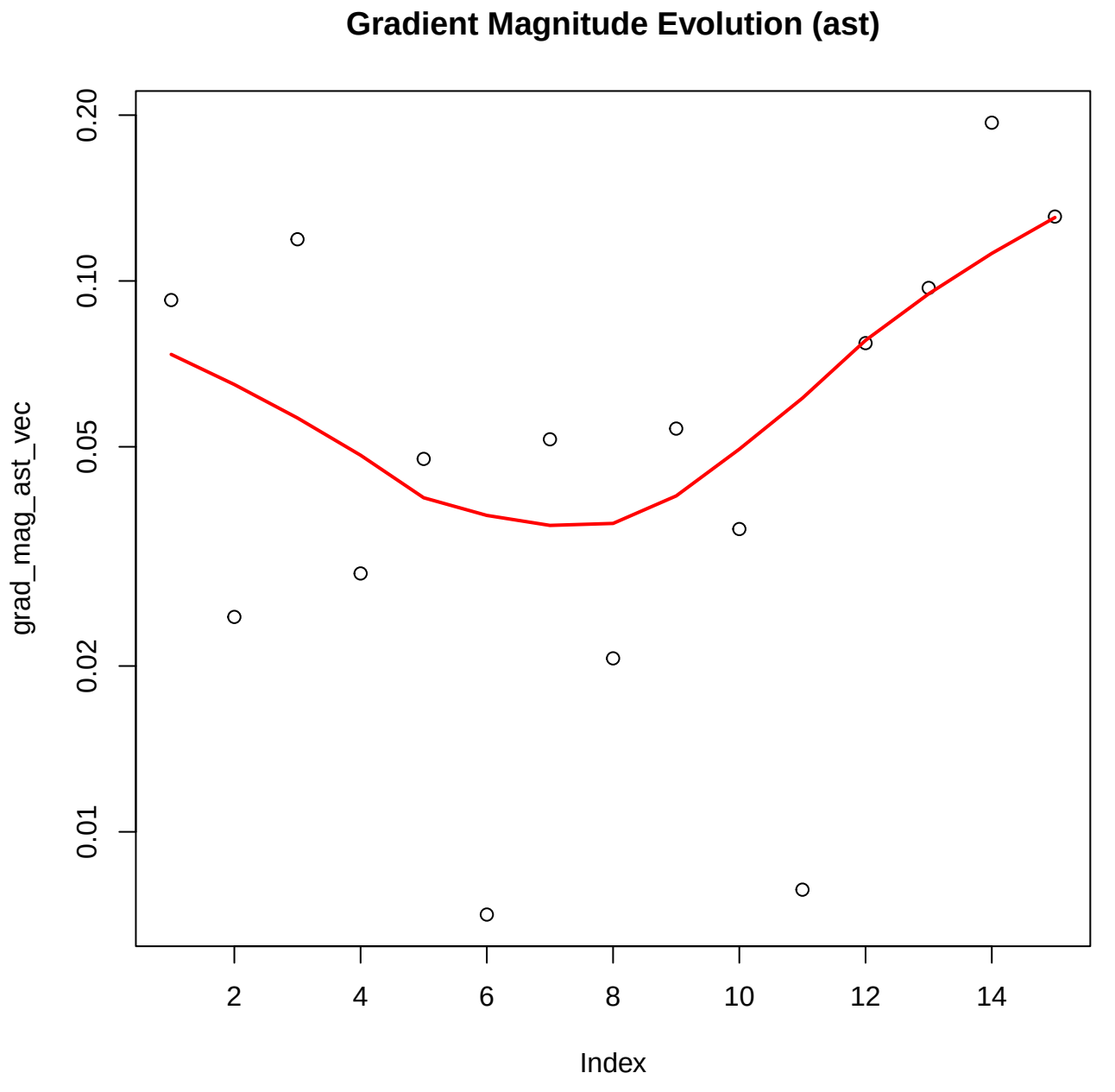
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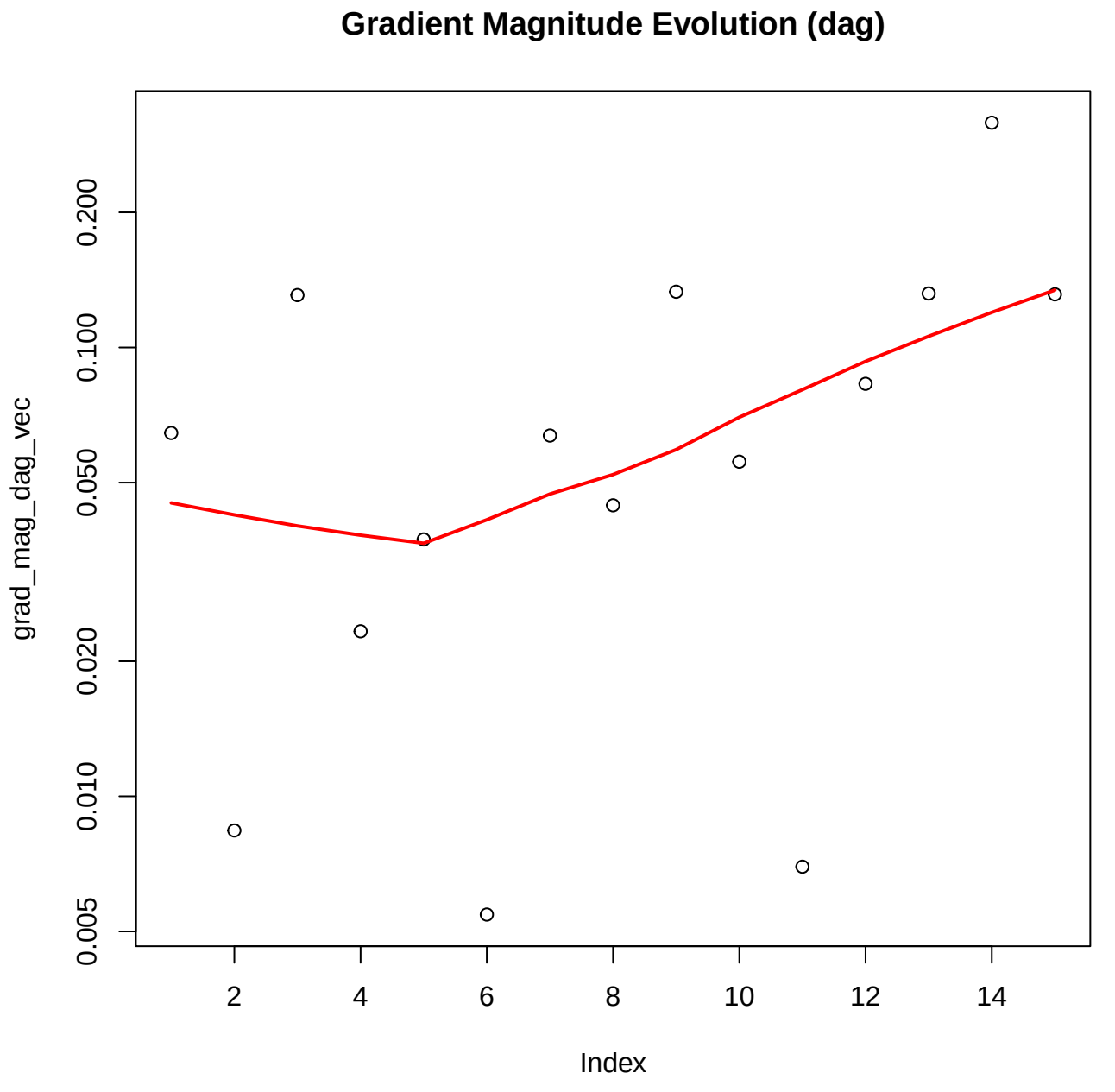
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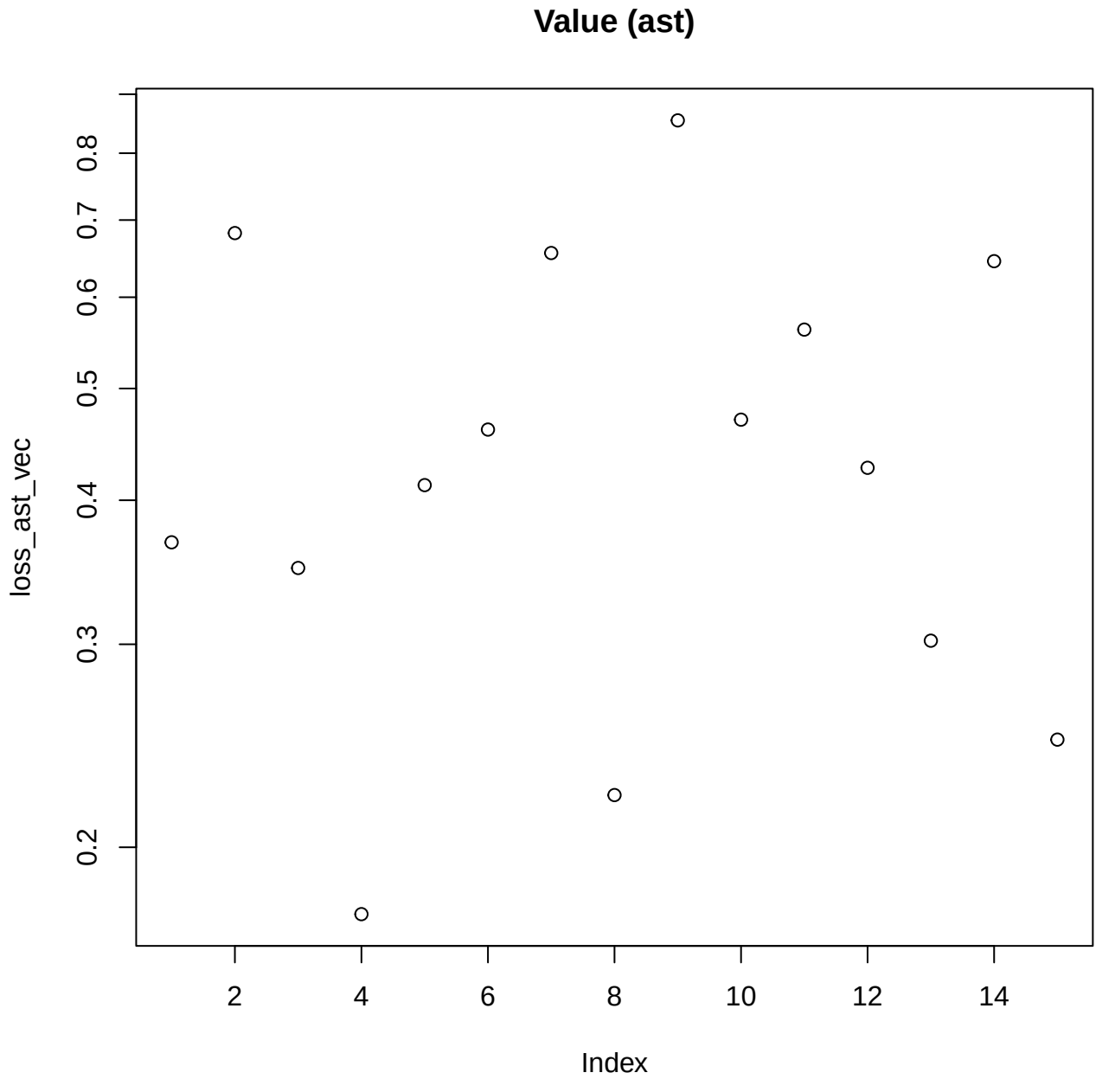
Value (ast)
loss_ast_vec
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0.6
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2
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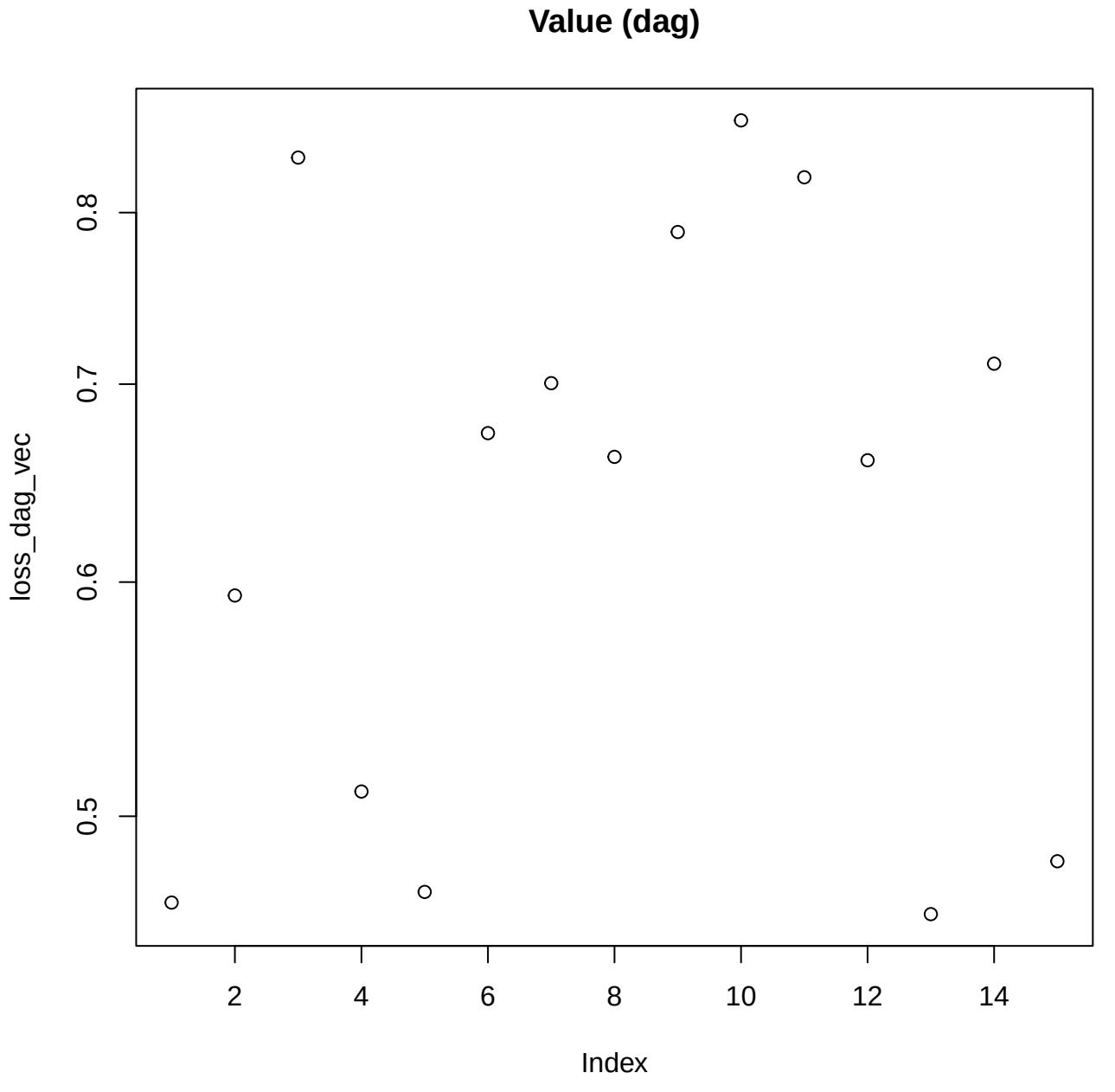
12
14
Index


Value (dag)
loss_dag_vec
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0.8
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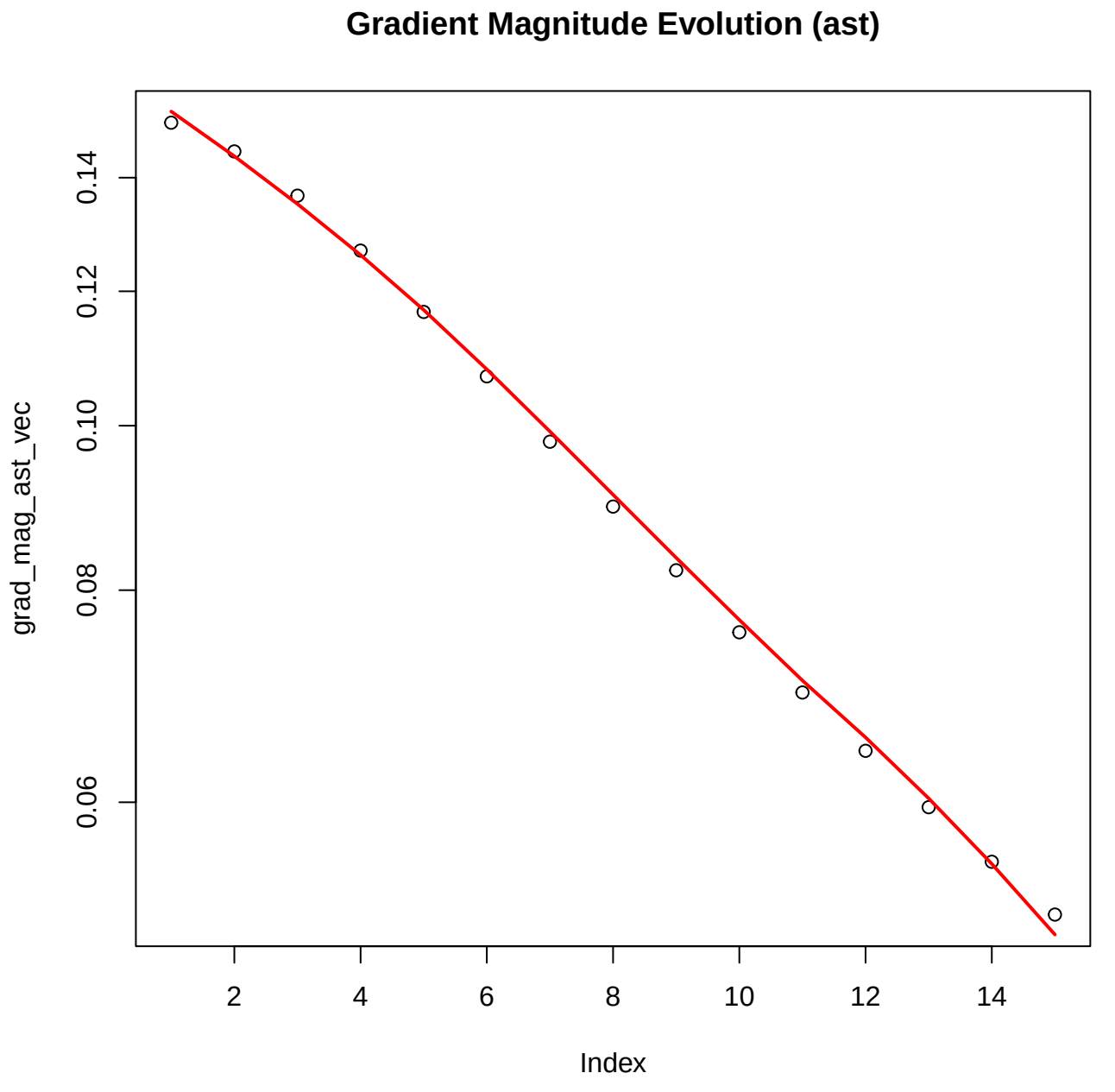
10
12
14
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.10
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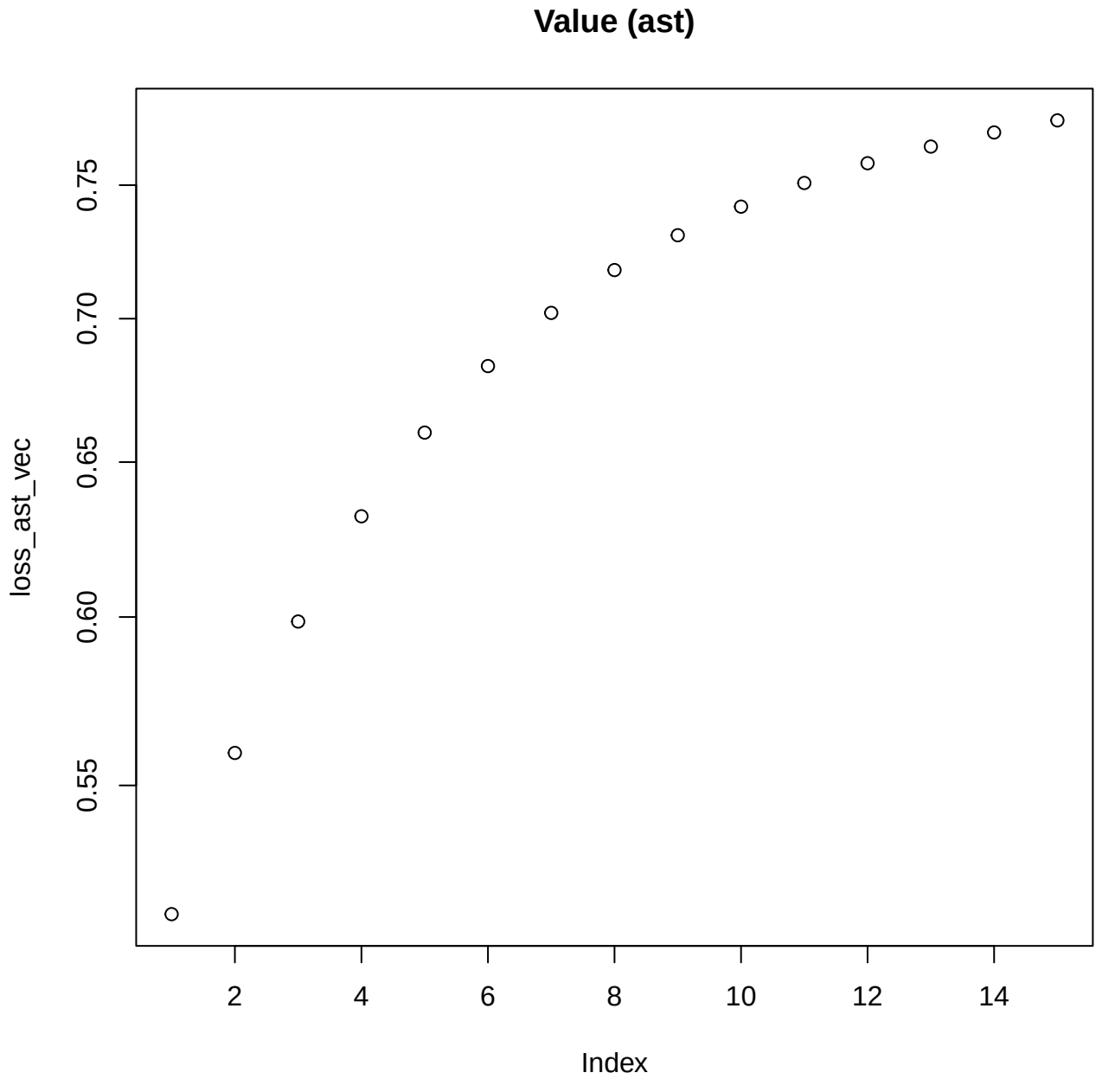
12
14
Index


Value (ast)
loss_ast_vec
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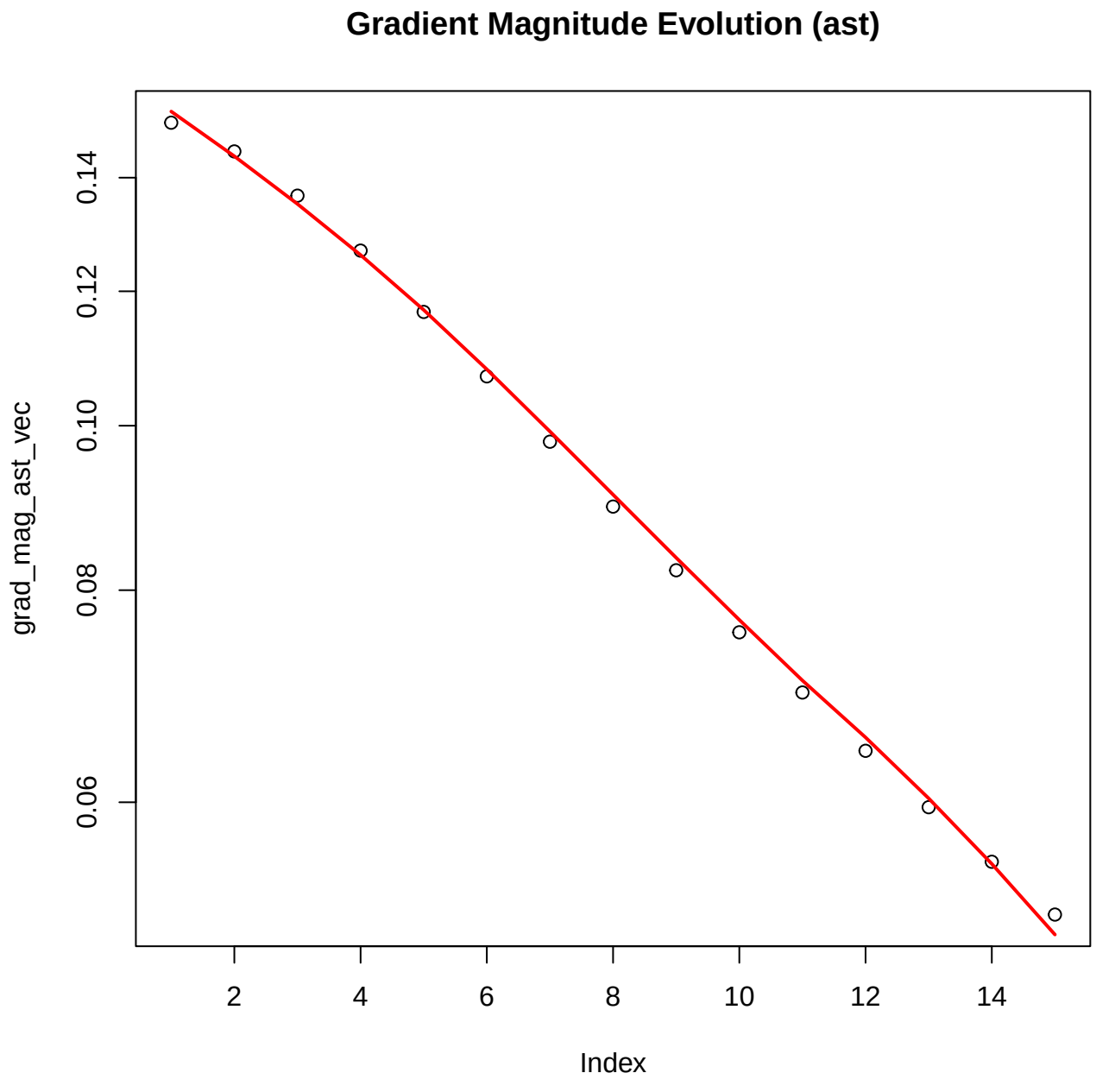
14
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.10
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14
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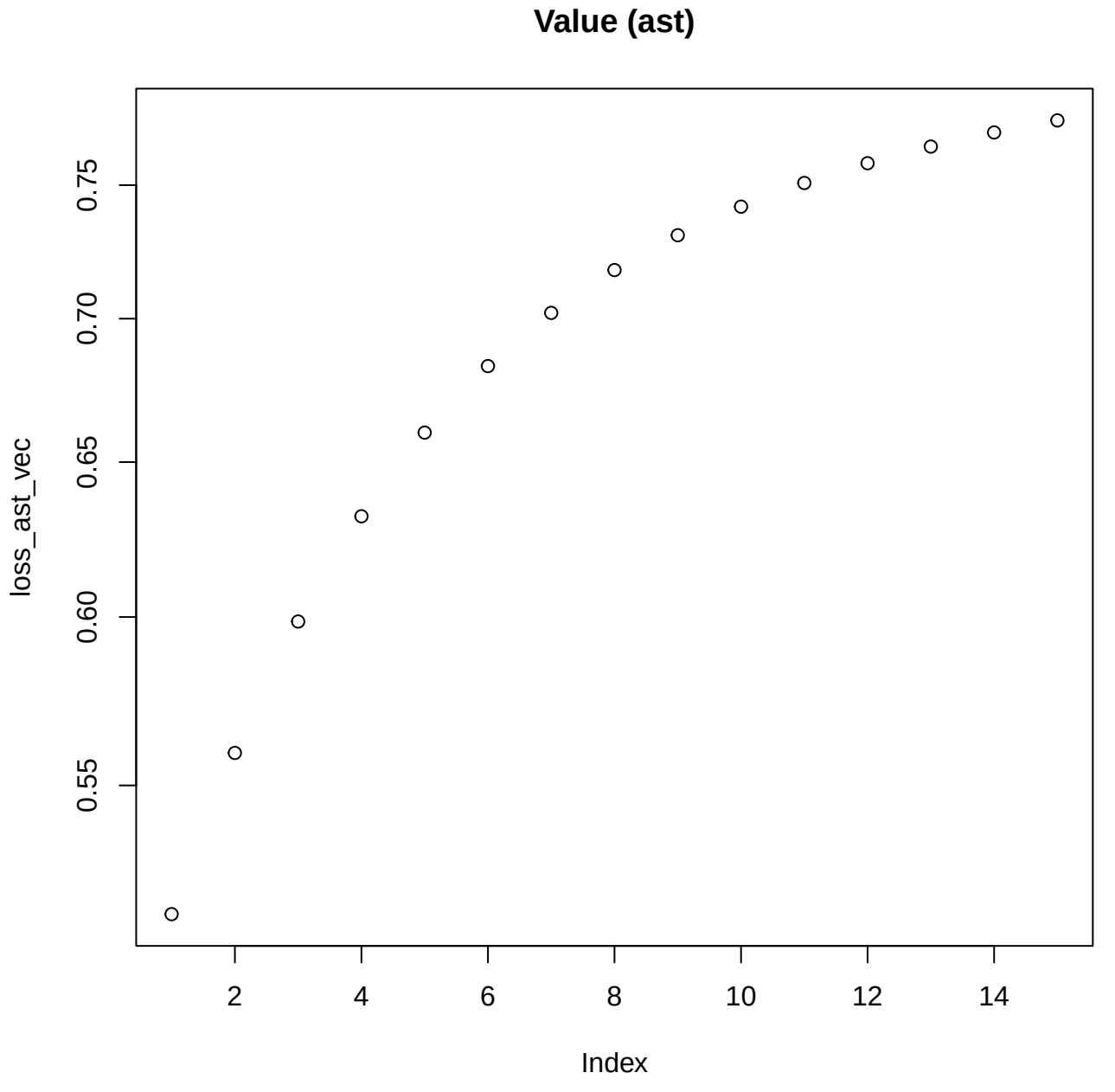
Value (ast)
loss_ast_vec
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0.65
0.70
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14
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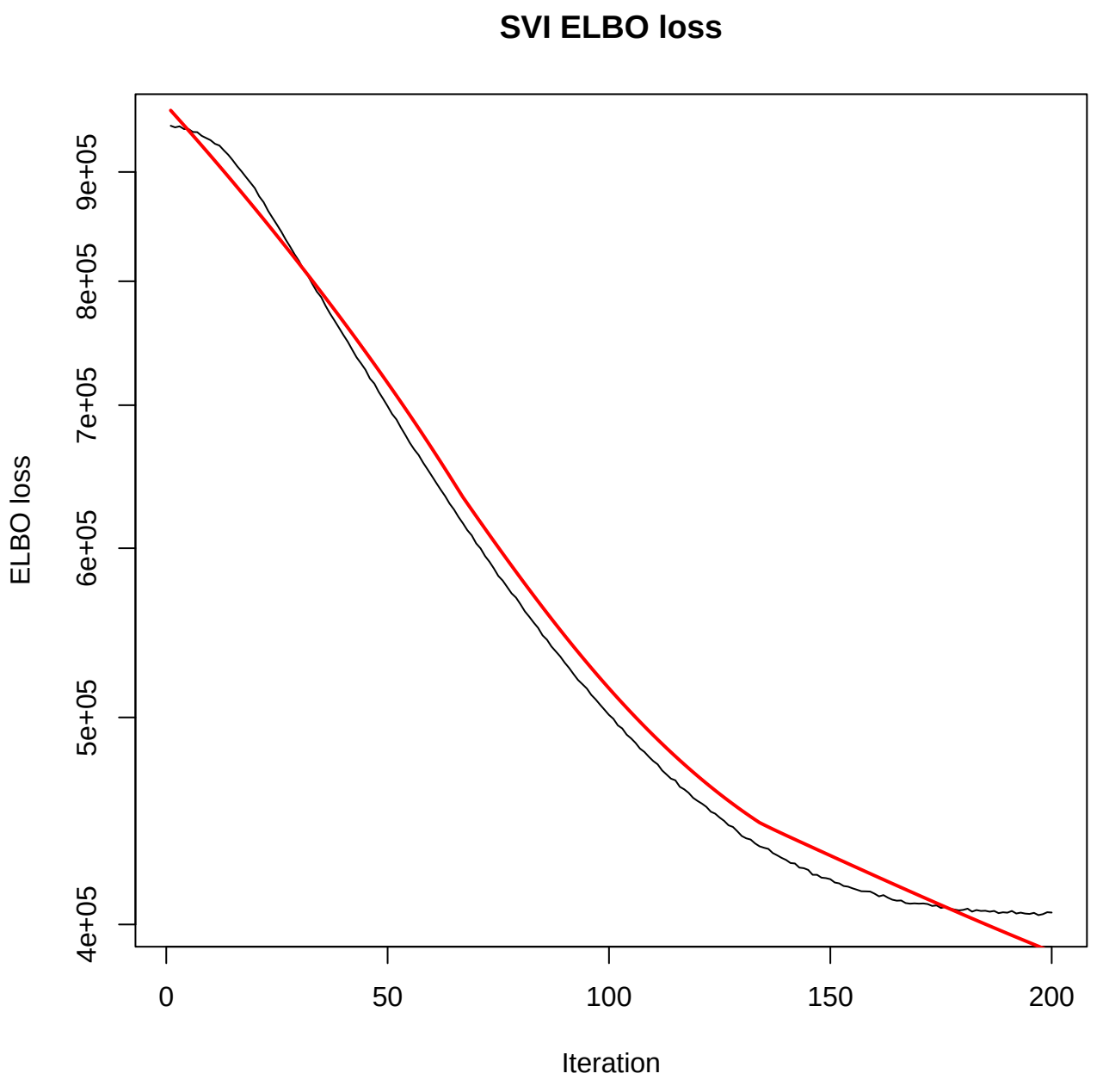
SVI ELBO loss
ELBO loss
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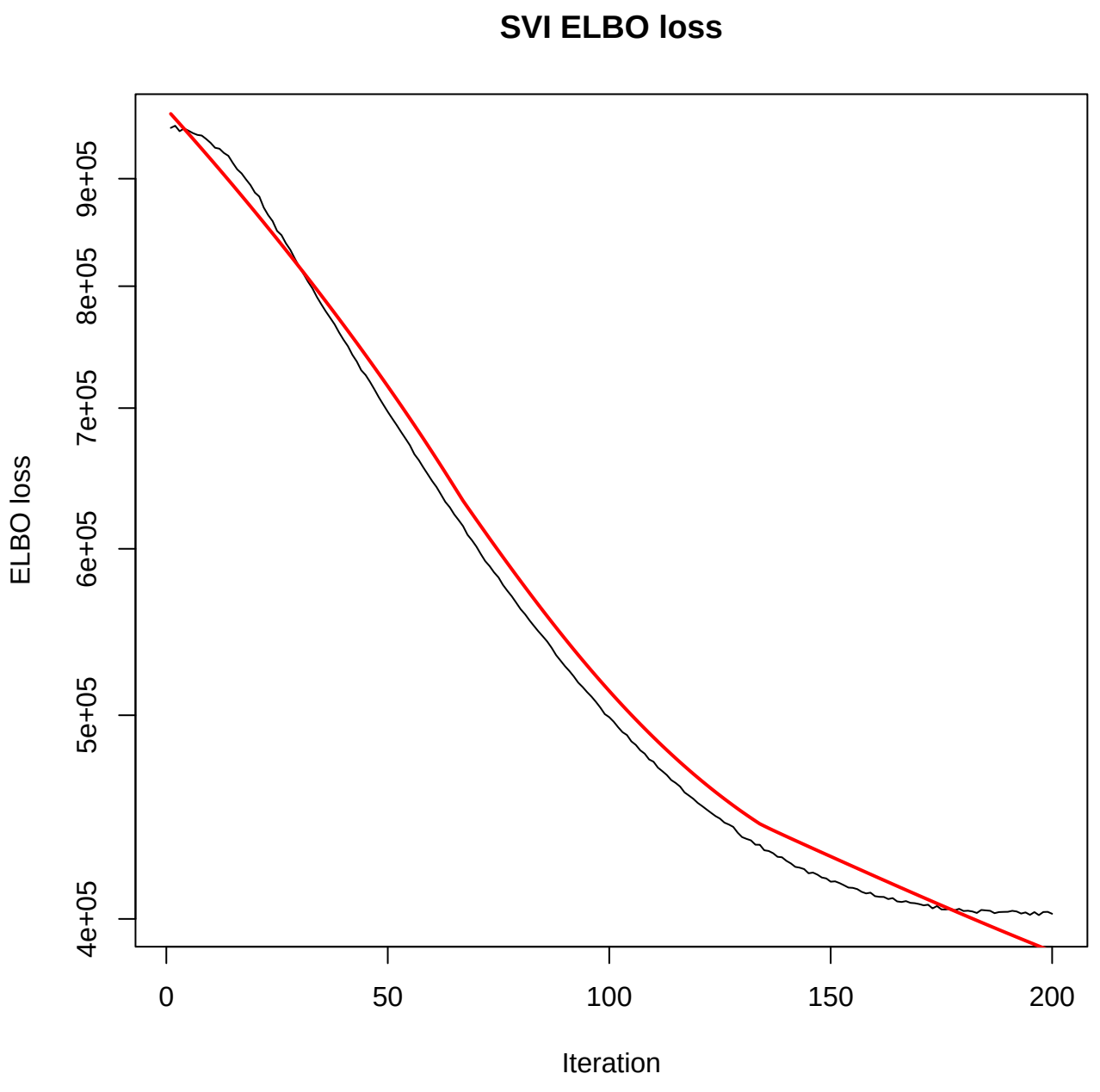
8e+05
7e+05
6e+05
5e+05
4e+05
0
50
100
150
200
Iteration


SVI ELBO loss
ELBO loss
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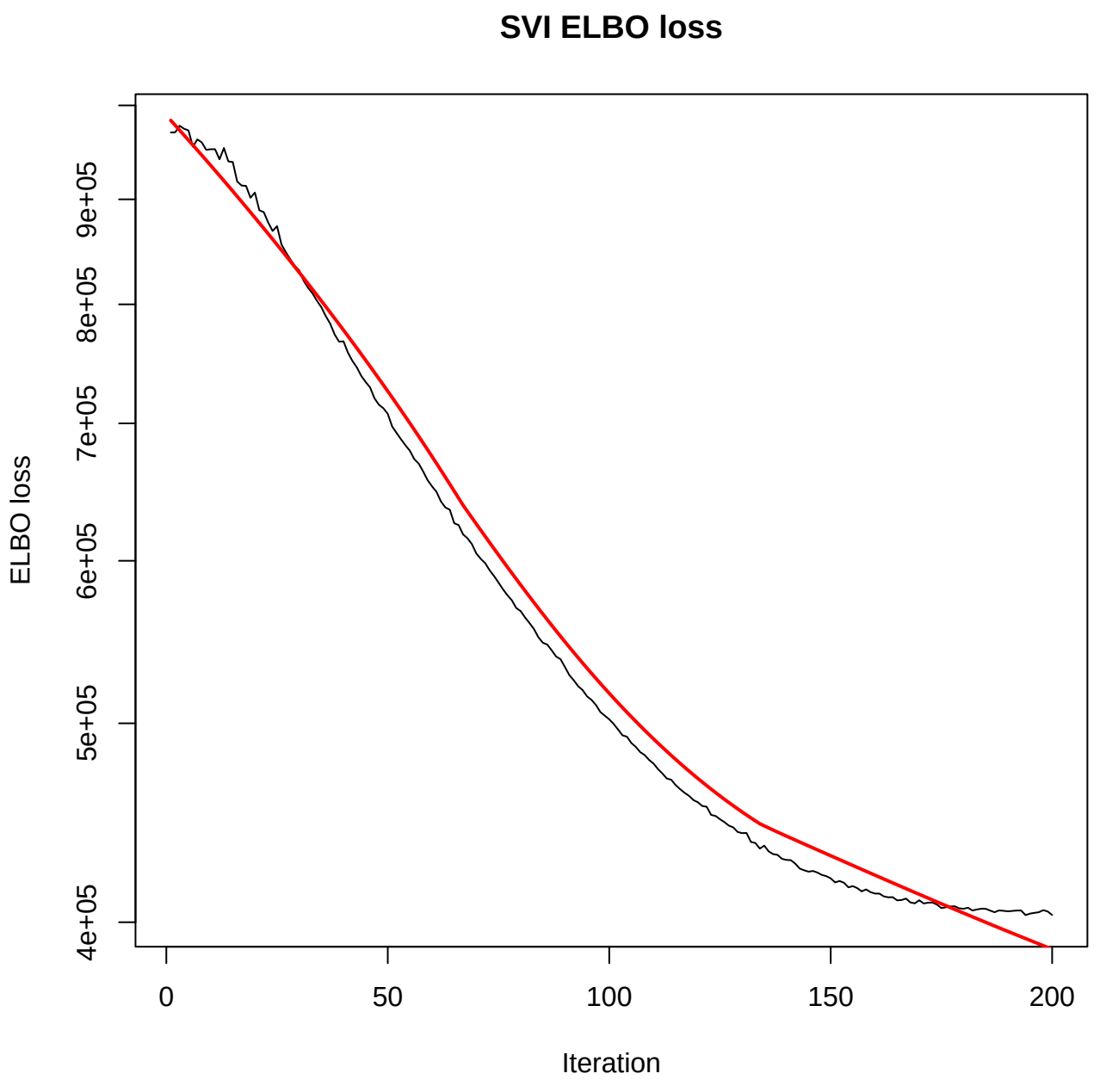
Iteration


SVI ELBO loss
ELBO loss
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100
150
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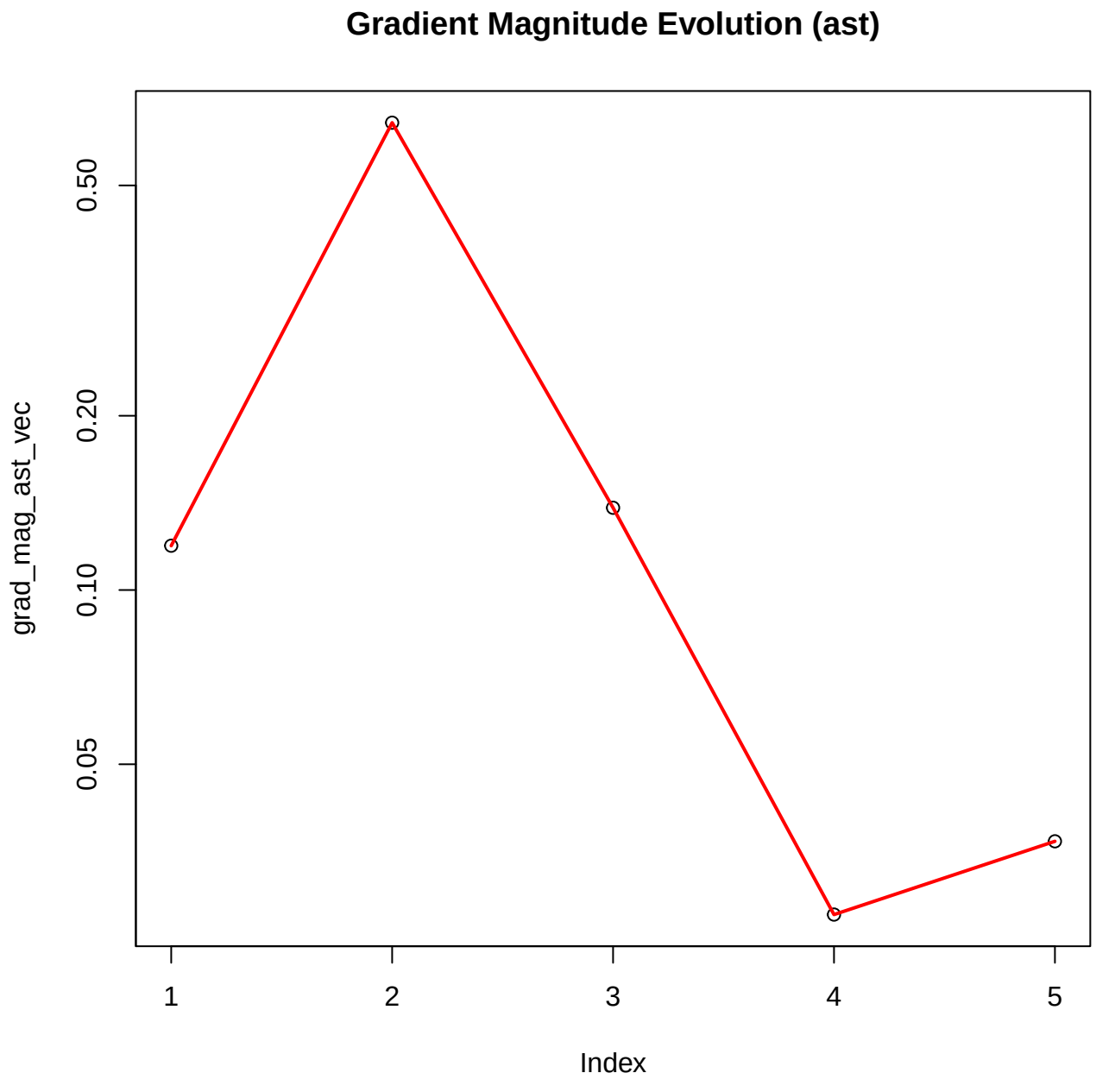
Iteration


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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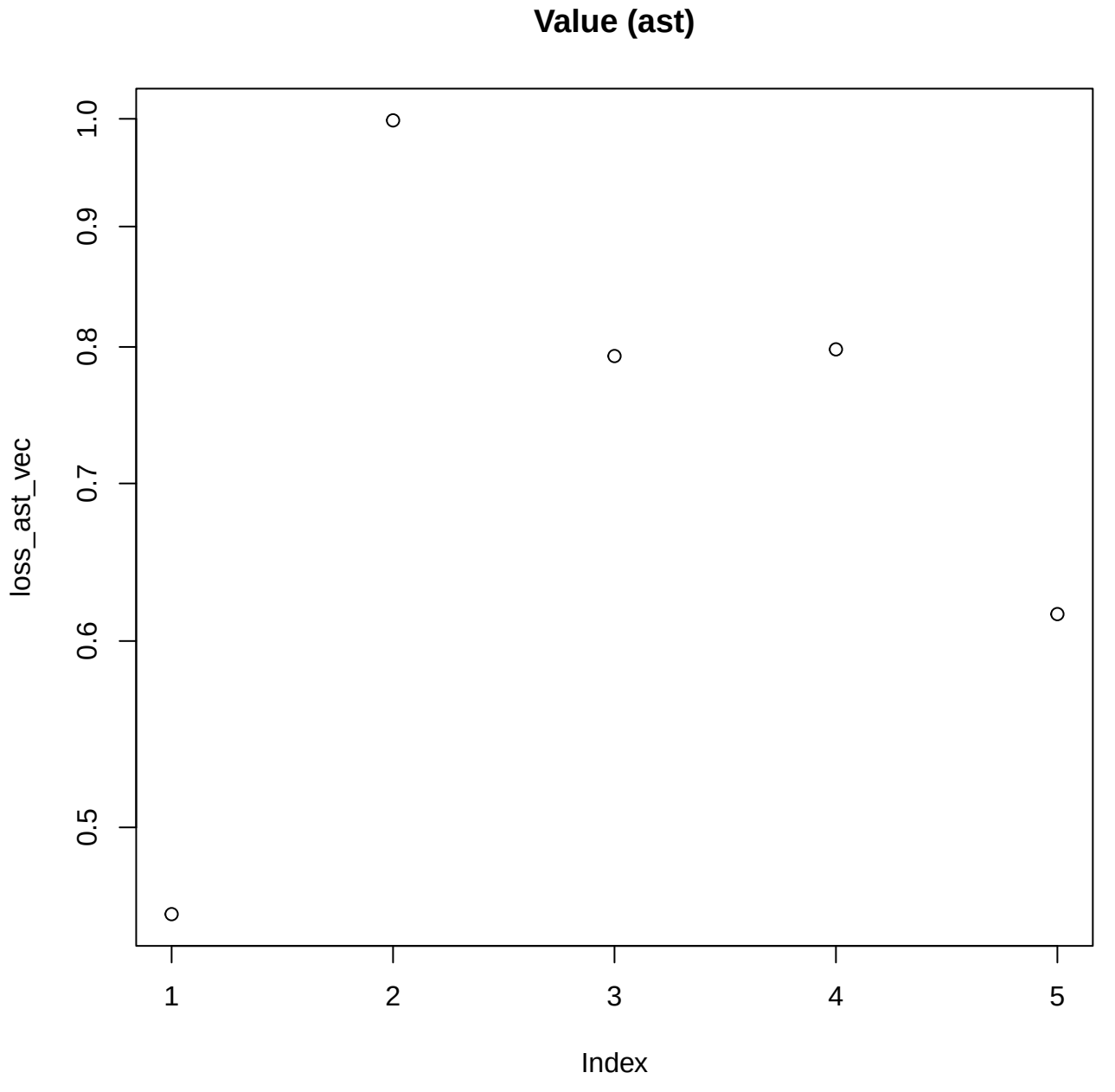
2
3
4
5
Index


Value (ast)
loss_ast_vec
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0.9
0.8
0.7
0.6
0.5
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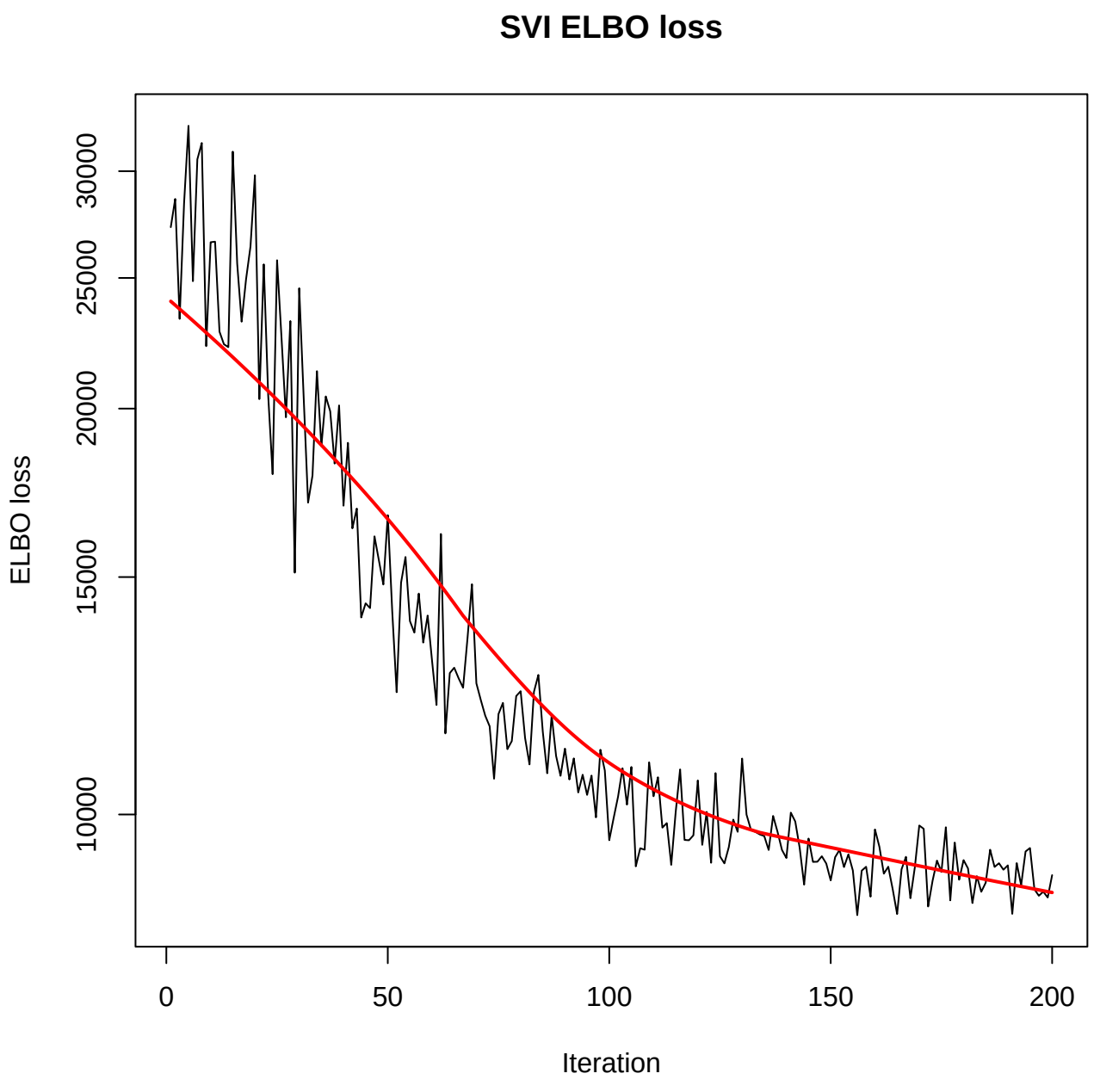
2
3
4
5
Index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ELBO loss
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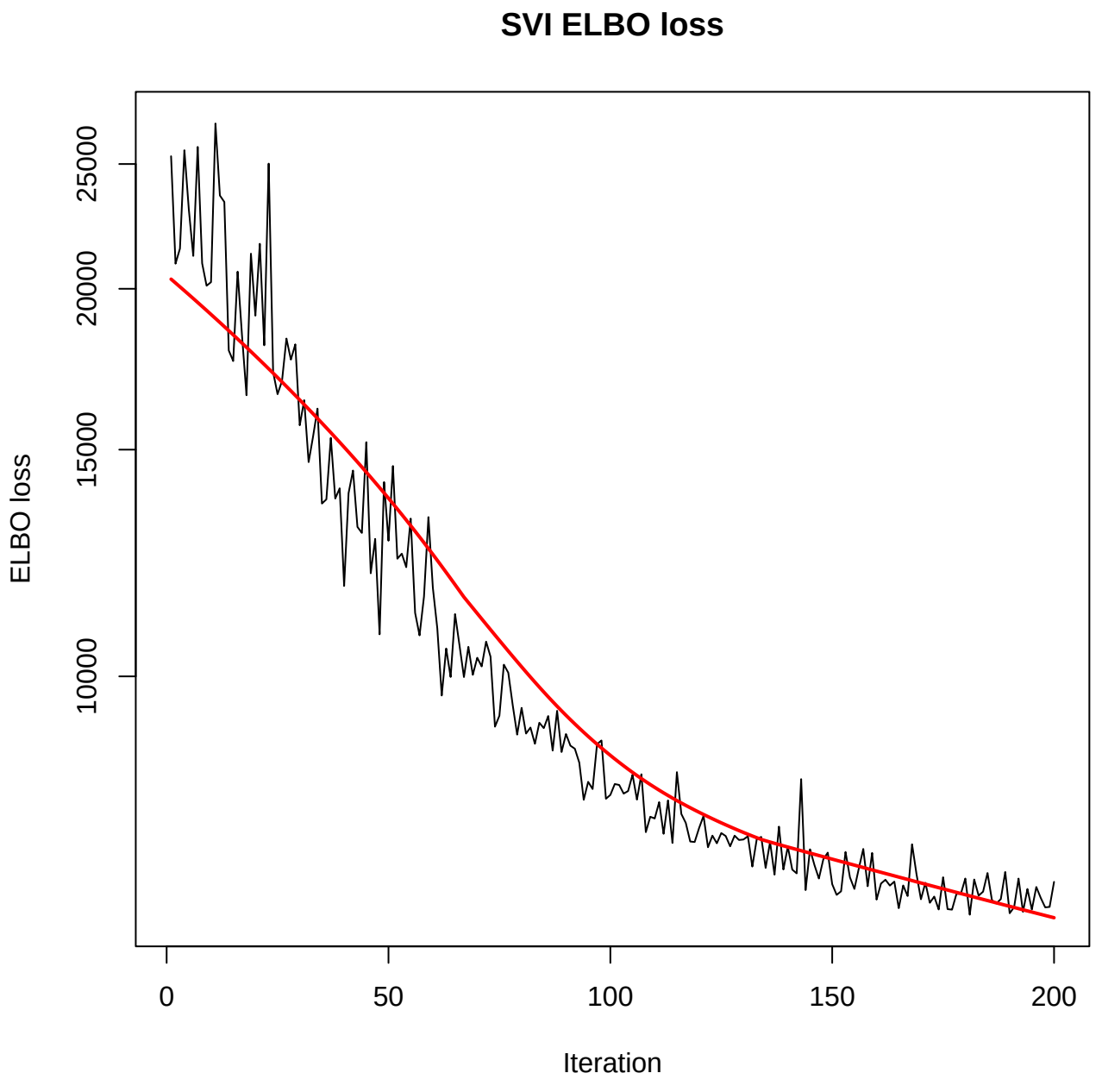
100
150
200
Iteration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ELBO loss
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100
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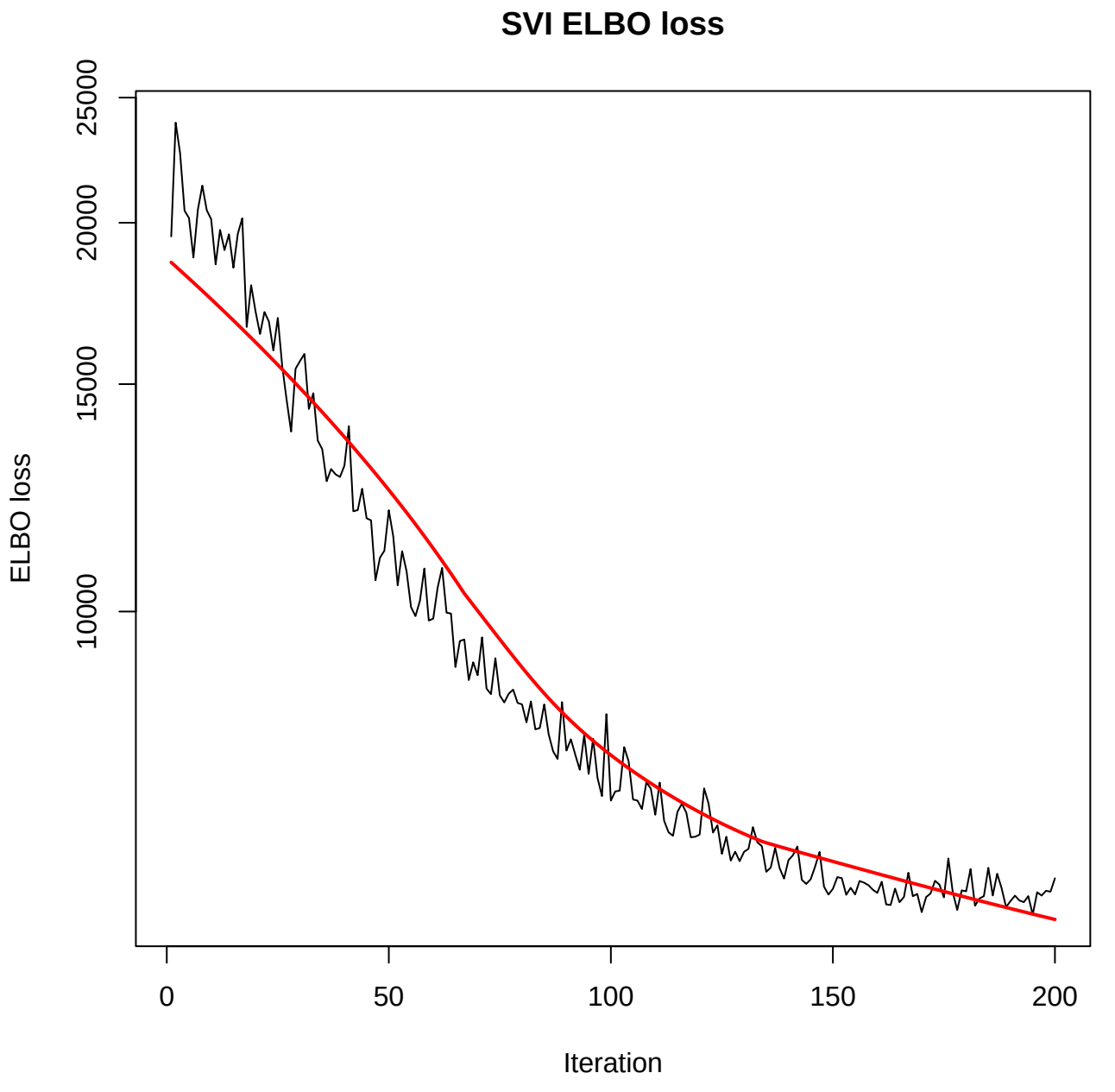
200
Iteration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ELBO loss
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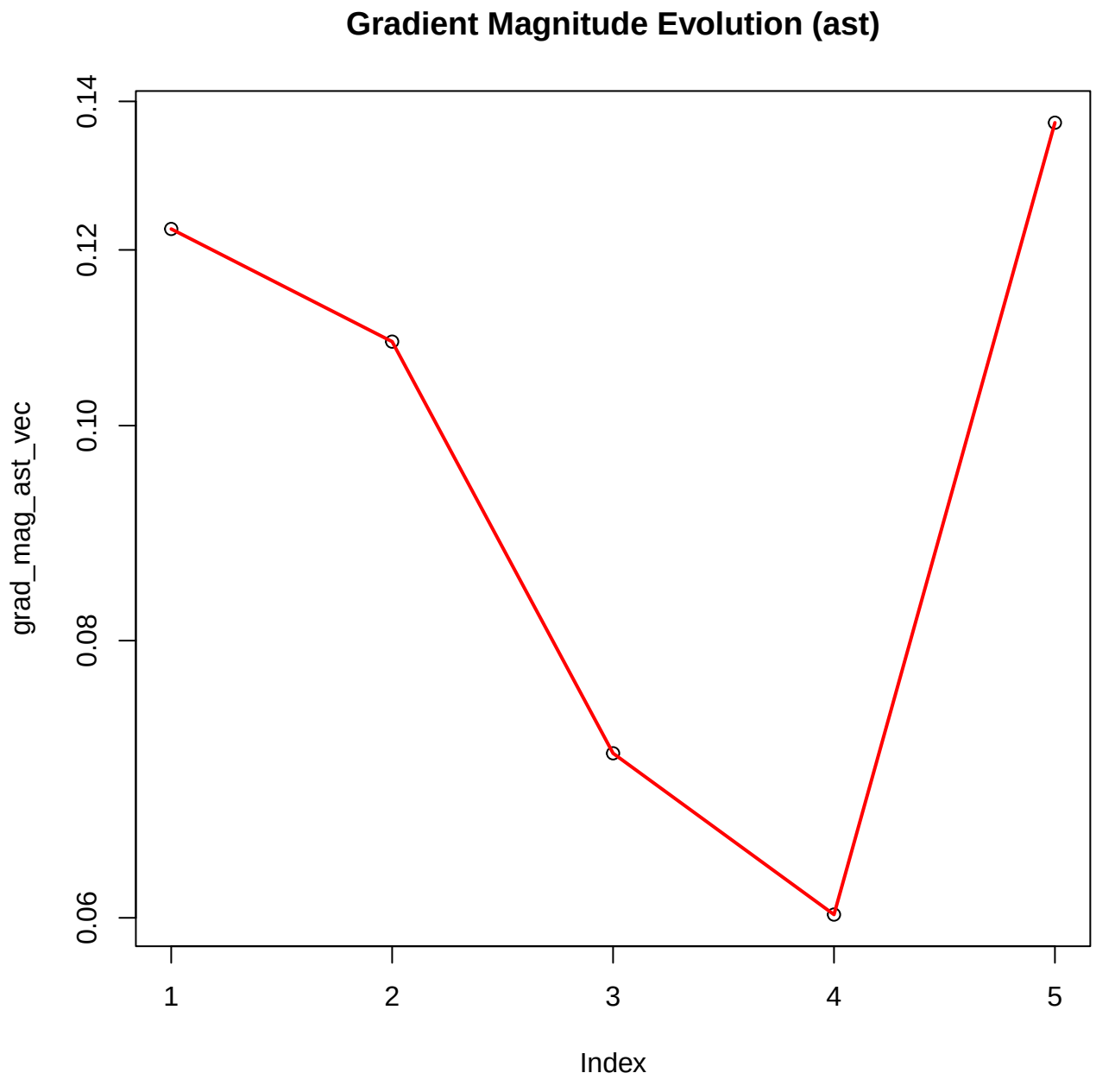
Iteration


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.08
0.10
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0.14
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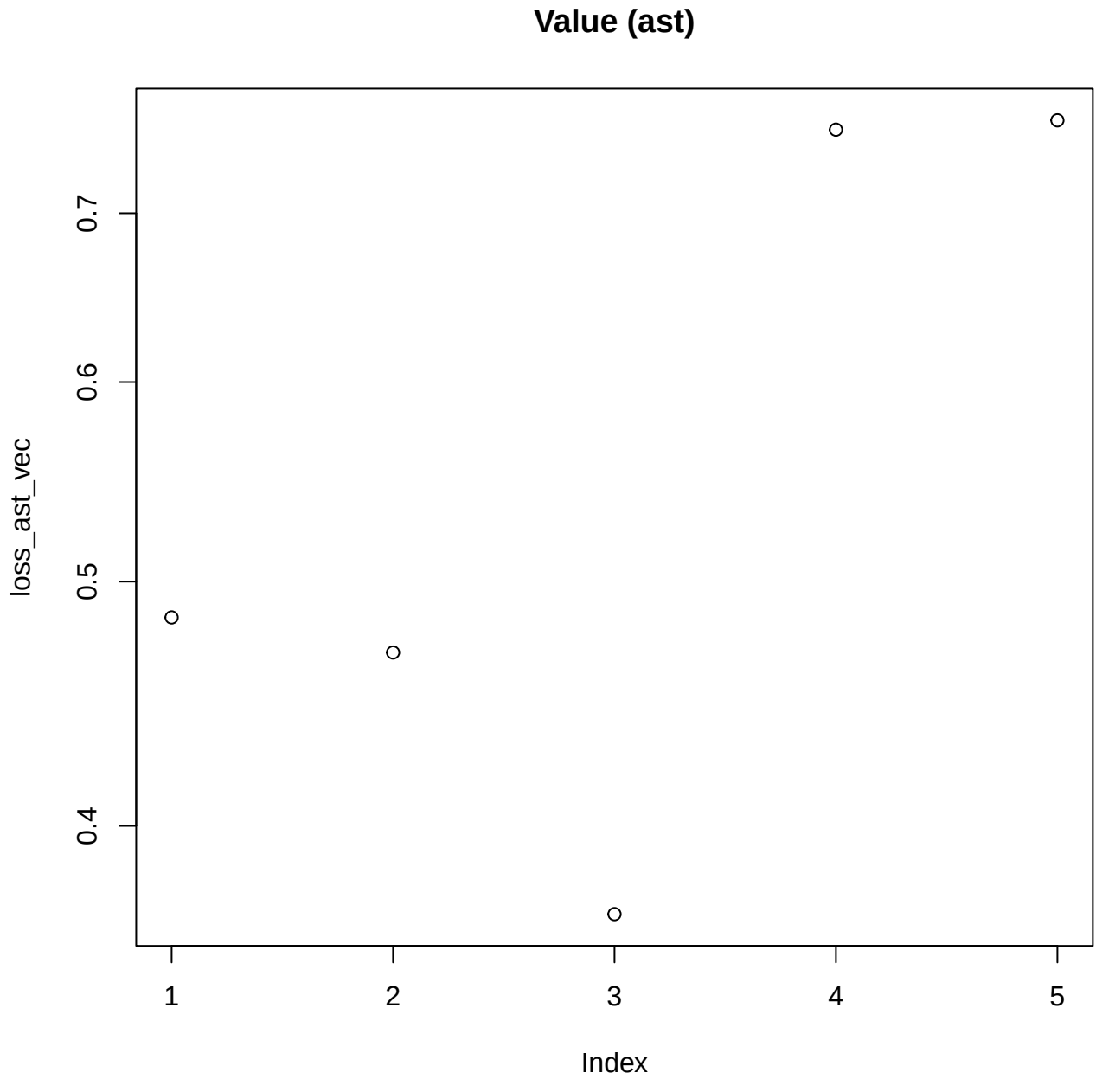
2
3
4
5
Index


Value (ast)
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0.7
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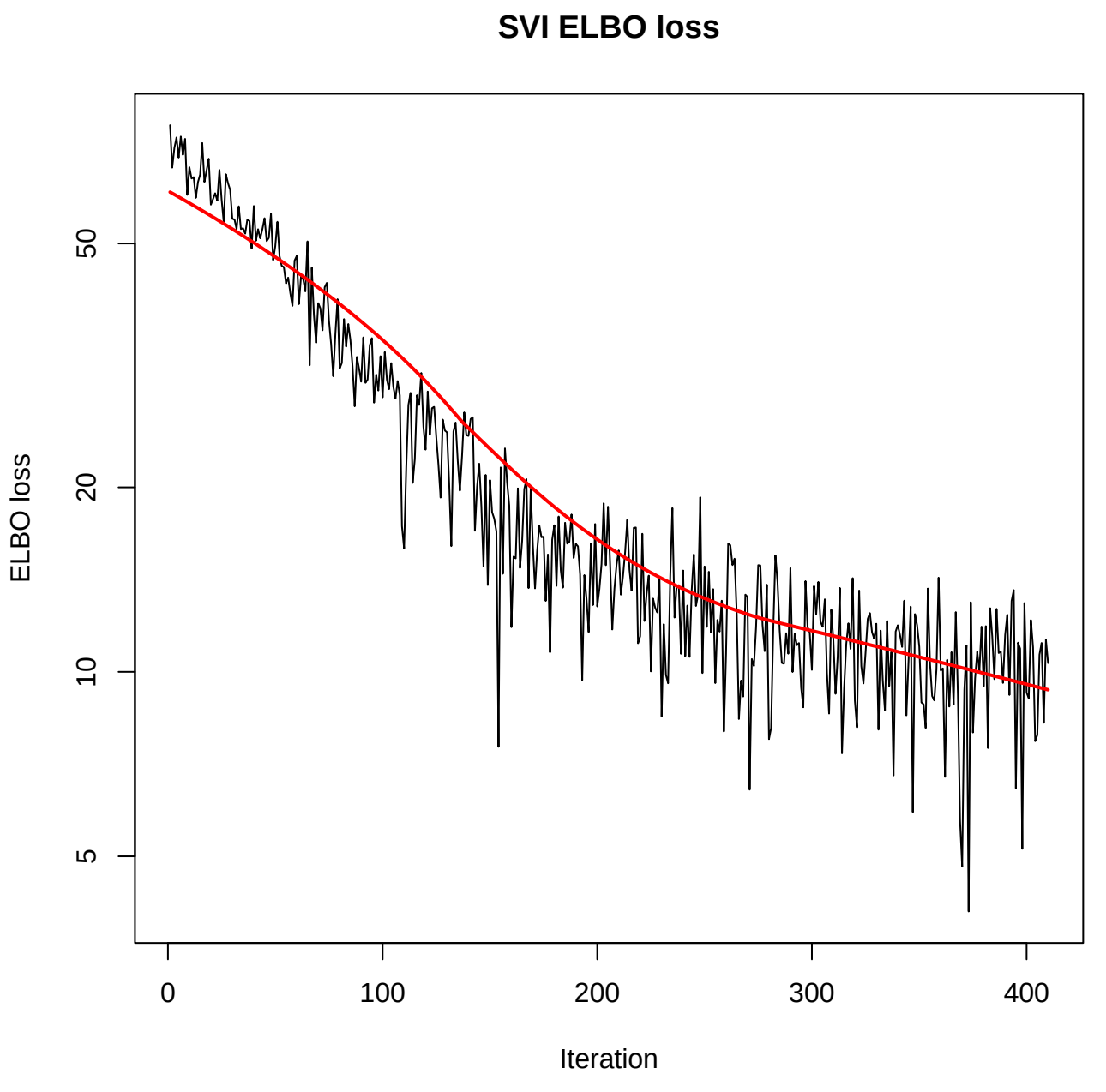
3
4
5
Index


SVI ELBO loss
ELBO loss
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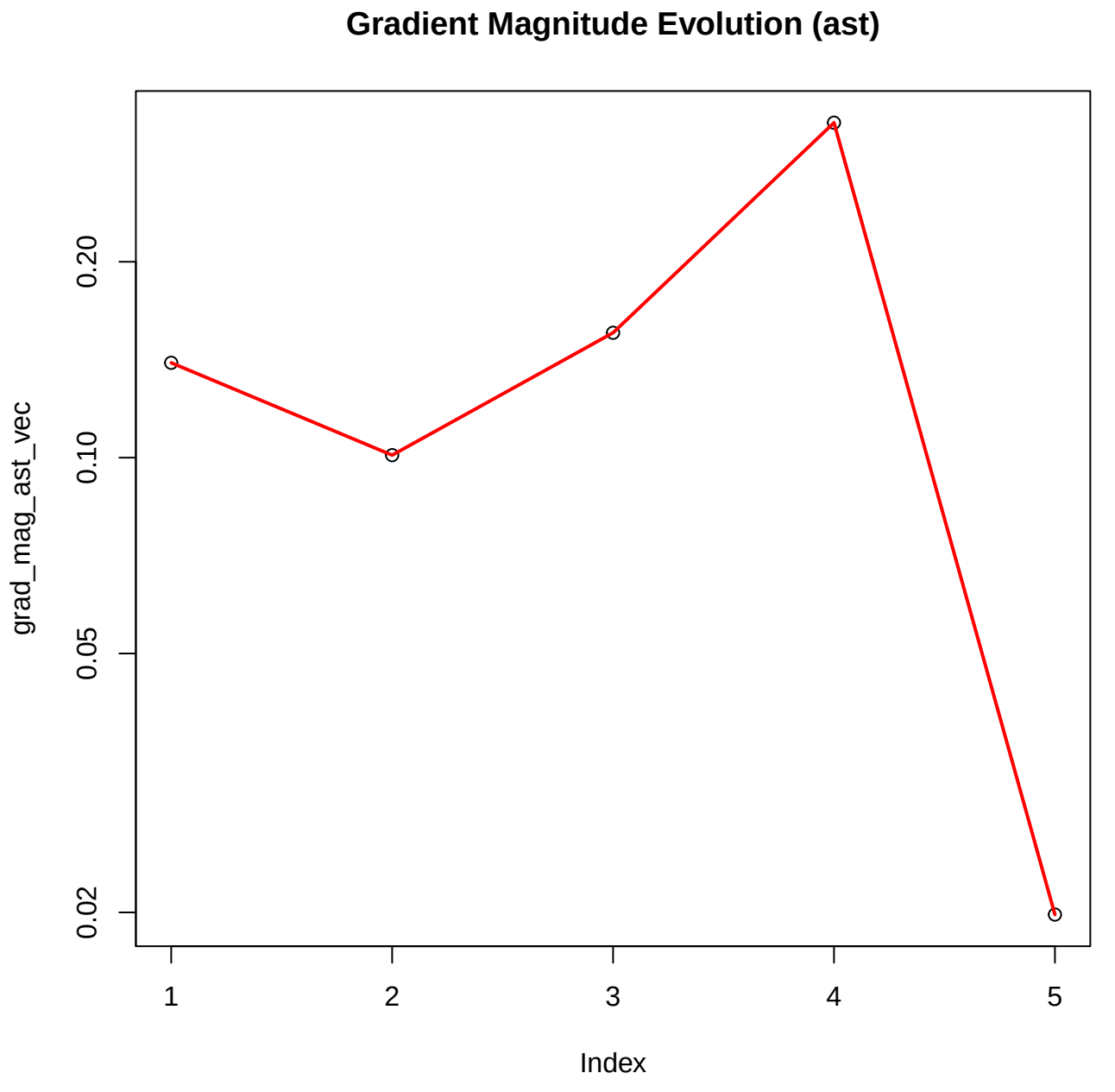
10
5
0
100
200
300
400
Iteration


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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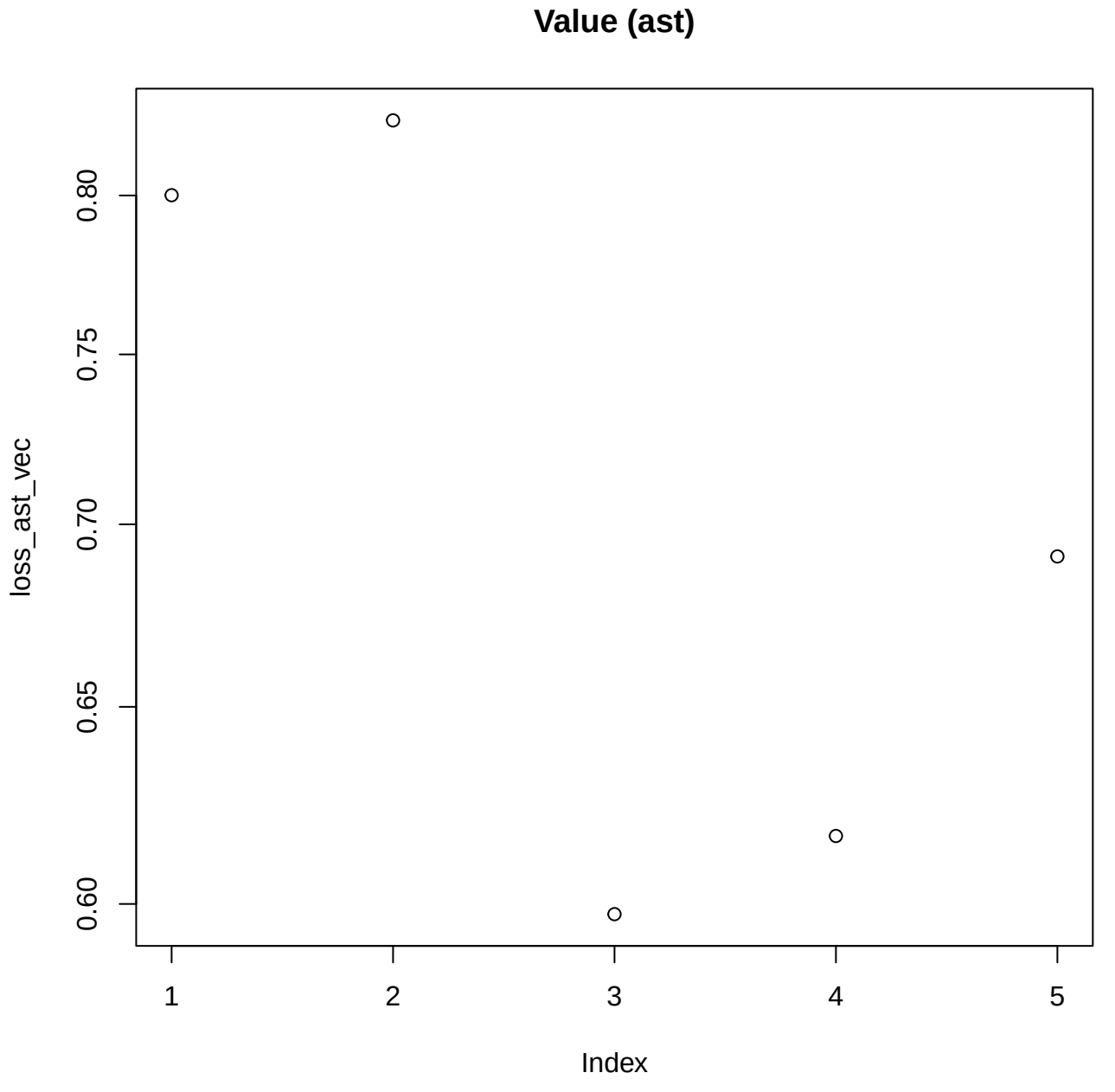
0.05
0.10
0.20
1
2
3
4
5
Index


Value (ast)
loss_ast_vec
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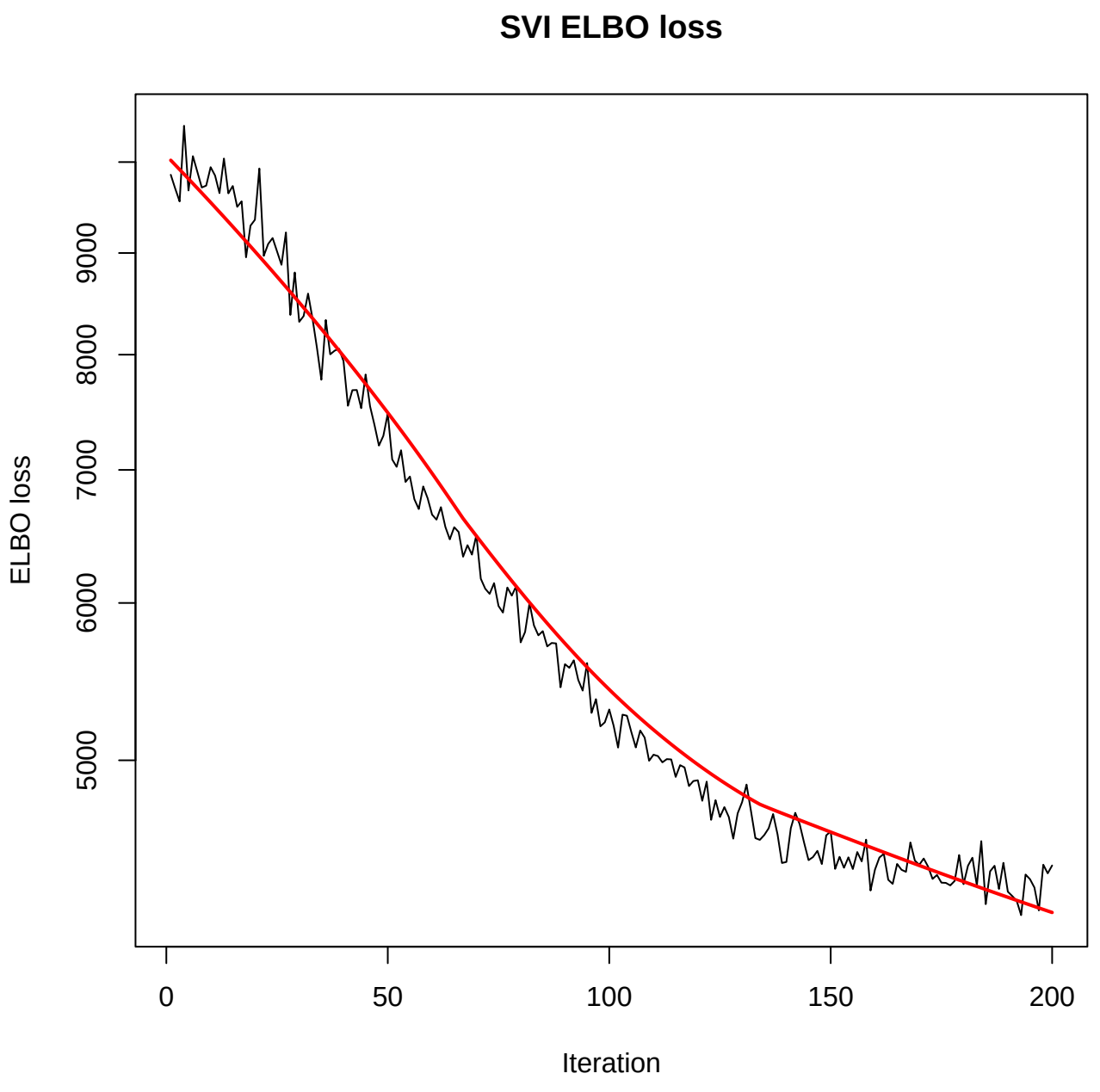
5
Index


SVI ELBO loss
ELBO loss
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6000
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9000
0
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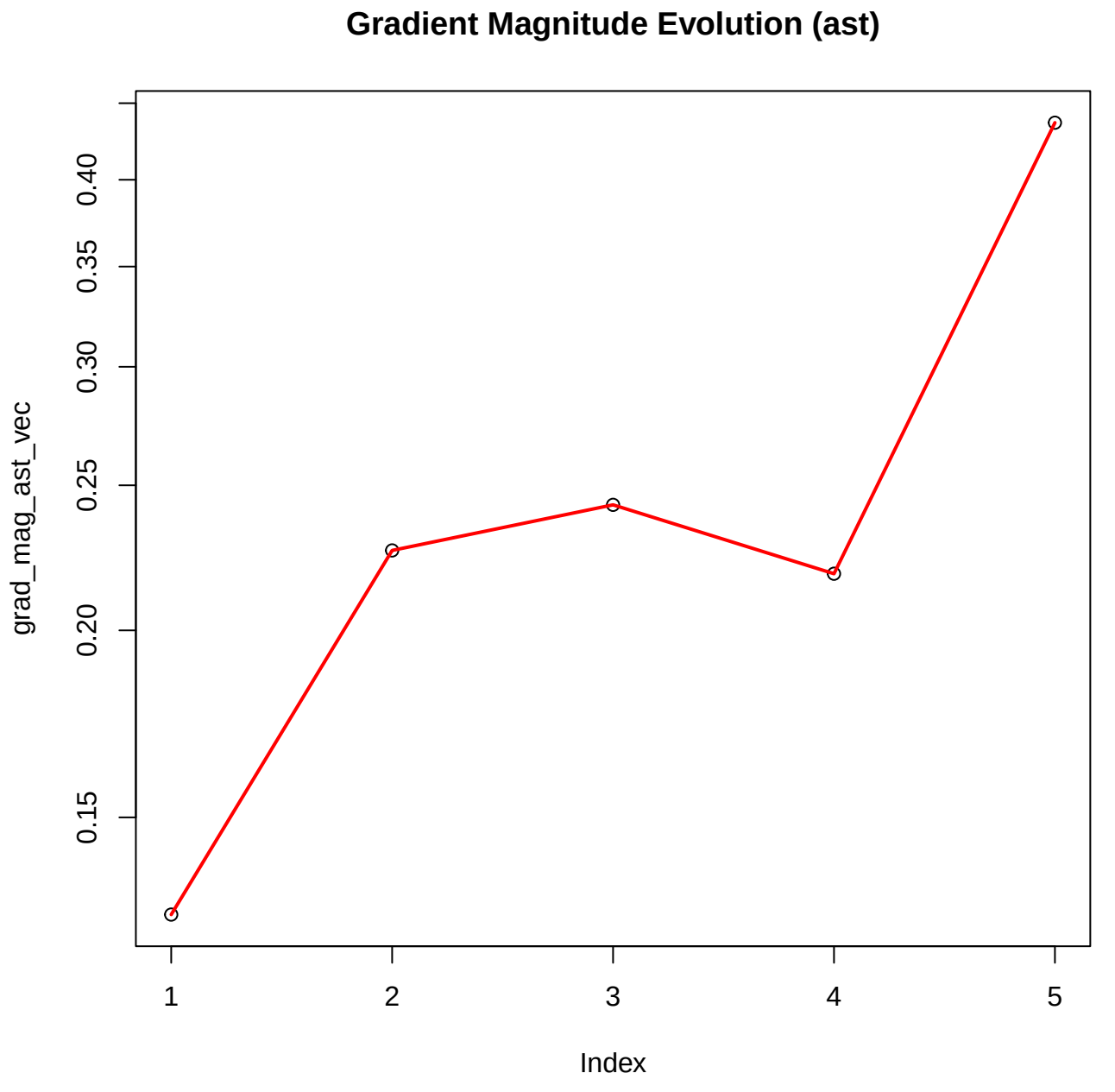
100
150
200
Iteration


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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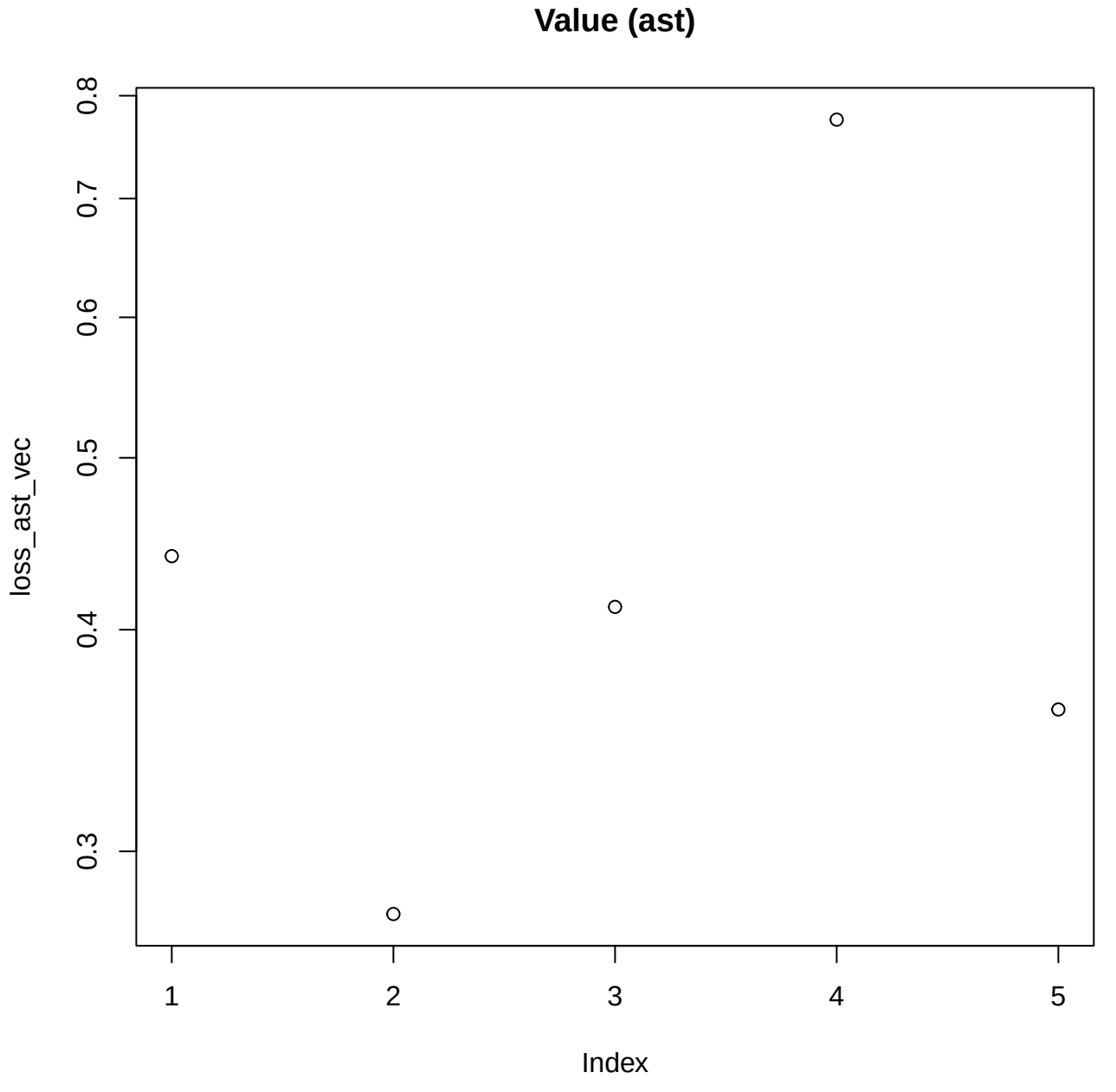
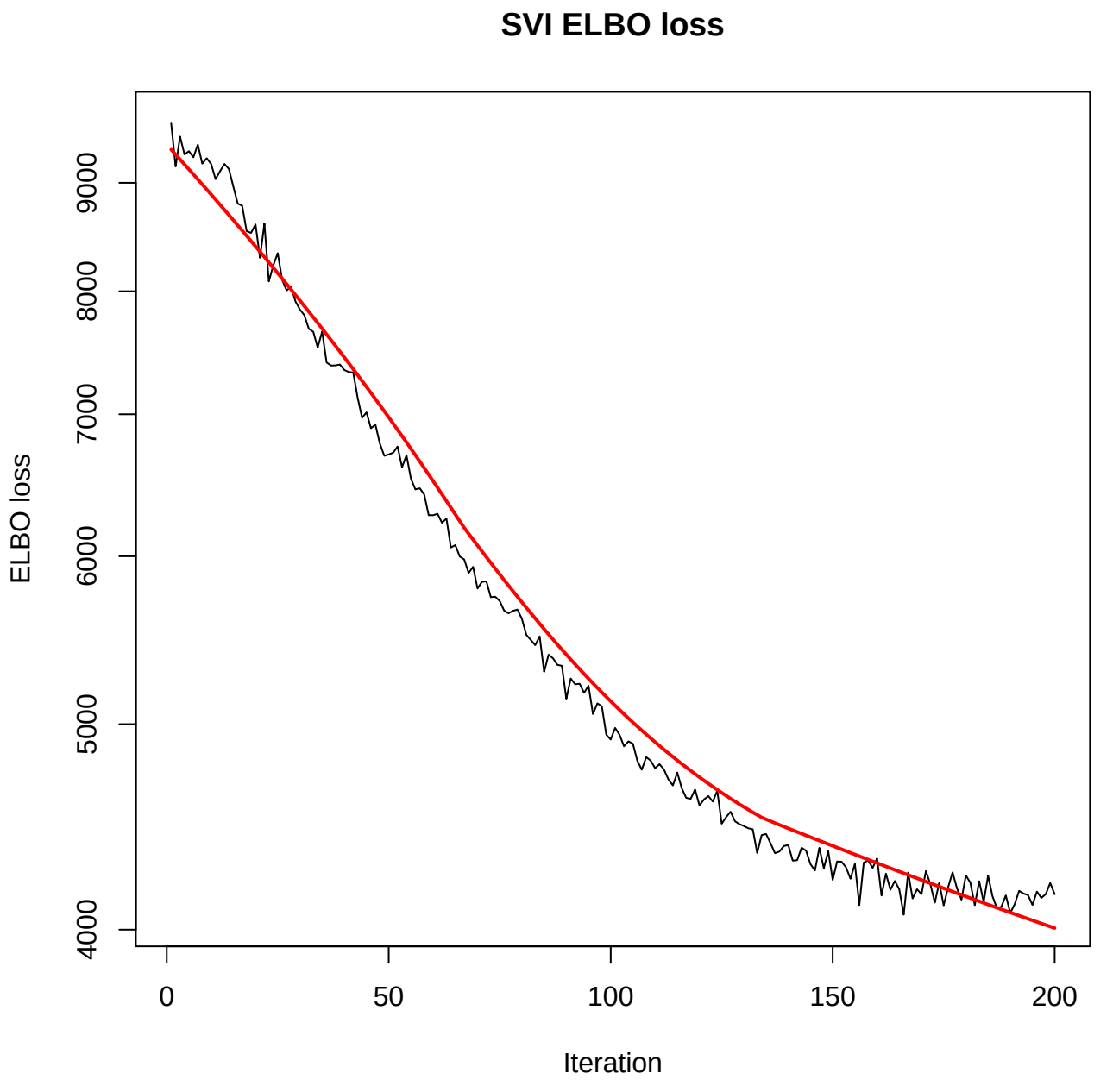
0.25
0.30
0.35
0.40
1
2
3
4
5
Index


Value (ast)
loss_ast_vec
0.3
0.4
0.5
0.6
0.7
0.8
1
2
3
4
5
Index


SVI ELBO loss
ELBO loss
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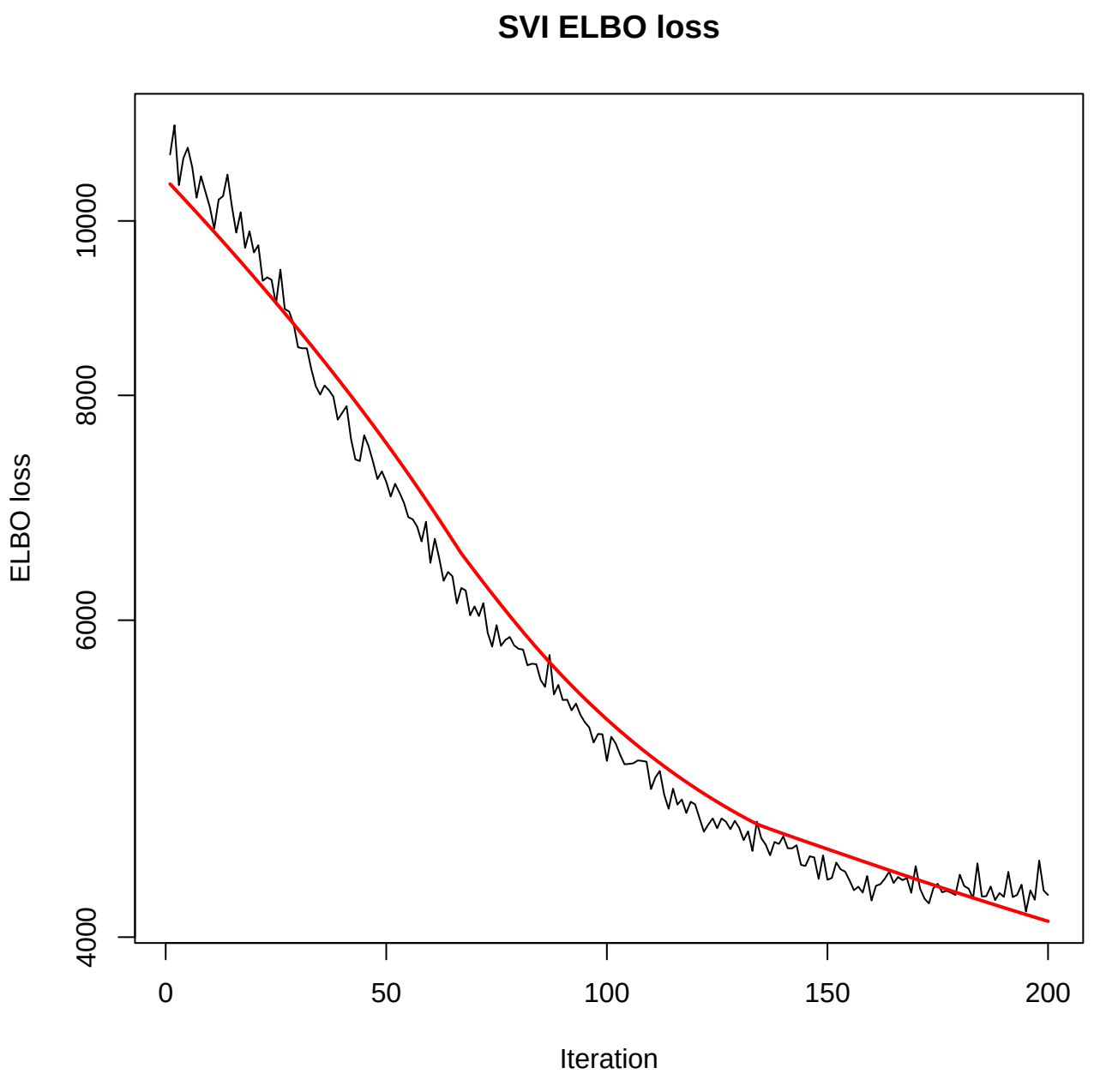
8000
7000
6000
5000
4000
0
50
100
150
200
Iteration


SVI ELBO loss
ELBO loss
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8000
6000
4000
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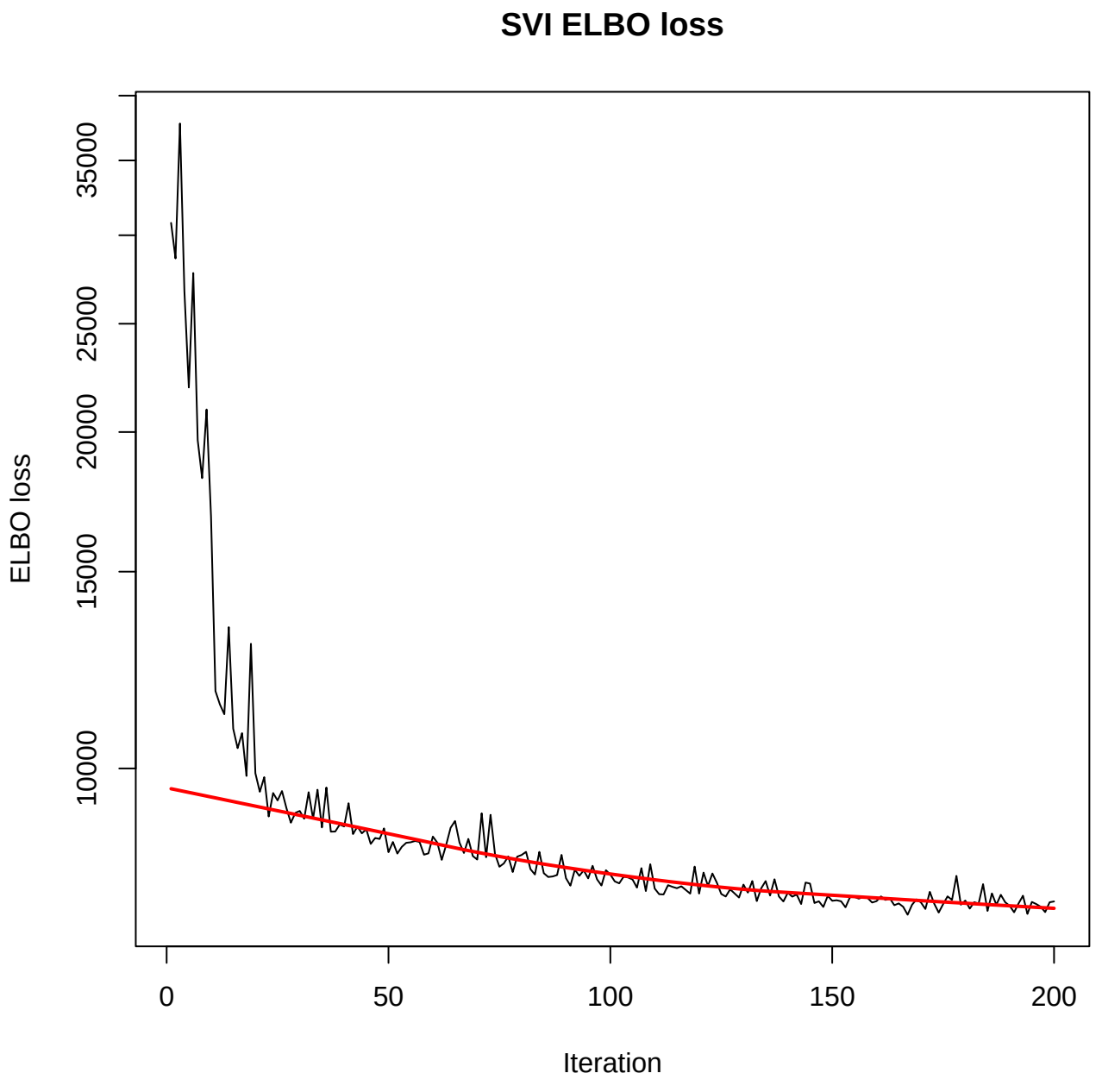
100
150
200
Iteration


SVI ELBO loss
ELBO loss
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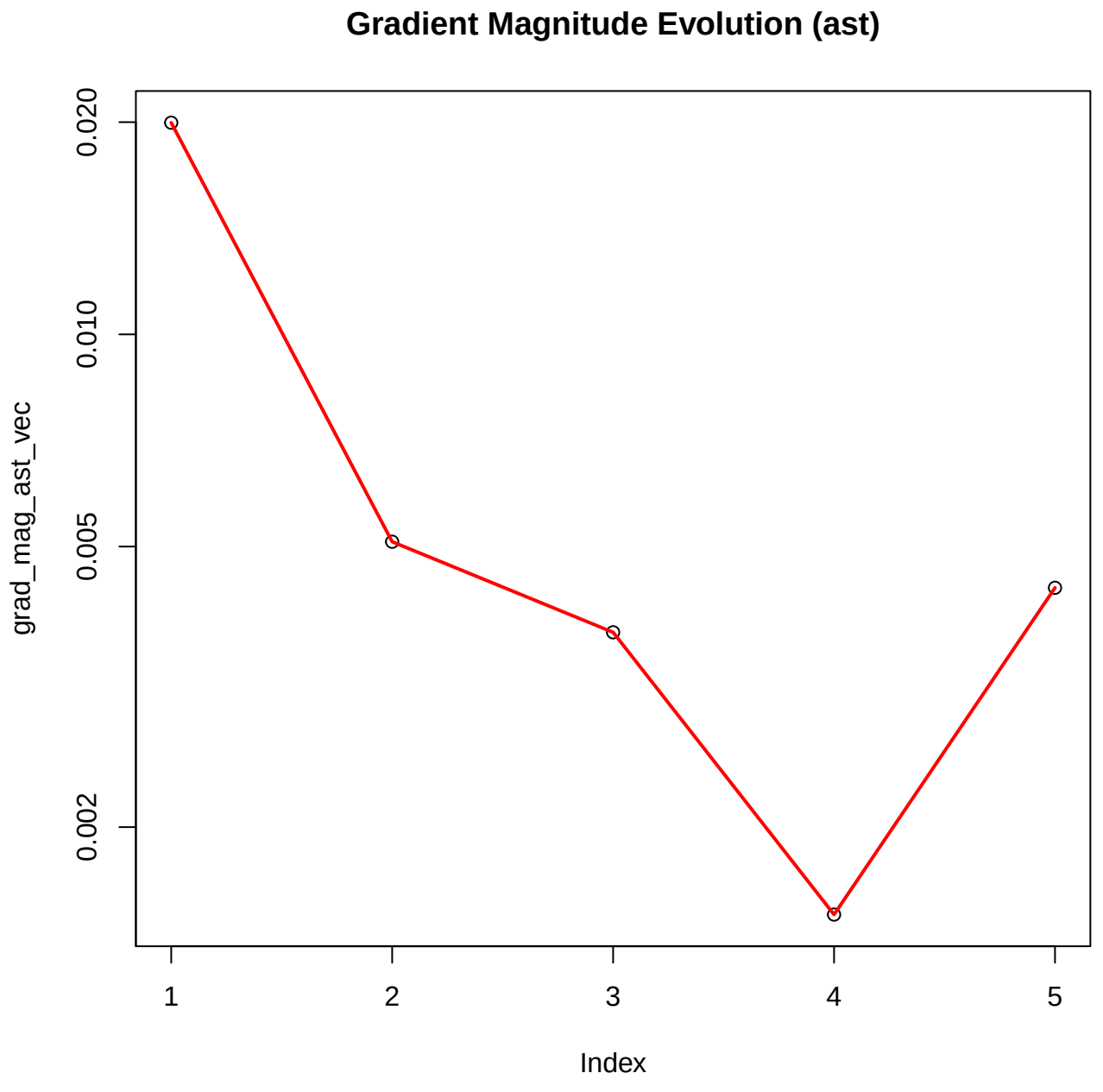
25000
20000
15000
10000
0
50
100
150
200
Iteration


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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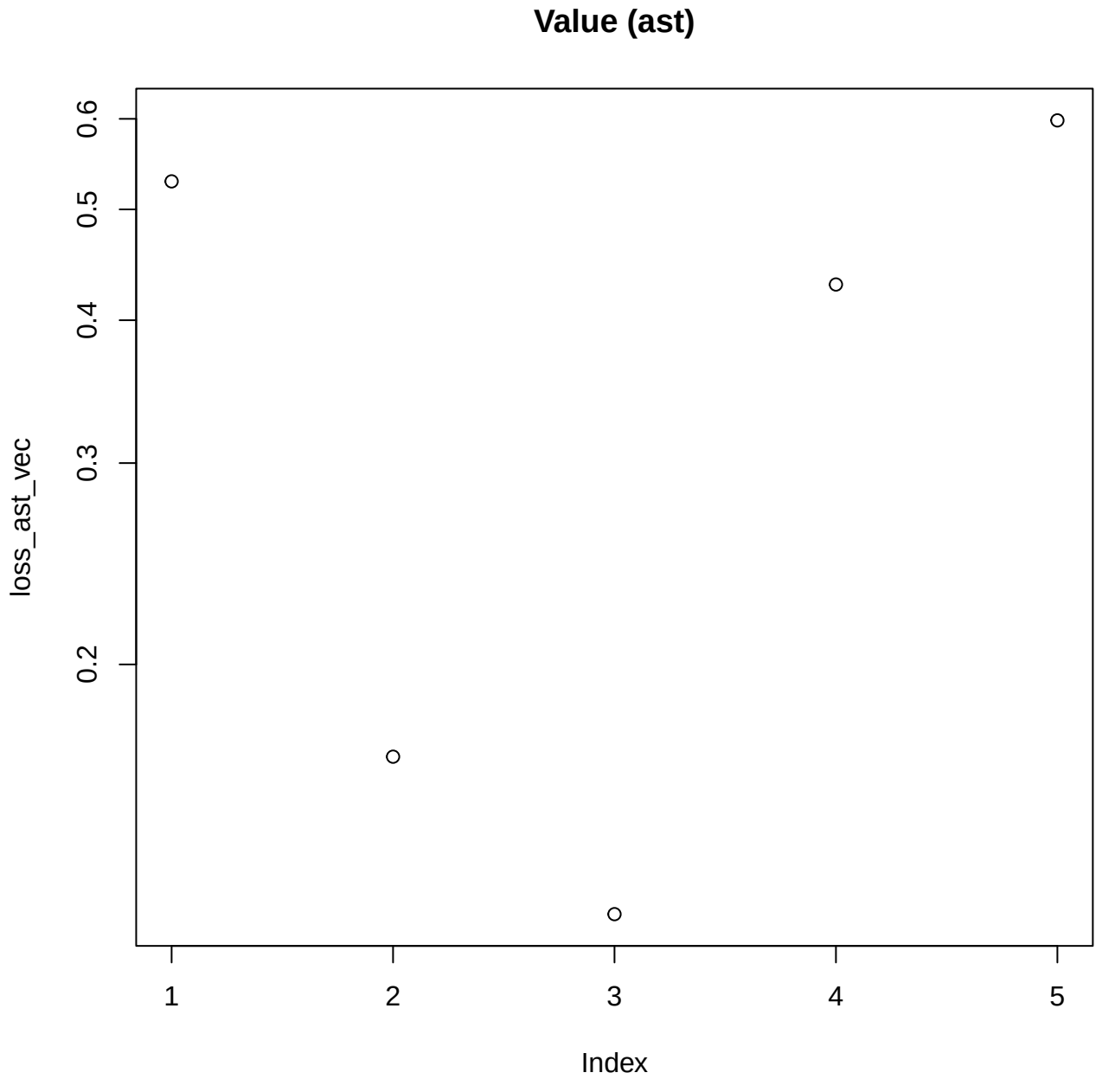
0.010
0.005
0.002
1
2
3
4
5
Index


Value (ast)
loss_ast_vec
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0.3
0.4
0.5
0.6
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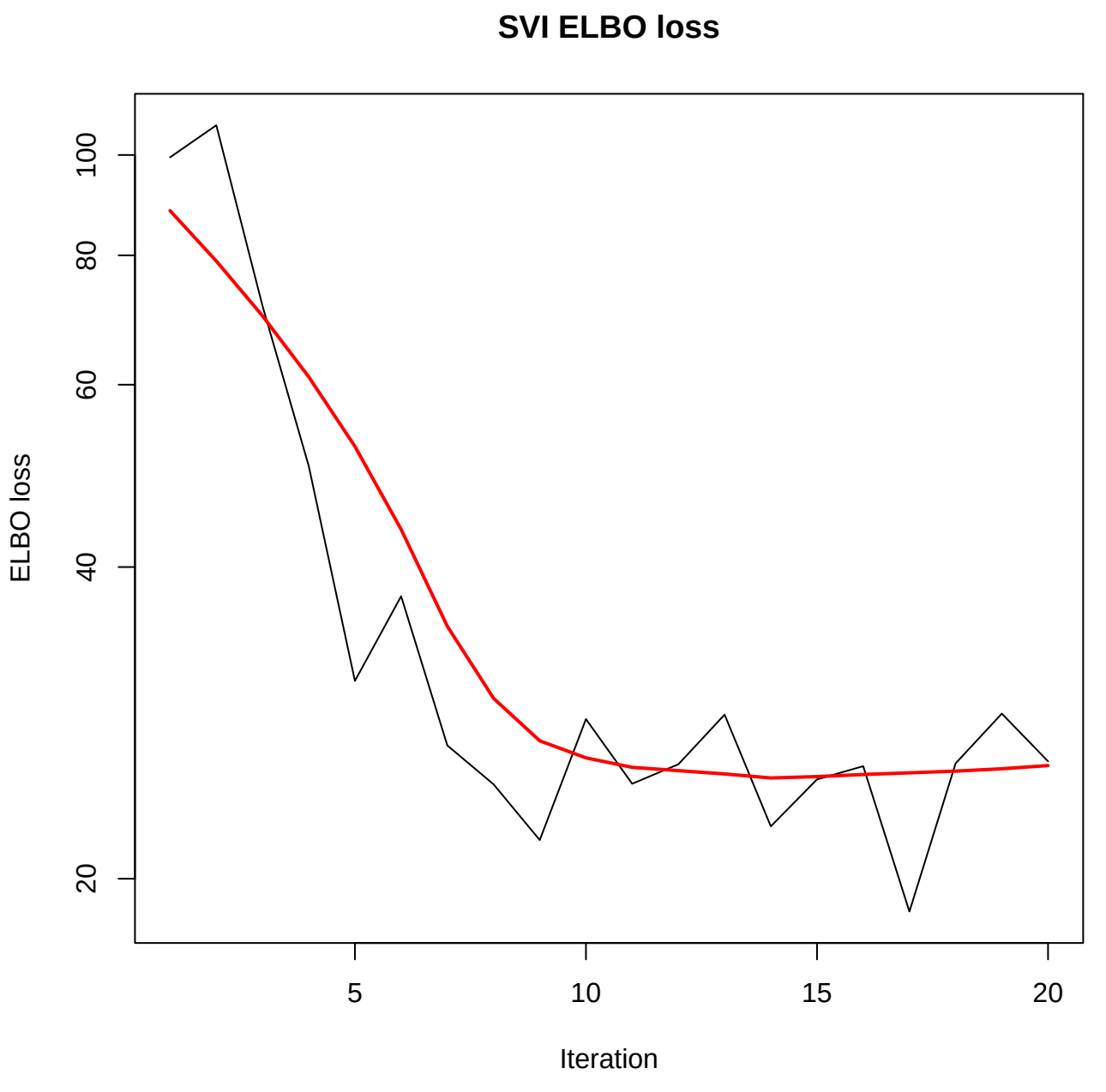
2
3
4
5
Index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ELBO loss
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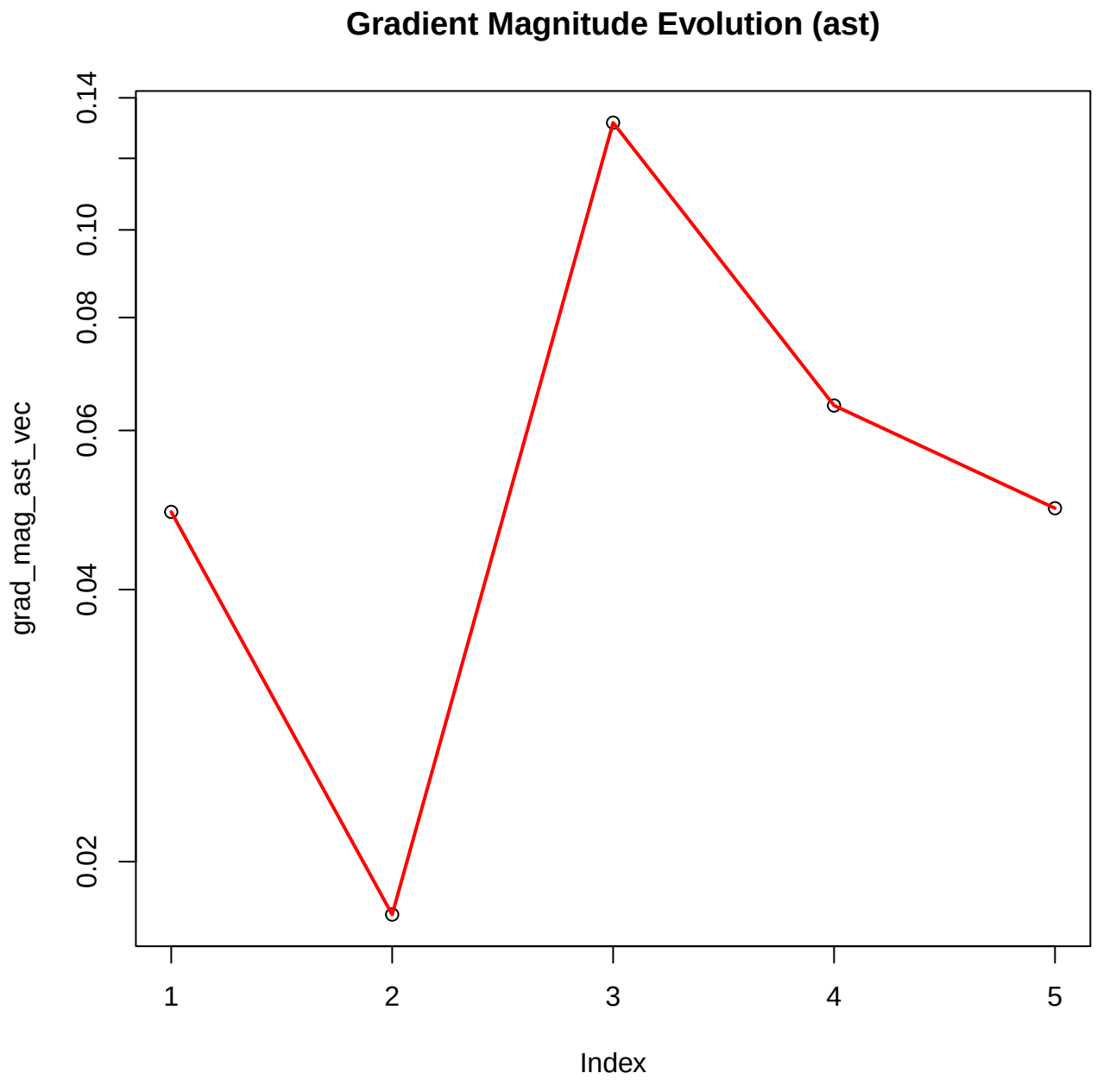
60
40
20
5
10
15
20
Iteration


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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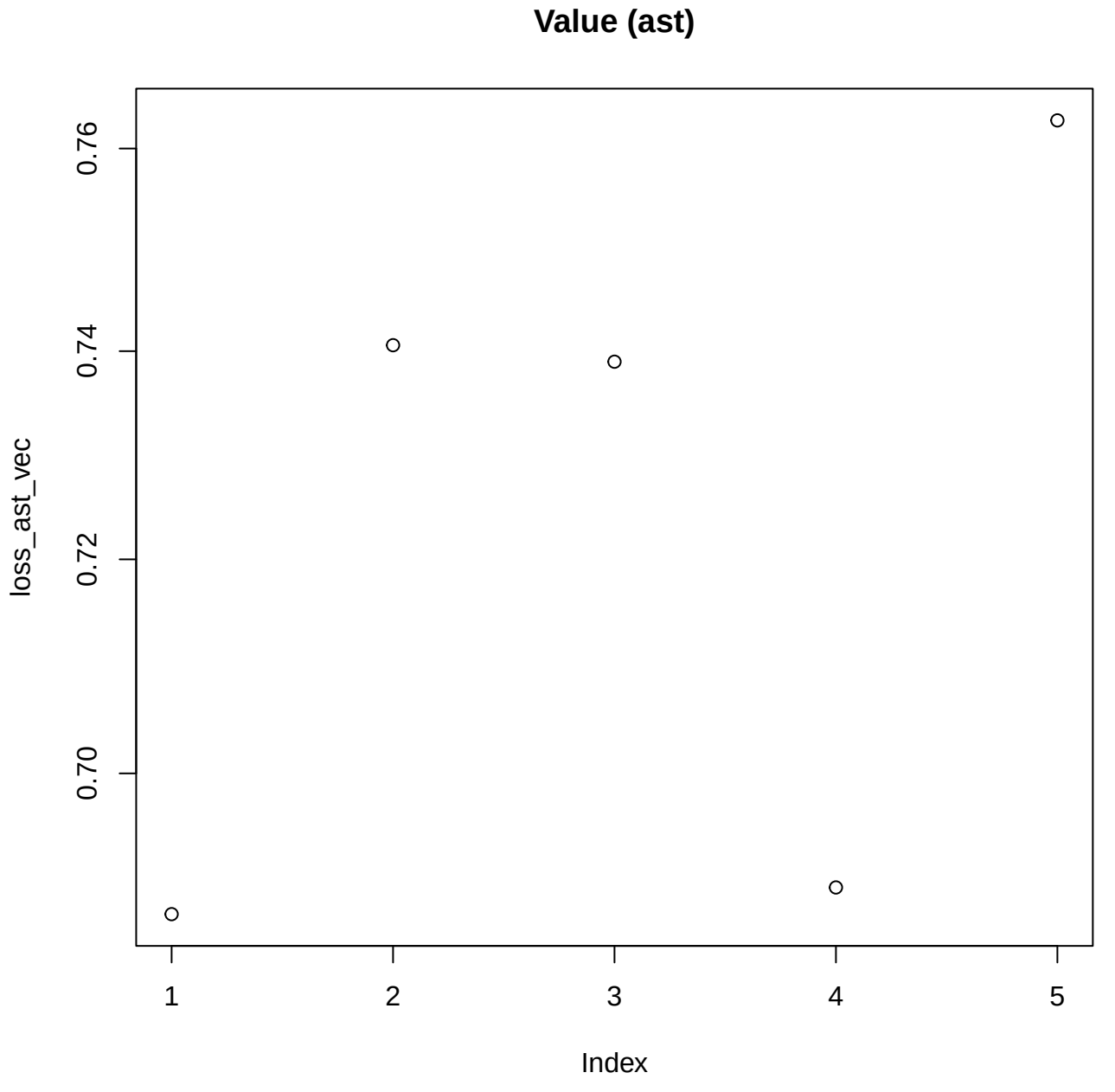
0.04
0.06
0.08
0.10
0.14
1
2
3
4
5
Index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0.76
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2
3
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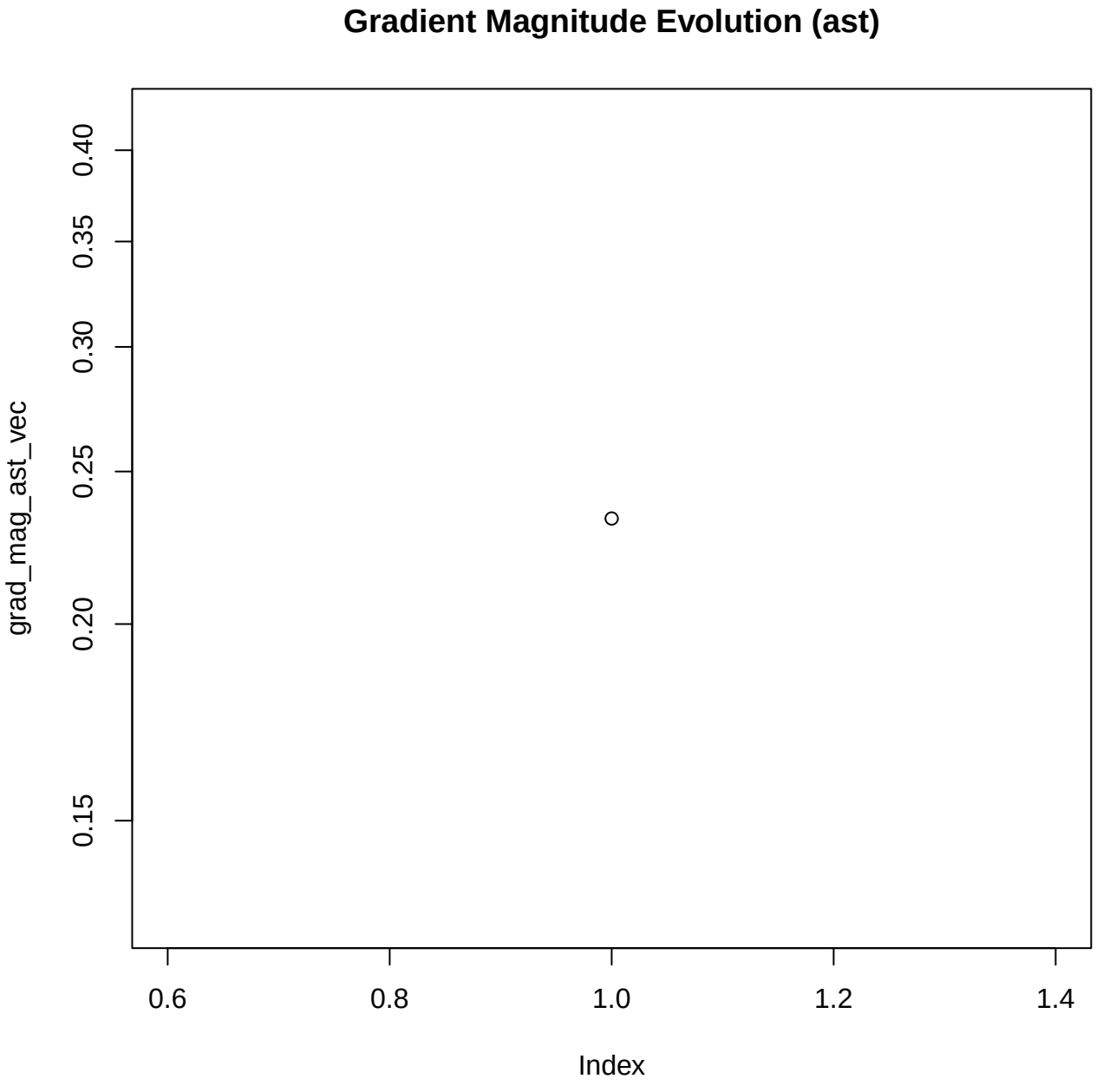
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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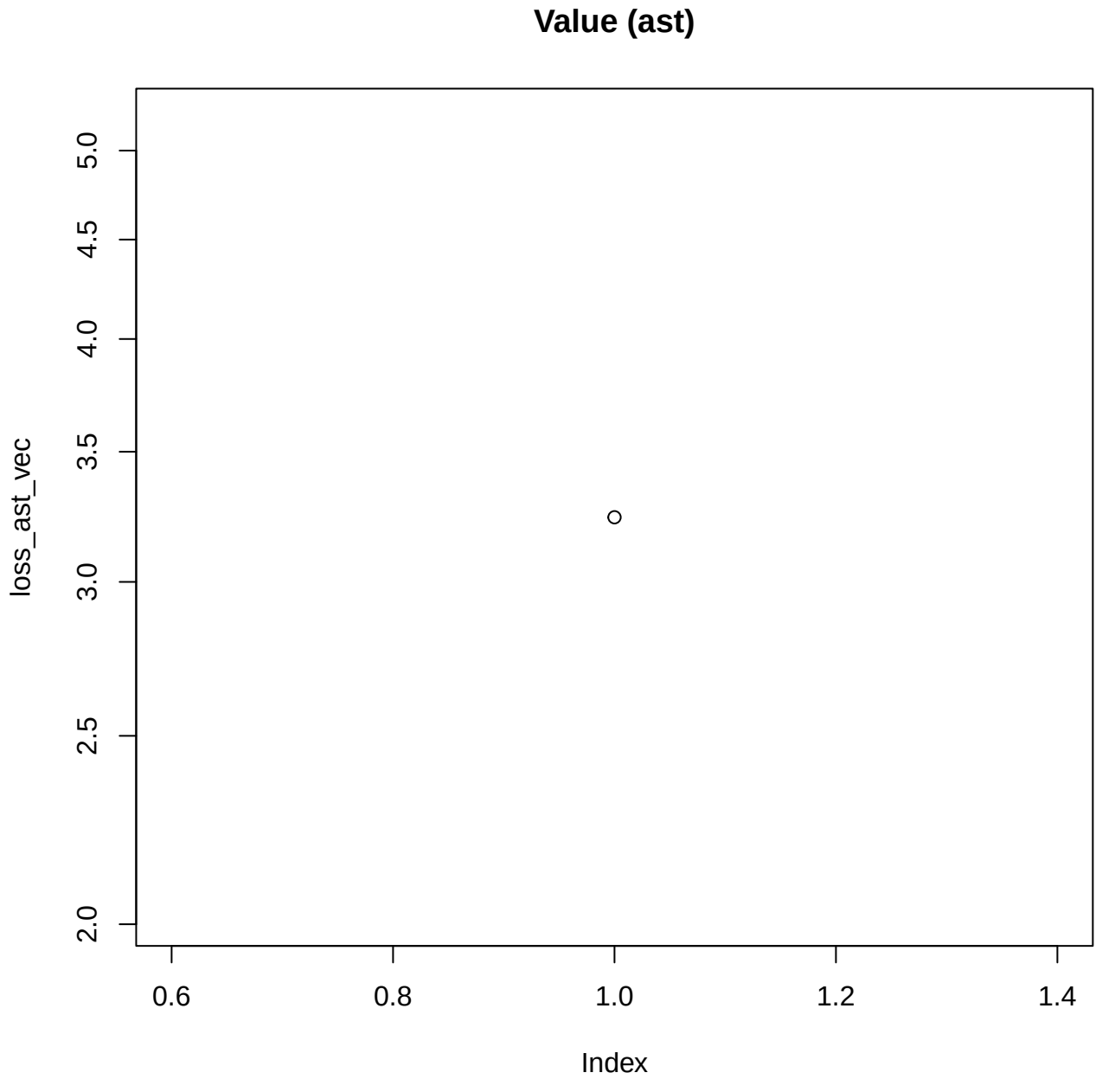
0.40
0.6
0.8
1.0
1.2
1.4
Index


Value (ast)
loss_ast_vec
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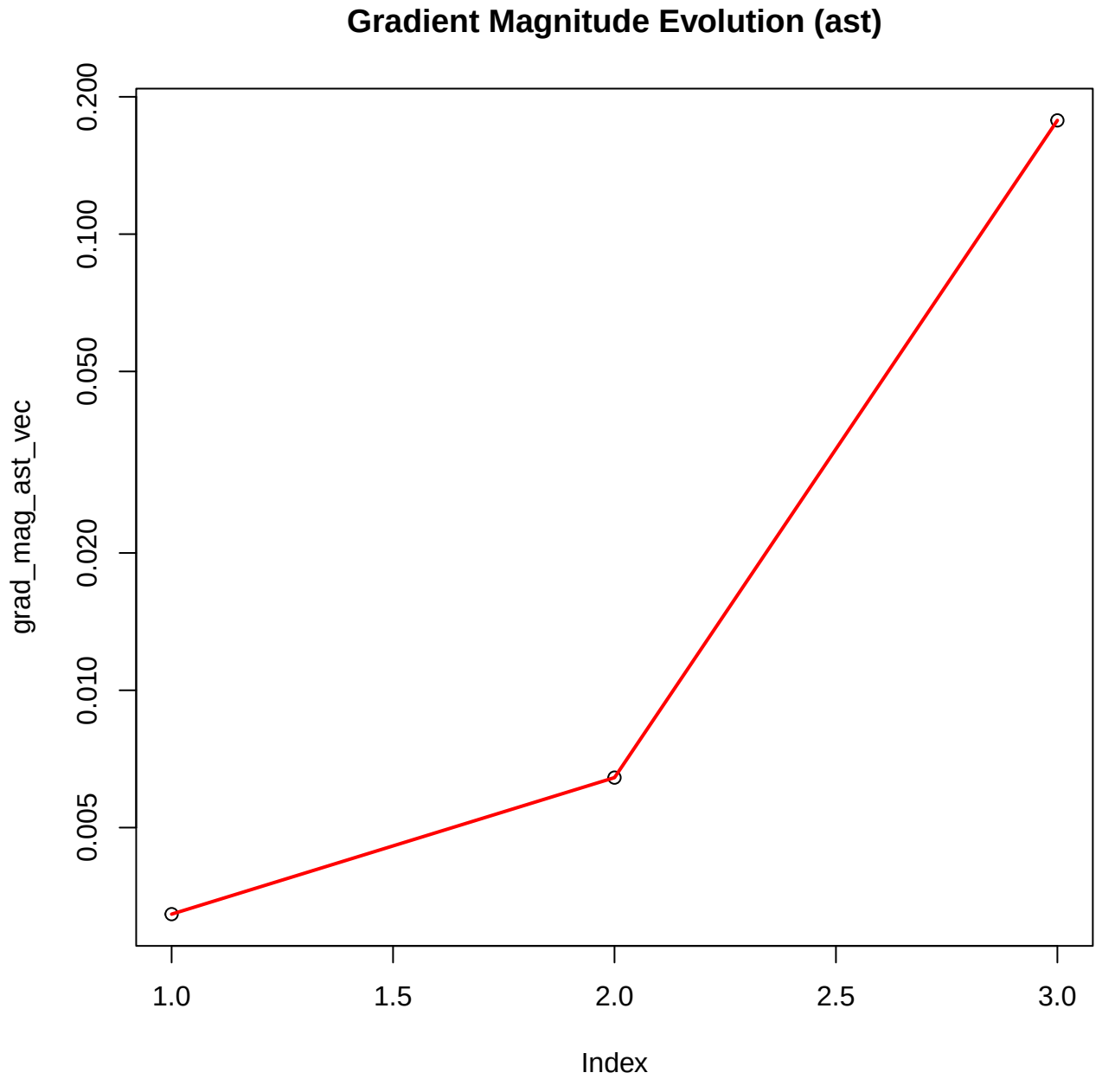
2.0
0.6
0.8
1.0
1.2
1.4
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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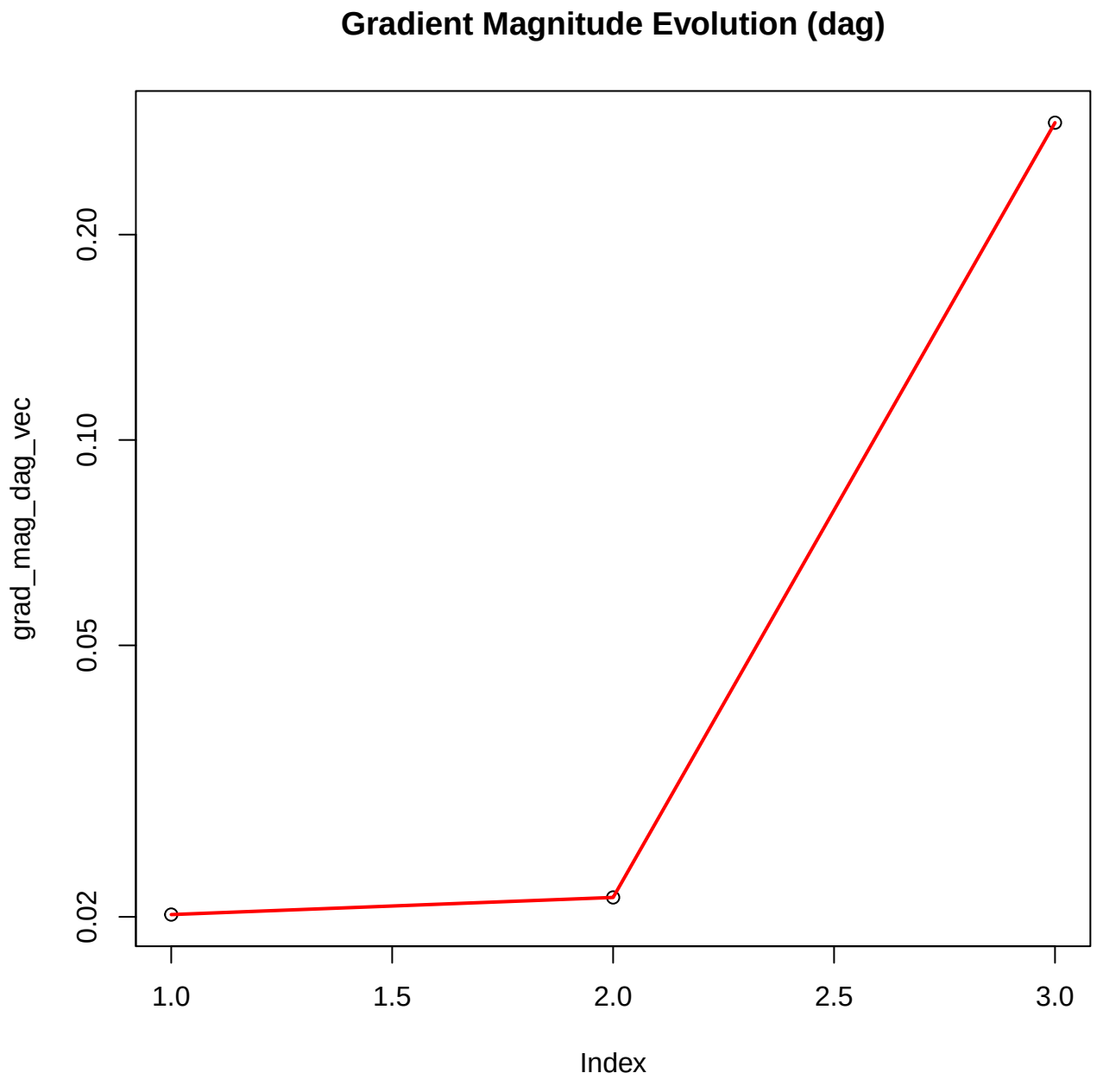
0.020
0.050
0.100
0.200
1.0
1.5
2.0
2.5
3.0
Index


Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
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2.5
3.0
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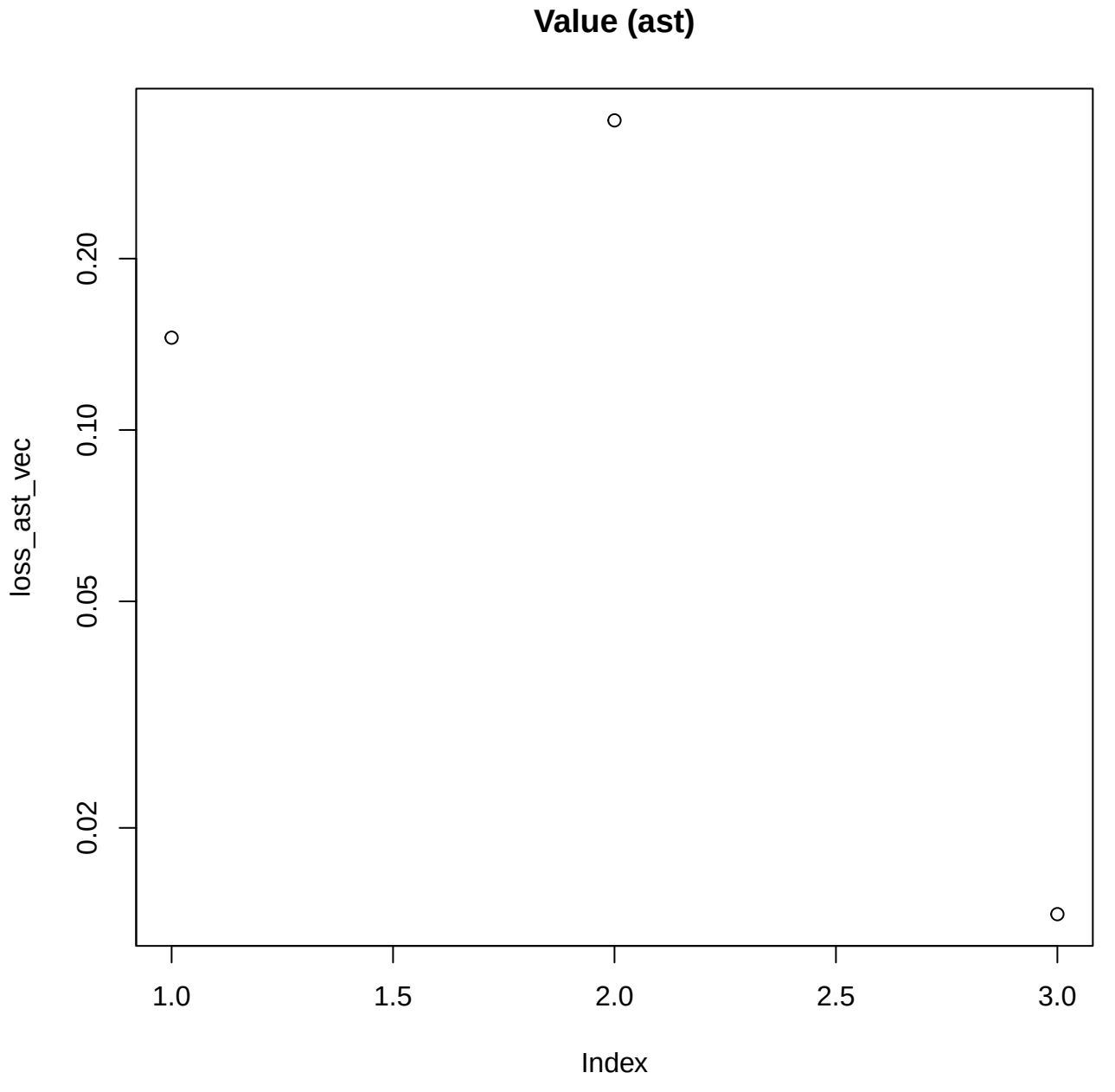
Value (ast)
loss_ast_vec
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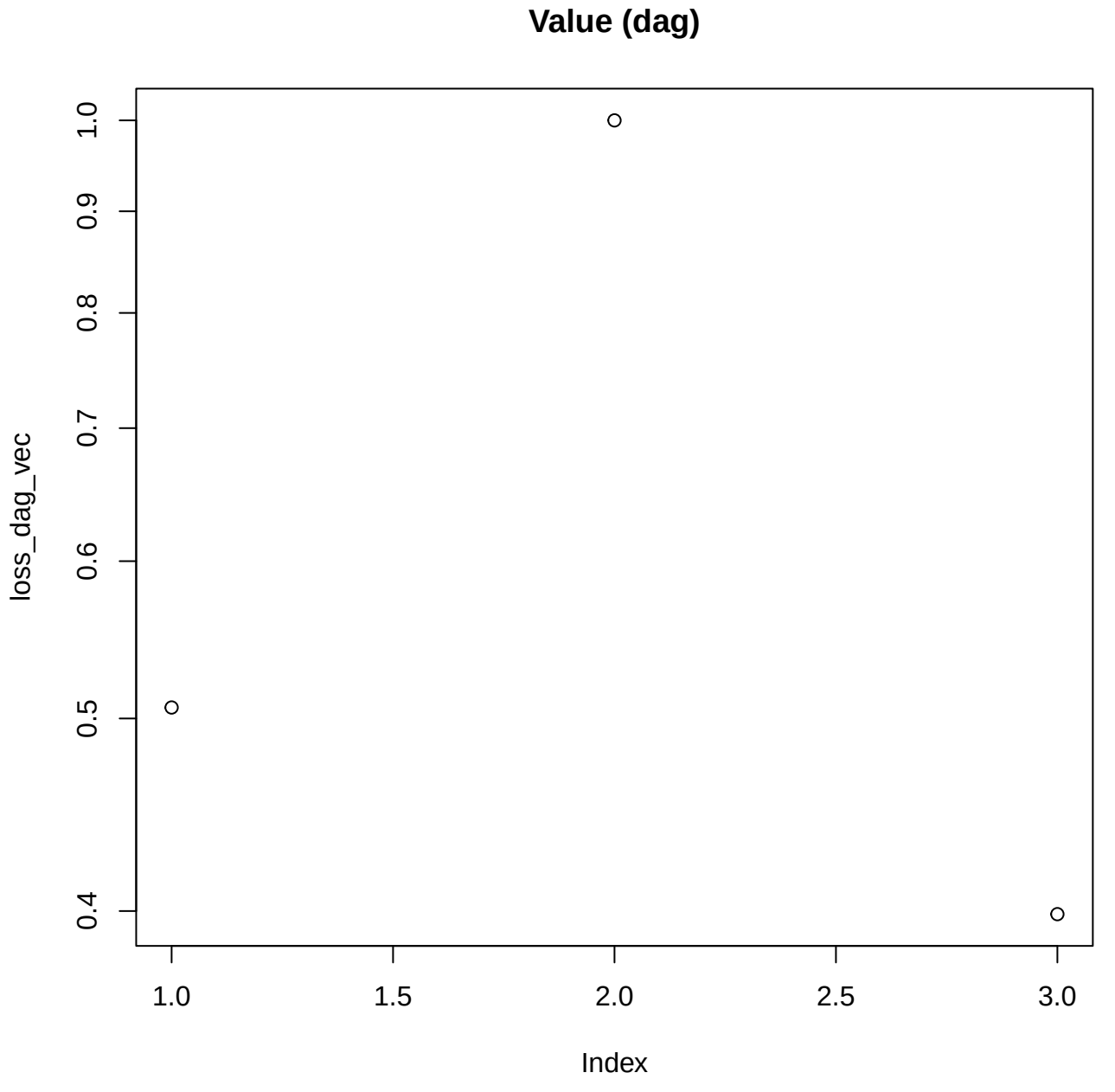
Value (dag)
loss_dag_vec
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0.5
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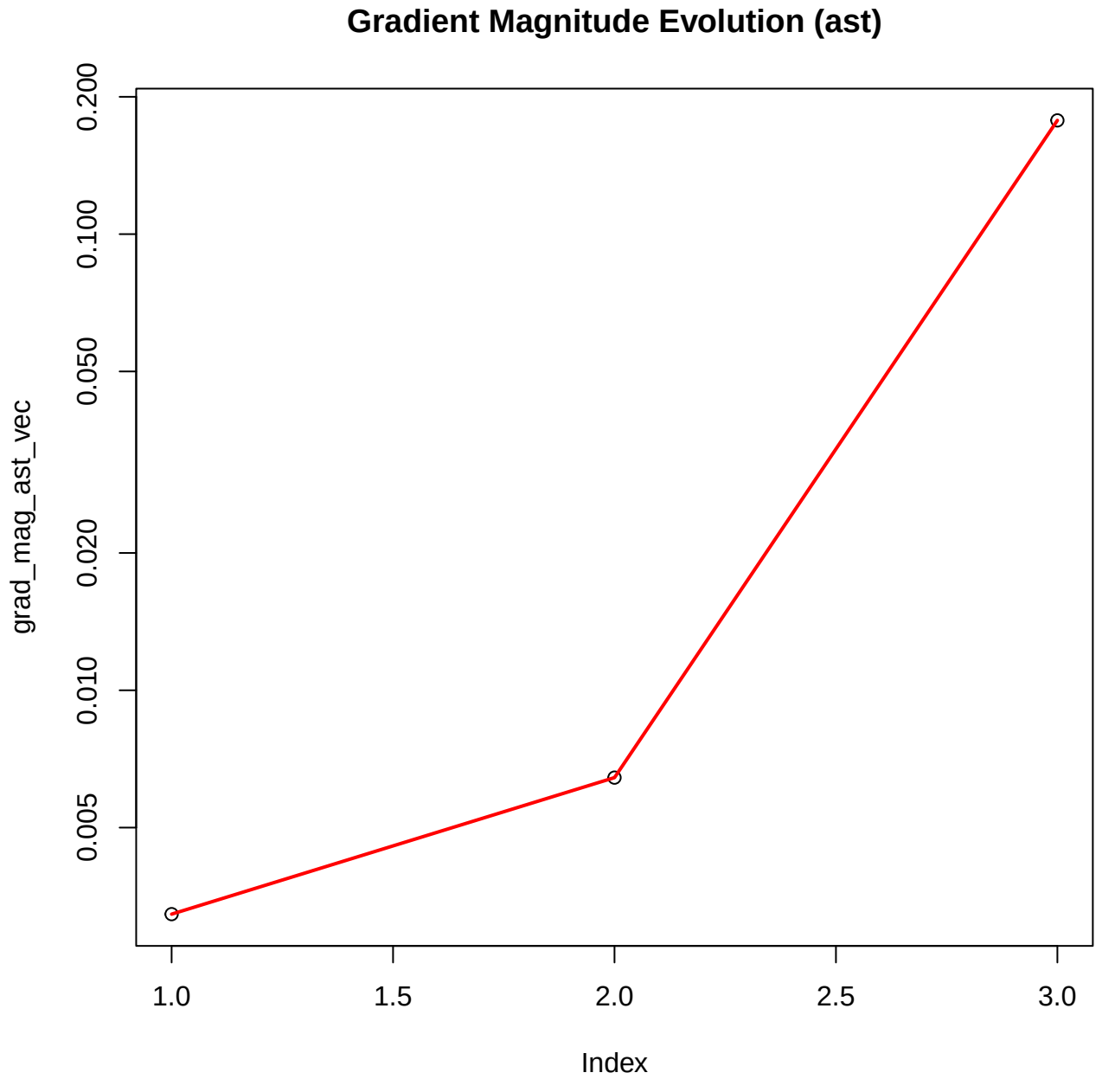
1.5
2.0
2.5
3.0
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.020
0.050
0.100
0.200
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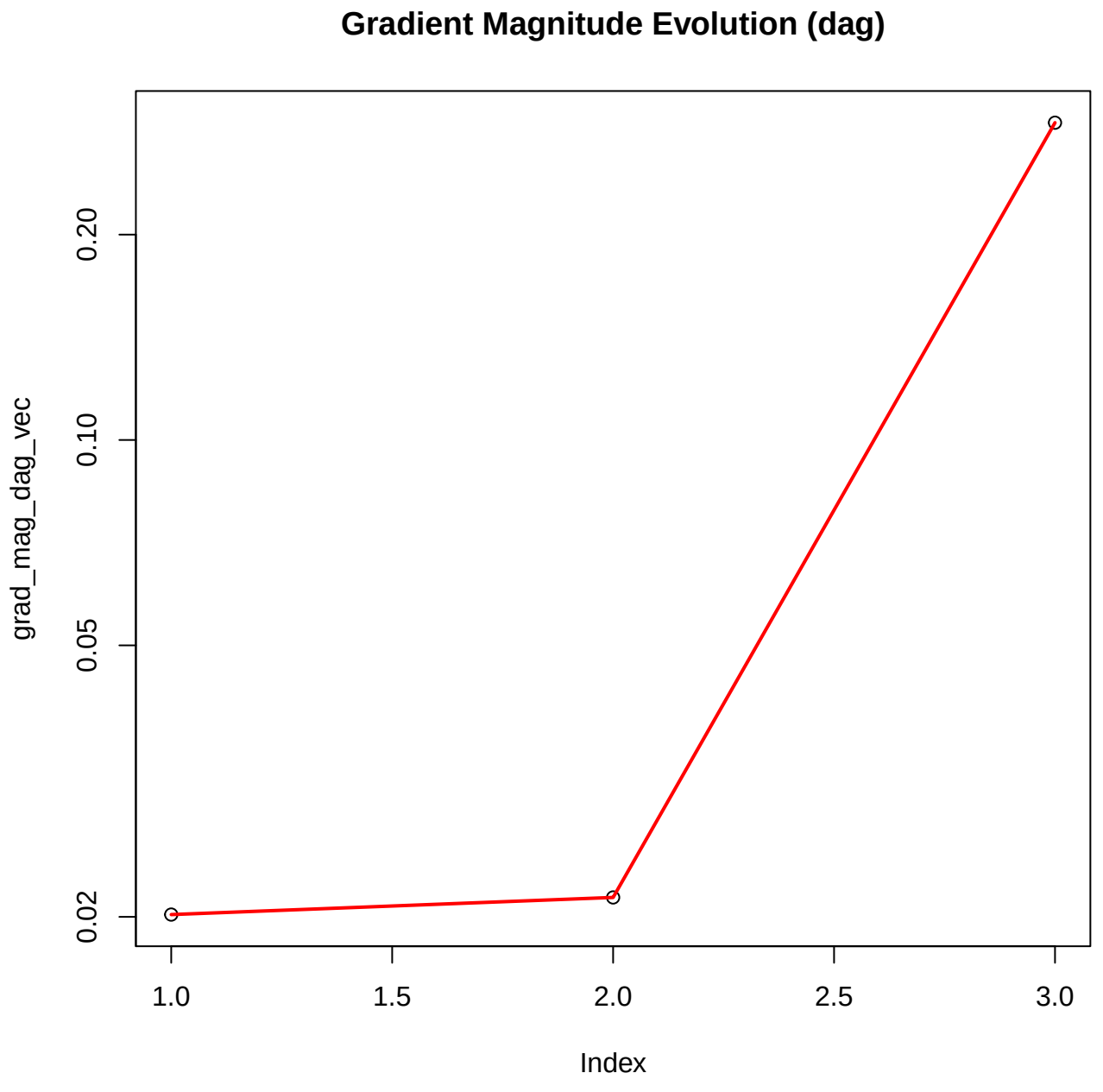
1.5
2.0
2.5
3.0
Index


Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
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0.10
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1.5
2.0
2.5
3.0
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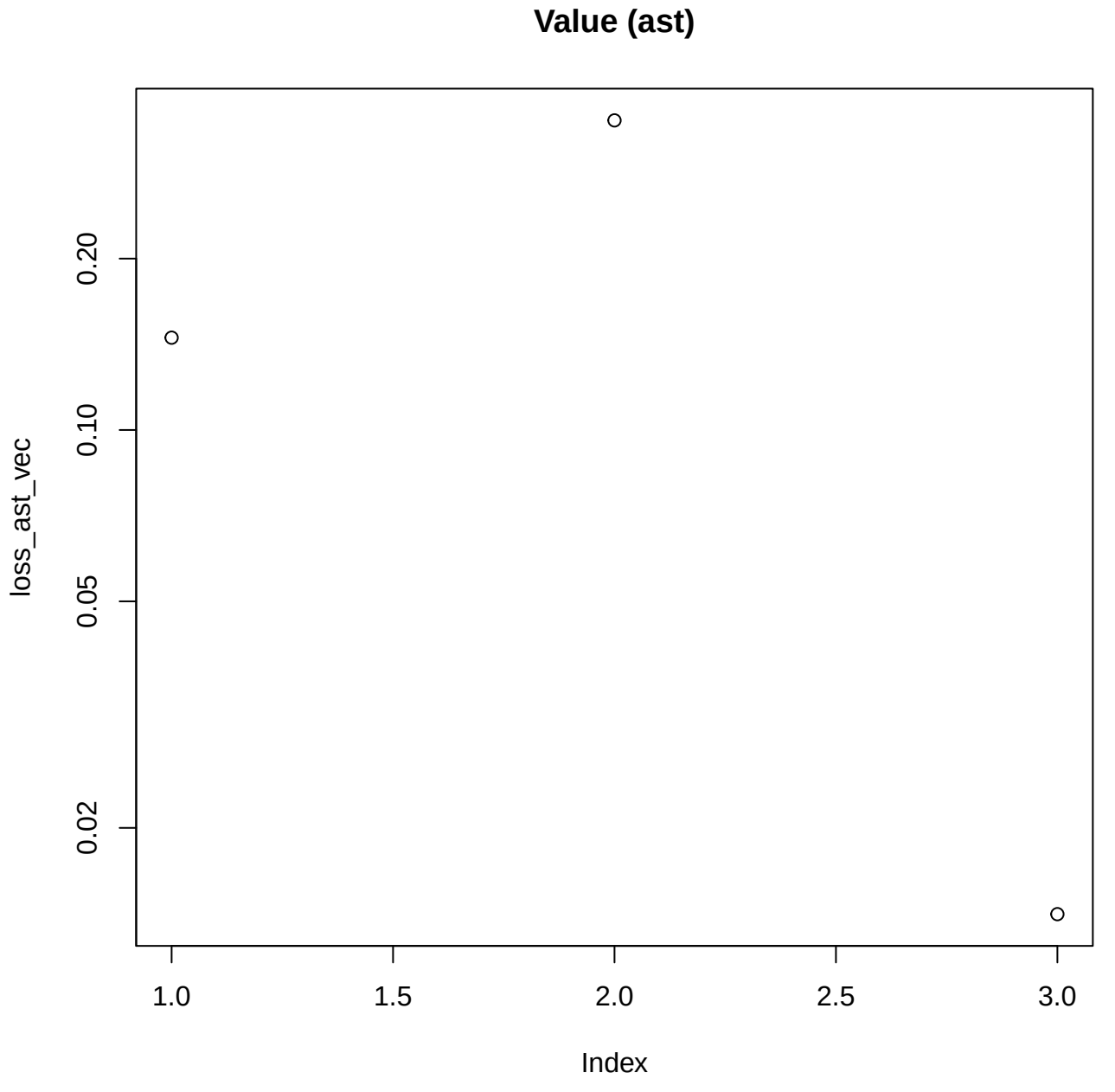
Value (ast)
loss_ast_vec
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0.05
0.10
0.20
1.0
1.5
2.0
2.5
3.0
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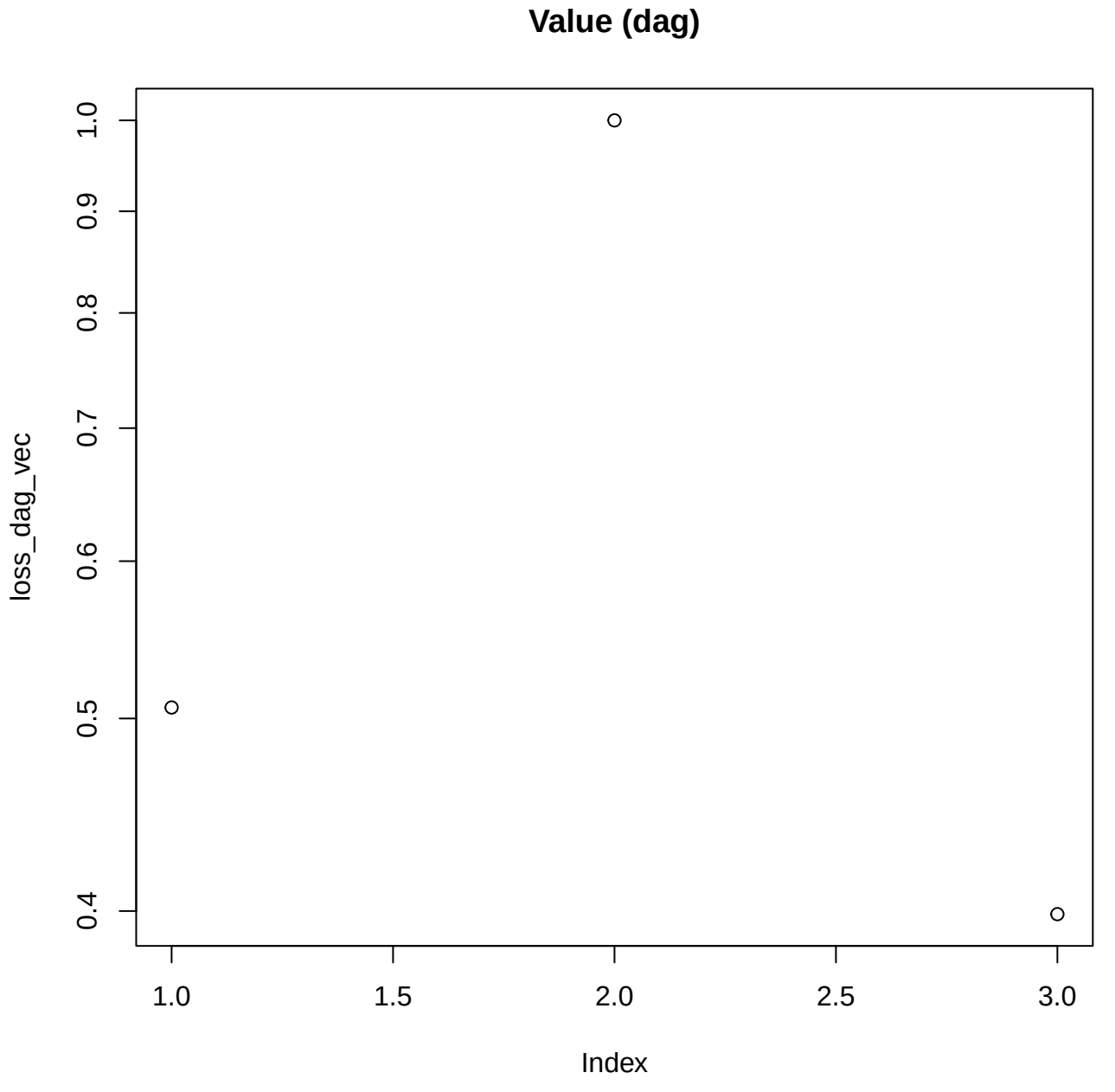
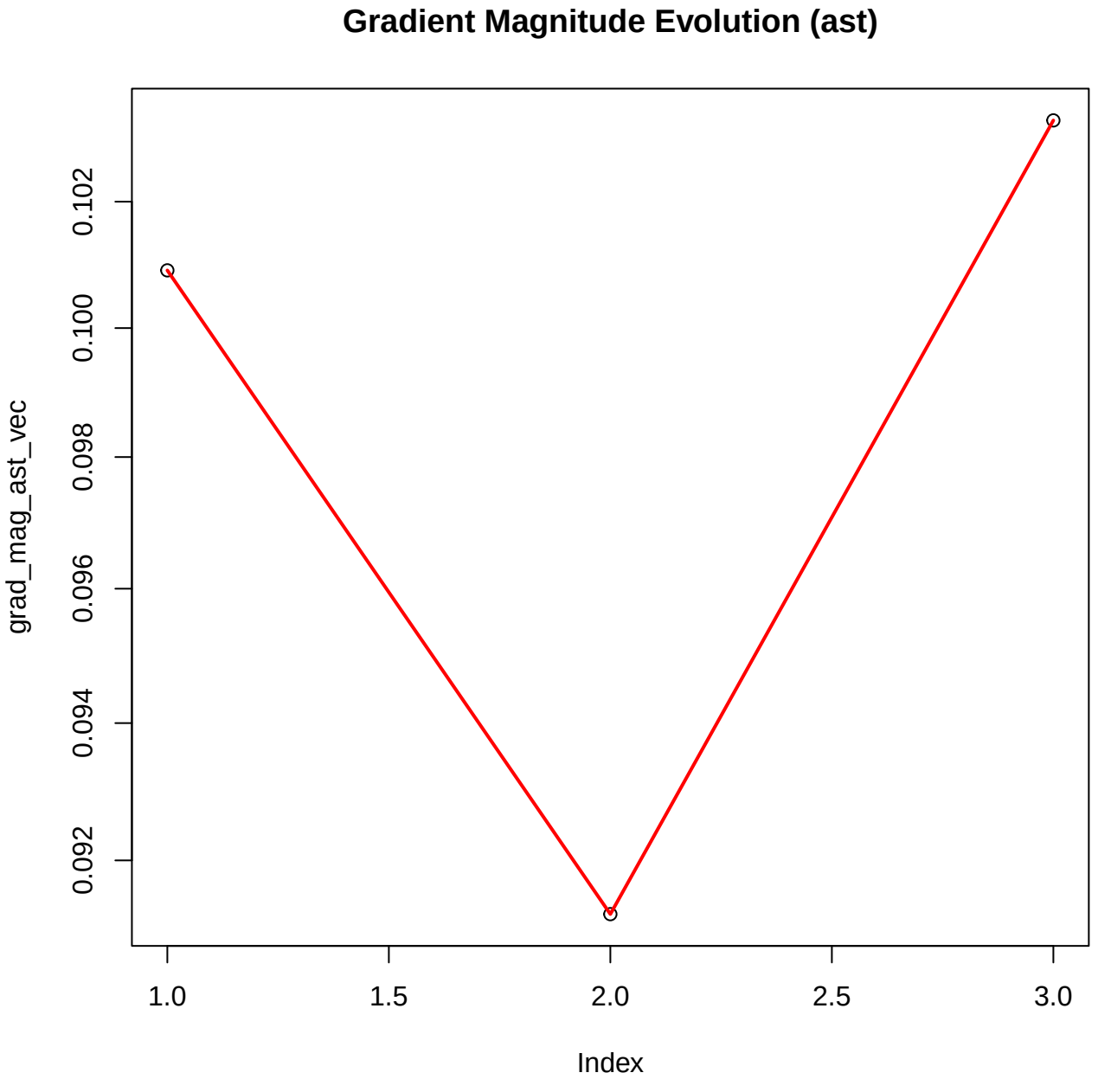

Value (dag)
loss_dag_vec
1.0
0.9
0.8
0.7
0.6
0.5
0.4
1.0
1.5
2.0
2.5
3.0
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.098
0.100
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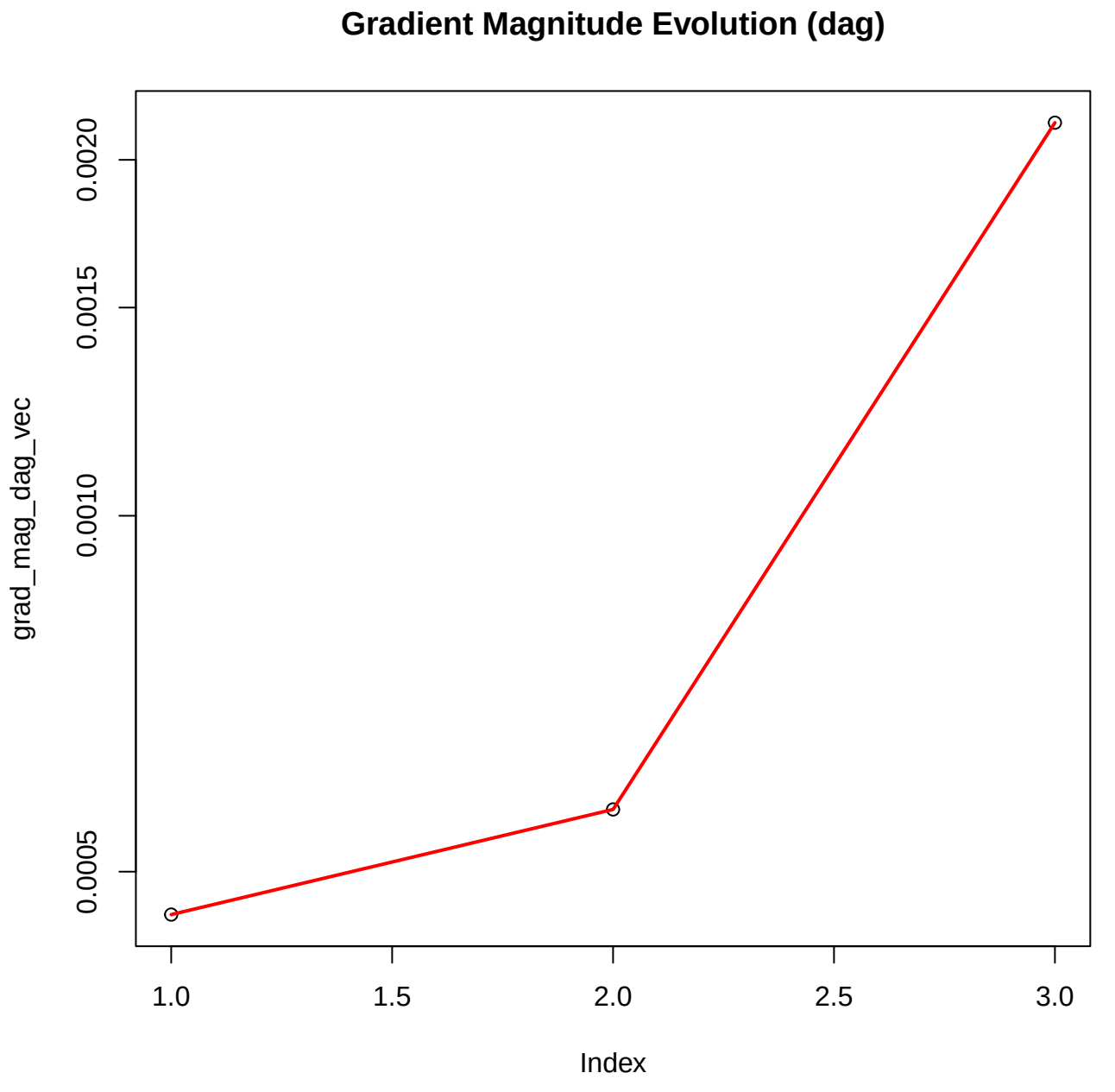
3.0
Index


Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
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0.0010
0.0015
0.0020
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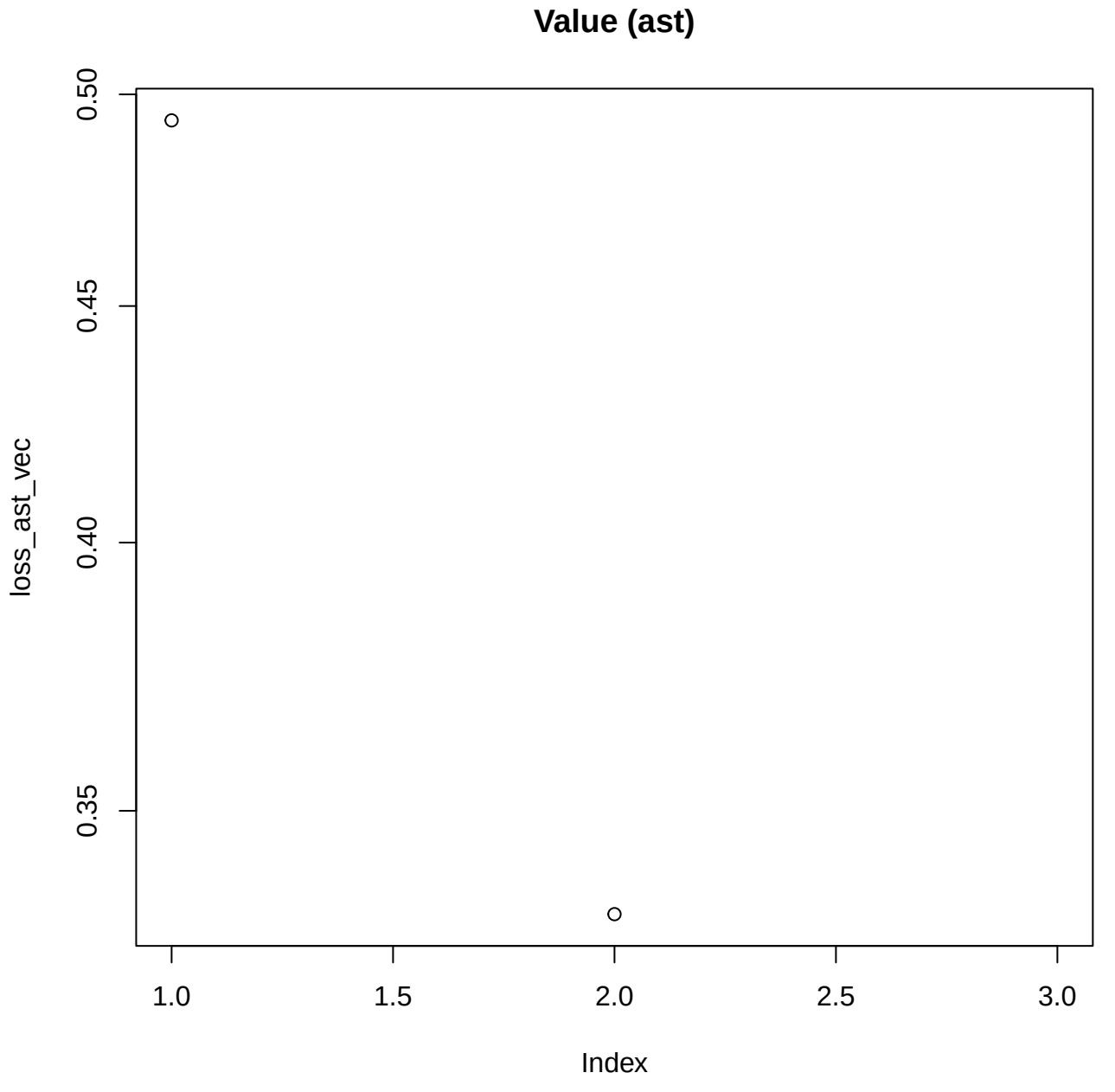
1.5
2.0
2.5
3.0
Index


Value (ast)
loss_ast_vec
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0.40
0.35
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2.5
3.0
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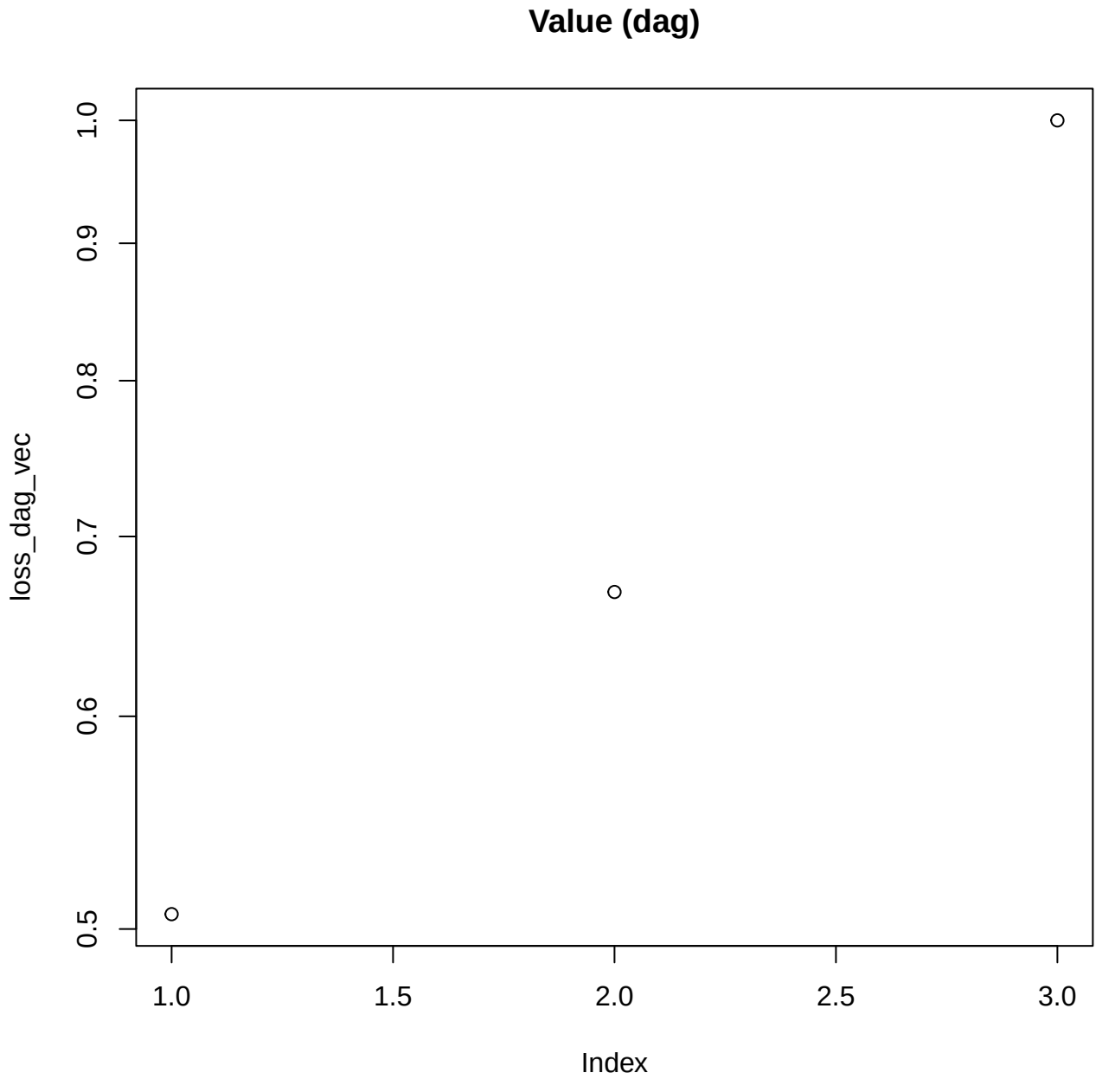
Value (dag)
loss_dag_vec
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0.8
0.9
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3.0
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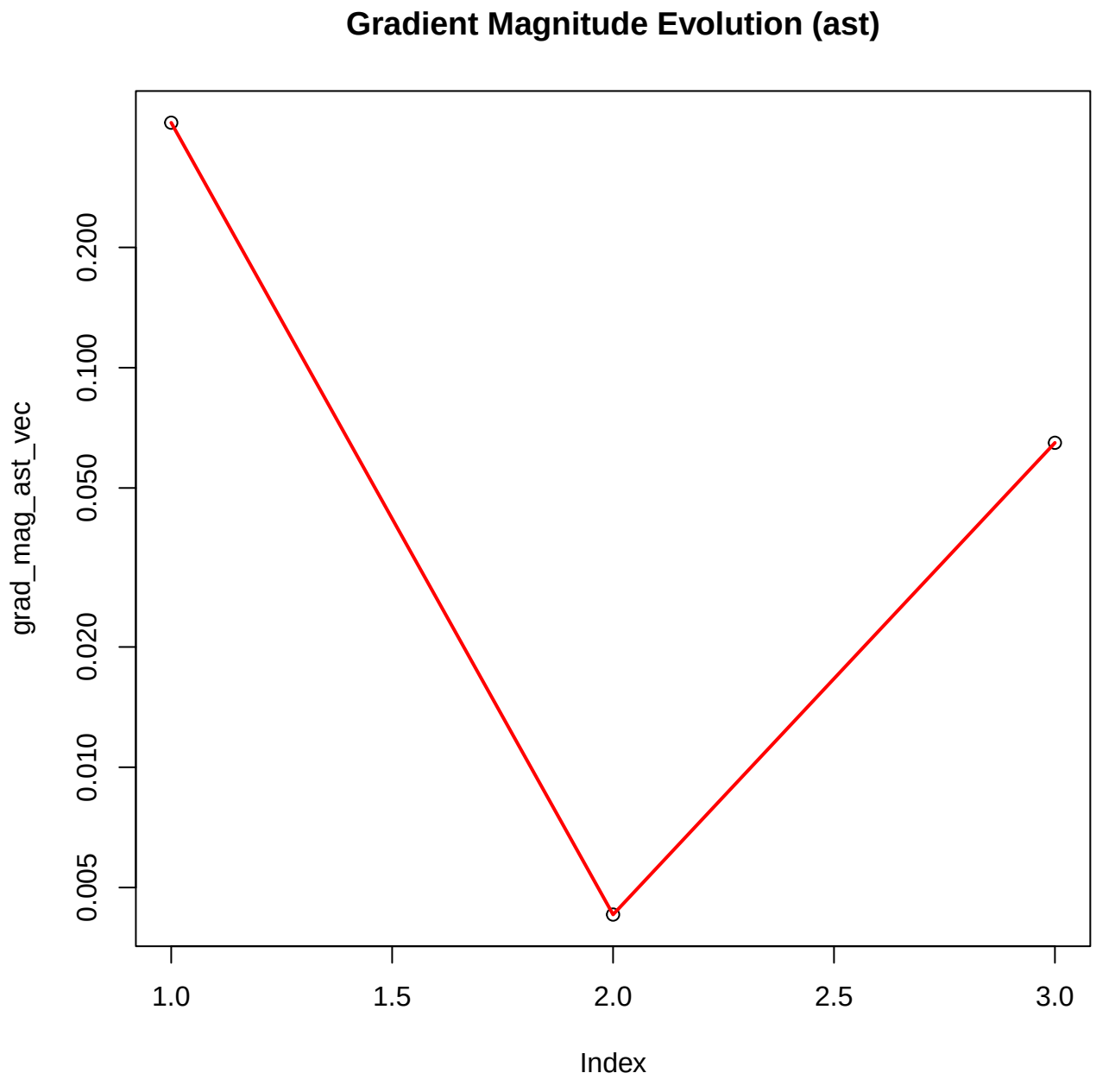
Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.020
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0.100
0.200
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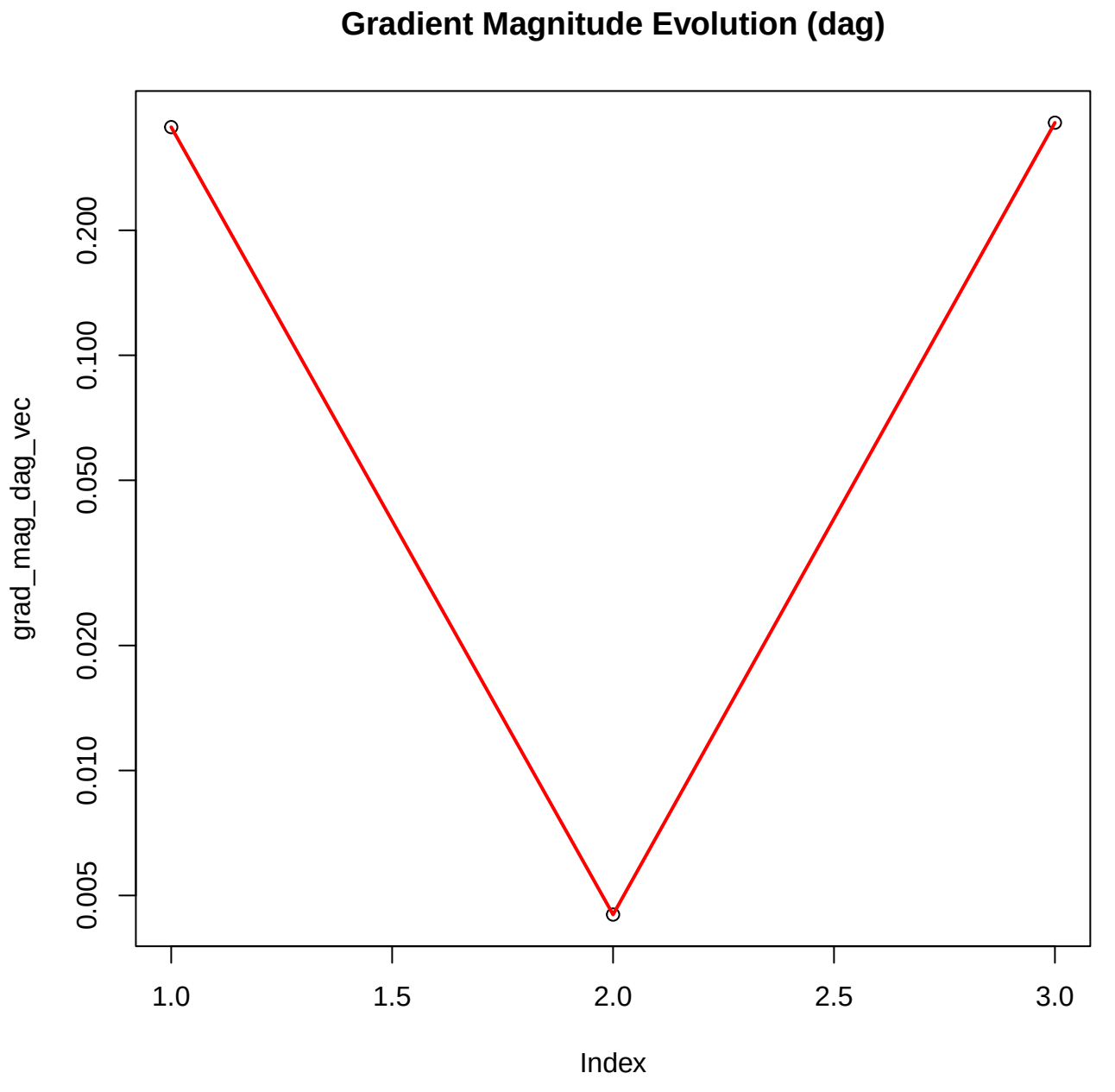
Gradient Magnitude Evolution (dag)
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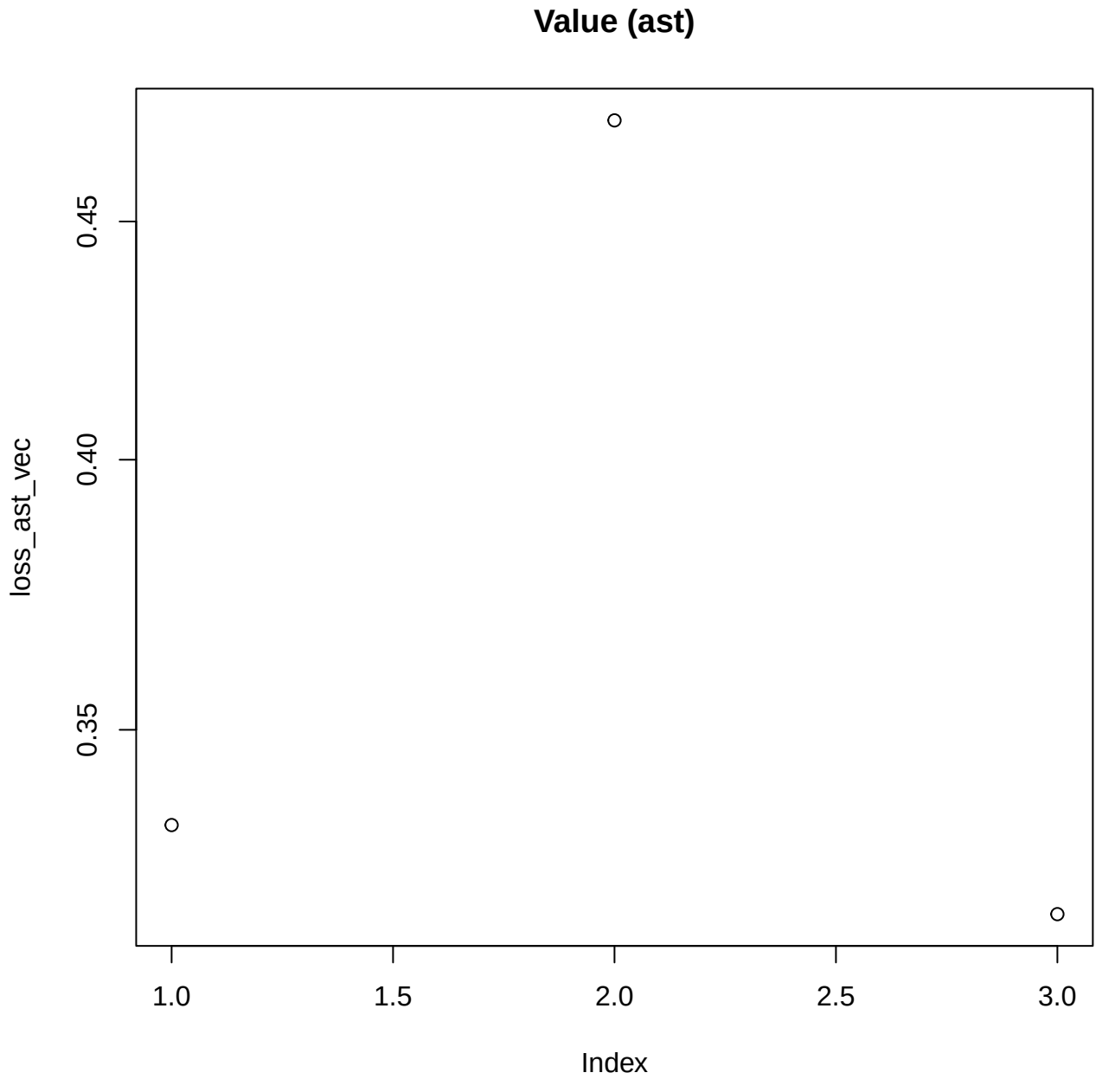
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loss_ast_vec
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2.5
3.0
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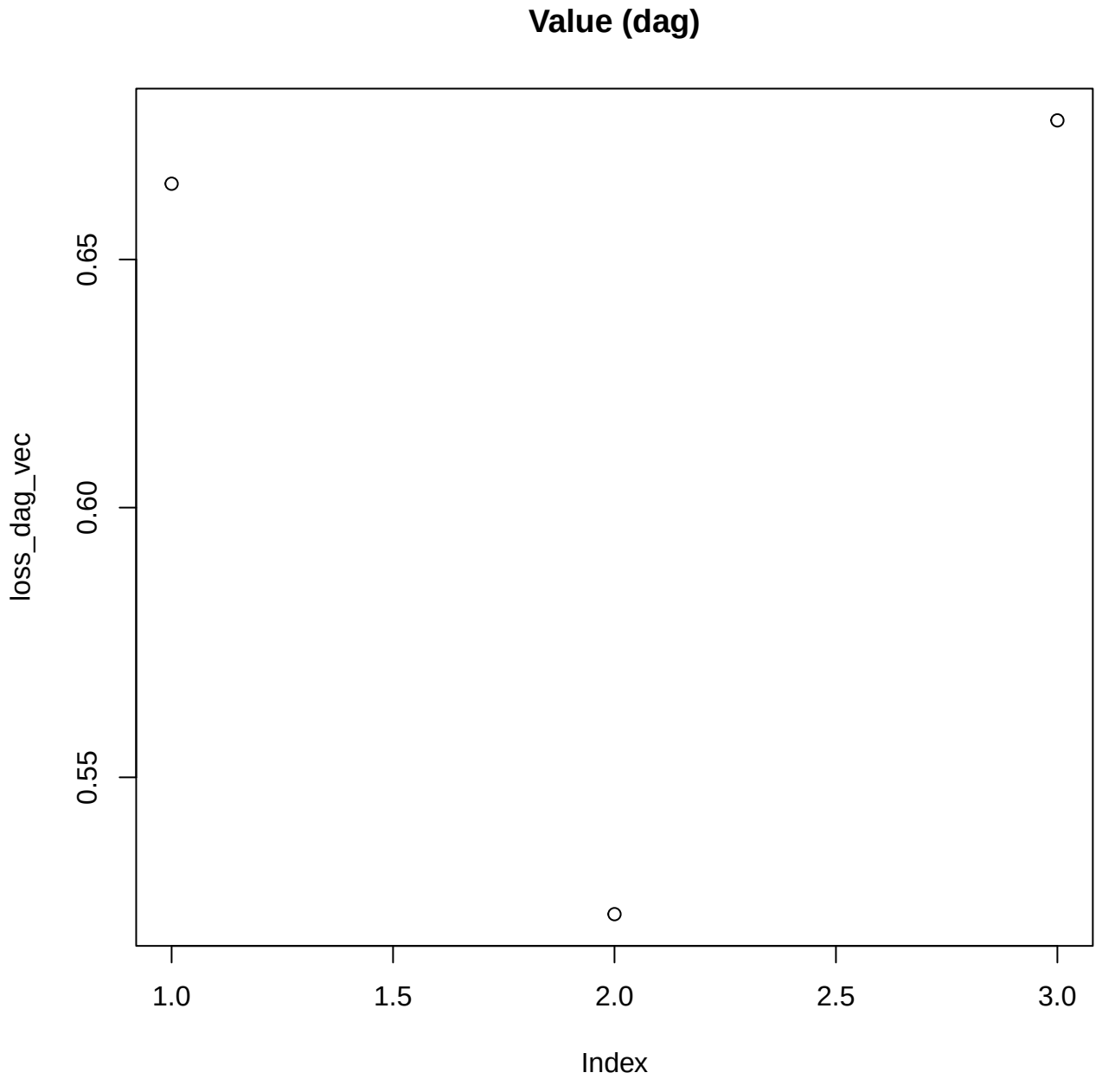
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0.65
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2.5
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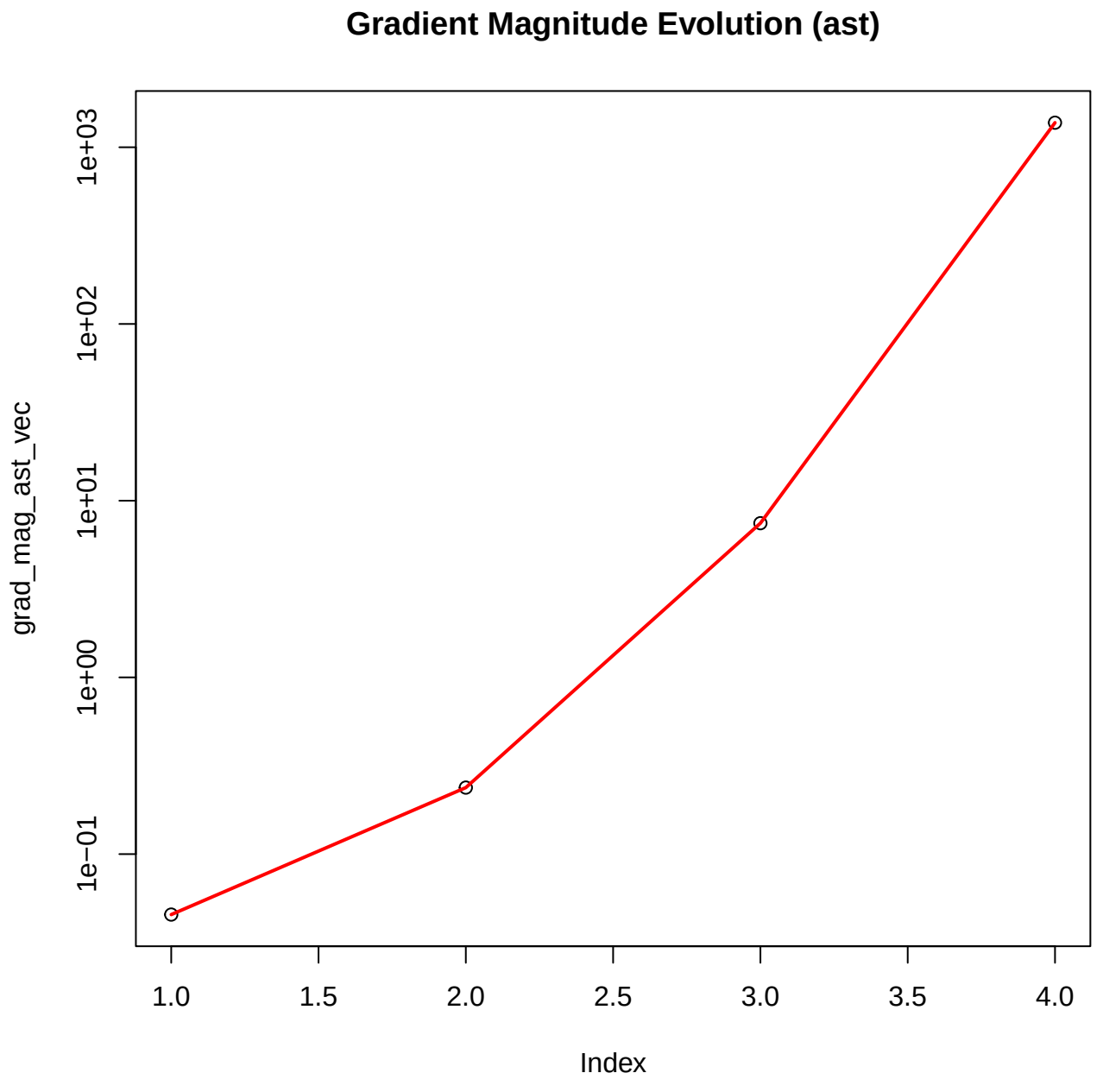
Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e−01
1e+00
1e+01
1e+02
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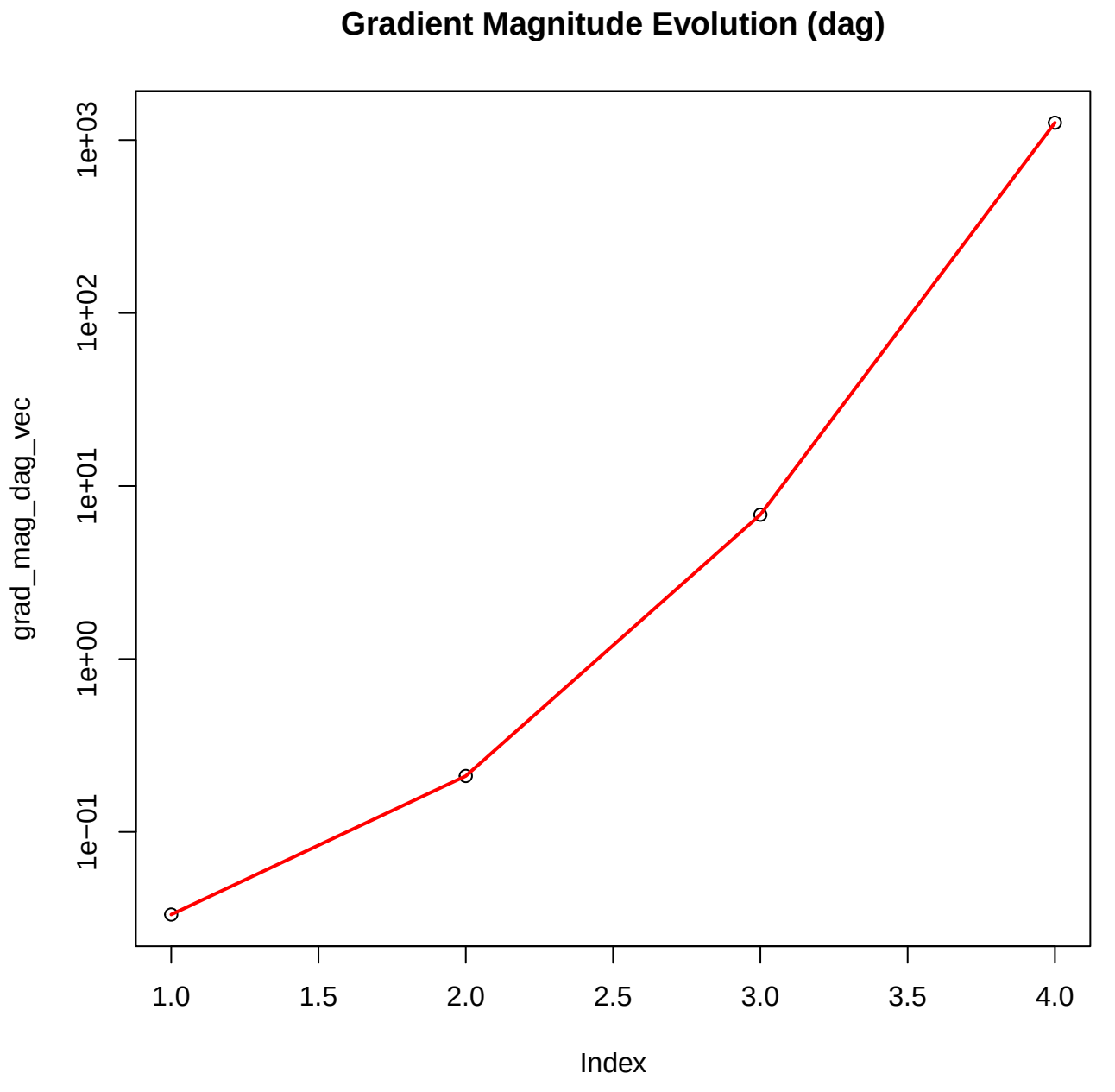
Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
1e−01
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1e+01
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4.0
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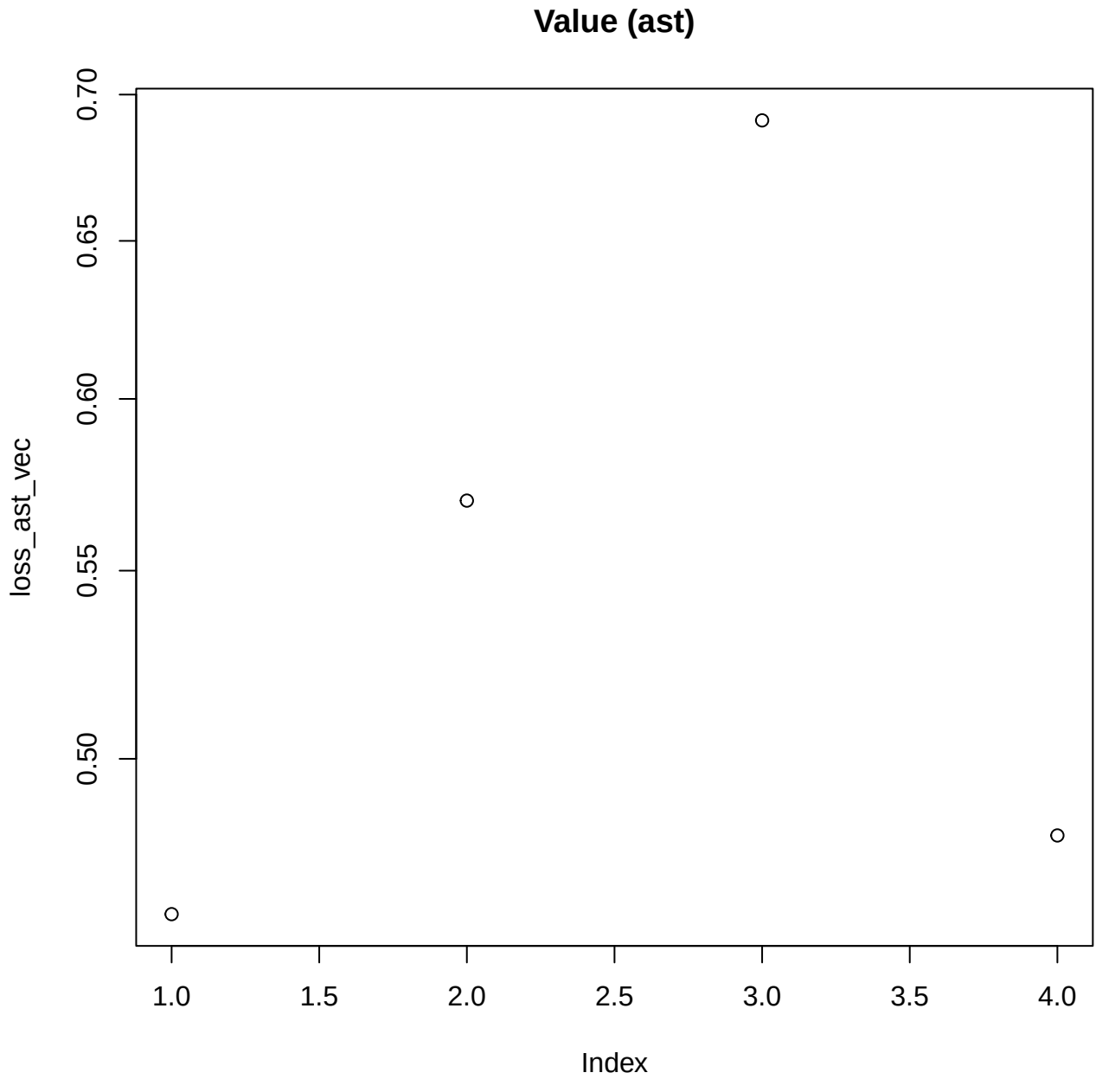
Value (ast)
loss_ast_vec
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2.5
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3.5
4.0
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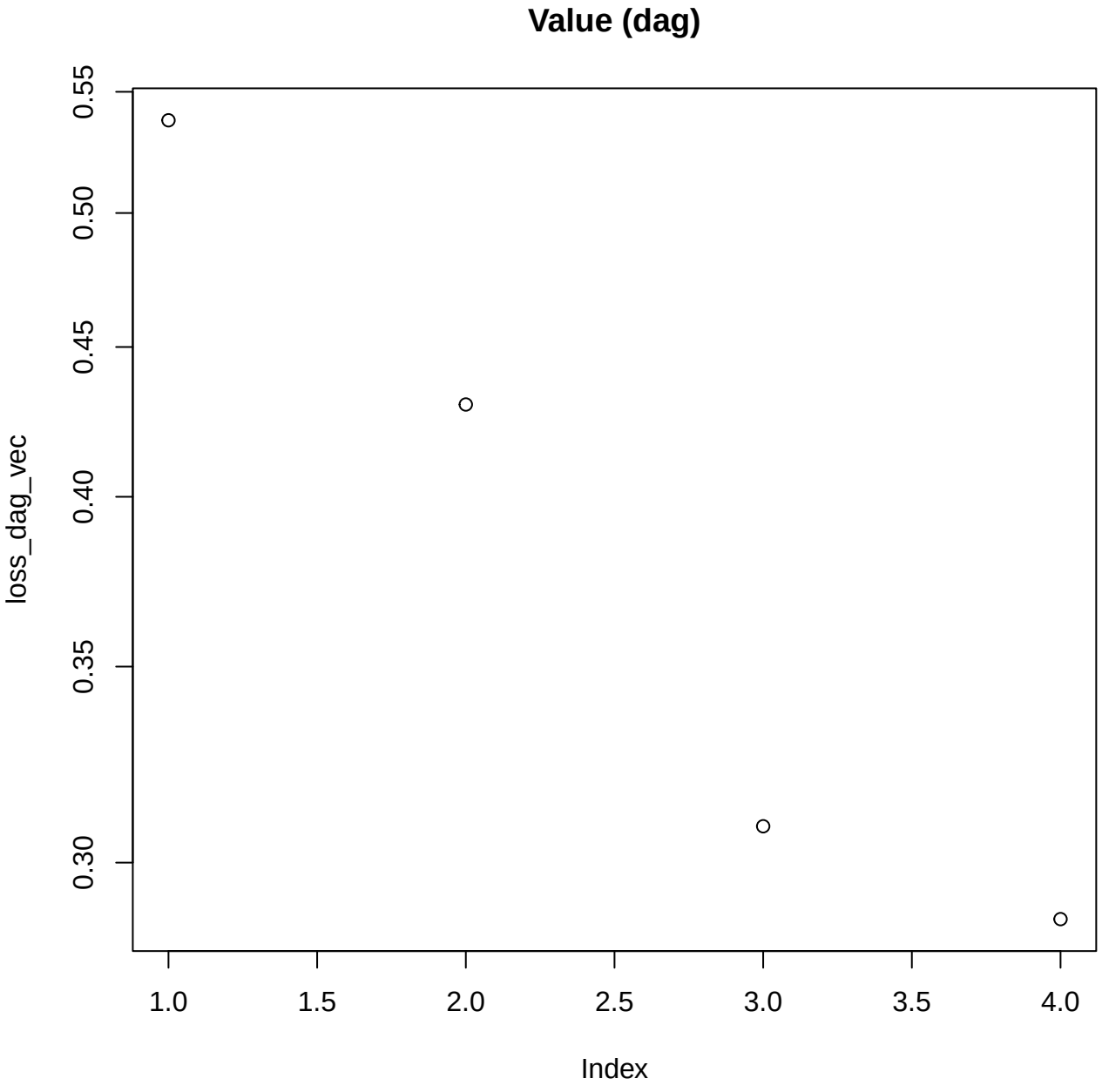
Value (dag)
loss_dag_vec
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0.35
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2.5
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3.5
4.0
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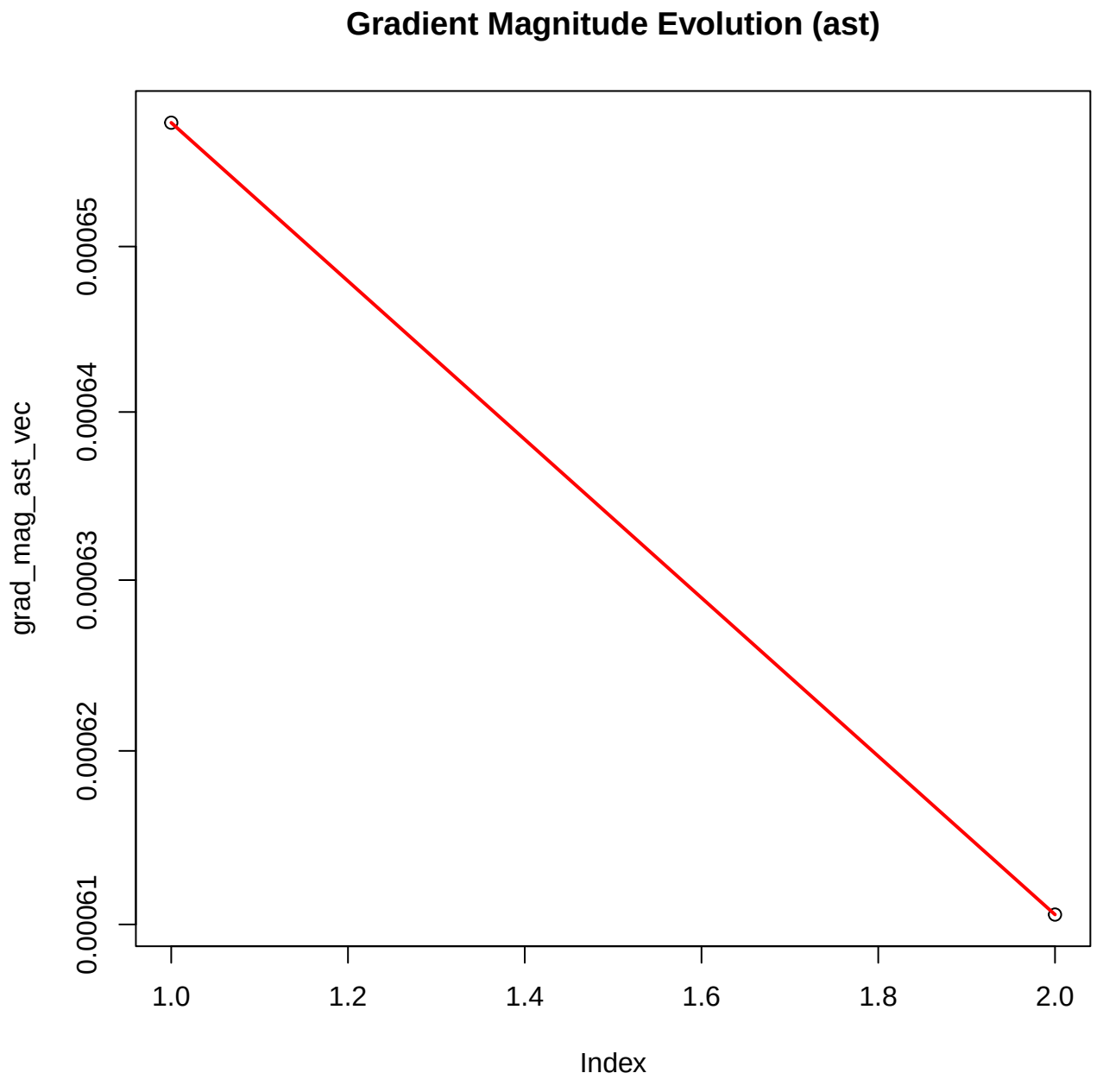
Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00061
0.00062
0.00063
0.00064
0.00065
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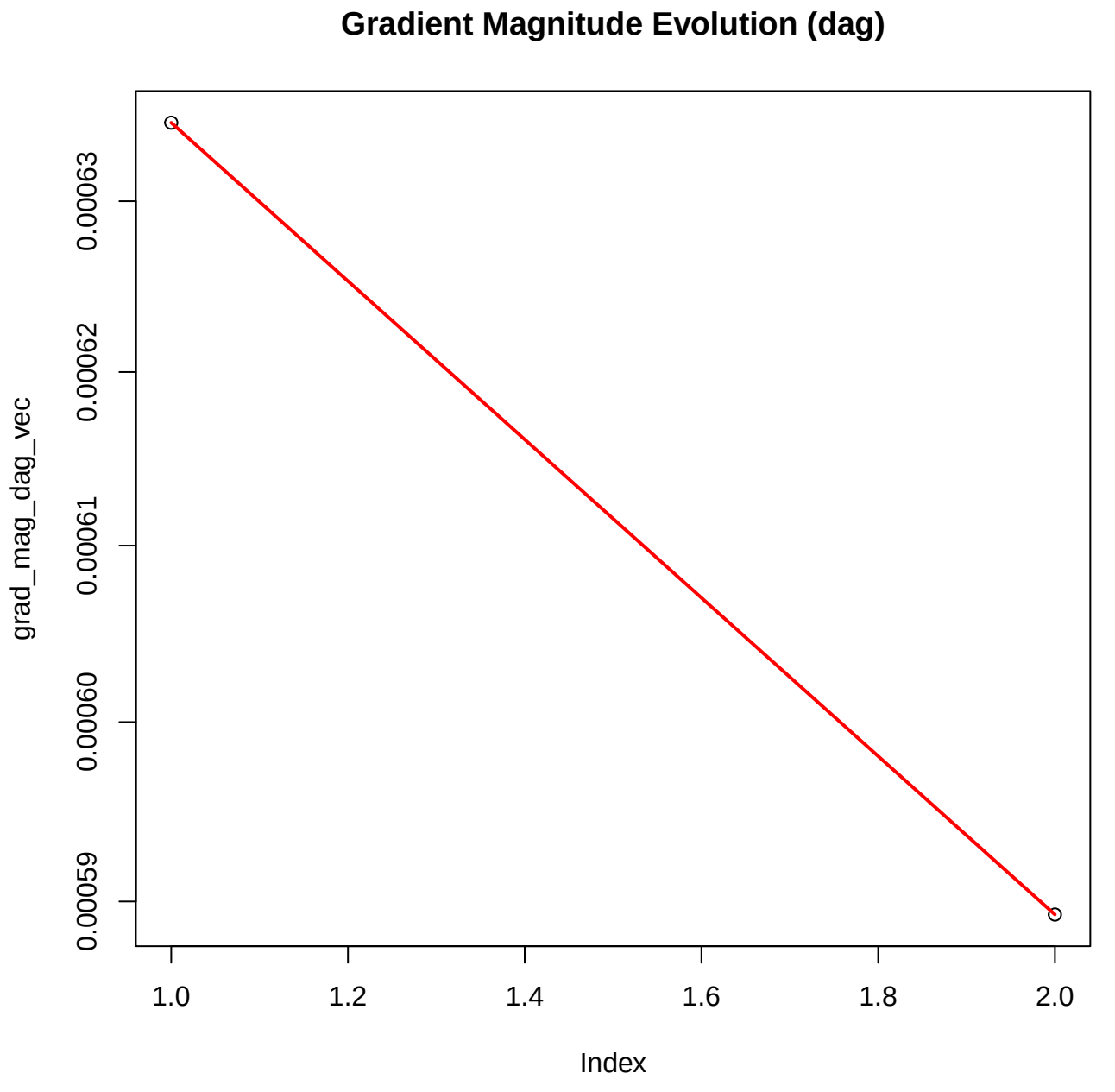
1.4
1.6
1.8
2.0
Index


Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
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0.00060
0.00061
0.00062
0.00063
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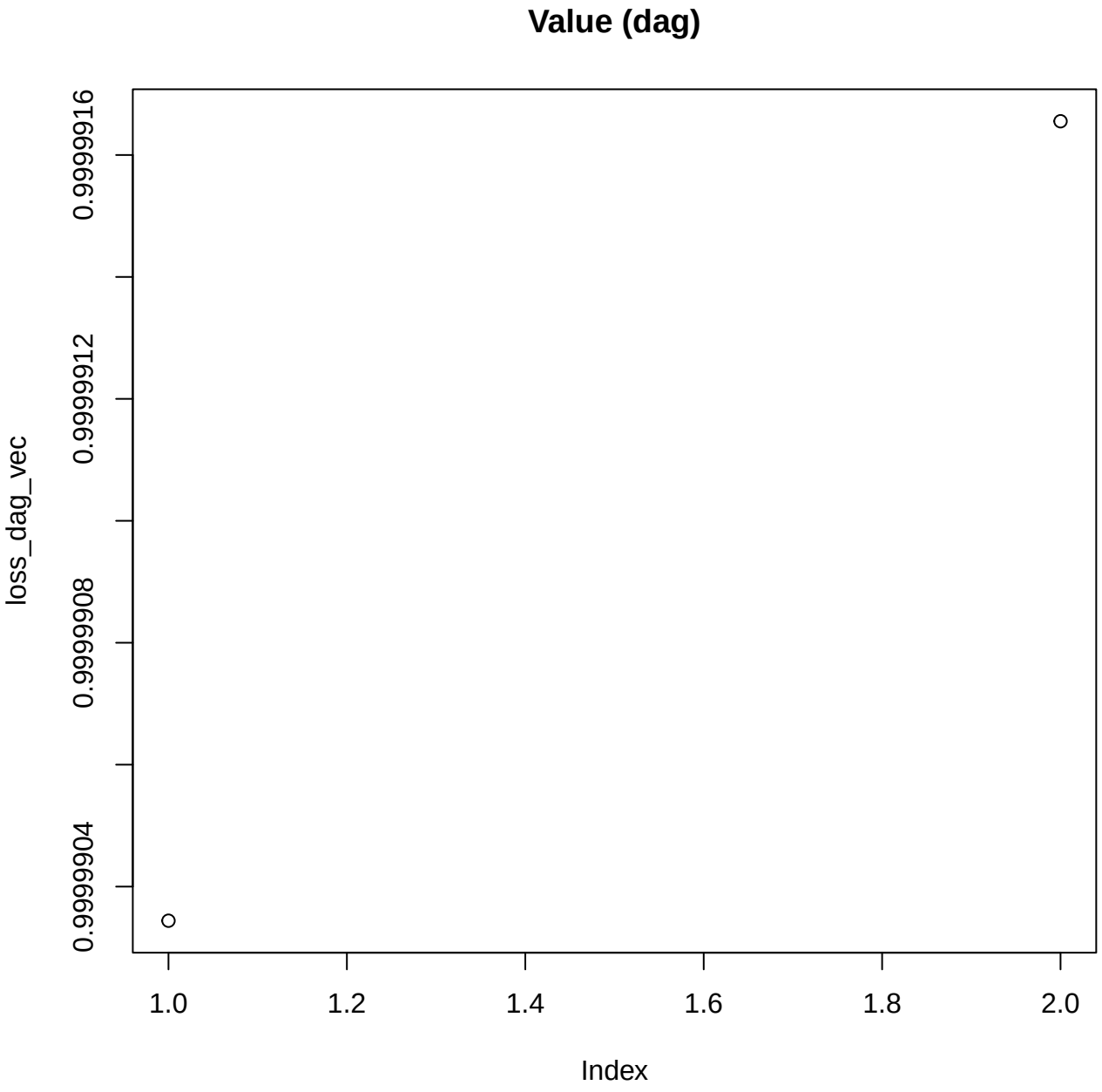
1.4
1.6
1.8
2.0
Index


Value (dag)
loss_dag_vec
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0.9999912
0.9999916
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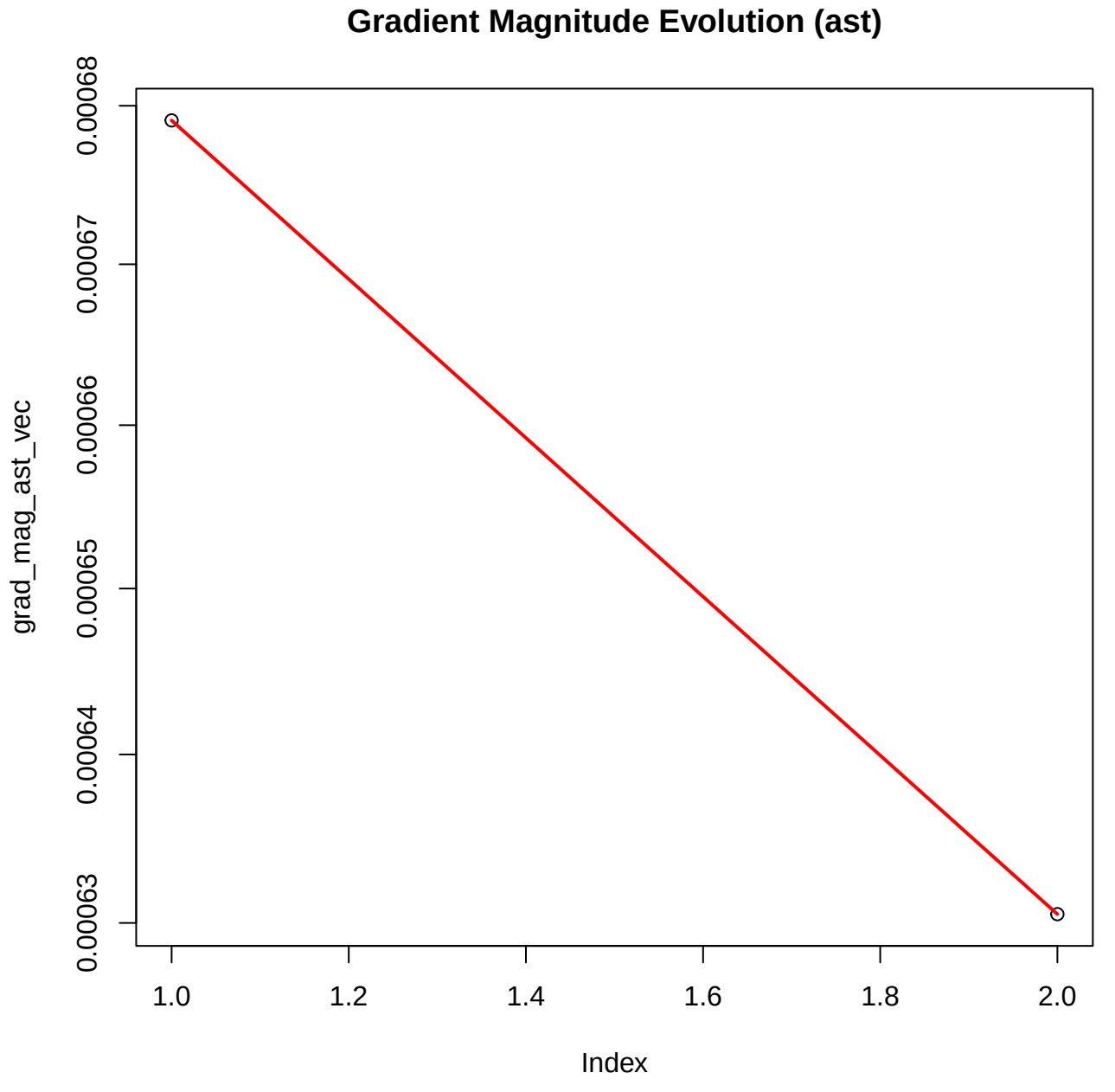
1.4
1.6
1.8
2.0
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.00064
0.00065
0.00066
0.00067
0.00068
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1.4
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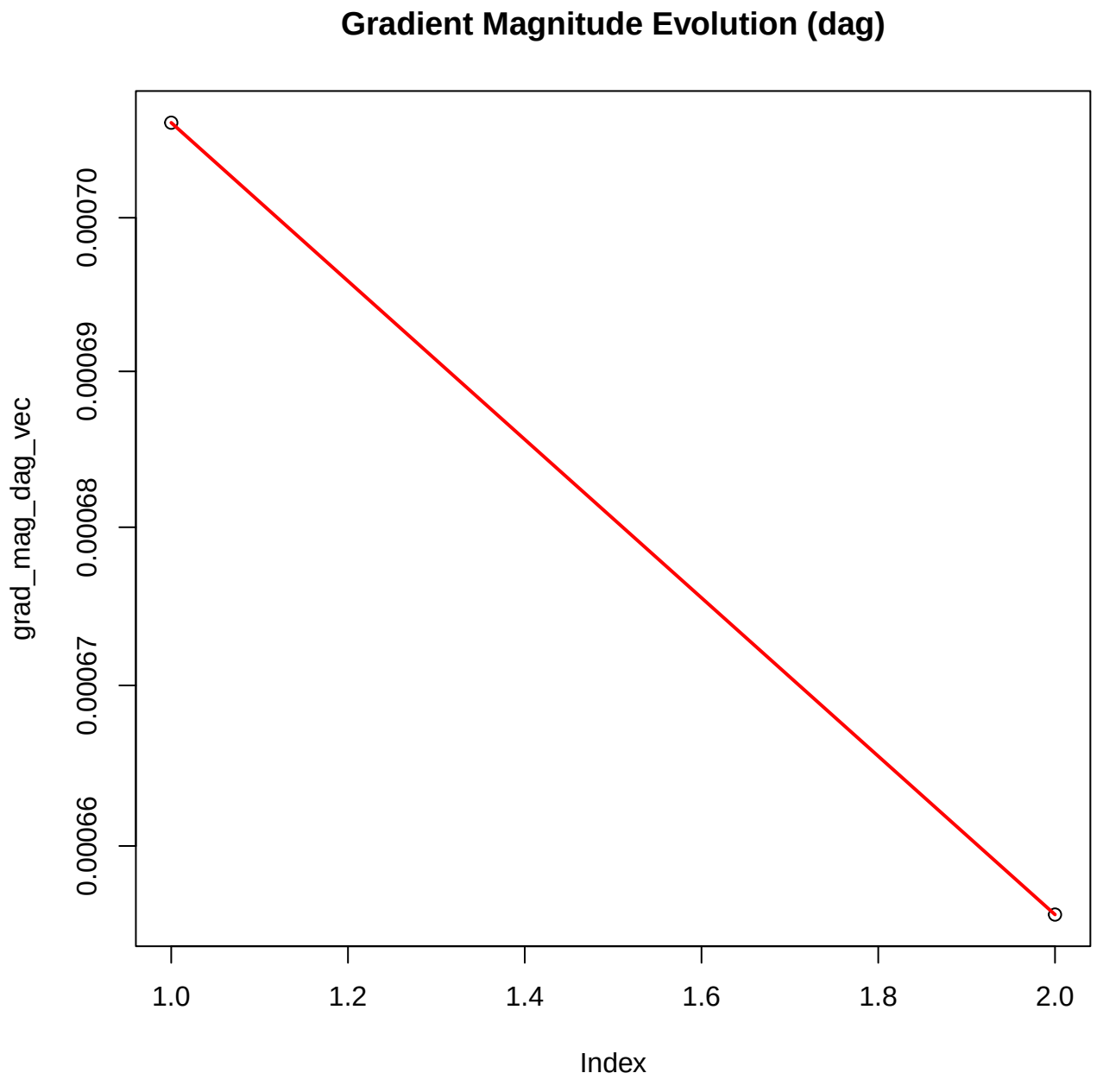
1.8
2.0
Index


Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
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0.00067
0.00068
0.00069
0.00070
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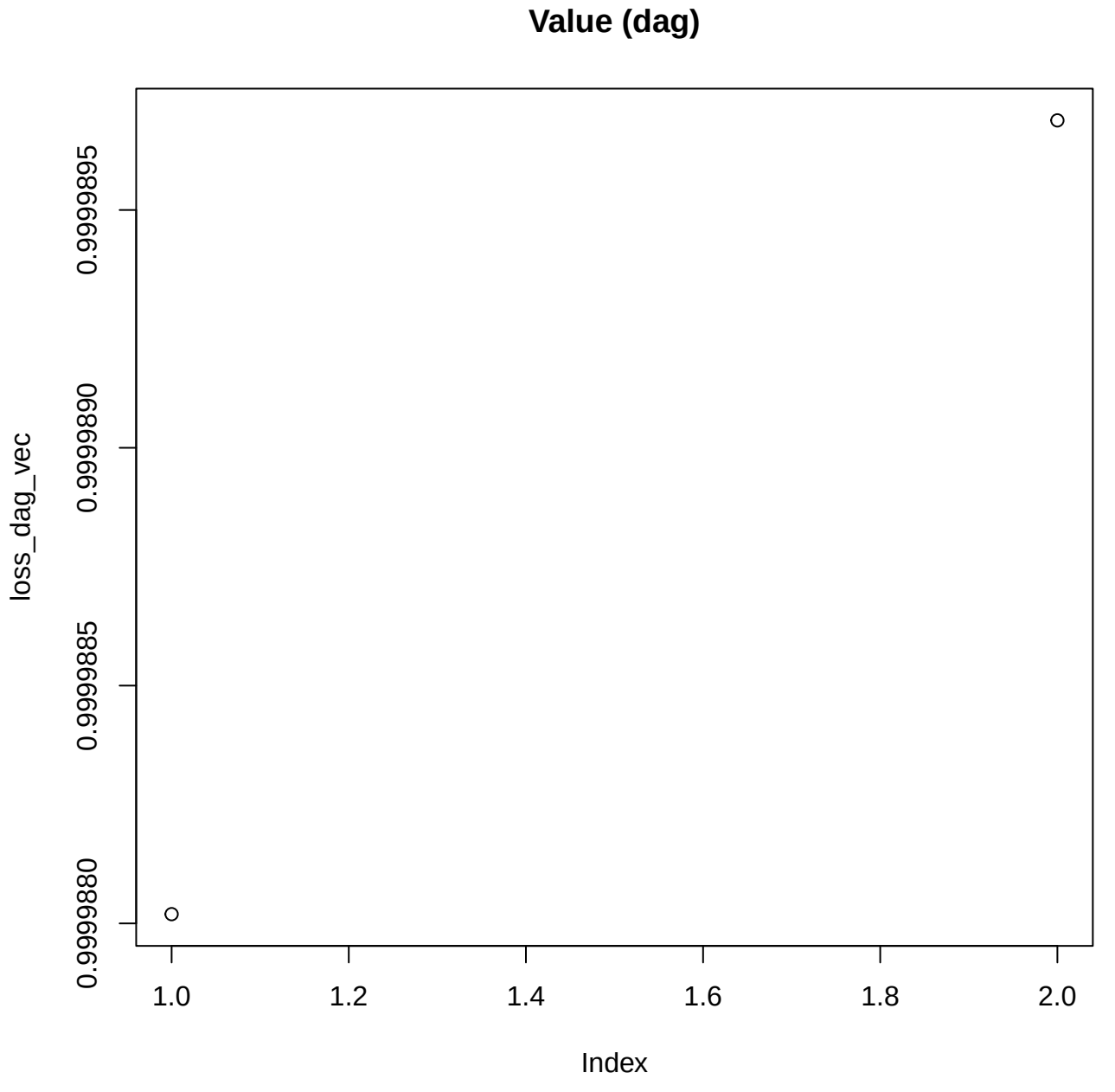
1.4
1.6
1.8
2.0
Index


Value (dag)
loss_dag_vec
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0.9999890
0.9999895
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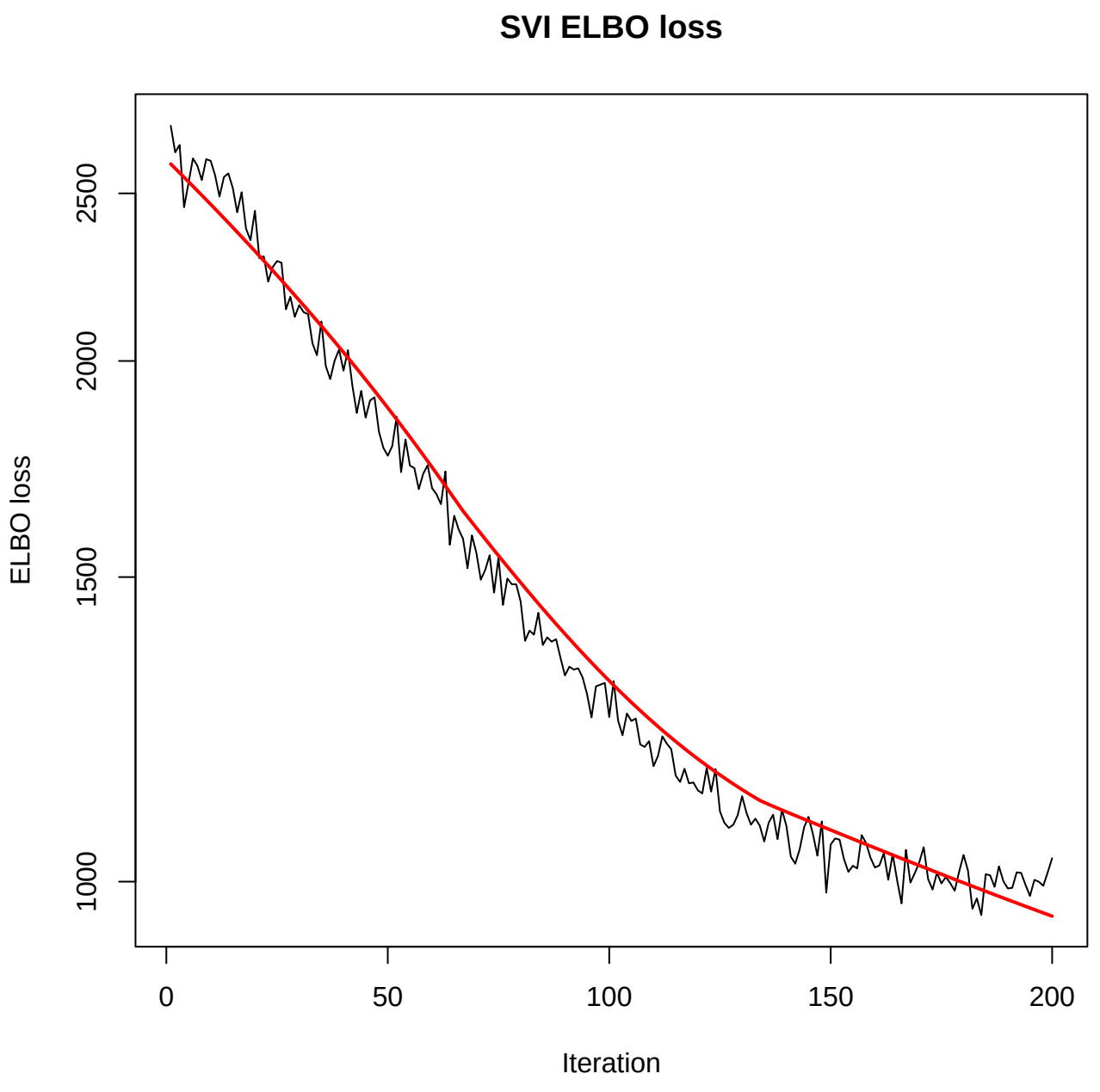
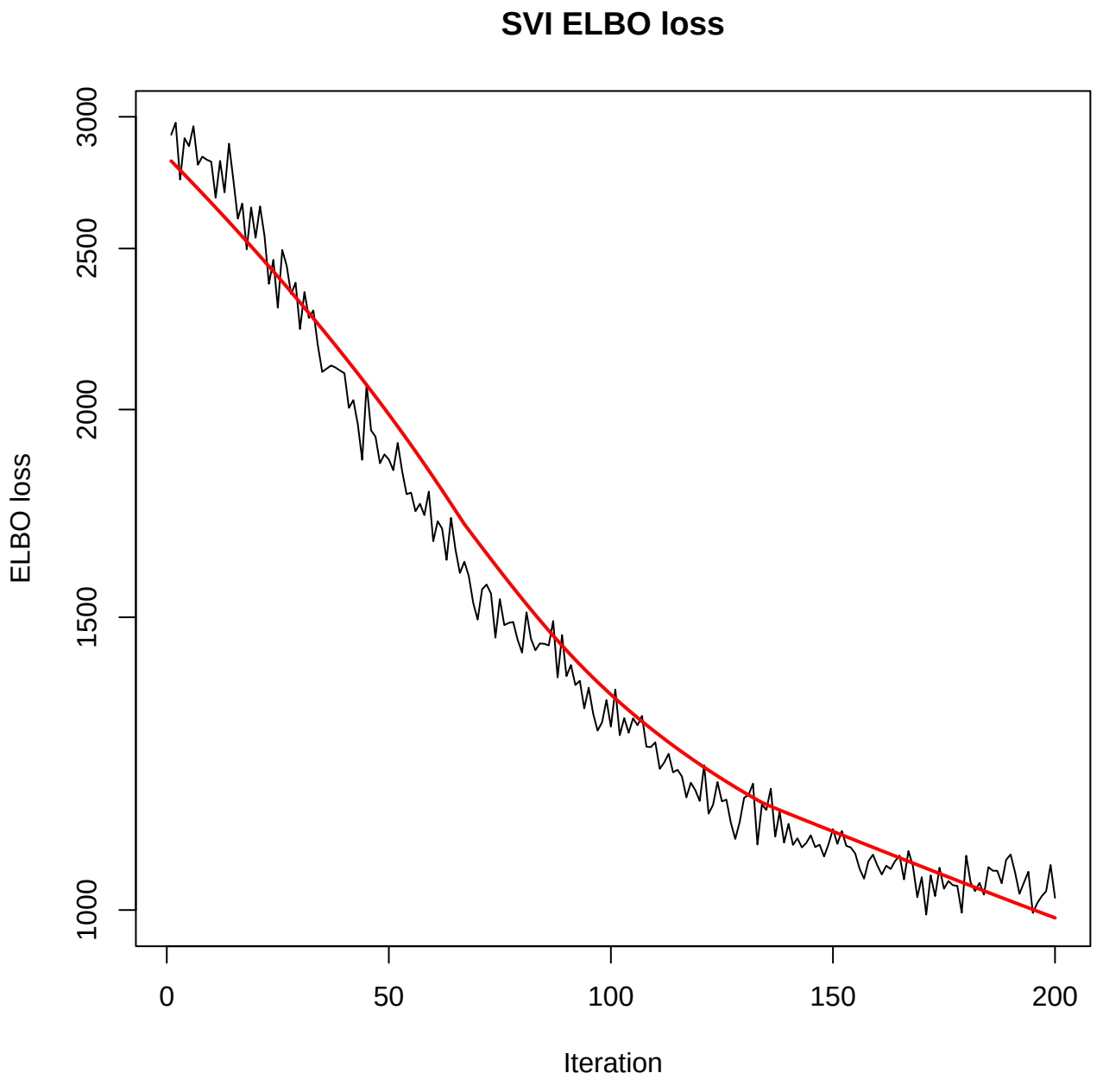
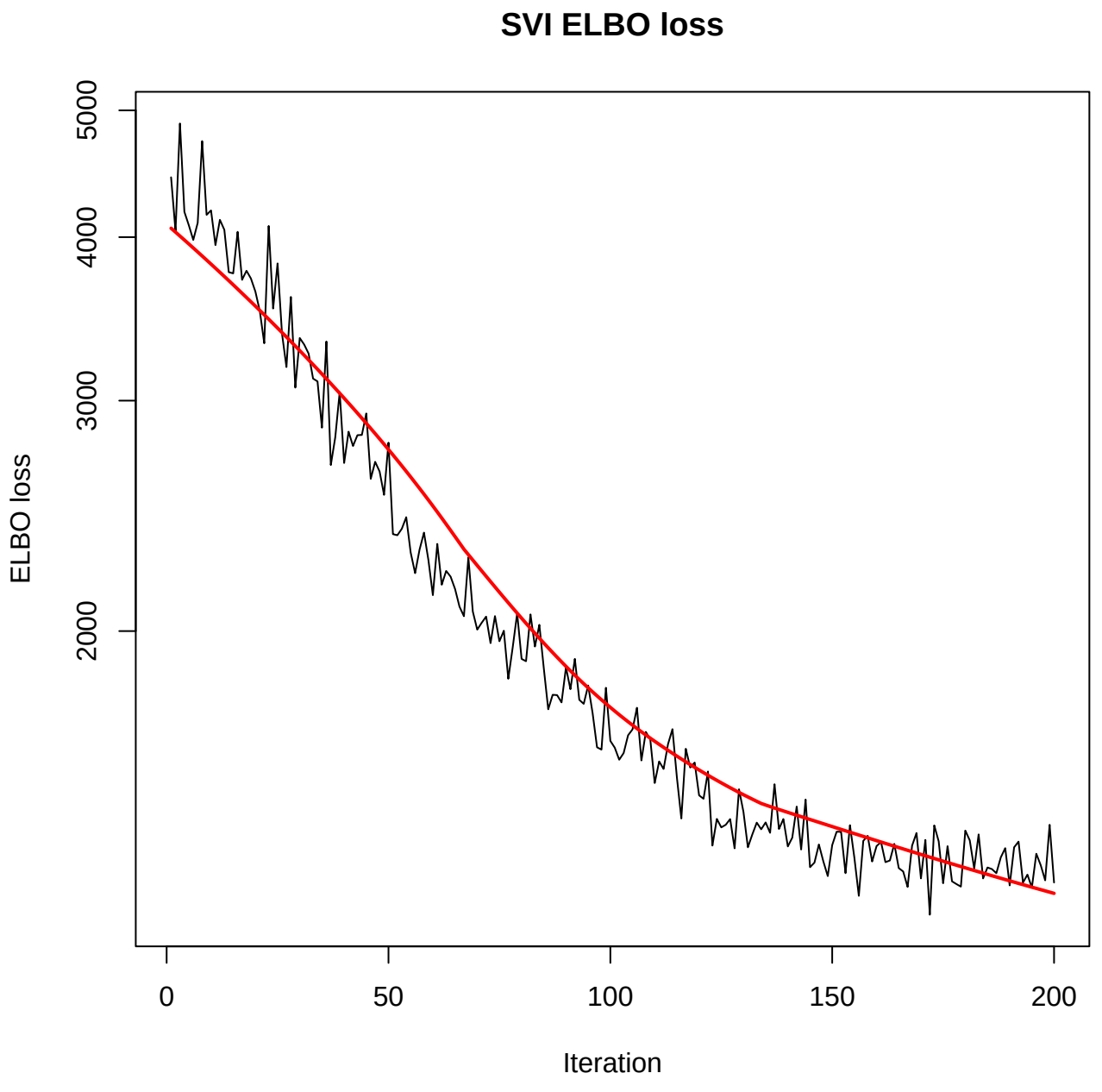
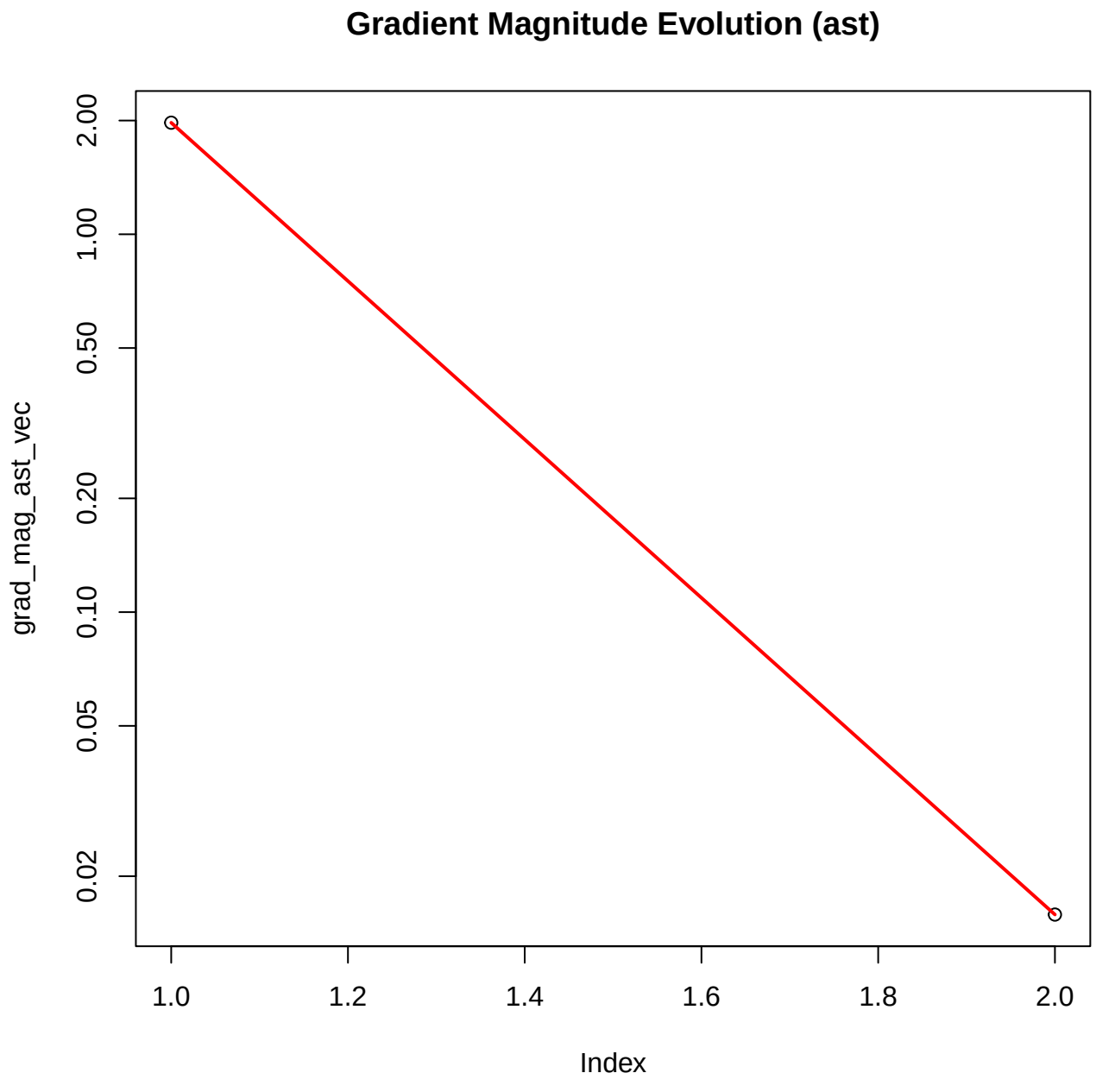
1.4
1.6
1.8
2.0
Index


SVI ELBO loss
ELBO loss
2500
2000
1500
1000
0
50
100
150
200
Iteration


SVI ELBO loss
ELBO loss
3000
2500
2000
1500
1000
0
50
100
150
200
Iteration


SVI ELBO loss
ELBO loss
5000
4000
3000
2000
0
50
100
150
200
Iteration


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
2.00
1.00
0.50
0.20
0.10
0.05
0.02
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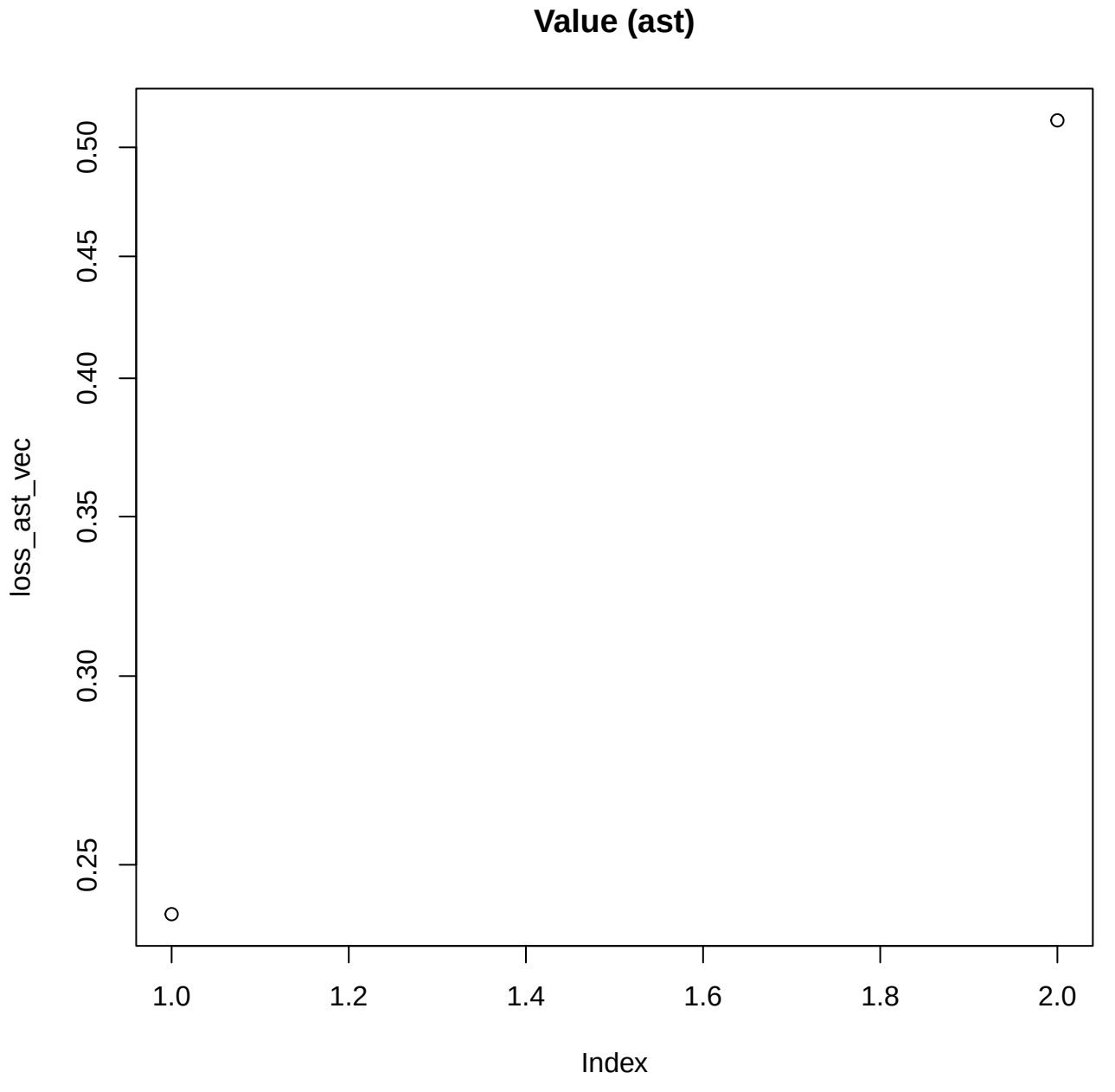
2.0
Index


Value (ast)
loss_ast_vec
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0.50
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1.4
1.6
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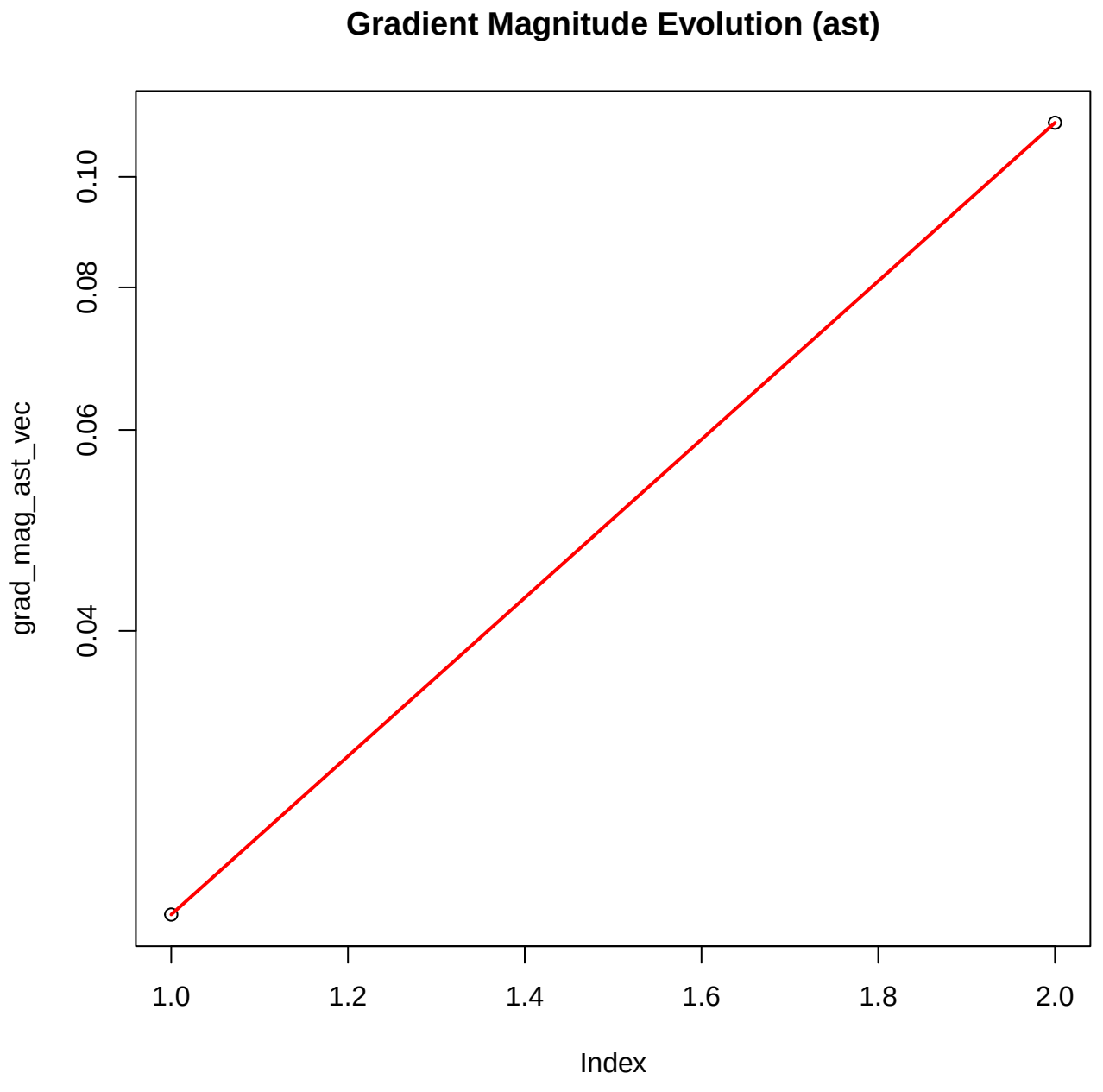
Gradient Magnitude Evolution (ast)
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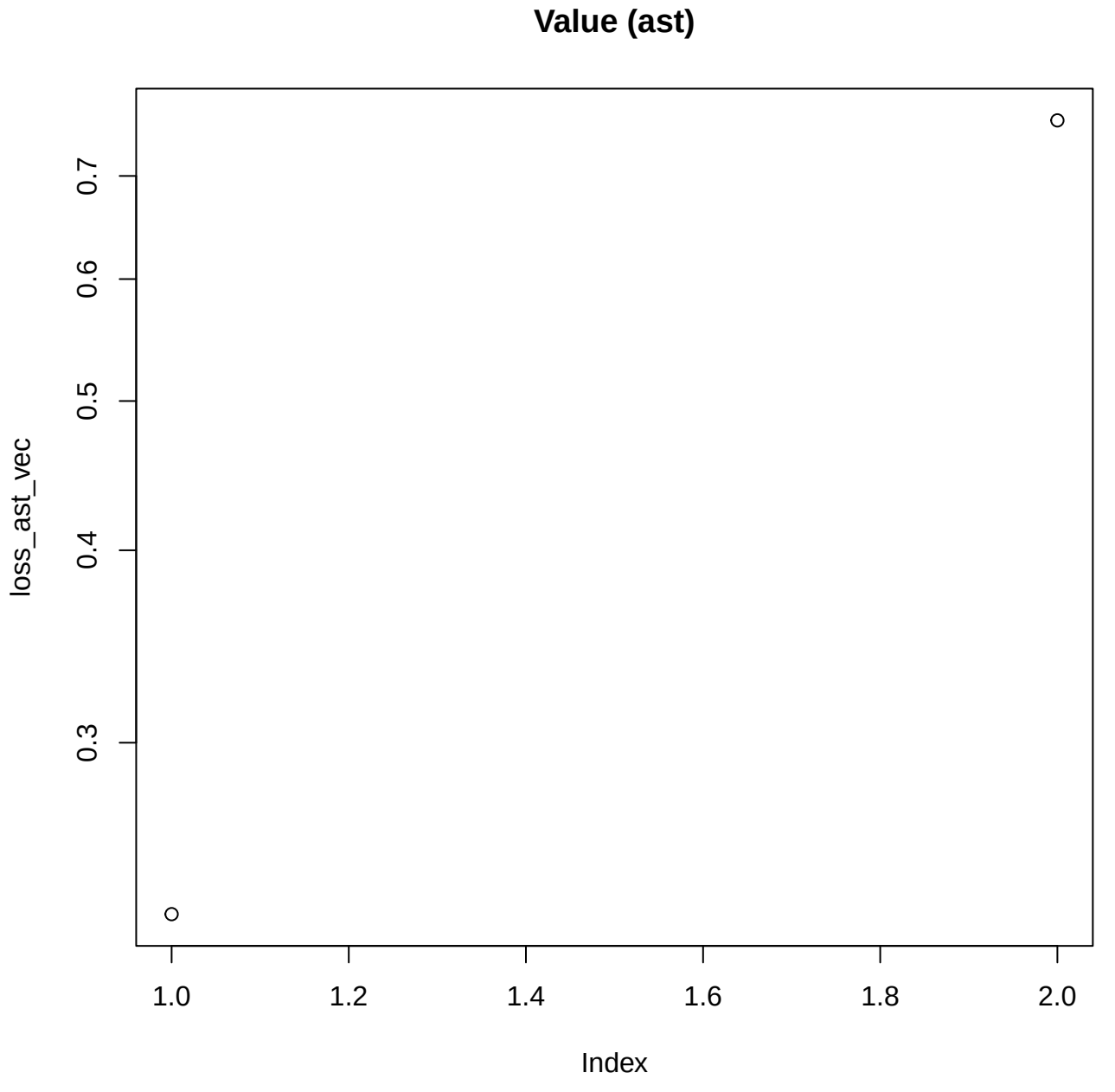
Value (ast)
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1.4
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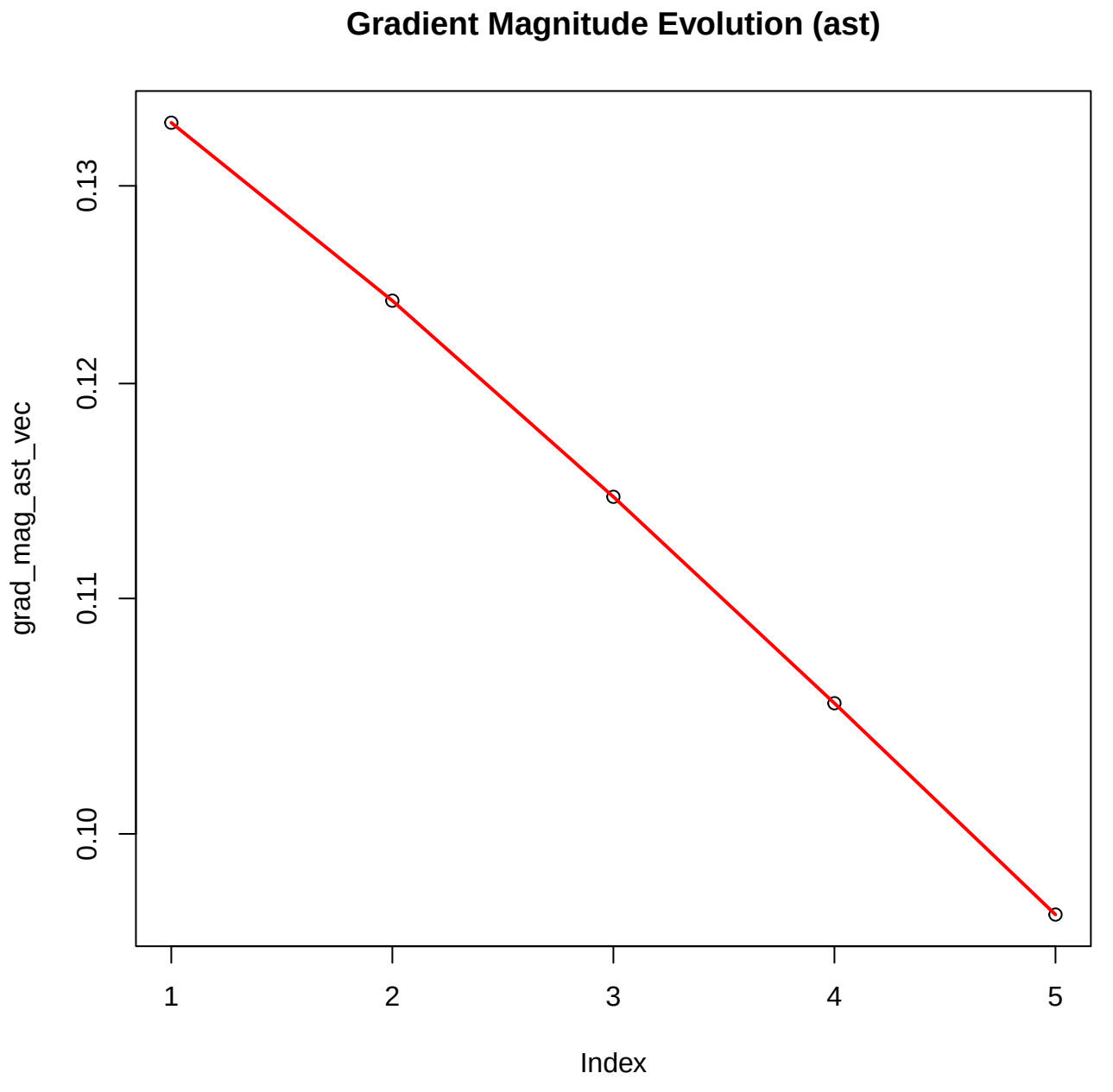
Index


Gradient Magnitude Evolution (ast)
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5
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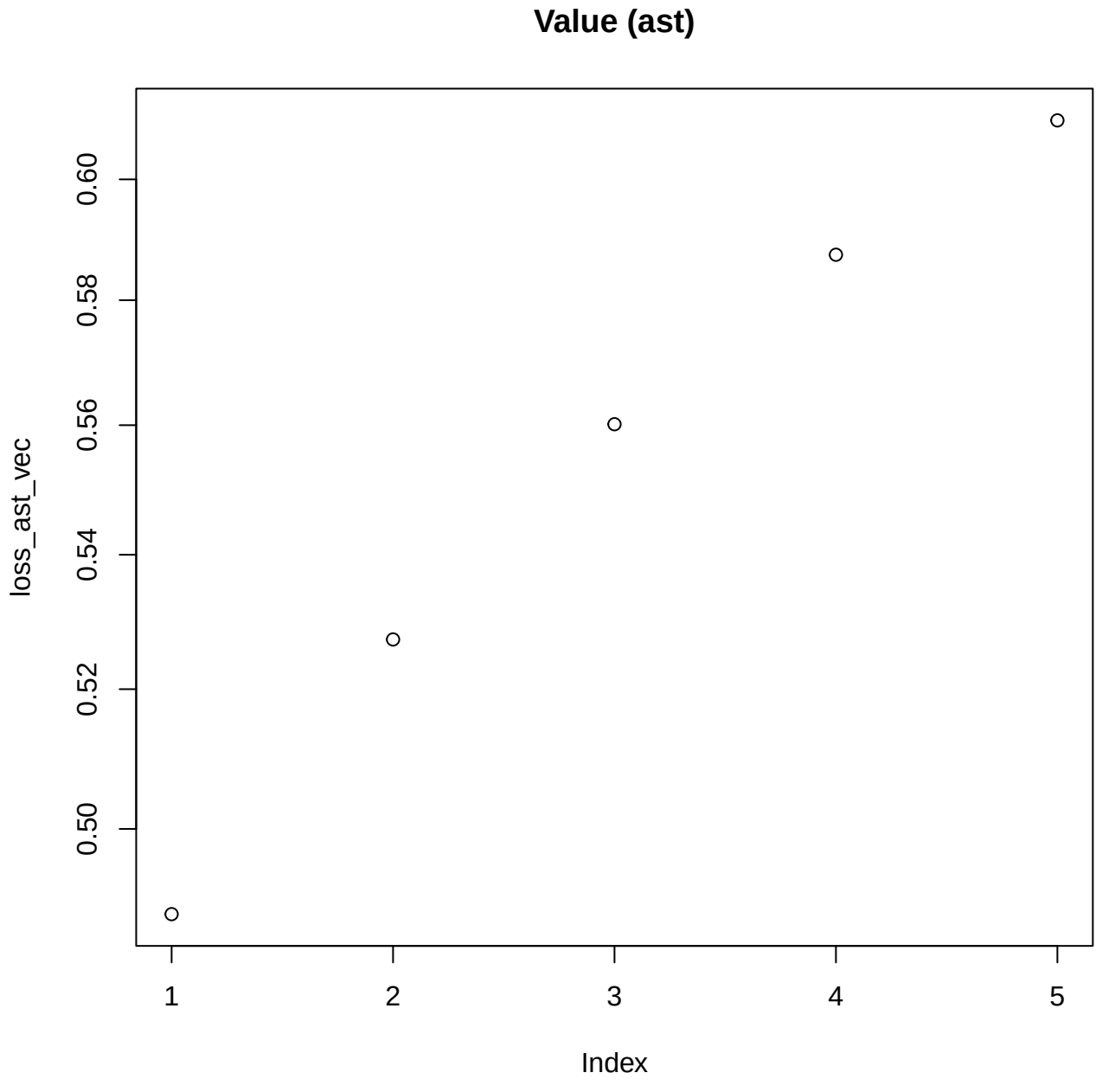
Value (ast)
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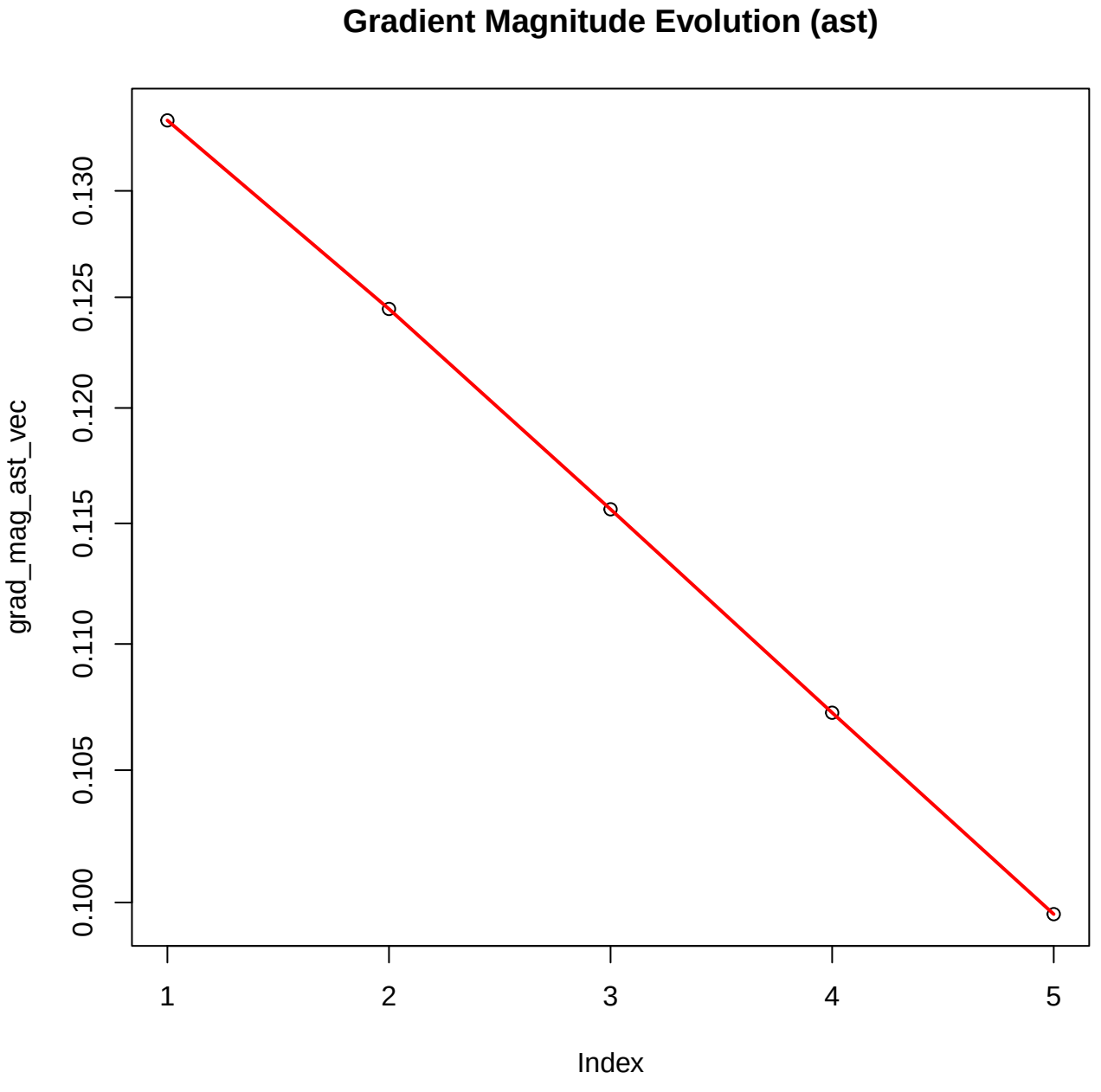
5
Index


Gradient Magnitude Evolution (ast)
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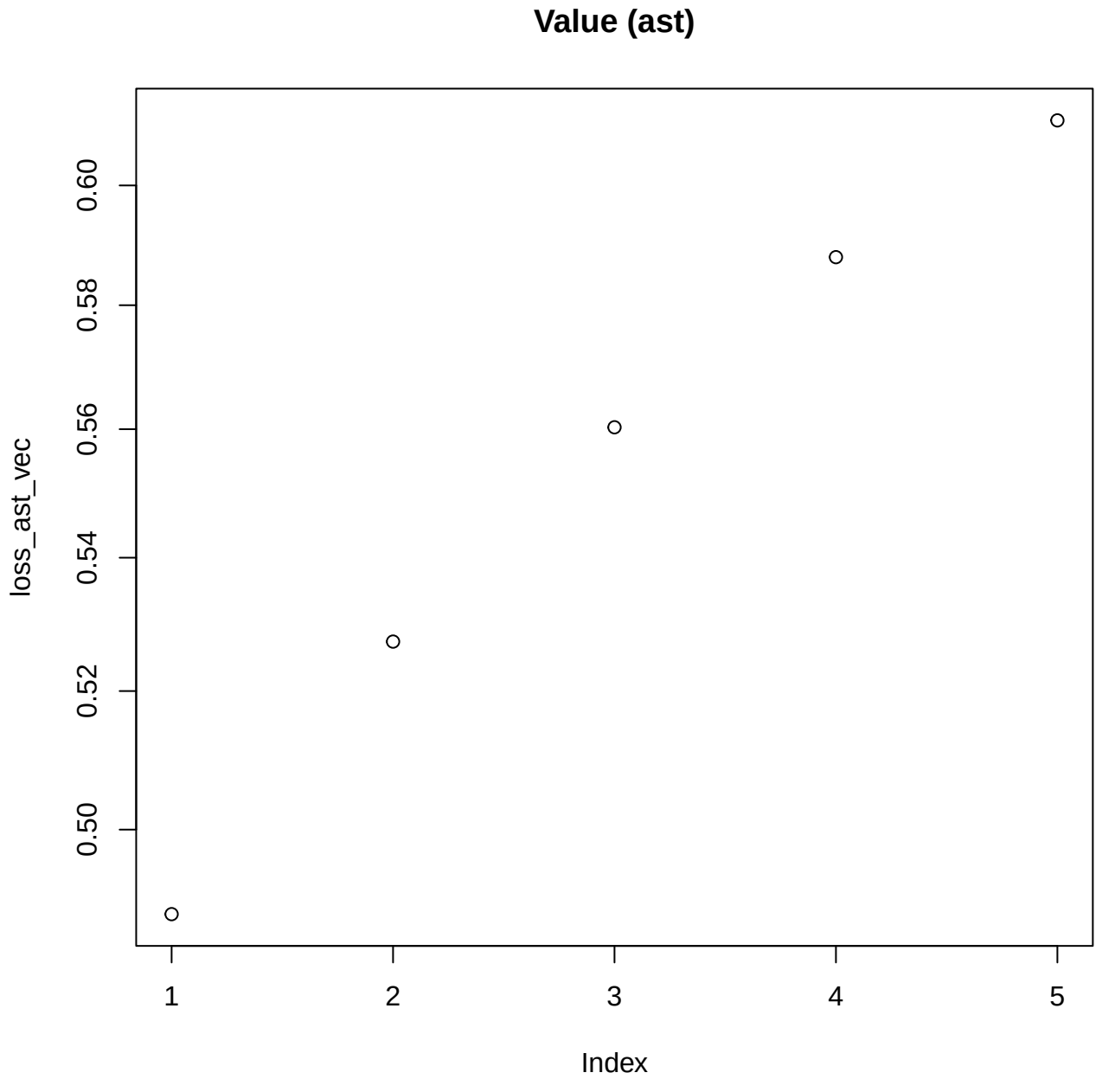
Index


Value (ast)
loss_ast_vec
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0.54
0.56
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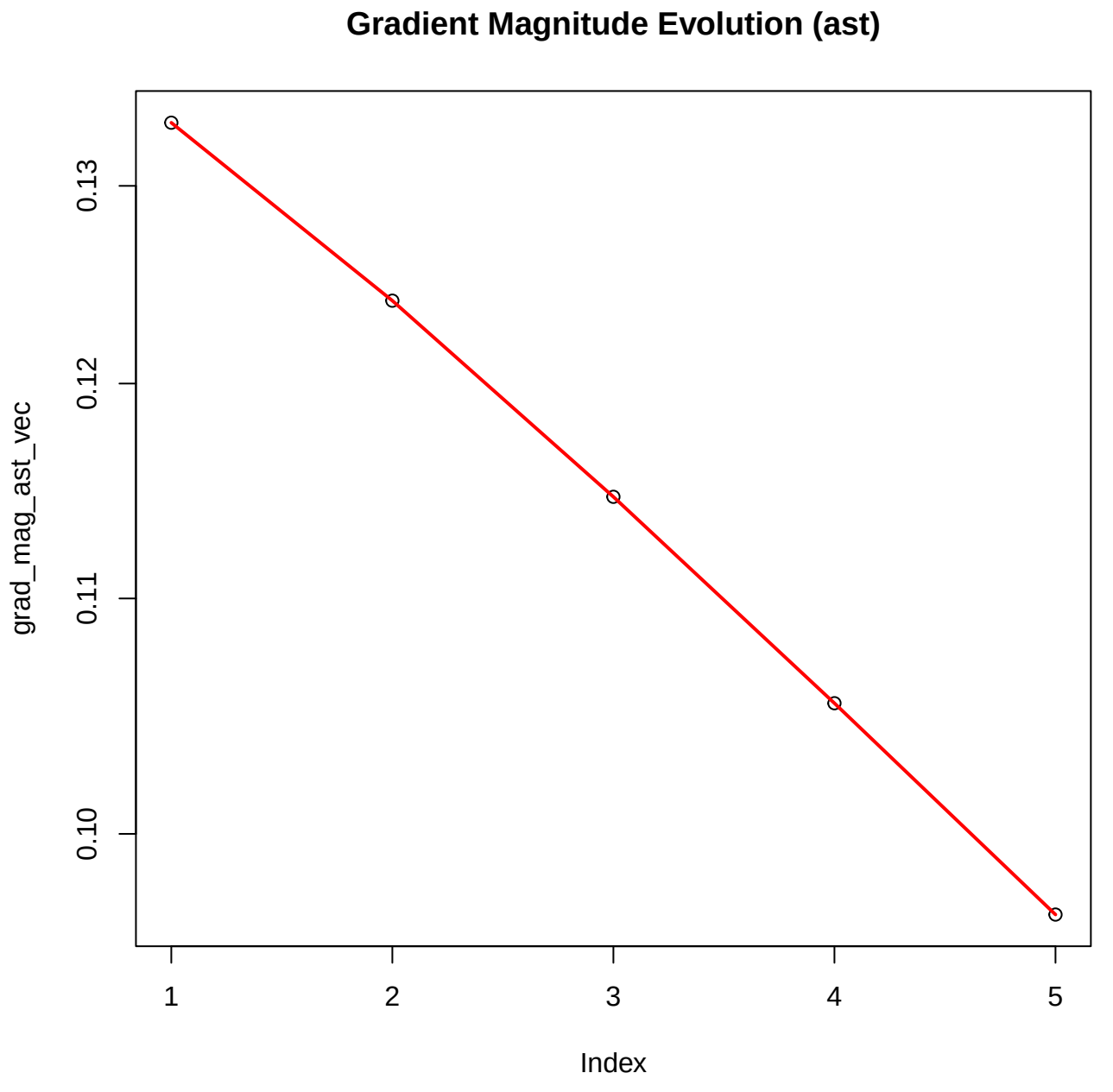
Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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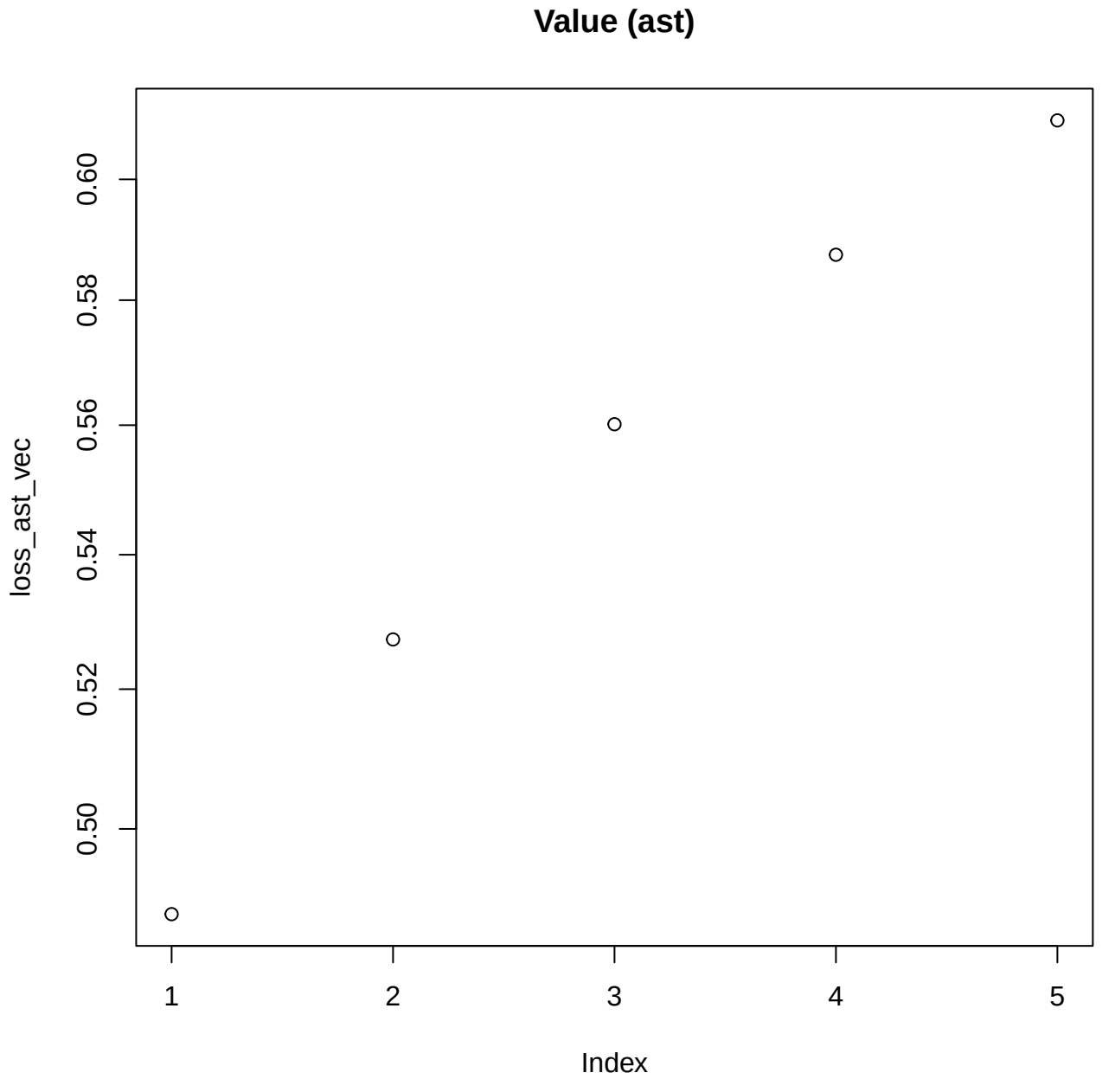
Value (ast)
loss_ast_vec
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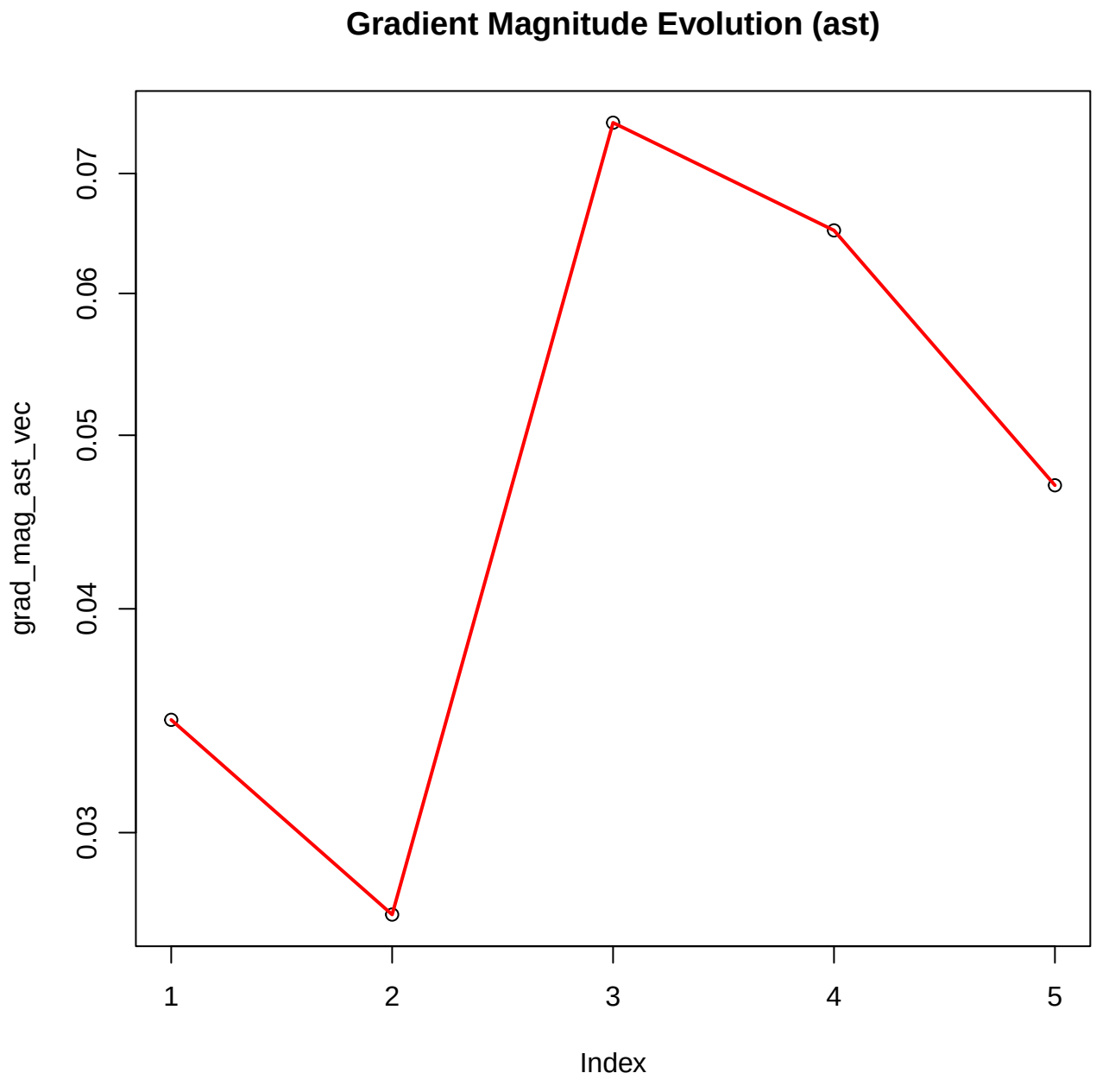
Gradient Magnitude Evolution (ast)
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5
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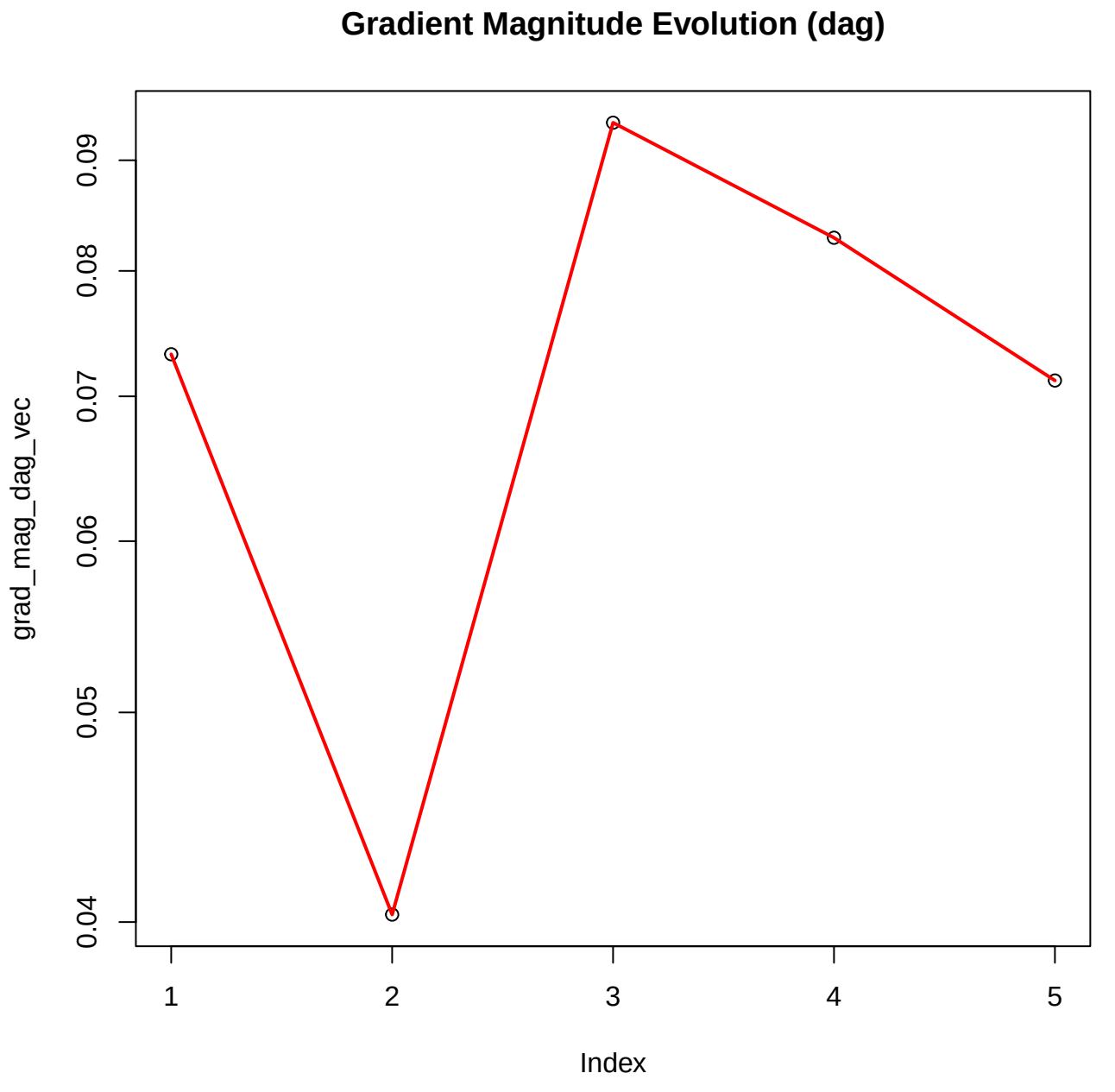
Gradient Magnitude Evolution (dag)
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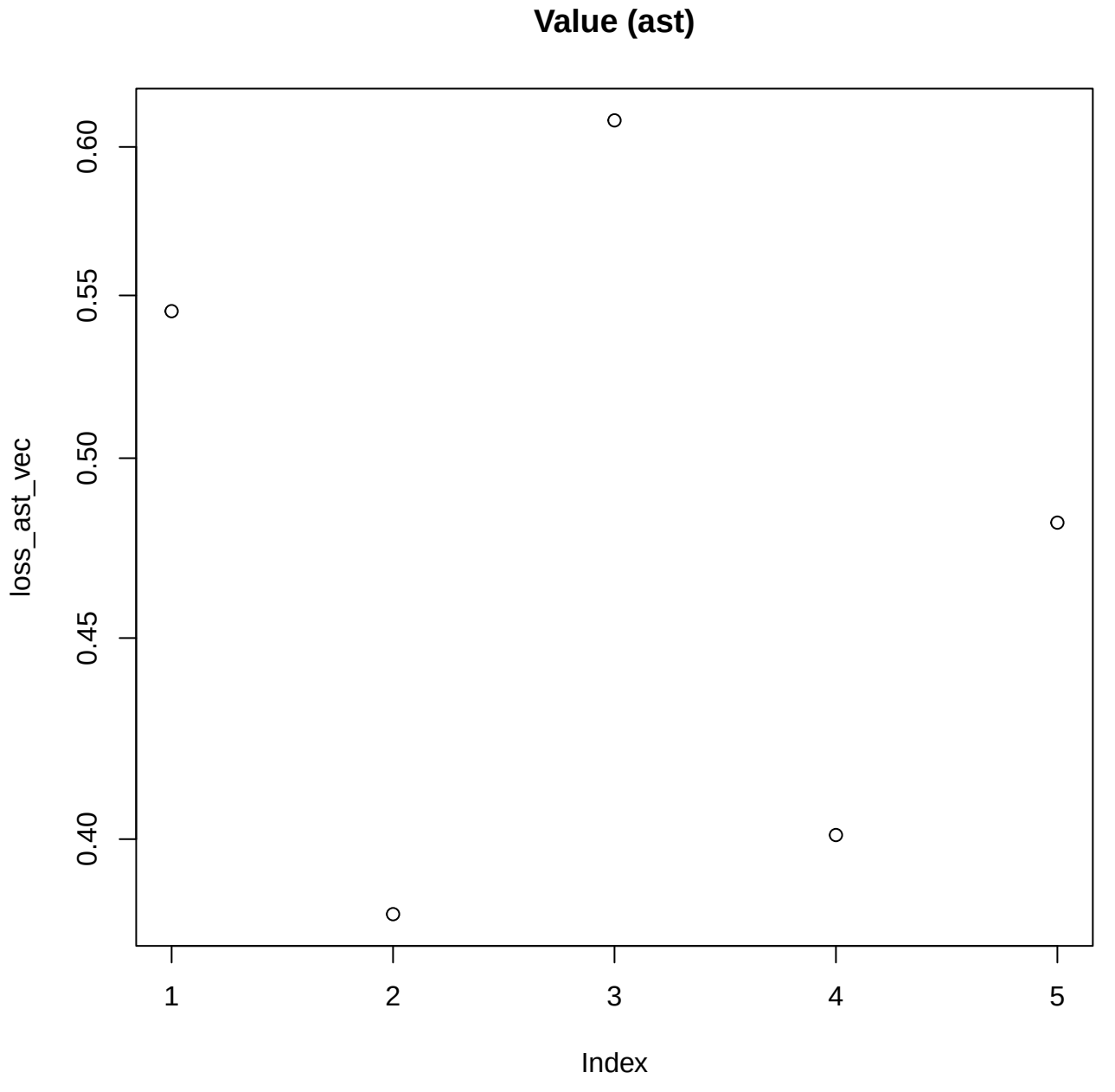
Index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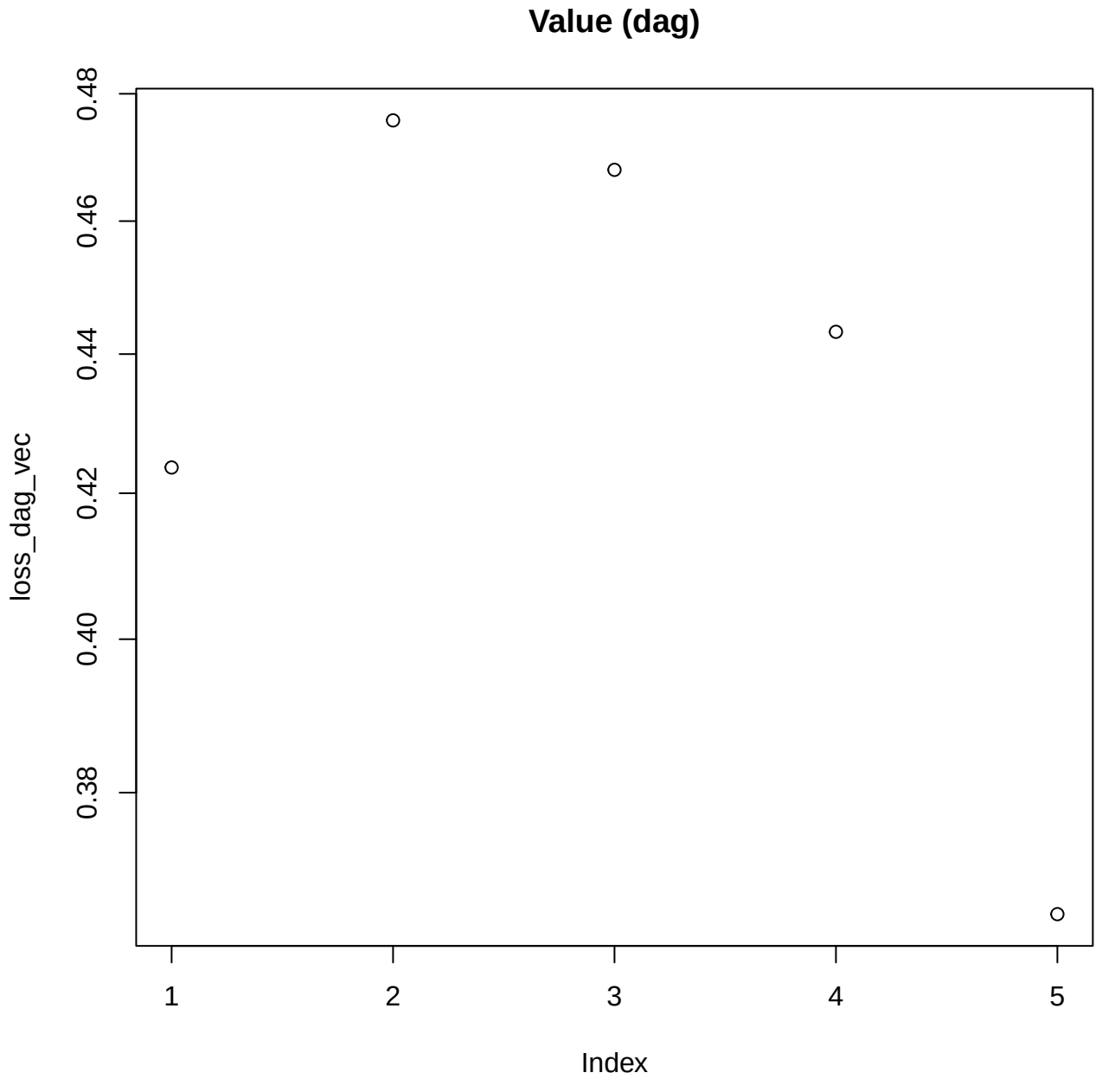
Value (dag)
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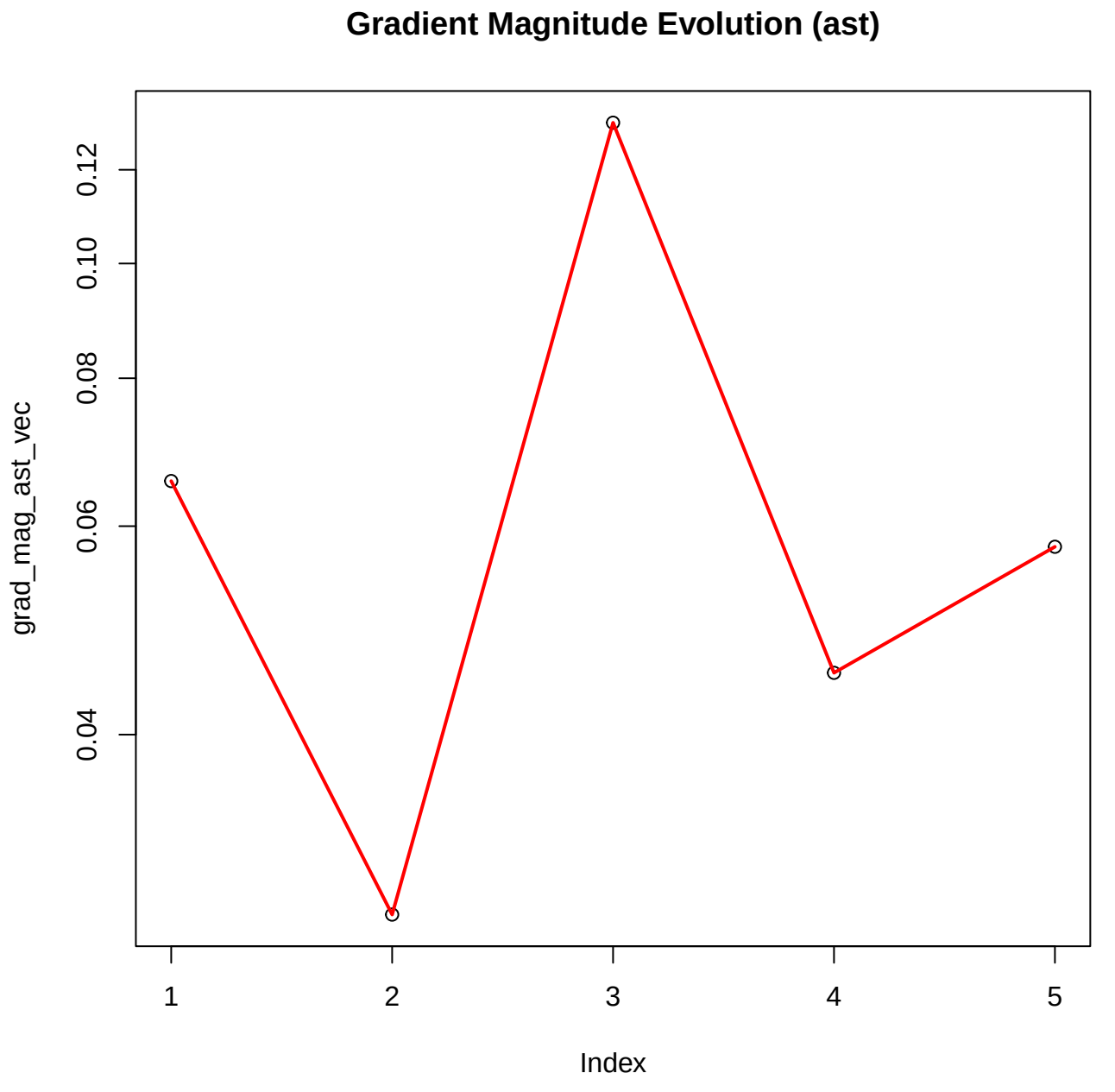
Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.08
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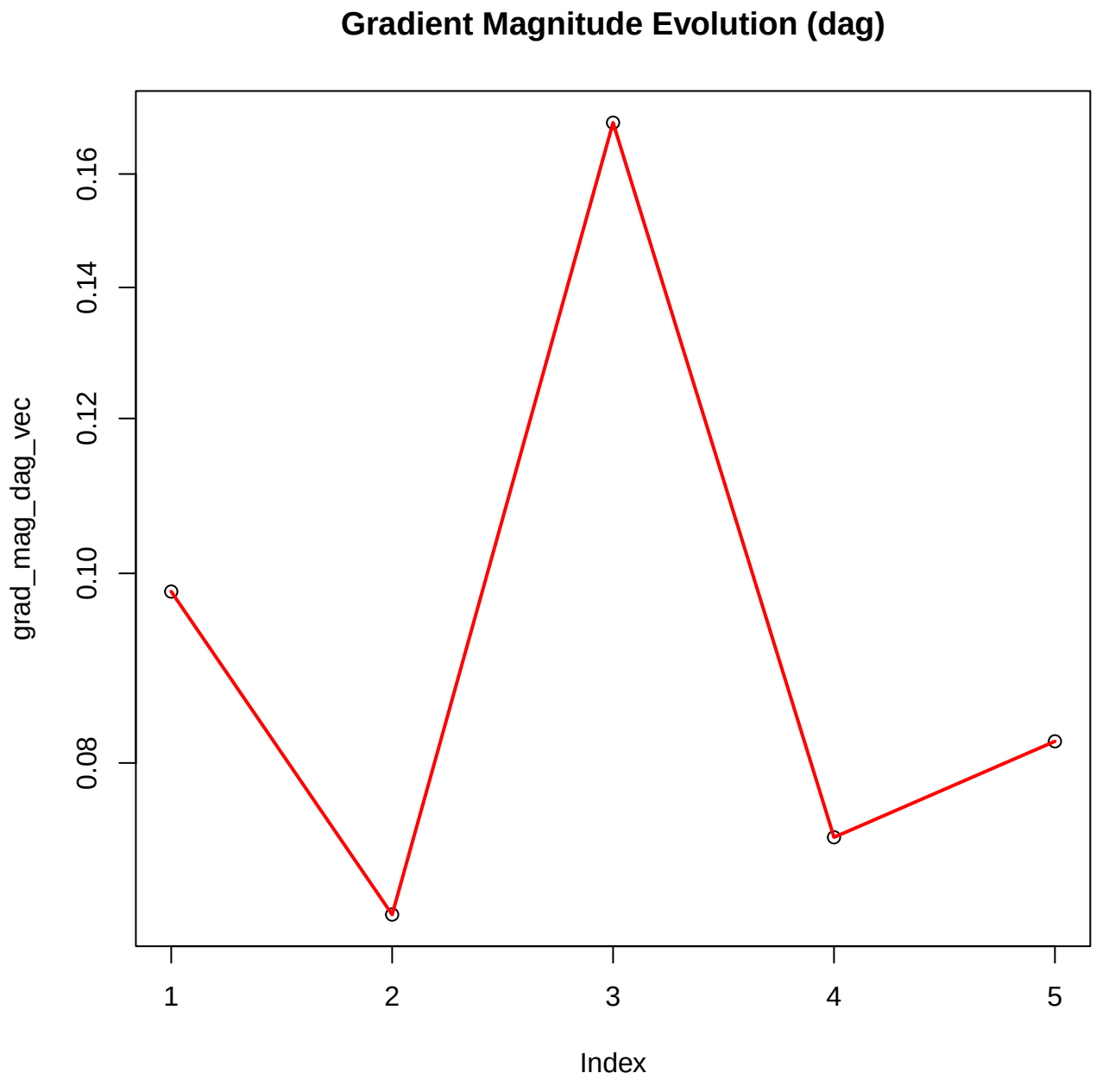
Gradient Magnitude Evolution (dag)
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2
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4
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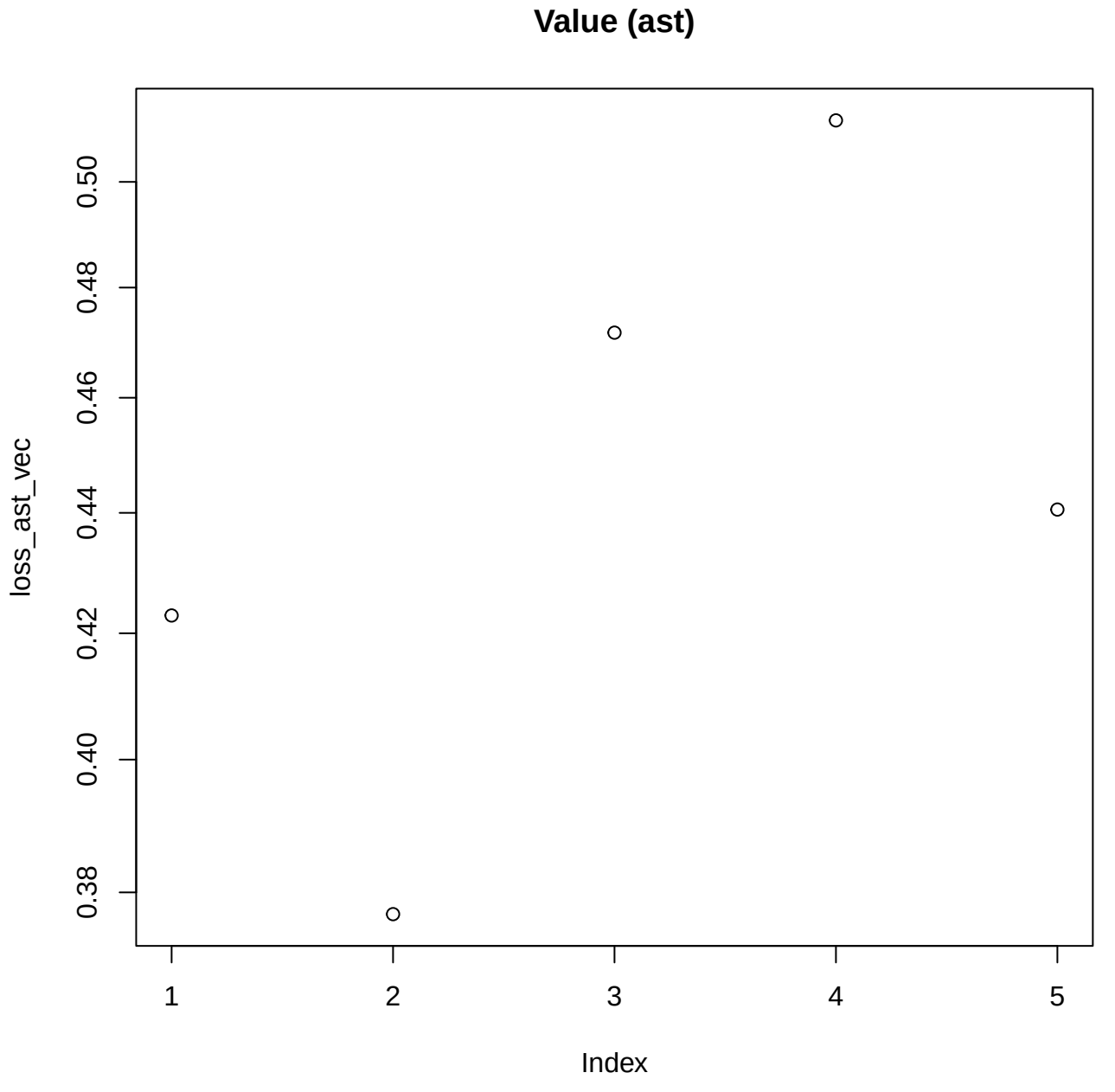
Value (ast)
loss_ast_vec
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0.44
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0.50
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2
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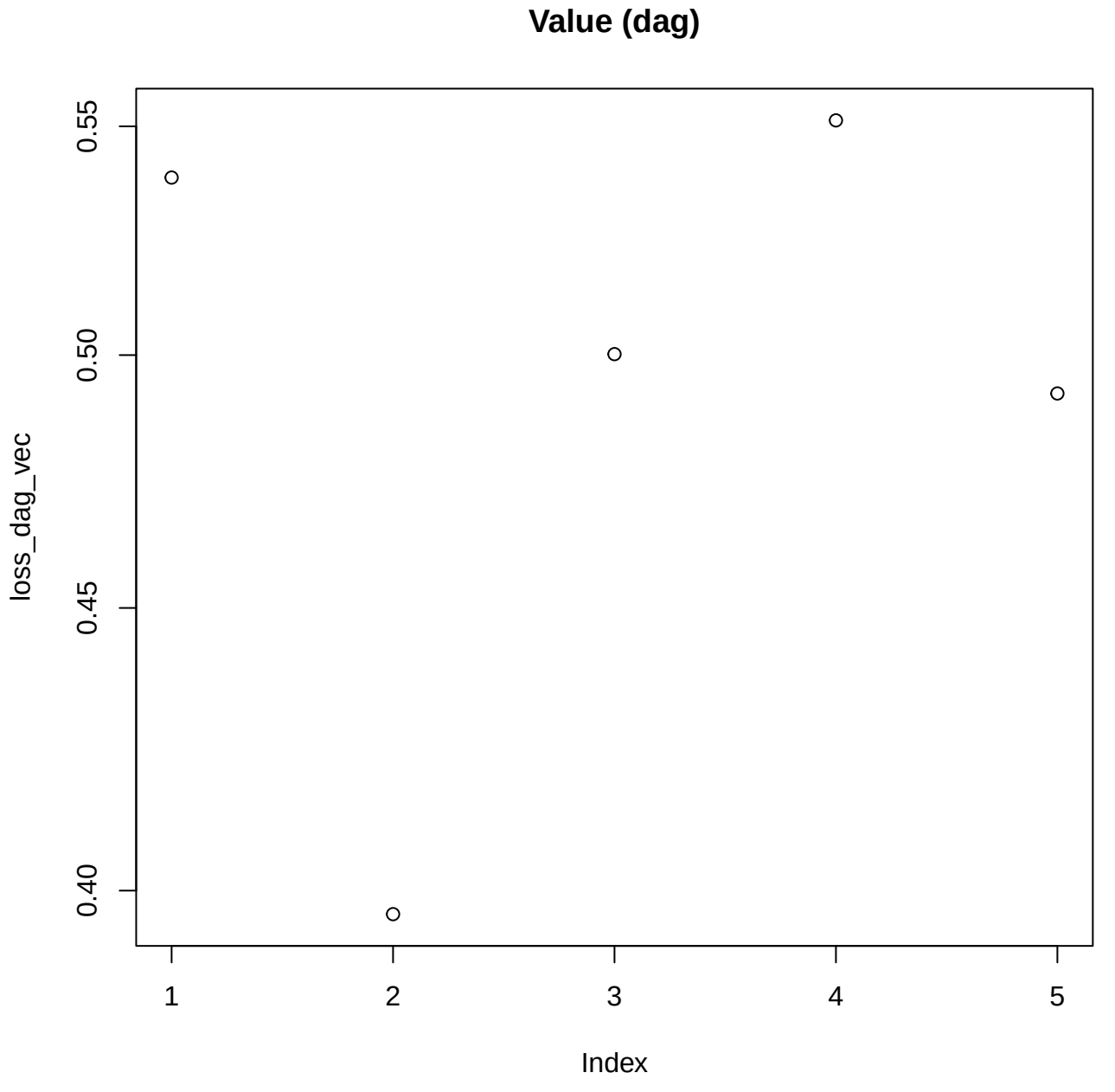
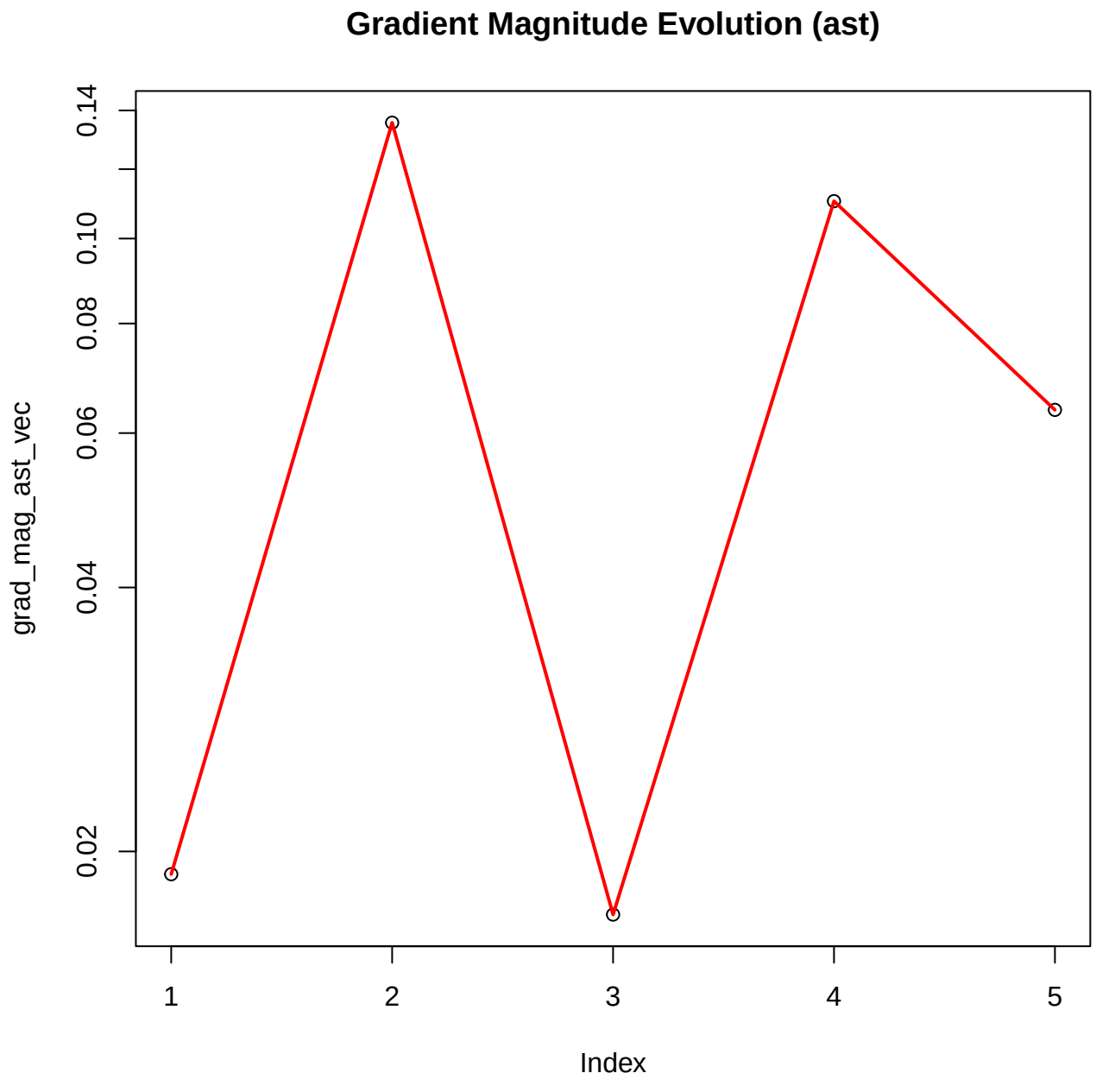
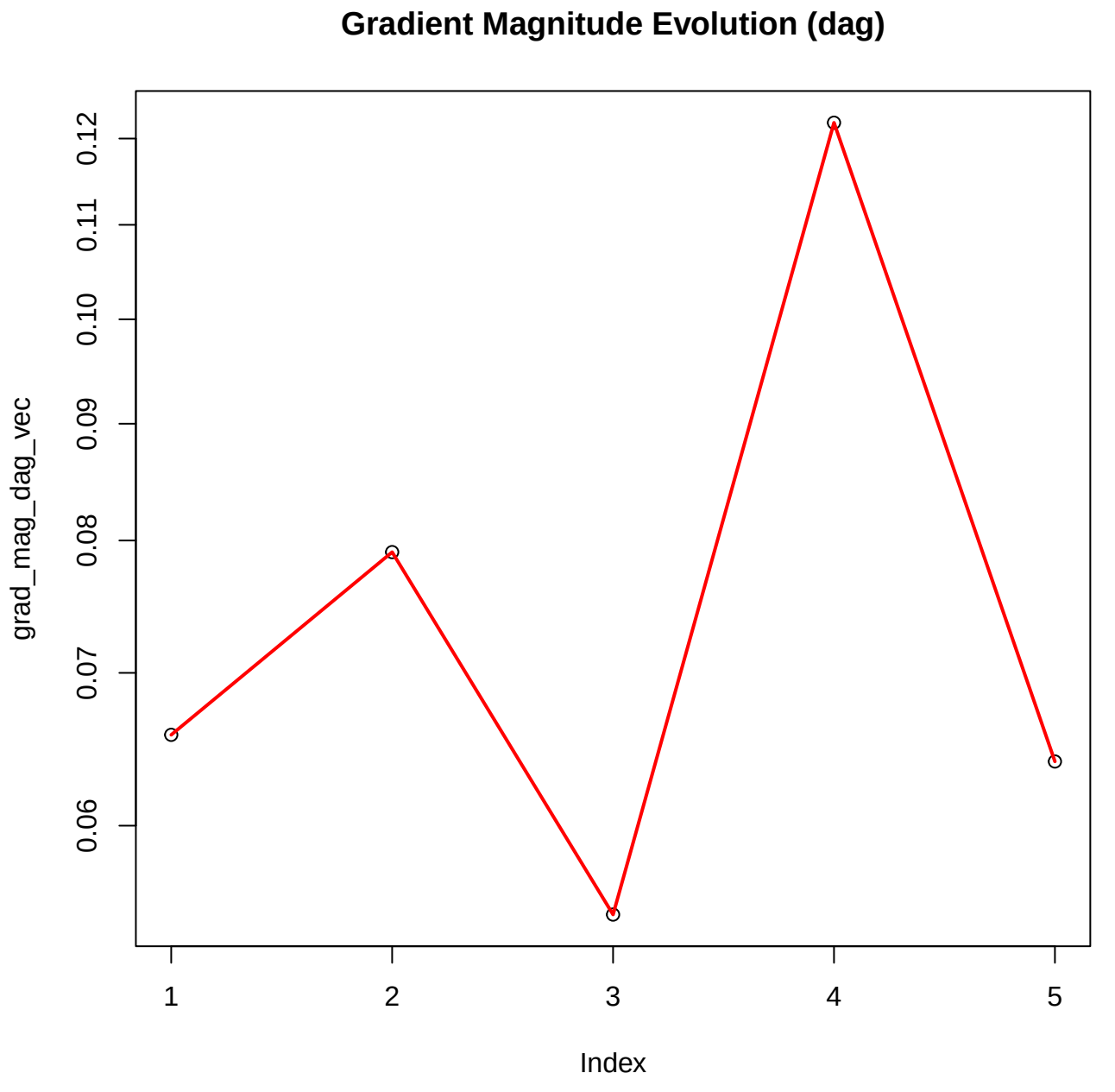
4
5
Index


Value (dag)
loss_dag_vec
0.40
0.45
0.50
0.55
1
2
3
4
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.02
0.04
0.06
0.08
0.10
0.14
1
2
3
4
5
Index


Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
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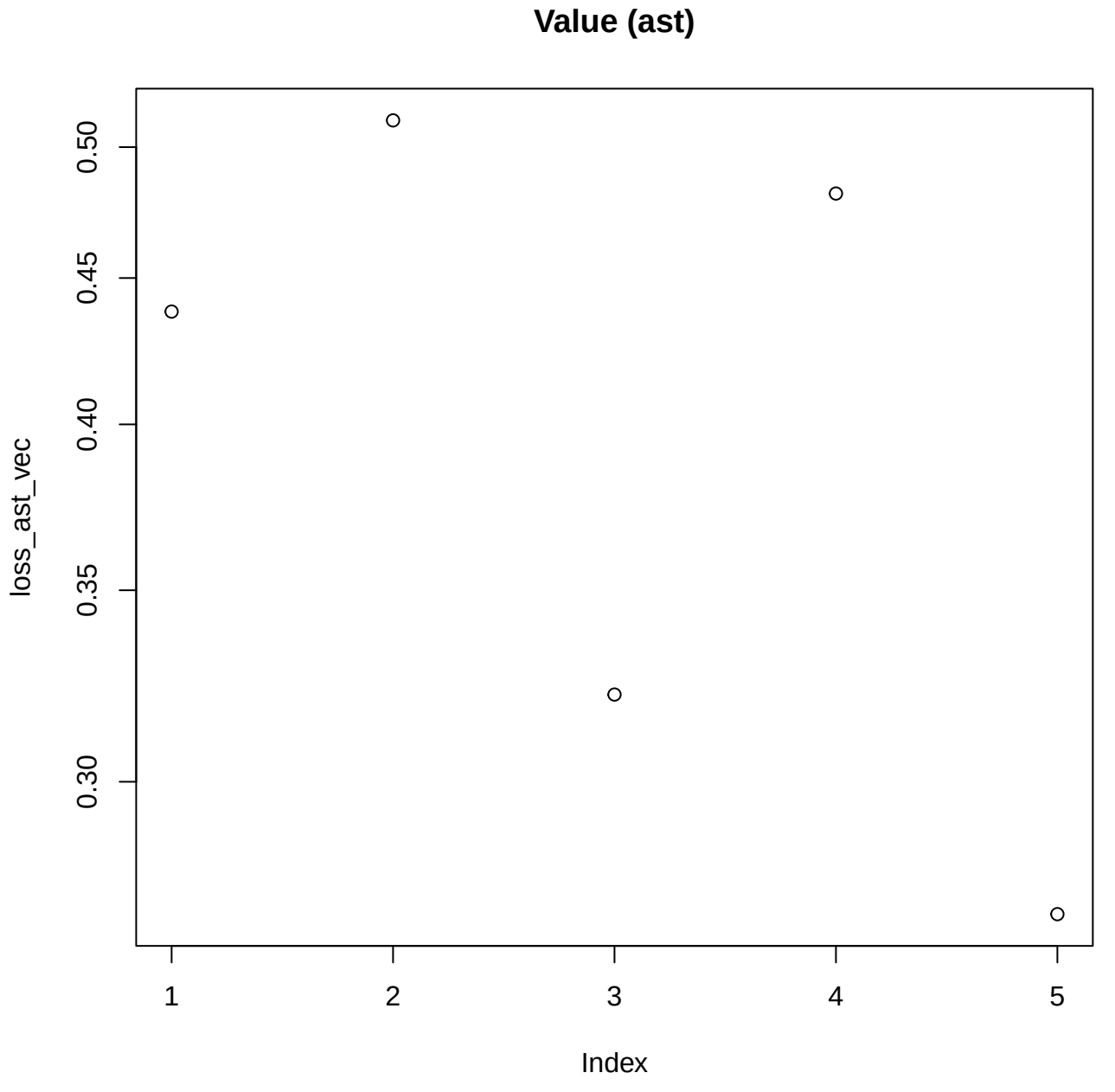
0.07
0.08
0.09
0.10
0.11
0.12
1
2
3
4
5
Index


Value (ast)
loss_ast_vec
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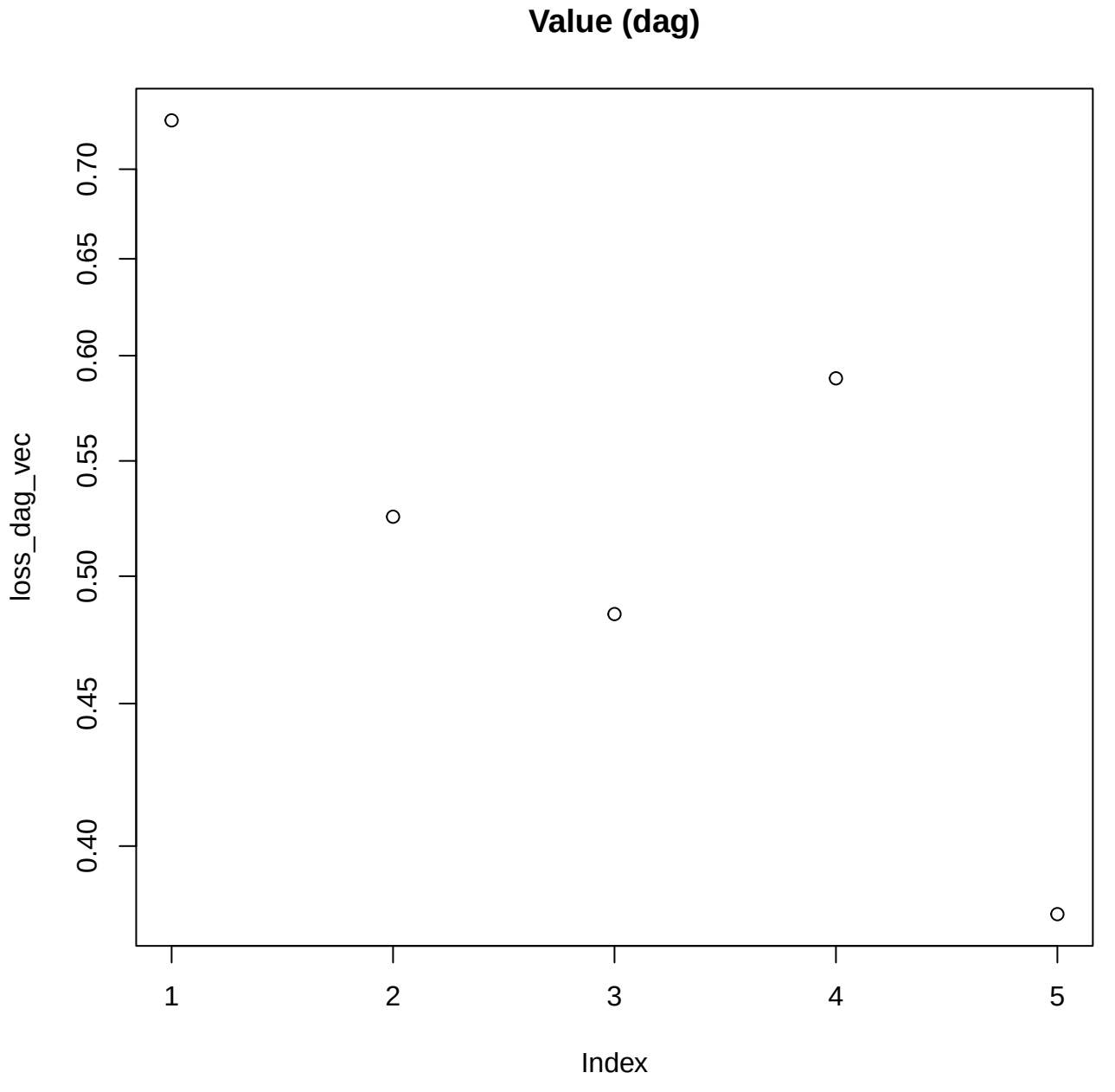
0.35
0.40
0.45
0.50
1
2
3
4
5
Index


Value (dag)
loss_dag_vec
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0.50
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0.60
0.65
0.70
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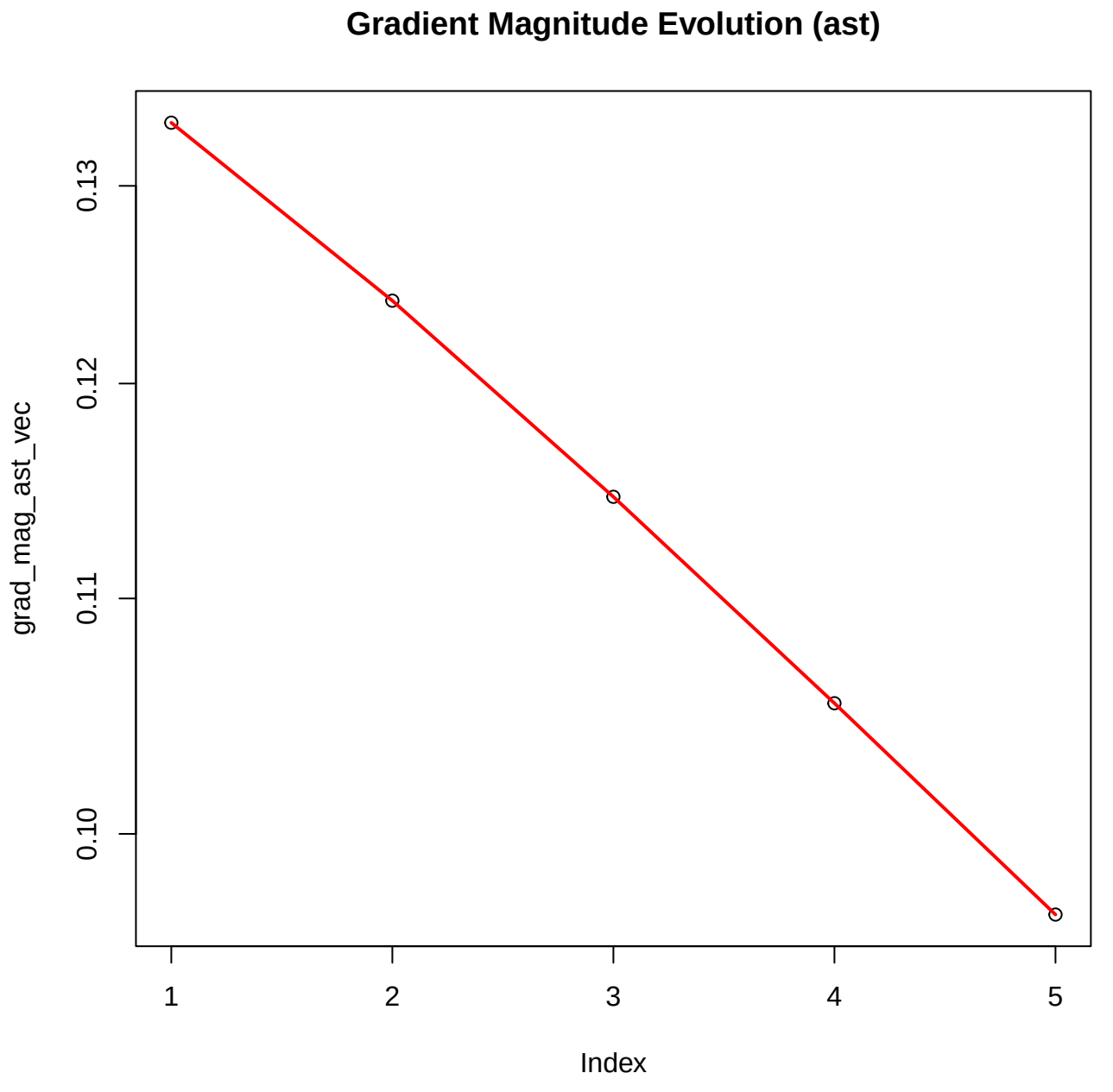
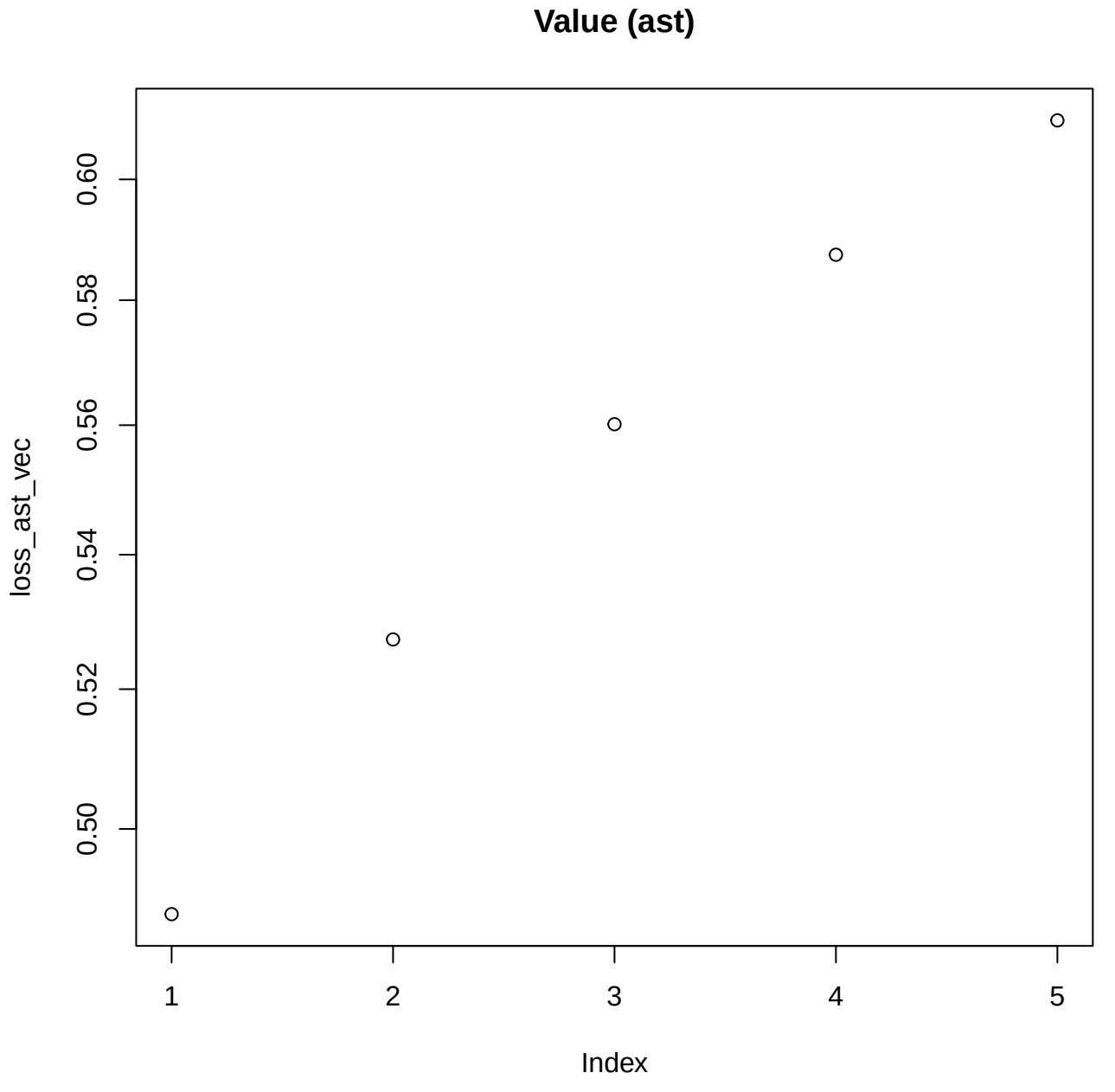
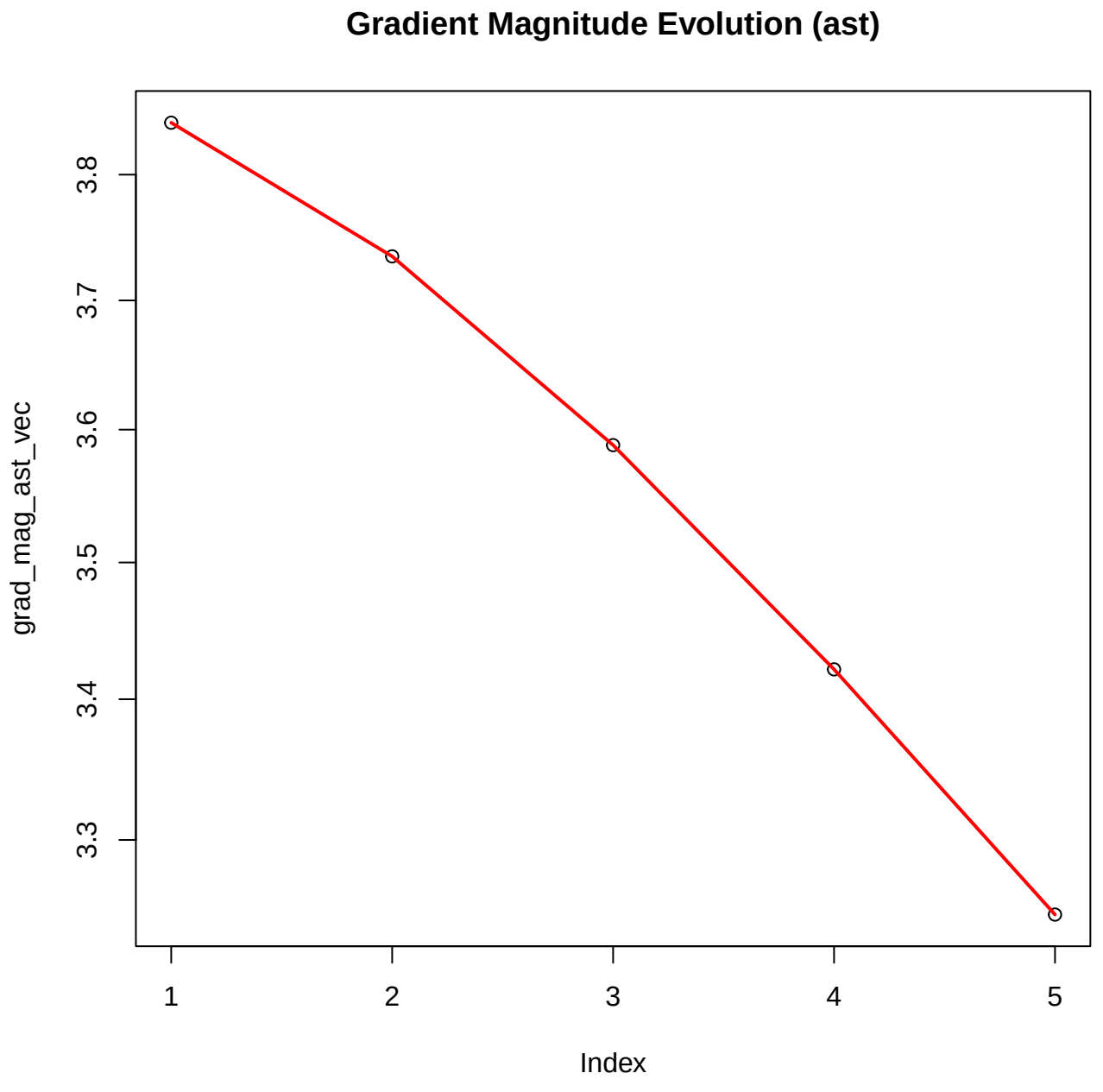
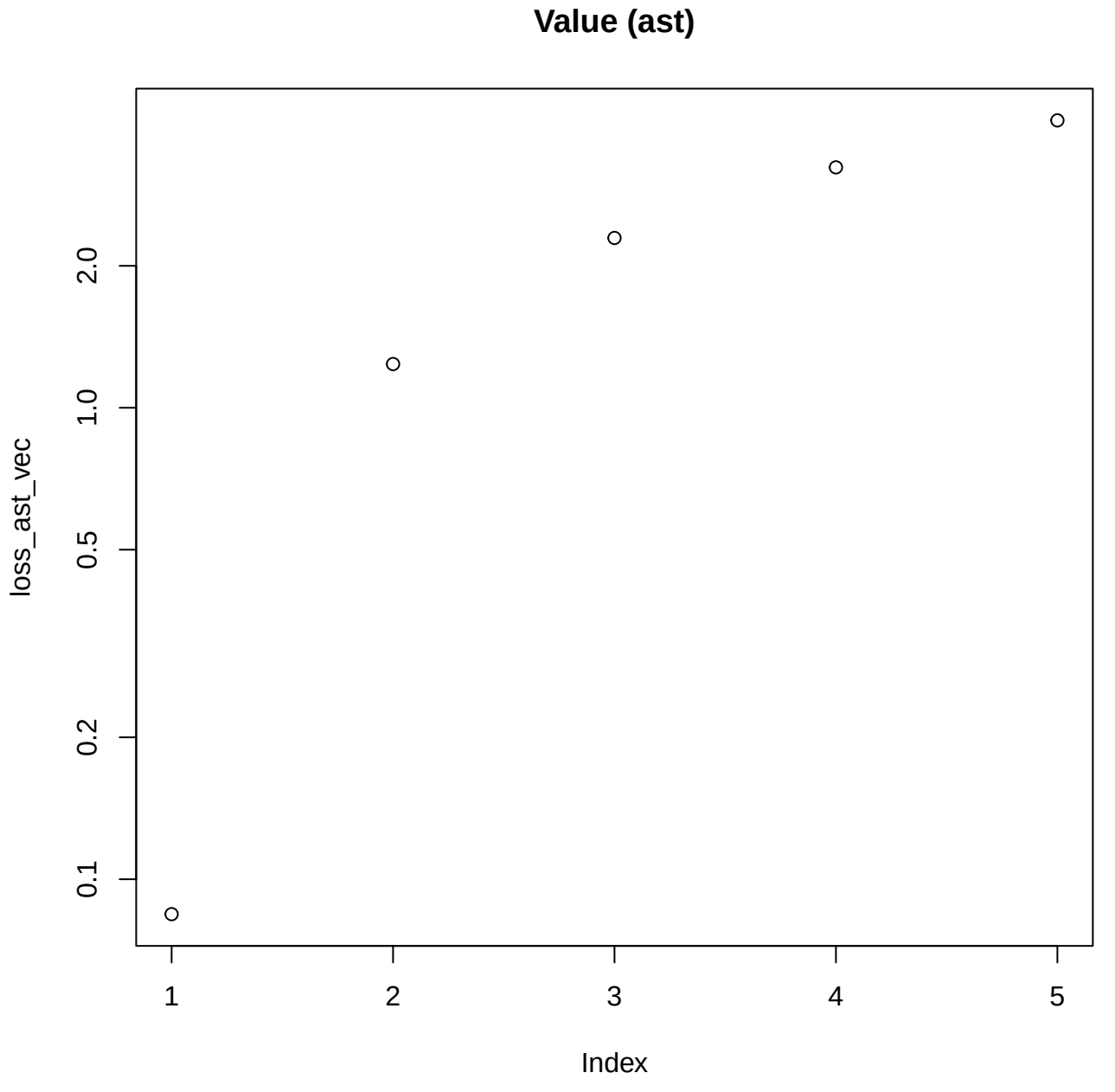
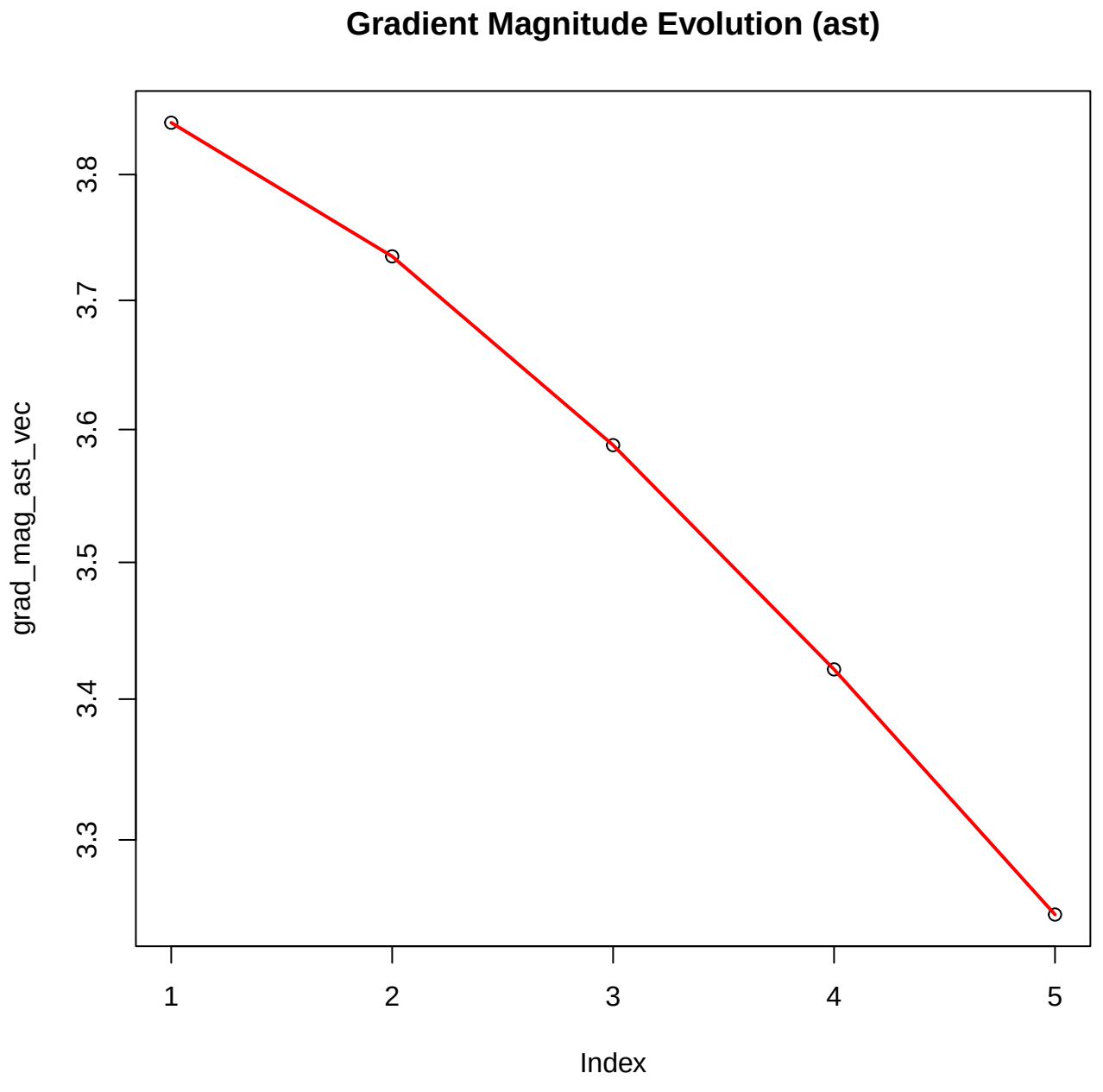
2
3
4
5
Index

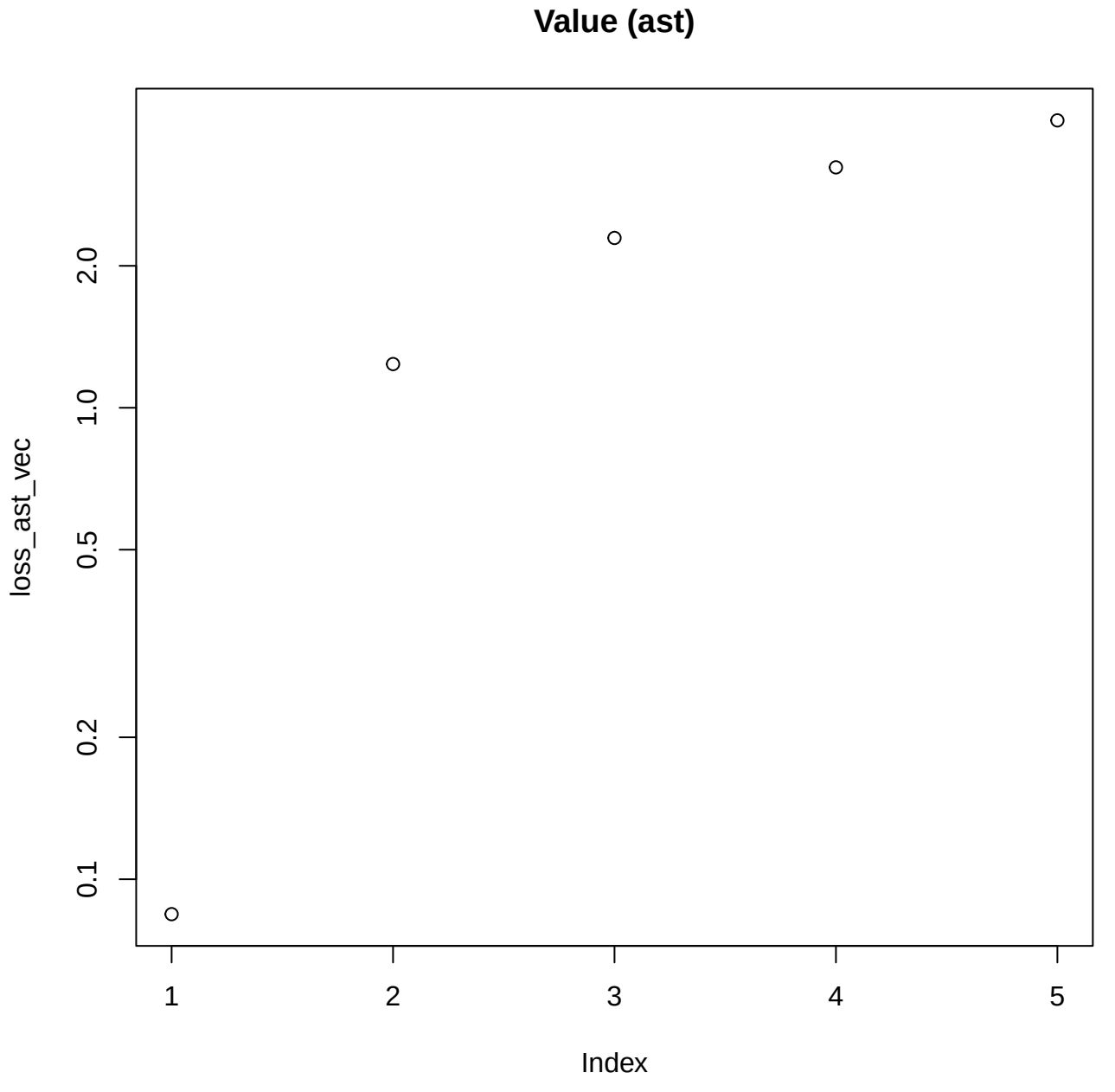

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index


Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
3.3
3.4
3.5
3.6
3.7
3.8
1
2
3
4
5
Index


Value (ast)
loss_ast_vec
0.1
0.2
0.5
1.0
2.0
1
2
3
4
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
3.3
3.4
3.5
3.6
3.7
3.8
1
2
3
4
5
Index


Value (ast)
loss_ast_vec
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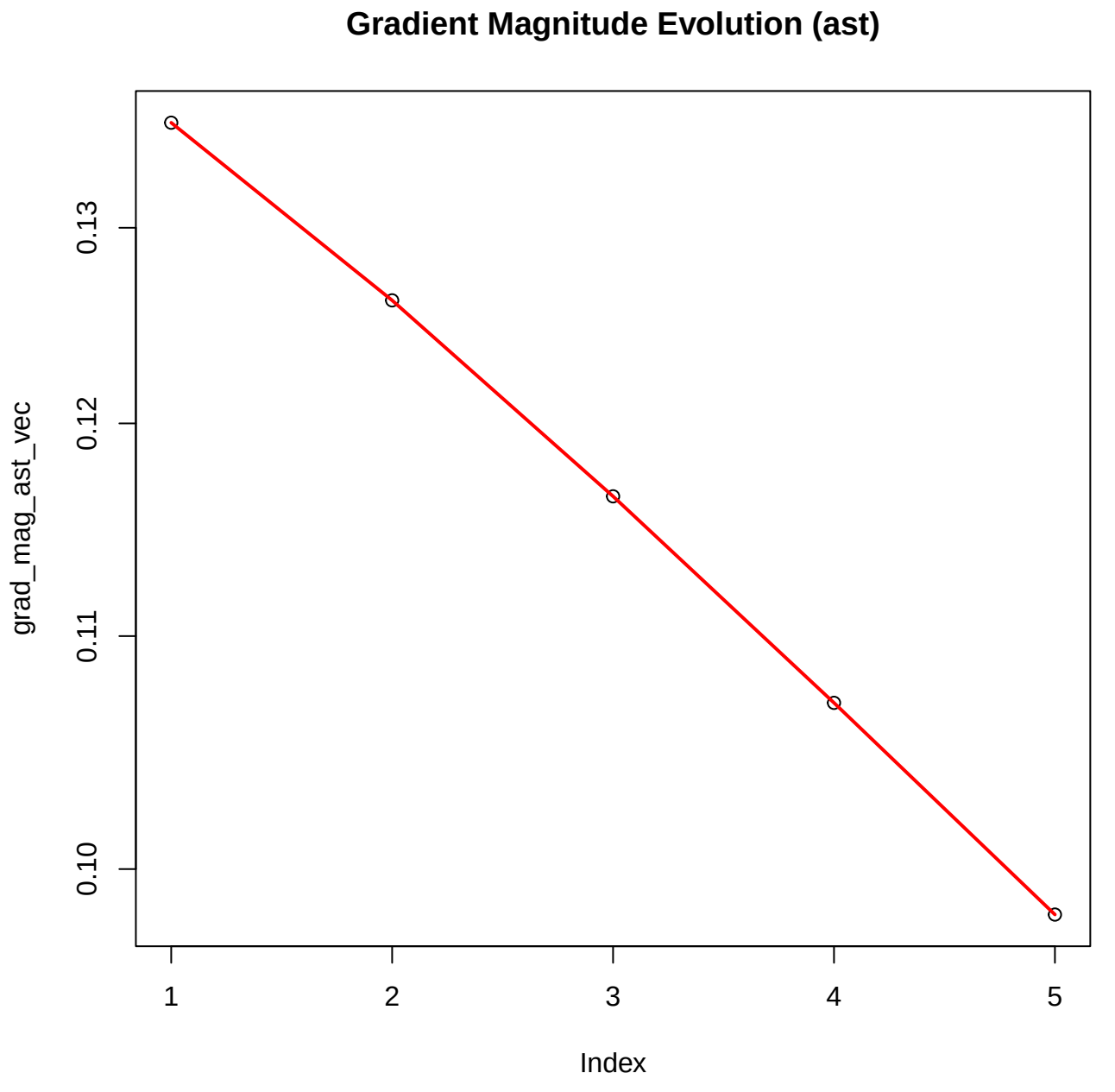
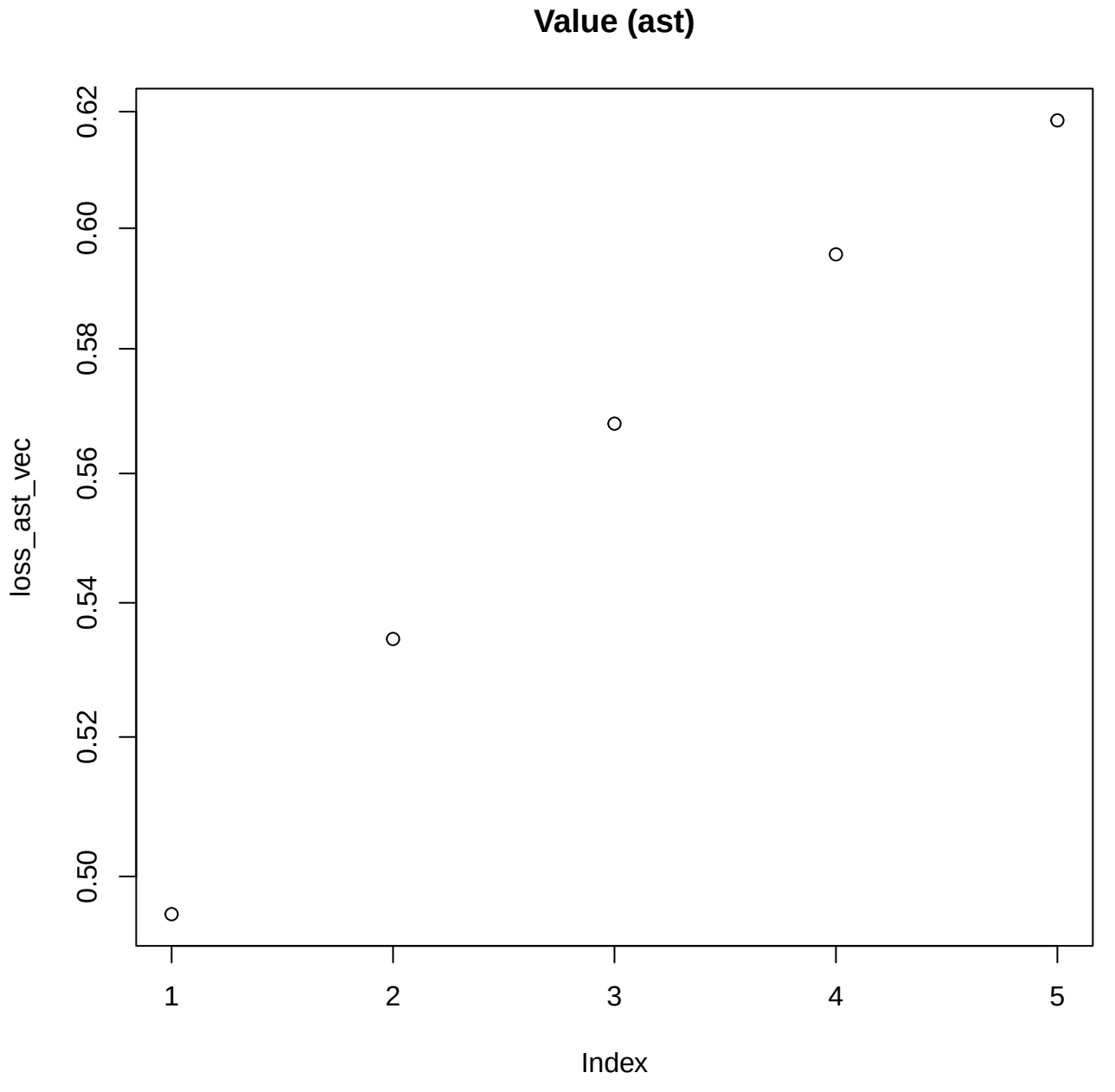
0.2
0.5
1.0
2.0
1
2
3
4
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index


Value (ast)
loss_ast_vec
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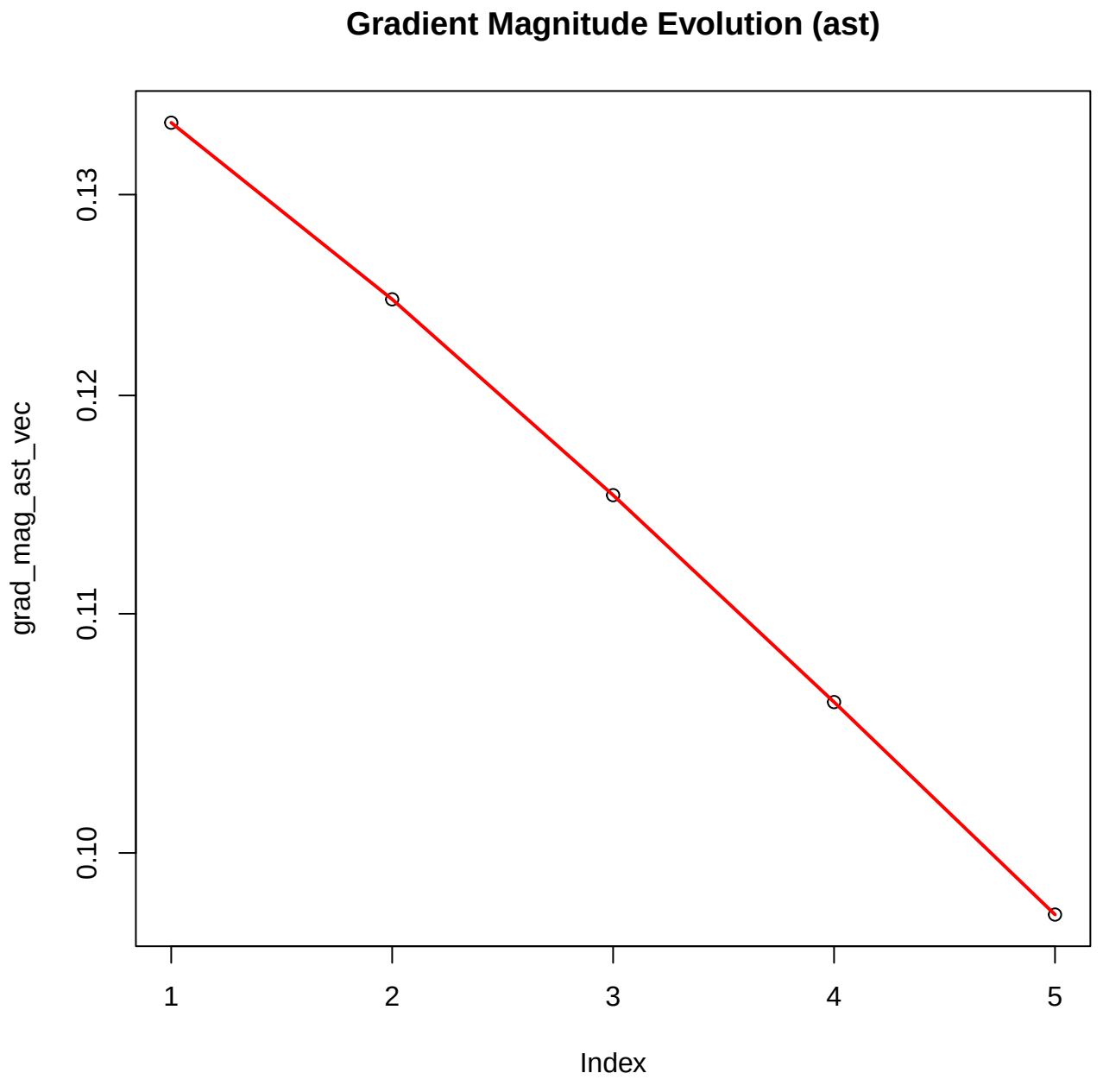
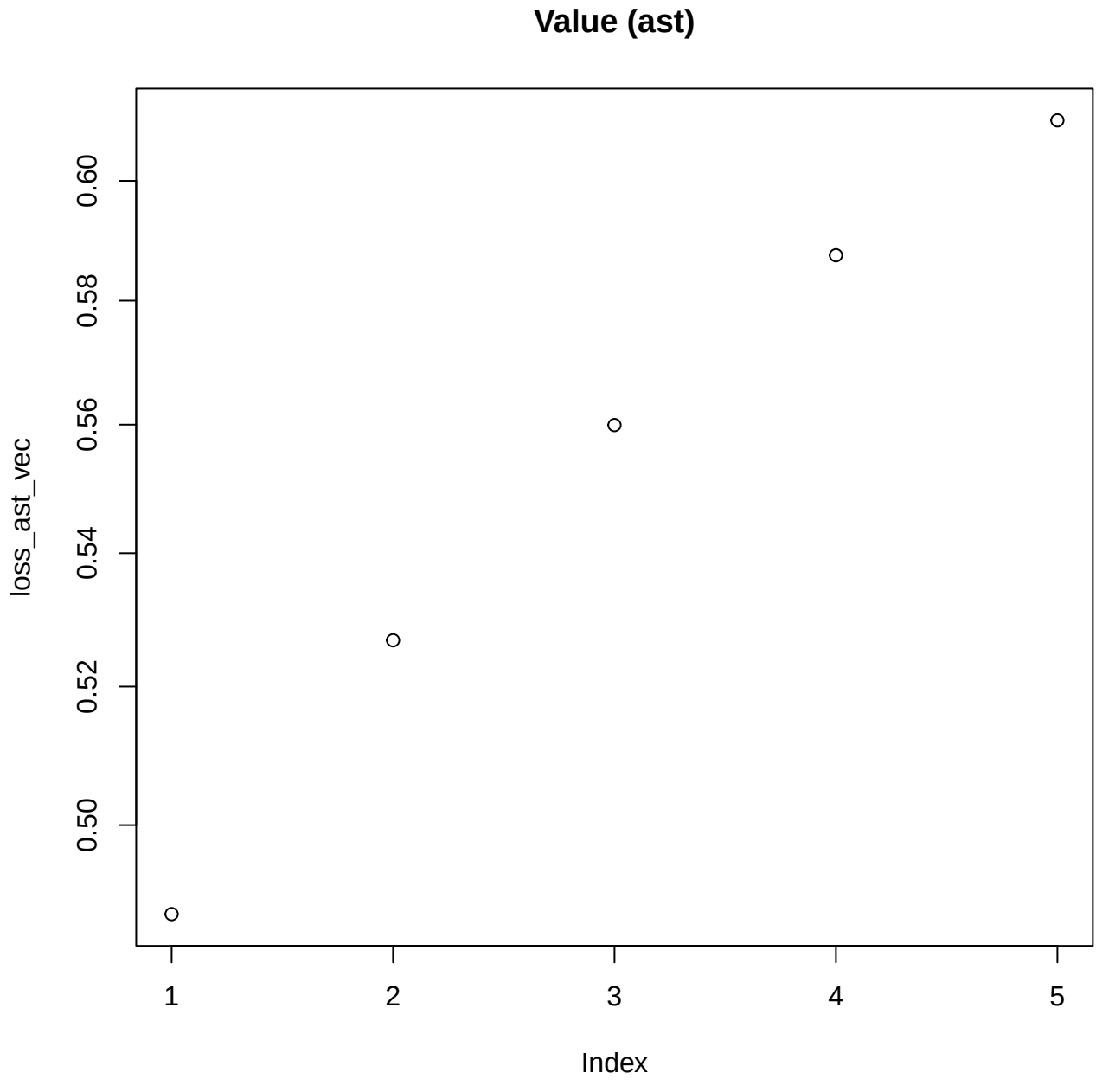
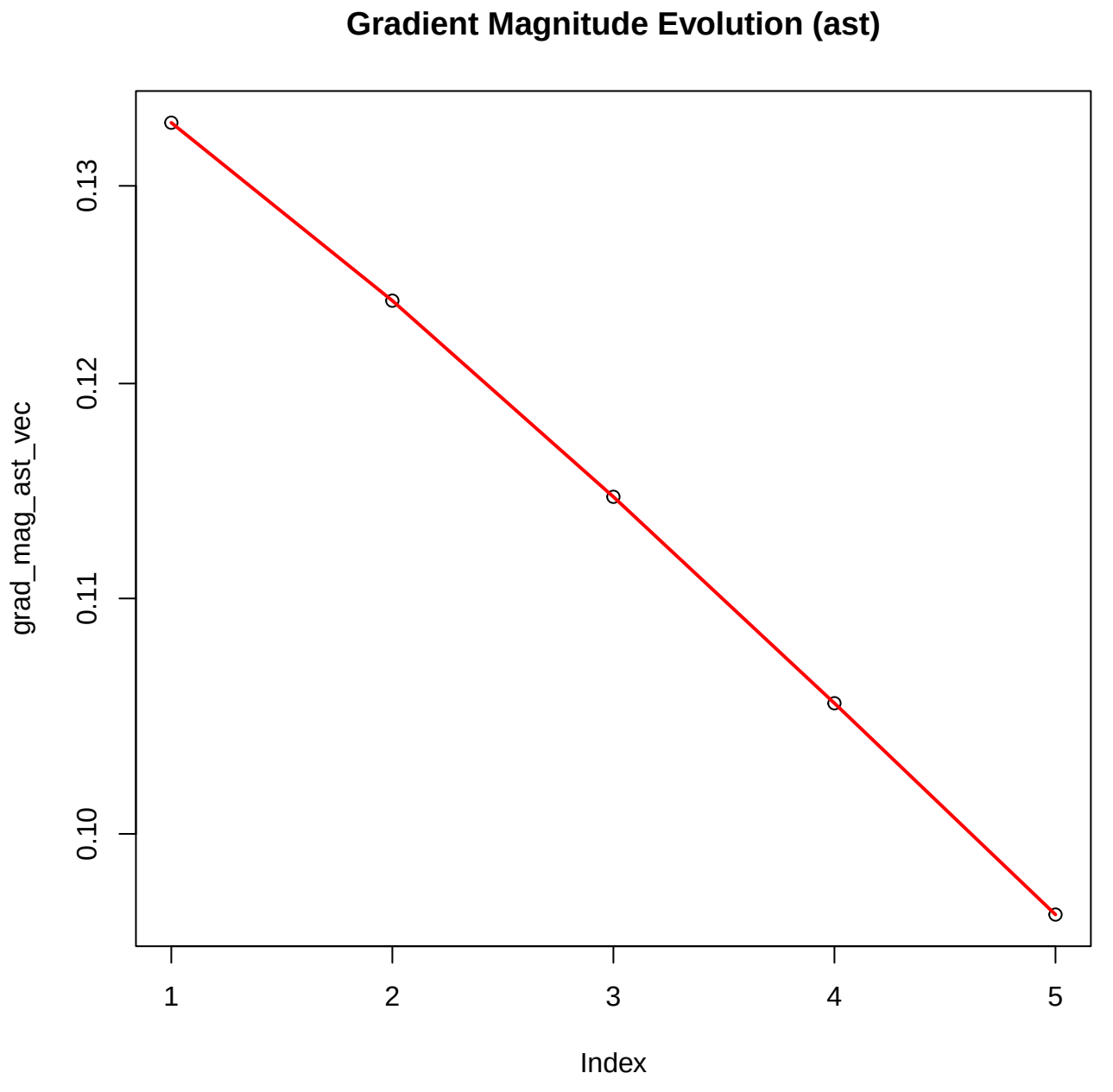
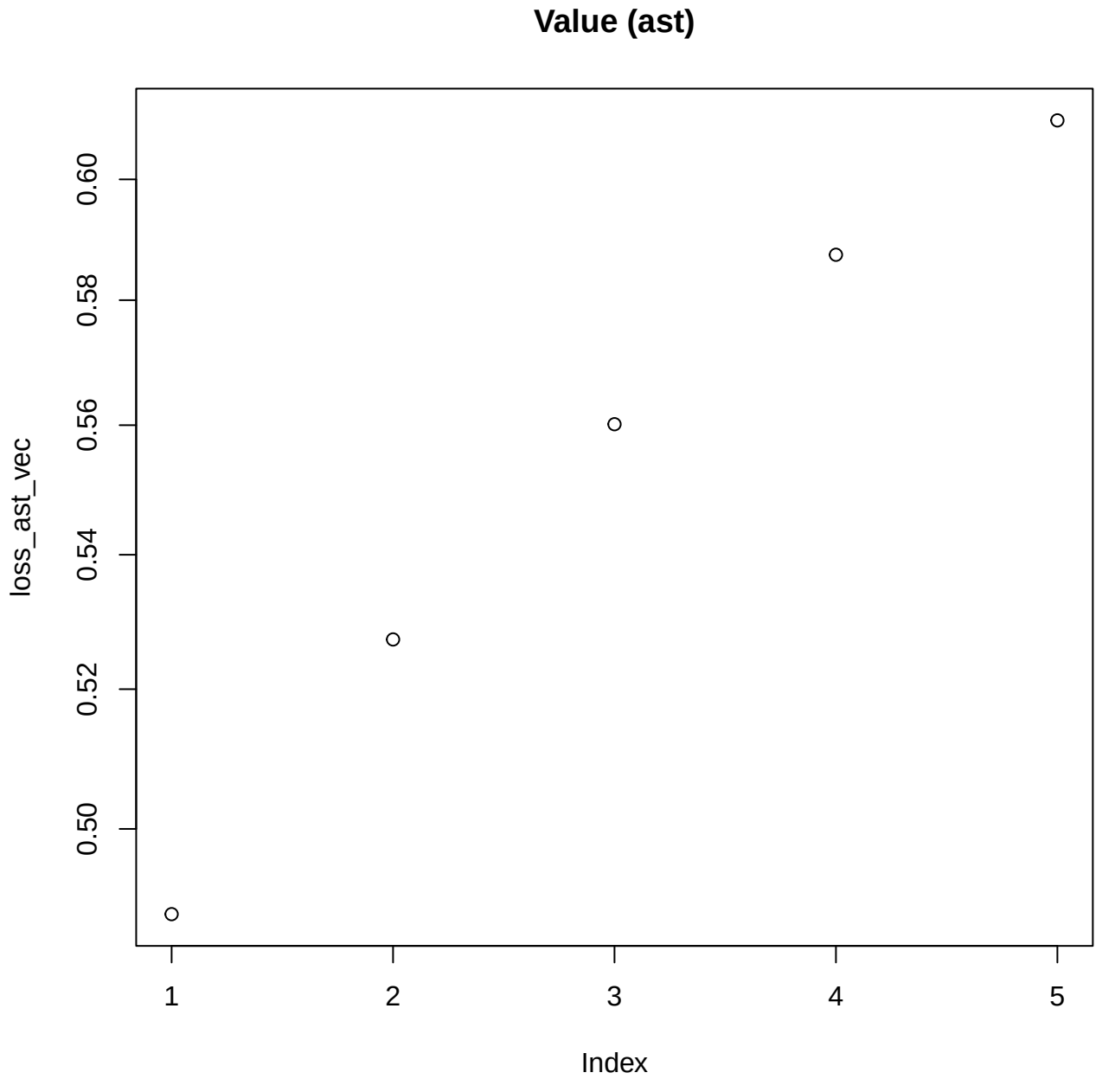
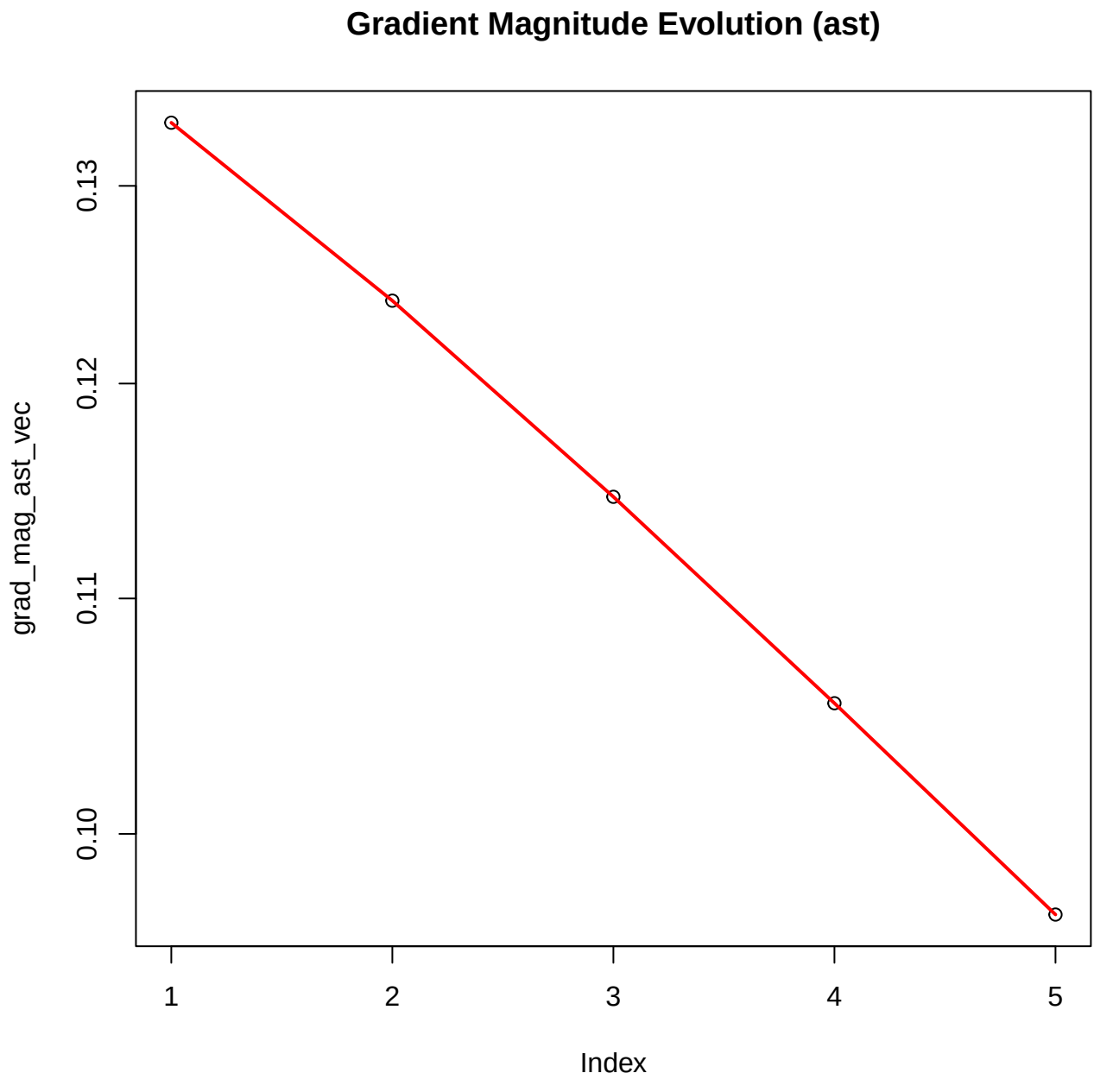
0.52
0.54
0.56
0.58
0.60
0.62
1
2
3
4
5
Index

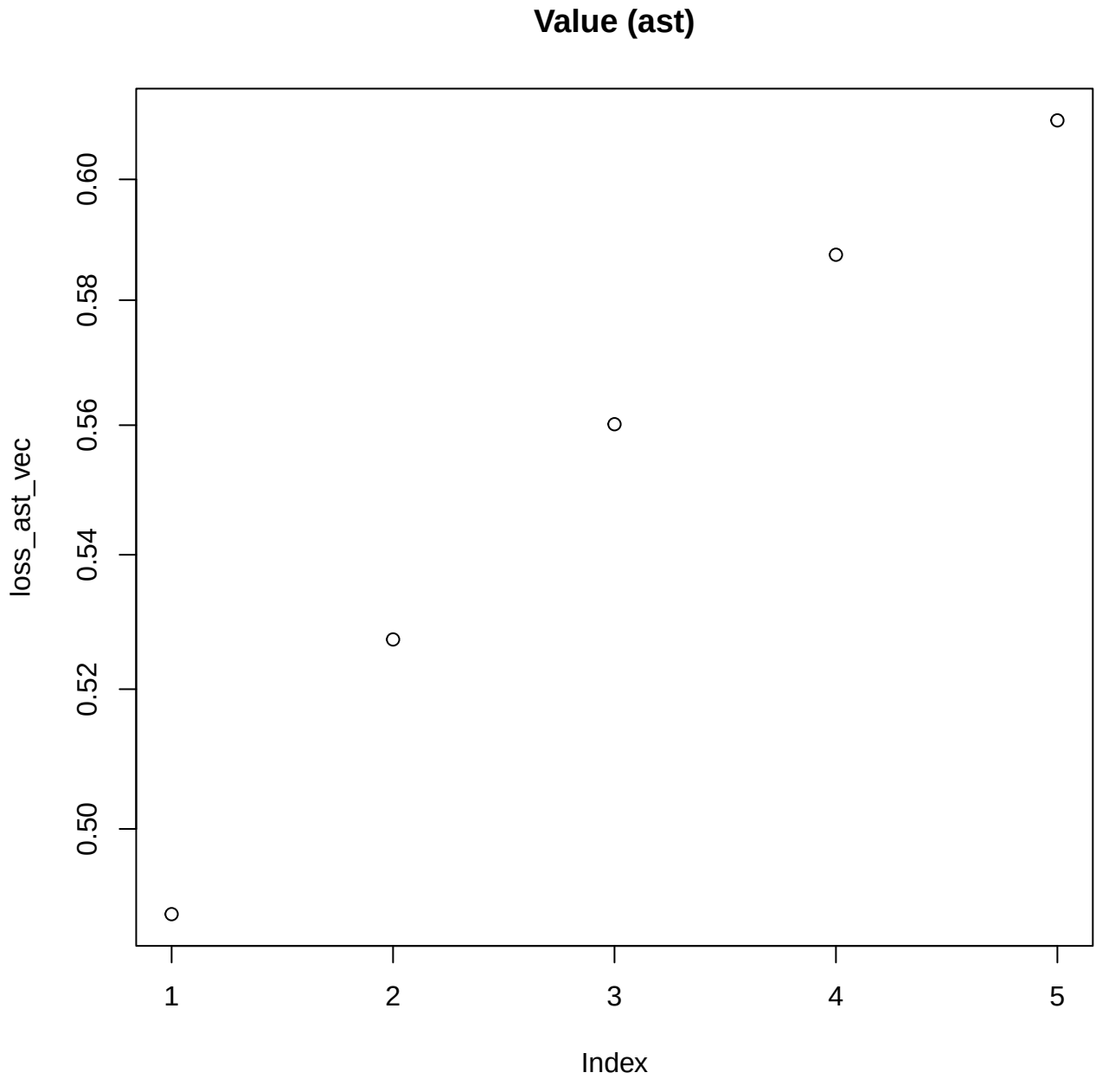

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index

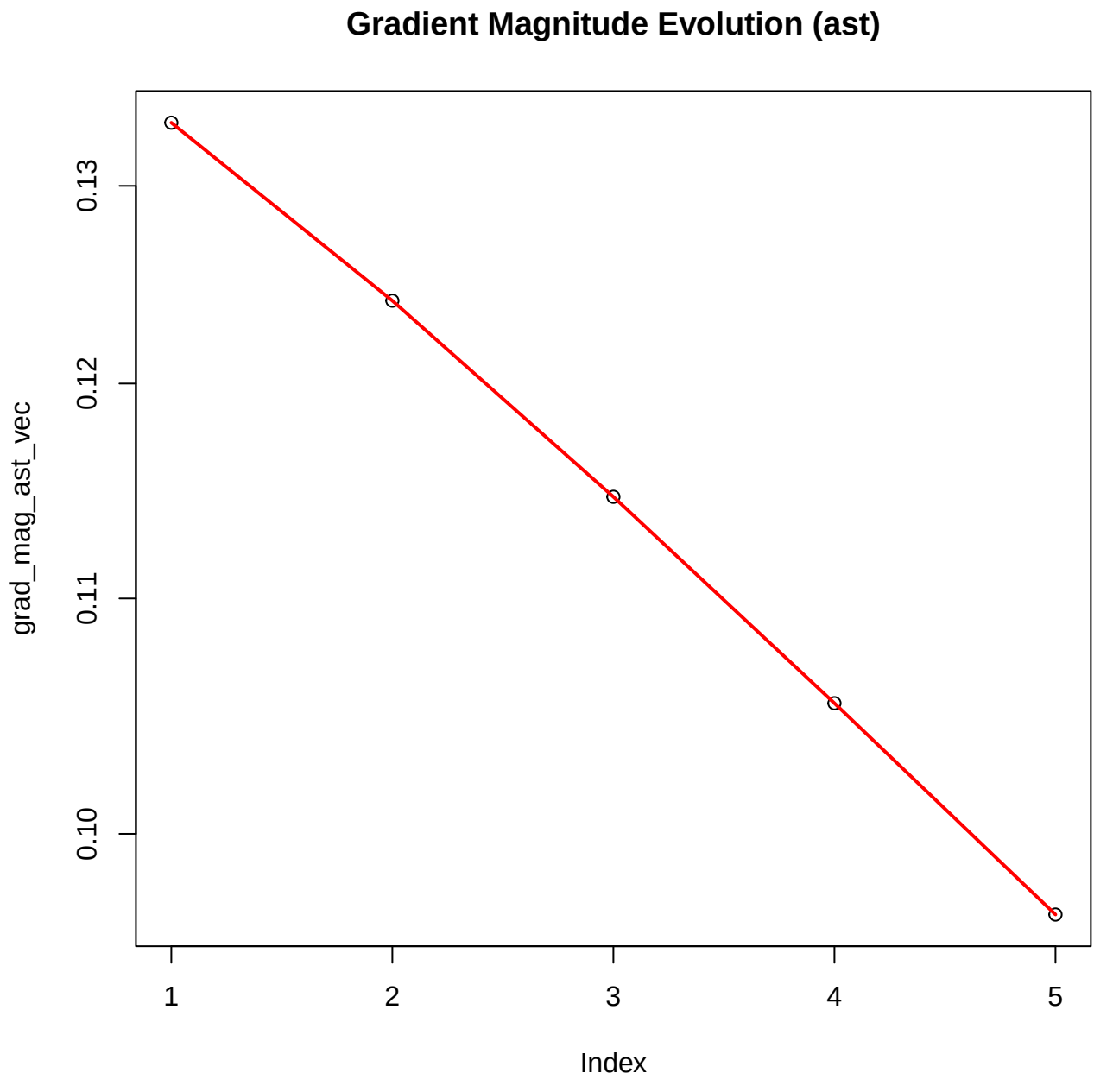

Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index

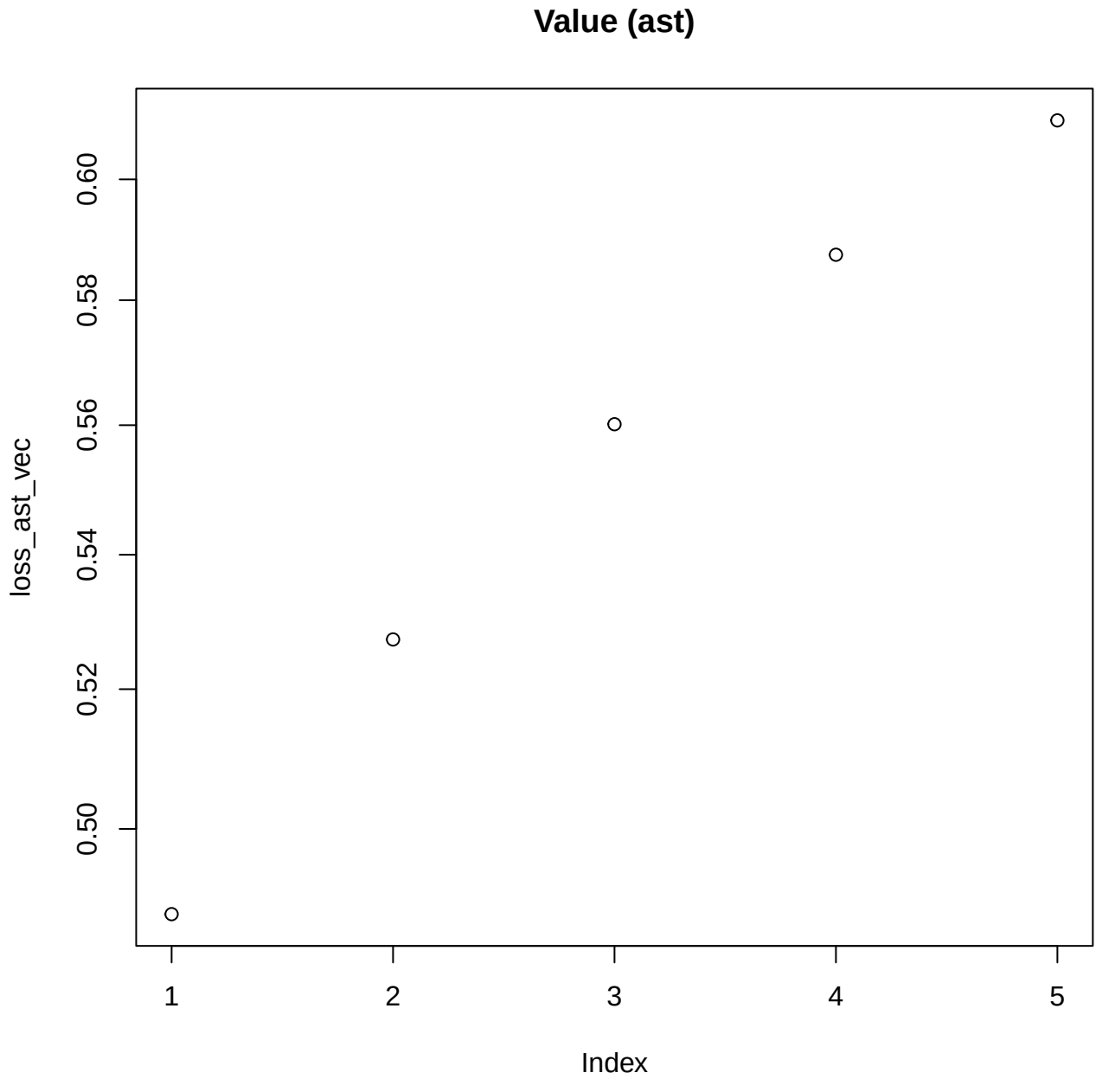

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index

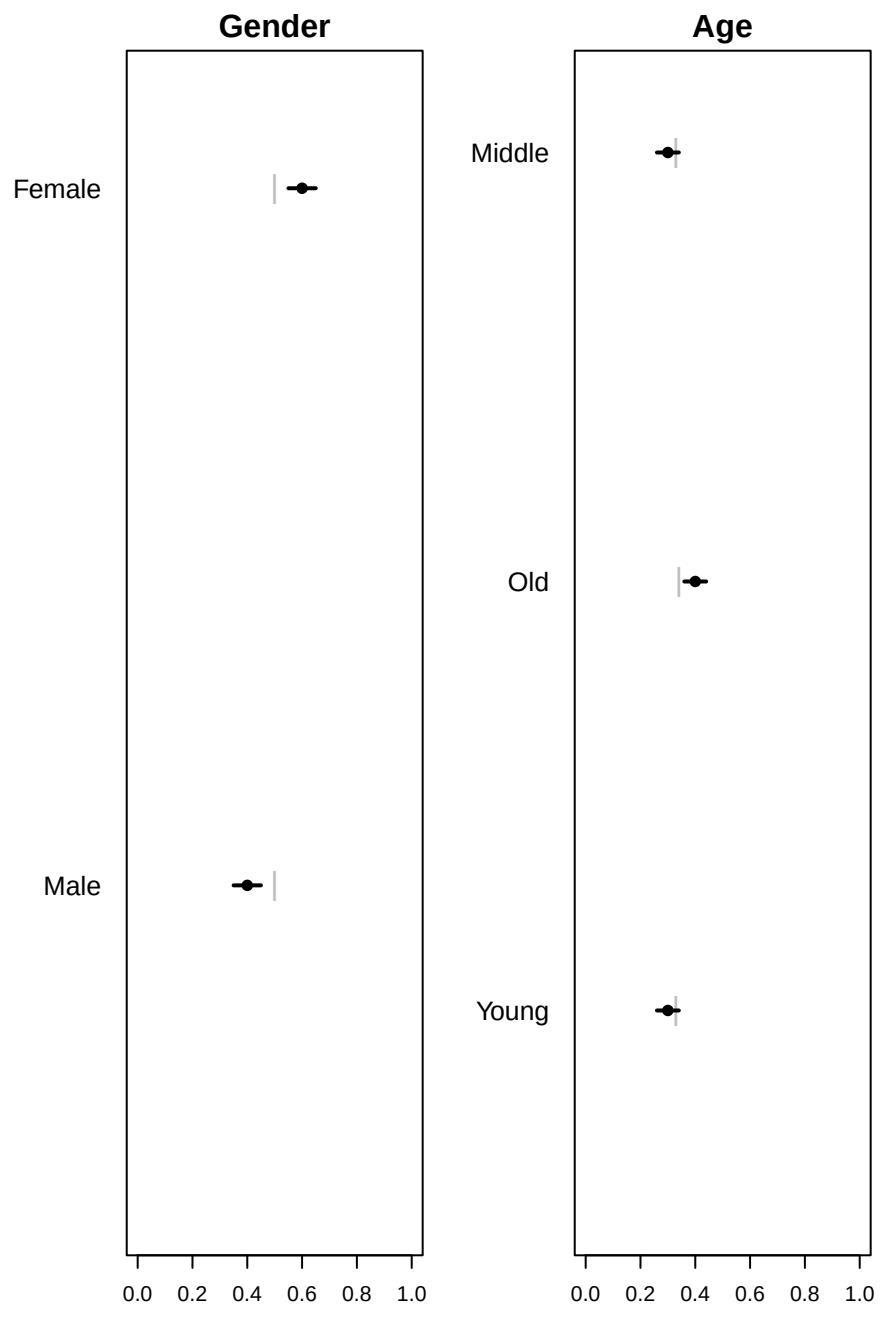

Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index

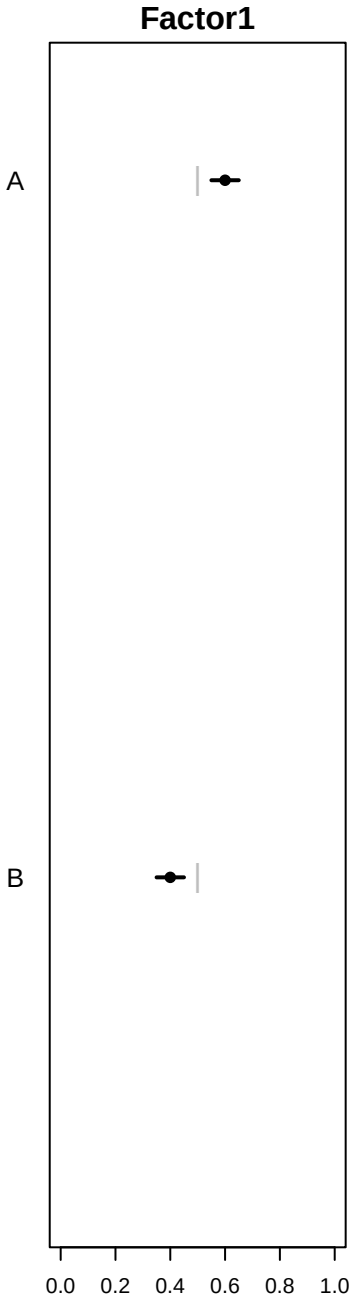

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index

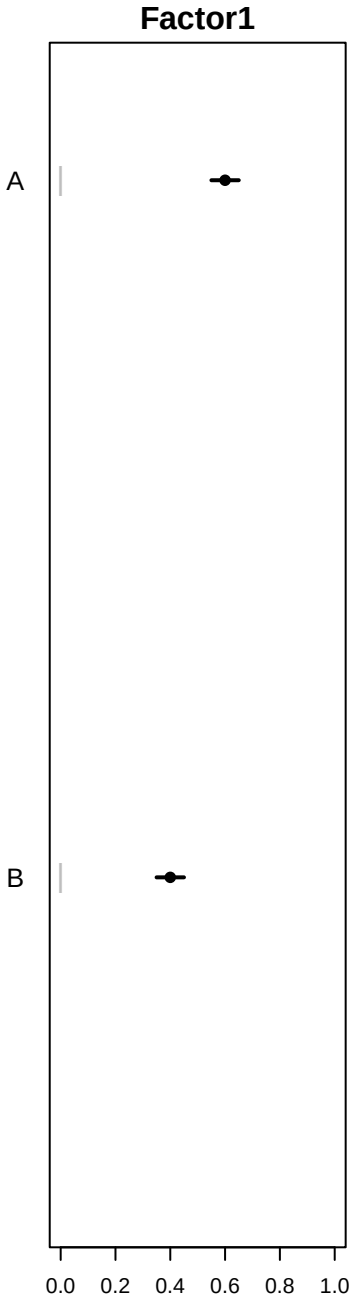

Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index

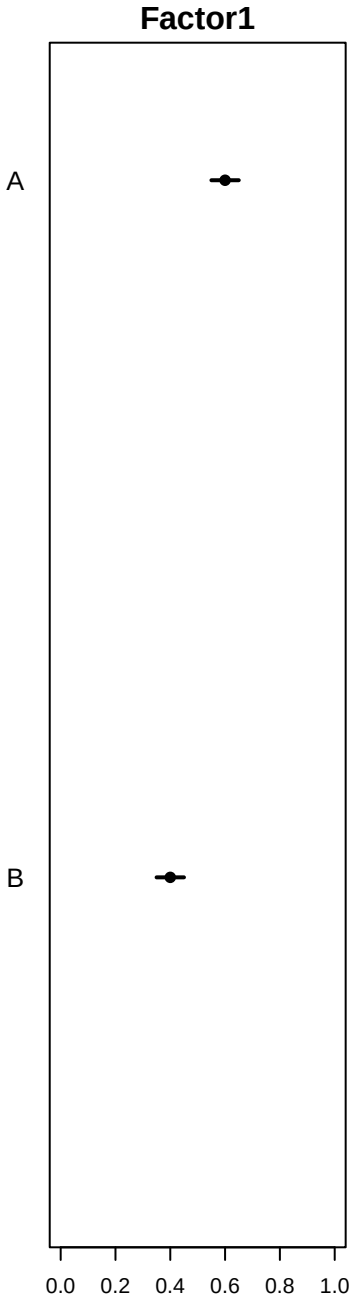

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index

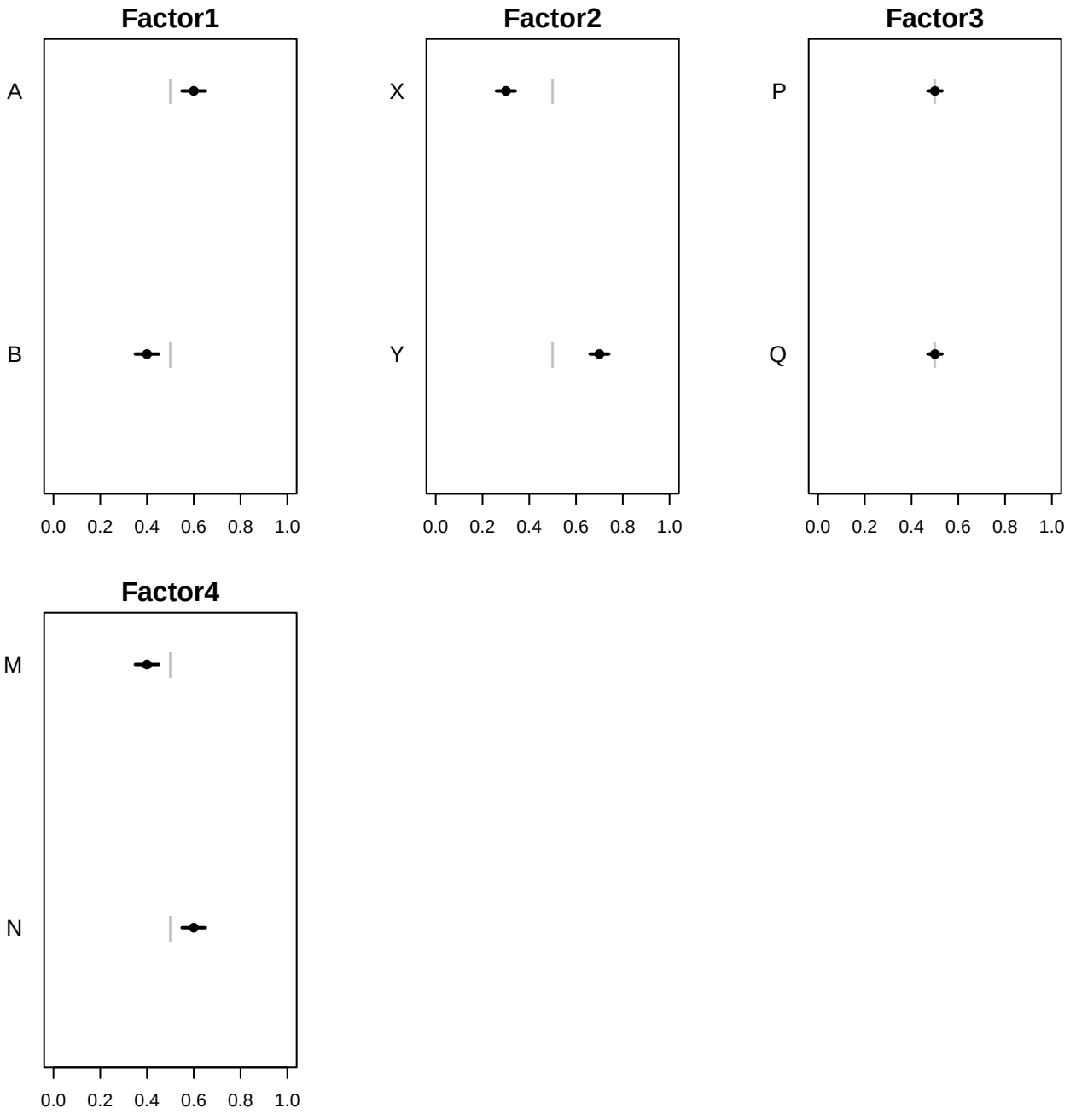

Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index


Gender
Female
Male
0.0
0.2
0.4
0.6
0.8
1.0
Age
Middle
Old
Young
0.0
0.2
0.4
0.6
0.8
1.0


Factor1
A
B
0.0
0.2
0.4
0.6
0.8
1.0


Factor1
A
B
0.0
0.2
0.4
0.6
0.8
1.0


Factor1
A
B
0.0
0.2
0.4
0.6
0.8
1.0


Factor1
A
B
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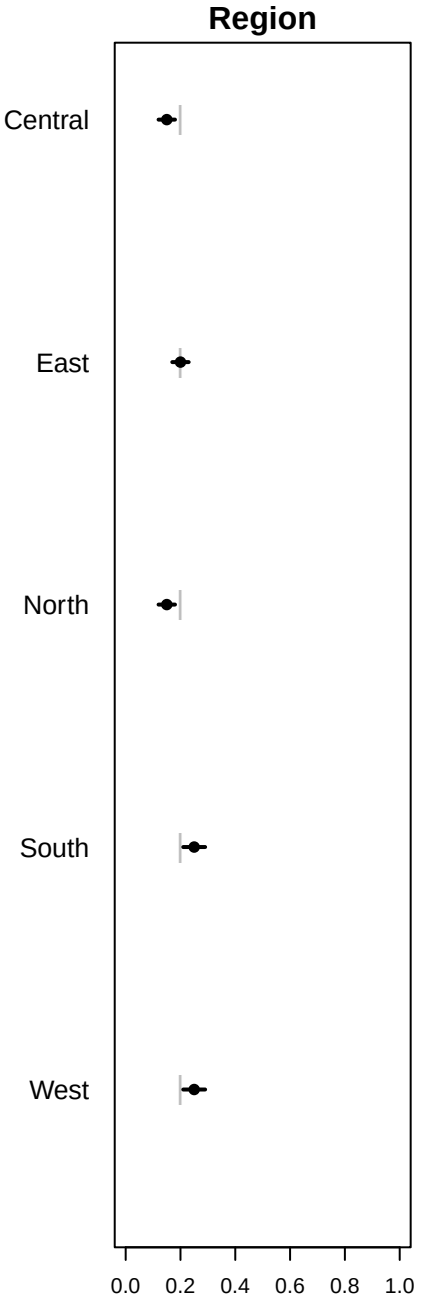
0.2
0.4
0.6
0.8
1.0
Factor2
X
Y
0.0
0.2
0.4
0.6
0.8
1.0
Factor3
P
Q
0.0
0.2
0.4
0.6
0.8
1.0
Factor4
M
N
0.0
0.2
0.4
0.6
0.8
1.0


Region
Central
East
North
South
West
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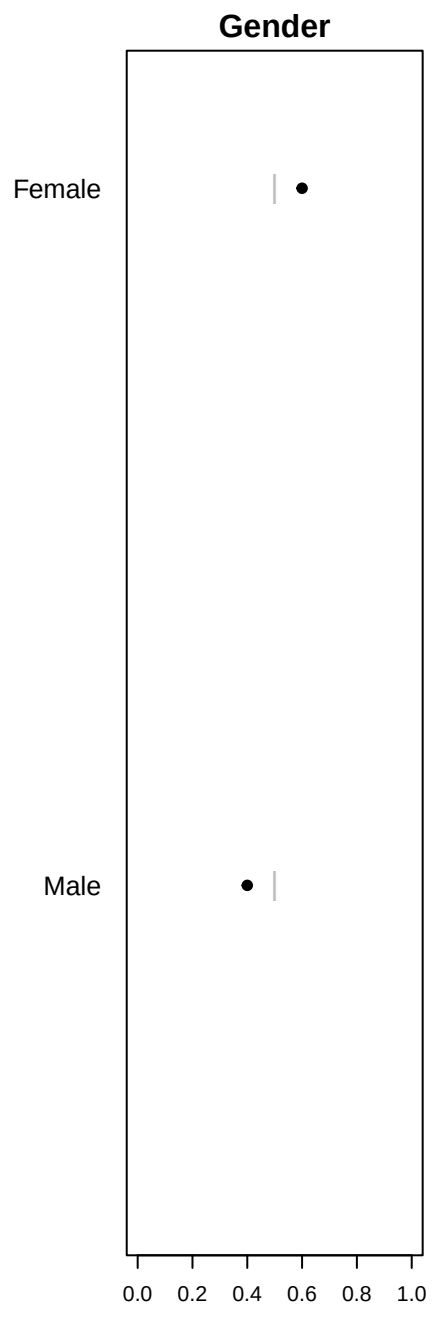
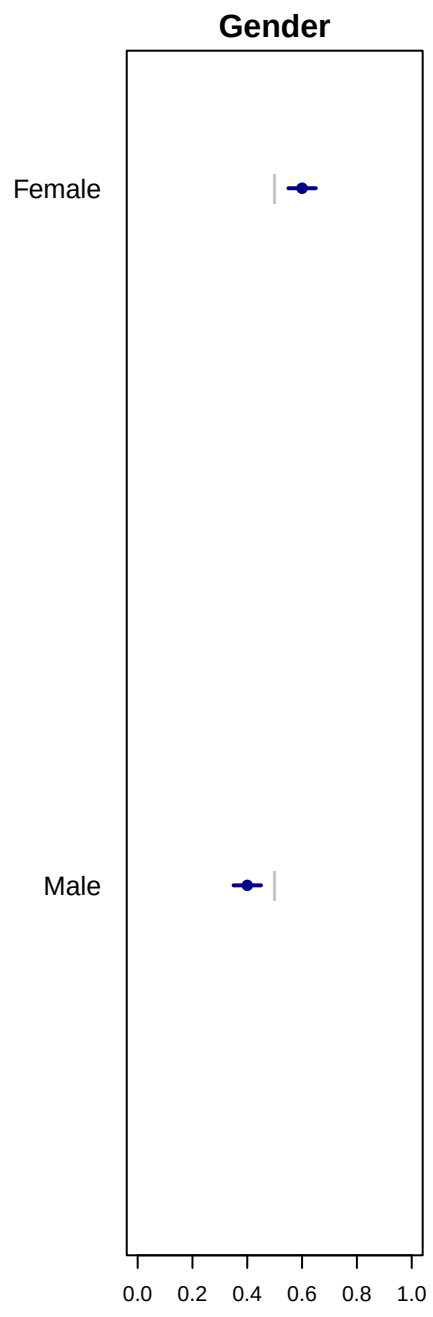
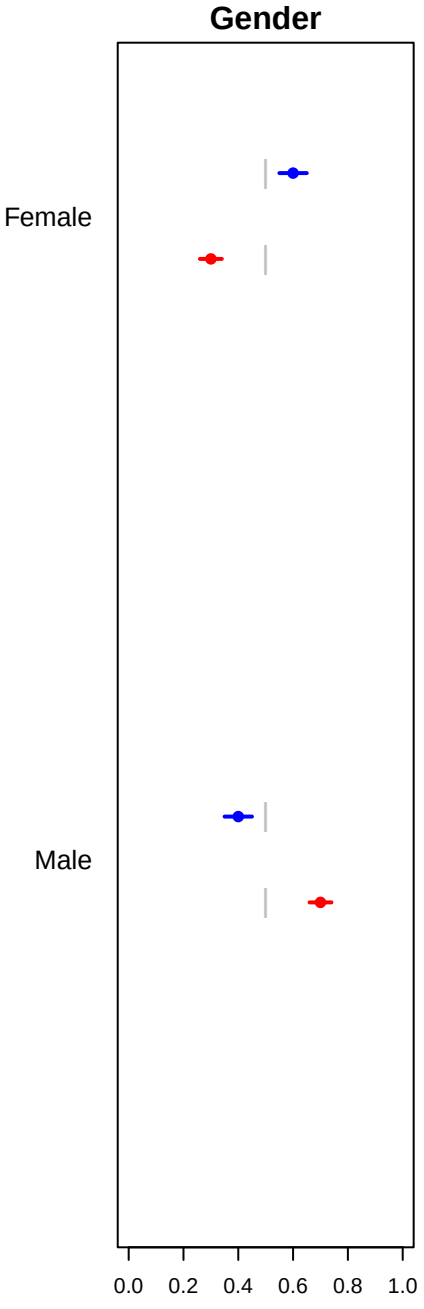
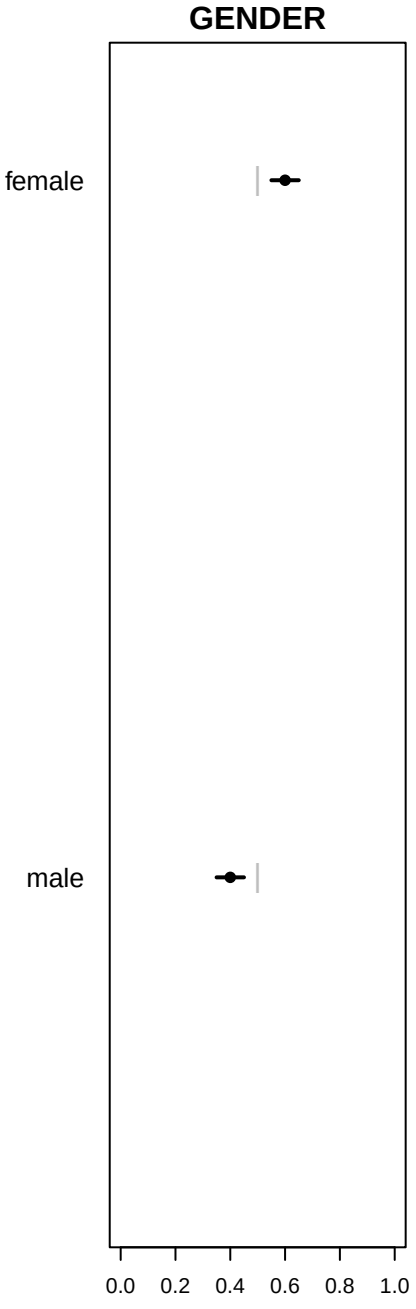
0.2
0.4
0.6
0.8
1.0


Gender
Female
Male
0.0
0.2
0.4
0.6
0.8
1.0


Gender
Female
Male
0.0
0.2
0.4
0.6
0.8
1.0


Gender
Female
Male
0.0
0.2
0.4
0.6
0.8
1.0


GENDER
female
male
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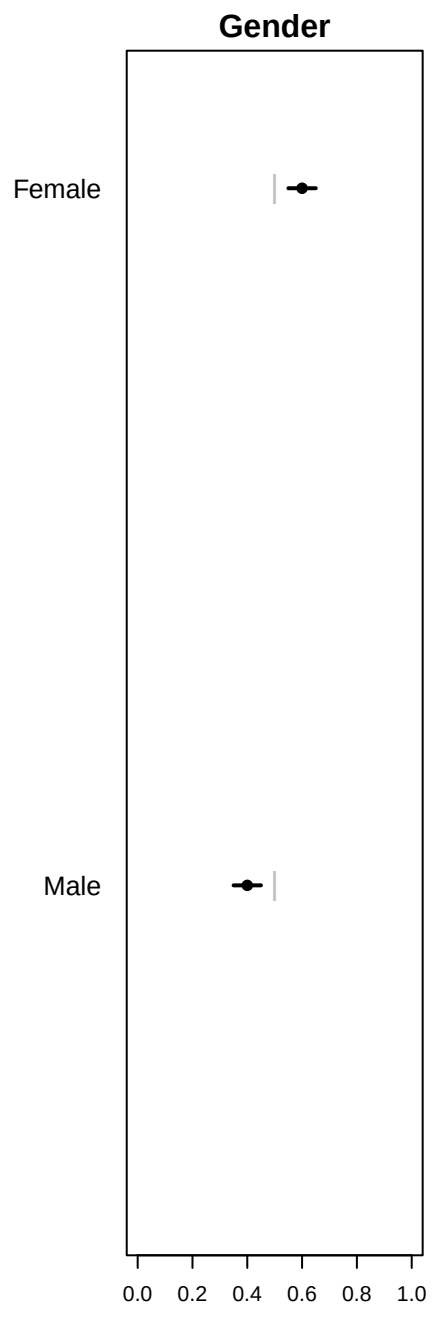
0.2
0.4
0.6
0.8
1.0


Gender
Female
Male
0.0
0.2
0.4
0.6
0.8
1.0